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LARGE-SCALE RESPONSE THEORY FOR
GAPLESS LATTICE FERMION SYSTEMS
IN LOW DIMENSIONS

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ਸਬਰੁ ਏਹੁ ਸੁਆਉ ਜੇ ਤੂੰ ਬੰਦਾ ਦਿਤੁ ਕਰਹਿ॥
ਵਧਿ ਥੀਵਹਿ ਦਰਿਆਉ ਟੁਟ ਨ ਥੀਵਹਿ ਵਾਹੜਾ॥੧੧੭॥

صبر ایہو سواو جے توں بندا دڑھ کریں
ودھ تھیوے دریا وٹ نہ تھیوے واہڑا

*Let patience be your purpose, implant this into your being
Thus, you shall grow into a great river, break you will not into a tiny stream.*

— Punjabi couplet, attributed to Farid Ganjshakar

Abstract

This thesis develops a rigorous large-scale response theory for gapless lattice Fermi systems in low dimensions, focusing on the mathematical validity of linear response – the Kubo formula – for transport in gapless systems, where the absence of a spectral gap prevents the direct use of standard adiabatic theorems.

After reviewing fundamental results on many-body lattice fermions, we first analyse the transport properties of non-interacting, gapless, one-dimensional quantum systems and of the edge modes of two-dimensional topological insulators. We prove the validity of the Kubo formula, in the zero-temperature and infinite-volume limit, for perturbations that are weak and slowly varying in space and time. The proof relies on expressing the real-time Duhamel series in imaginary time, which ensures its convergence uniformly in the scaling parameter and system size, at low temperatures. This representation also reveals a key cancellation in the scaling limit, linked to the emergent anomalous chiral symmetry of relativistic one-dimensional fermions. As a consequence, in the combined low-temperature and scaling limit, the linear response provides the full physical response. In particular, the method yields a dynamical proof of the quantisation of edge conductance in two-dimensional quantum Hall systems.

The analysis is then extended to weakly interacting, gapless one-dimensional fermionic systems, for which we obtain explicit expressions for the response functions in terms of renormalised Fermi velocities. Here, sharp estimates for interacting Euclidean correlation functions are established via rigorous renormalisation group methods. The asymptotic exactness of linear response persists due to a cancellation in the scaling limit of the correlations – reminiscent of bosonisation – derived rigorously from emergent chiral Ward identities.

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Chapter 1

Introduction

Conventions. In this work, we shall always assume to have chosen units such that the electric charge, the reduced Planck's constant and Boltzmann's constant are dimensionless and equal to 1, *i.e.*

$$e^2 = \hbar = k_B = 1 .$$

Reading map. The chapters are organised in chronological order. Alternatively, after reading Chapters 1 and 2, the reader may go through Chapters 3 and 4 independently of each other.

1.1 Motivations and literature overview

Most measurements in condensed matter physics probe how a many-body system reacts to a weak external perturbation: a small electric field produces a current, a magnetic field yields a (local) magnetisation, a weak temperature gradient provokes a heat flow, and so on. In this weak regime, the change in an intensive physical observable is often linear at leading order in the perturbation.

For many-body quantum systems, linear response theory provides a systematic bridge between a *microscopic*, equilibrium description of the system, seen as a collection of electrons and ions, and the *macroscopic* transport coefficients reported by experiments: conductivities, electric and magnetic susceptibilities, specific heats, and beyond. Therefore, linear response approximation is a widely used tool in the *dynamical* derivation of transport coefficients in condensed matter physics, allowing one to determine the response of systems initially at equilibrium after the introduction of a weak and time-dependent perturbation.

Let us informally describe the setting. For $d = 1, 2, 3$, we consider d -dimensional systems in the tight-binding approximation, *i.e.* whose microscopic degrees of freedom can be parameterised by fermions moving on a lattice $\Gamma_L = (\mathbb{Z}/L\mathbb{Z})^d$, in the grand-canonical picture. In the following we shall describe the system in Fock space, which will be reviewed in Section 2.1. Let $\mathcal{H}$ be the Fock space Hamiltonian of the system, and let $\rho_{\beta,\mu,L}$ be the associated equilibrium Gibbs state, at inverse temperature β and chemical potential μ :

$$\rho_{\beta,\mu,L} = \frac{e^{-\beta(\mathcal{H}-\mu\mathcal{N})}}{\mathrm{Tr} e^{-\beta(\mathcal{H}-\mu\mathcal{N})}} , \quad (1.1.1)$$

with $\mathcal{N}$ the number operator. We will always suppose that the Hamiltonian $\mathcal{H}$ is local, *i.e.* it is given by the sum of even monomials in the fermionic creation and annihilation operators

that only involve subsets X of Γ_L with diameter bounded uniformly in L . Suppose that, far in the past, the system is exposed to an external, local perturbation $\mathcal{P}$, which is turned on slowly in time; we describe this setting by introducing the time-dependent Hamiltonian:

$$\mathcal{H}(\eta t) = \mathcal{H} + \varepsilon e^{\eta t} \mathcal{P}, \quad t \leq 0. \quad (1.1.2)$$

The parameter $\eta > 0$ is called the adiabatic parameter, and it describes the rate of variation of the perturbation over time, while the parameter ε describes the strength of the perturbation. In what follows, we will be interested in the regime in which β and L are large, and eventually will be taken to infinity. We will always suppose that ε and η are small, uniformly in β, L .

The evolution of the state is defined by the Liouville-von Neumann equation

$$i\partial_t \rho(t) = [\mathcal{H}(\eta t), \rho(t)], \quad \rho(-\infty) = \rho_{\beta, \mu, L}, \quad (1.1.3)$$

and we will be interested in the variation of physical observables computed on the solution of (1.1.3):

$$\text{Tr } \mathcal{O}_X \rho(t) - \text{Tr } \mathcal{O}_X \rho(-\infty), \quad (1.1.4)$$

with $\mathcal{O}_X$ a lattice observable spatially supported on $X \subset \Gamma_L$. In order to understand the effect of the perturbation on (1.1.4), it is natural to try to set up a perturbative expansion for ε small. By Duhamel iteration one has, neglecting higher-order terms in ε :

$$\text{Tr } \mathcal{O}_X \rho(0) - \text{Tr } \mathcal{O}_X \rho(-\infty) = -i\varepsilon \int_{-\infty}^0 dt e^{\eta t} \text{Tr } [\mathcal{O}_X, \tau_t(\mathcal{P})] \rho_{\beta, \mu, L} + \text{higher orders in } \varepsilon \mathcal{P}, \quad (1.1.5)$$

with $\tau_t(\mathcal{A}) := e^{i\mathcal{H}t} \mathcal{A} e^{-i\mathcal{H}t}$. The explicit term on the right-hand side of (1.1.5) is called the Kubo formula [55], and it describes the linear response of the system at time $t = 0$.

The first natural problem to address is about the evaluation of the Kubo formula, for specific models, and for interesting perturbations $\mathcal{P}$. This is the starting point of the theory of the integer quantum Hall effect (IQHE) in $d = 2$, where the Kubo formula is used to describe the variation of the transverse current after introducing a weak external electric field, for gapped fermionic models at zero temperature and subject to a strong, orthogonal magnetic field. Already proving that the Kubo formula gives a finite result as $\eta \rightarrow 0$ is nontrivial: in the case of the IQHE, this is ultimately due to the chemical potential being chosen to lie in a spectral¹ gap, which makes the time integral convergent uniformly in η , thanks to oscillations. In fact, one can prove much more than just the finiteness of the Kubo formula: the transverse conductivity of the system, at zero temperature and for $L \rightarrow \infty$, is quantised (see Fig. 1.1) in integer multiples of $1/2\pi$ (recall that we have chosen units such that $e^2 = \hbar = 1$), provided that the chemical potential μ is chosen in a spectral gap of $\mathcal{H}$. The mathematically rigorous literature on the subject is by now very large, and covers a large class of models, see *e.g.* [1, 47] for mathematical reviews. Recently, the proof of quantisation of the Hall conductivity has been extended to gapped many-body systems, [6, 42, 49], see [68] for a recent review.

¹An open interval $I \subset \mathbb{R}$ is called a spectral gap for an Hamiltonian H if it does not contain any spectrum, namely $I \cap \sigma(H) = \emptyset$, and it separates $\sigma(H)$ in (at least) two connected components. More generally, the finiteness and quantisation of the Hall conductivity can be proved to hold for systems whose chemical potential lies in a mobility gap. An interval $I \subset \mathbb{R}$ is said to be a mobility gap for an Hamiltonian H if it is densely filled by the pure-point spectrum of H , with eigenstates which are exponentially localised in space. This is the relevant setting for systems in presence of strong random disorder.

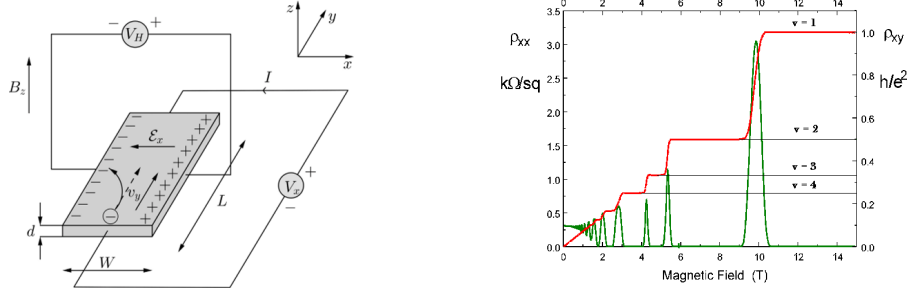


Figure 1.1: Setup for the Quantum Hall Effect: a transverse voltage is observed. Additionally, a plot with the famous plateaus for the transverse resistivity for a Quantum Hall sample.

The IQHE is a particularly successful application of linear response to explain a striking physical phenomenon. More in general, the Kubo formula is widely used to predict the behaviour of semi-metallic and metallic systems as well, where no spectral gap is available. This is the case of graphene, for instance, where the longitudinal conductivity defined from the Kubo formula is equal to $1/4$. This remarkable phenomenon has been observed in [65]. A first explanation starting from a non-interacting model has been given in [71]; more recently, the universality of the longitudinal conductivity of graphene has been proved in [41], starting from weakly interacting fermions on the honeycomb lattice. In recent times, rigorous results, concerning the Kubo formula for a wide class of interacting gapless systems, have been obtained; specifically, one-dimensional metals [19, 20, 26, 60], two-dimensional Hall systems at criticality [39, 43] and edge modes of $2d$ systems [2, 59, 61], three-dimensional Weyl semimetals [44]. See [68] for an overview.

The next natural question is about the validity of the Kubo formula, in the sense of rigorously proving that higher-order terms in (1.1.5) are indeed subleading, uniformly in β and in L . For gapped systems at zero temperature, the validity of the Kubo formula can be explained using the adiabatic theorem. This has been done in [5] for non-interacting systems on the lattice, in [31] for the Landau Hamiltonian and more recently in [7, 64, 72] for interacting lattice models. See [50] for a review and additional references. In the case of the quantum Hall effect, one can actually show that linear response is exact, for non-interacting [54, 57] and also for interacting system [8, 73]: all power-law corrections after the linear response vanish. Recently, the derivation of the Kubo formula for the quantum Hall effect has been extended to the case of $2d$ disordered, non-interacting systems, where the spectral gap is replaced by a mobility gap [29]. A different proof of the validity of the Kubo formula, which also applies to small positive temperatures, has been given in [48], for weakly interacting fermions, using cluster expansion methods. There, the higher-order terms in the Duhamel expansion are controlled using an extension of the argument behind the stability of KMS states [23]; thanks to a suitable complex deformation argument, all terms in the Duhamel expansion can be represented in terms of imaginary time correlations, whose good decay properties for gapped systems can be proven using cluster expansion techniques. This method can also be used to prove the validity of the many-body adiabatic theorem [48], and will play an important role in this work.

It is an open problem to understand the validity of the Kubo formula in a metallic phase starting from a microscopic model. This is of obvious physical relevance, in view of the ubiquitous applications of linear response in cases in which no spectral gap is present.

In the case of a finite system coupled to external leads at thermal equilibrium, the related problem of proving convergence to a non-equilibrium steady state (and proving the validity of the Kubo formula) has been widely studied in the literature. Rigorous results have been proved under suitable spectral assumptions, for non-interacting [3, 4, 66] and also interacting [27, 35, 53, 63] systems.

In this work, we shall address the validity of the Kubo formula for closed, extended systems, in the presence of time-dependent perturbations. In order to prove the validity of the Kubo formula for metallic systems, one has to face two main technical problems: the first is about the control of the time integration, that for gapped systems was ultimately possible thanks to the oscillations introduced by the spectral gap; the second is about the slow space decay of correlations, needed in order to control expressions of the type

$$\sum_{X_1, \dots, X_n} \text{Tr} [[\dots [[\mathcal{O}_X, \tau_{t_1}(\mathcal{P}_{X_1})], \tau_{t_2}(\mathcal{P}_{X_2})], \dots], \tau_{t_n}(\mathcal{P}_{X_n})] \rho_{\beta, \mu, L}, \quad (1.1.6)$$

which appear naturally in the Duhamel expansion of (1.1.5), after representing the perturbation $\mathcal{P}$ as a sum of local terms, $\mathcal{P} = \sum_Y \mathcal{P}_Y$. Due to the absence of a spectral gap, one cannot expect the correlation functions of the system to be integrable in the times t_i , uniformly in β, L .

This thesis endeavours to contribute to the understanding of the validity of the Kubo formula for low-dimensional, gapless systems. Specifically, it summarises the work carried out in the following papers:

- [69] M. Porta and H. P. Singh. Large Scale Response of Gapless 1d and Quasi-1d Systems. [arXiv:2411.04023](#) (2024), to appear in *Annales Henri Poincaré*;
- [70] M. Porta, G. Scola and H. P. Singh. Large scale dynamical response of interacting 1d Fermi systems. [arXiv:2509.08665](#) (2025).

1.2 Response theory of non-interacting systems

In paper [69], or equivalently in Chapter 3 of the present work, we focus on gapless cases relevant for condensed matter applications. Therein, we consider non-interacting Fermi gases in one dimension, and edge modes for two-dimensional topological insulators, that can be viewed as quasi-1d metallic systems.

For the moment, let us focus on one-dimensional systems. We shall denote by Γ_L a one-dimensional lattice with length L and periodic boundary conditions. The Hamiltonian $\mathcal{H}$ takes the form:

$$\mathcal{H} = \sum_{x, y \in \Gamma_L} \sum_{\rho, \rho' \in S_M} a_{x, \rho}^* H_{\rho \rho'}(x; y) a_{y, \rho'}, \quad (1.2.1)$$

where $\rho \in S_M = \{1, \dots, M\}$ labels internal degrees of freedom, $a_{x, \rho}^*, a_{x, \rho}$ are the fermionic creation and annihilation operators, and H is a single-particle Hamiltonian, which we assume to be translation-invariant: $H_{\rho \rho'}(x; y) \equiv H_{\rho \rho'}(x - y)$. This allows us to introduce the Bloch Hamiltonian $\hat{H}(k)$, $k \in \mathbb{T}$, an $M \times M$ Hermitian matrix, whose eigenvalues as a function of k describe the energy bands. We will suppose that the energy bands cross the Fermi level μ at N distinct points, called the Fermi points, k_F^ω , $\omega = 1, \dots, N$. We will assume that the intersections are transversal, which is generically true; this allows us to approximate the dispersion relation in proximity of the Fermi level μ by an asymptotically linear law, with slope v_ω , also called the Fermi velocity.

Let us now describe the class of perturbations $\mathcal{P}$ we will consider. Let $\mu(\cdot)$ be (the restriction to Γ_L of) a smooth and fast-decaying function. We shall consider perturbations of density type:

$$\varepsilon e^{\eta t} \mathcal{P} = \theta e^{\eta t} \sum_{x \in \Gamma_L} \sum_{\rho \in \mathcal{S}_M} \mu(\theta x) a_{x,\rho}^* a_{x,\rho} \quad (1.2.2)$$

where $0 < \theta \ll 1$ and $0 < \eta \ll 1$, independently of β, L . From (1.2.2), we see that the perturbation lives in a spatial region with diameter of order $1/\theta$, and its strength is θ . We will be interested in the following order of limits: first $L \rightarrow \infty$; then $\beta \rightarrow \infty$; and then $\eta, \theta \rightarrow 0$. More precisely, we will consider an Euler-like scaling, in which $\theta = a\eta$, with $a \leq |\log \eta|$. This leaves room to have $\eta \gg \theta$ or $\eta \ll \theta$; the two regimes will give rise to different outcomes (already at a qualitative level), as we will see in Chapter 3.

The Euler scaling is the natural regime in which one expects that the long-time behaviour of a large class of integrable systems is effectively described by hydrodynamic models, see [30, 32] for recent reviews. Proving that this is indeed the case is a hard mathematical problem, which partially motivates our work.

In order to clarify the picture, let us give some heuristics on the large-scale properties of the model. Let us consider the Euclidean two-point function, defined as:

$$\langle \mathbf{T} \gamma_{x_0}(a_{x,\rho}) \gamma_{y_0}(a_{y,\rho'}^*) \rangle_{\beta,L}, \quad (1.2.3)$$

with $0 \leq x_0, y_0 < \beta$, $\gamma_t(A) = e^{t(\mathcal{H} - \mu N)} A e^{-t(\mathcal{H} - \mu N)}$ the imaginary time evolution, and $\mathbf{T}$ the fermionic time-ordering (defined below in (2.2.13)). Let $\underline{x} = (x_0, x)$ and $\underline{y} = (y_0, y)$. Then, for $|\underline{x} - \underline{y}| = O(1/\theta)$ and $\theta \ll 1$, the zero temperature and infinite volume two-point function is given by, up to faster decaying terms:

$$\langle \mathbf{T} \gamma_{x_0}(a_{x,\rho}) \gamma_{y_0}(a_{y,\rho'}^*) \rangle_{\infty} \simeq \sum_{\omega=1}^N e^{ik_F^\omega(x-y)} P_{\rho\rho'}^\omega \int \frac{d\underline{q}}{(2\pi)^2} \frac{e^{i\underline{q} \cdot (\underline{x} - \underline{y})}}{iq_0 + v_\omega q_1} \chi(|\underline{q}|/\theta), \quad (1.2.4)$$

where P^ω is the Bloch projector associated with the energy band crossing the Fermi level at k_F^ω , and $\chi(\cdot)$ is a smooth cutoff function, equal to 1 if the argument is less than 1 and equal to zero if the argument is larger than 2. That is, the two-point function is a superpositions of ultraviolet-regularised two-point functions of $1+1$ dimensional chiral fermions, with propagators given by $(\partial_0 - iv_\omega \partial_1)^{-1}$.

Thus, it makes sense to define $\underline{X} = \theta \underline{x}$ and $\underline{Y} = \theta \underline{y}$. In terms of these rescaled variables,

$$\langle \mathbf{T} \gamma_{X_0/\theta}(a_{X/\theta,\rho}) \gamma_{Y_0/\theta}(a_{Y/\theta,\rho'}^*) \rangle_{\infty} \simeq \theta \sum_{\omega=1}^N e^{ik_F^\omega(X-Y)/\theta} P_{\rho\rho'}^\omega \int \frac{d\underline{q}}{(2\pi)^2} \frac{e^{i\underline{q} \cdot (\underline{X} - \underline{Y})}}{iq_0 + v_\omega q_1} \chi(|\underline{q}|); \quad (1.2.5)$$

that is, up to a fast oscillating phase, the two-point function decays as $\theta \times |\underline{X} - \underline{Y}|^{-1}$. Concerning the perturbation (1.2.2), it can be rewritten as, defining $T = \theta t$, and recalling that $\eta = (1/a)\theta$:

$$\varepsilon e^{\eta t} \mathcal{P} = \theta e^{(1/a)T} \sum_{X \in \theta \Gamma_L} \sum_{\rho=1}^M \mu(X) a_{X/\theta,\rho}^* a_{X/\theta,\rho}; \quad (1.2.6)$$

hence, defining $\tilde{\rho}(T) := \rho(T/\theta)$ with $\rho(\cdot)$ the solution of (1.1.3), we get:

$$i\partial_T \tilde{\rho}(T) = \left[\tilde{\mathcal{H}}((1/a)T), \tilde{\rho}(T) \right], \quad (1.2.7)$$

where:

$$\tilde{\mathcal{H}}((1/a)T) = \sum_{X,Y} \sum_{\rho,\rho'} a_{X/\theta,\rho}^* \tilde{H}_{\rho\rho'}(X-Y) a_{Y/\theta,\rho'} + e^{(1/a)T} \sum_{X \in \theta\Gamma_L} \sum_{\rho} \mu(X) a_{X/\theta,\rho}^* a_{X/\theta,\rho} \quad (1.2.8)$$

and $\tilde{H}(X-Y) = (1/\theta)H((X-Y)/\theta)$. The leading-order behaviour of the two-point function of the rescaled, unperturbed Hamiltonian in (1.2.8), computed at imaginary times X_0, Y_0 , is given by (1.2.5). Thus, in order to understand the contribution of the different chiralities to the solution of (1.2.7), as $\theta \rightarrow 0$, one is tempted to consider the dynamics of the following time-dependent, continuum model:

$$\mathcal{H}_\omega((1/a)T) = \int_{\mathbb{R}} dX \Psi_X^* v_\omega i \partial_X \Psi_X + e^{(1/a)T} \int_{\mathbb{R}} dX \mu(X) \Psi_X^* \Psi_X, \quad (1.2.9)$$

where Ψ_X is now a fermionic field, understood as an operator-valued distribution. It is reasonable to expect that, as $\theta \rightarrow 0$, the leading contribution to the transport properties associated with (1.2.7) can be represented as a superposition of terms that evolve according to the dynamics generated by (1.2.9). Eq. (1.2.9) can be viewed as describing an adiabatic perturbation of relativistic 1 + 1 dimensional chiral fermions. Of course, the expression (1.2.9) is formal, in view of the unboundedness from above and from below of the dispersion relation. Also, observe that, since we are making no assumption on the size of $\mu(\cdot)$, after rescaling time, no small parameter is left in front of the perturbation.

Putting aside the issue of mathematical rigor, the advantage of this representation is that the model (1.2.9) can be studied via bosonisation: the density operator $\Psi_X^* \Psi_X$ is known to behave like a free bosonic field, see *e.g.* [33, 37, 62]. Since the Duhamel series associated with (1.2.9) is expressed in terms of correlation function of density operators, this observation is expected to introduce a drastic simplification in the perturbation theory for the lattice model: for quasi-free bosonic systems, all connected correlation functions of order higher than two are zero. This property is expected to hold also in presence of weak density-density interactions, and it is an instance of the integrability of the emergent QFT; interacting systems will be the focus of Chapter 4 of this work. The last few years saw the rise of an enormous interest in the application of methods from the theory of integrable systems to study non-equilibrium quantum dynamics, starting from [21, 67]. This led to the derivation of generalised hydrodynamic models for the long-time dynamics of quantum systems; see *e.g.* [30, 32] for reviews.

In [69], we rigorously studied the transport properties of the family of lattice models described after (1.2.1), for perturbations in the Euler-like scaling regime, recall (1.2.2). Concerning the observable $\mathcal{O}_X$ in (1.1.4), we will restrict the attention to the density operator and to the current density operator. They are related by a lattice continuity equation, and this will play a crucial role in our analysis. Due to the absence of a spectral gap, the validity of the Kubo formula cannot be inferred from the adiabatic theorem, since the latter is not available in a quantitative form suitable for our purposes. Instead, our method relies on the application of [48] to the case of gapless systems. This approach, based on a complex deformation argument in time, allows us to obtain an exact representation for the real time Duhamel series in terms of Euclidean correlation functions, where the real time evolution is replaced by an imaginary time dynamics, as in (1.2.3). This allows to translate the oscillations introduced by the real time dynamics into (slow) decay properties of Euclidean correlations, which can be studied in a much more efficient way.

Furthermore, this representation is particularly convenient for comparing the perturbation theory of the original lattice model with the perturbation theory of the effective

continuum model (1.2.9), in the presence of a suitable ultraviolet cutoff. This allows in particular to detect a cancellation in the perturbation theory for the lattice model, at every order, which is related to the bosonic behaviour of the density operators of the effective model. The proof of this cancellation however does not rely on bosonisation, but rather on a careful analysis of the regularised loop integrals arising in the perturbation theory of the lattice model. This type of cancellation is well-known in the QFT literature, see [34]. It can also be viewed as a consequence of the emergent, anomalous symmetry of $1 + 1$ dimensional relativistic chiral fermions. In our framework, this cancellation is used up to a high, θ -dependent order; to control the higher orders, we show that the Duhamel series is actually convergent, and this allows to show that very high orders do not matter as $\theta \rightarrow 0$. All together, we show that the linear response term dominates, up to terms that are subleading as $\theta \rightarrow 0$: that is, in the scaling limit the linear response is exact. Concerning the linear response term, it can be computed explicitly in terms of the Fermi velocities, following the strategy already used in [2, 19, 20, 60, 61], based on lattice conservation laws.

More in detail, the work [69] proves convergence of the Duhamel expansion without a smallness assumption on $\mu(\cdot)$: without the improvement guaranteed by the aforementioned cancellation, the m -th order expansion in the Duhamel series (1.1.5) would be estimated proportionally to $\theta^{m-1} \times \|\hat{\mu}\|_{\ell_1}^{m-1} \times (1/\eta)^{2-m}$ (with $\hat{\mu}(\cdot)$ being the Fourier transform of $\mu(\cdot)$), which implies convergence for θ/η bounded and for $\|\hat{\mu}\|_{\ell_1}$ small enough. Instead, the improved analysis of [69] allows to extract a factor $1/m!$ in the estimate of the correlator, and a factor η^γ with $\gamma > 0$ due to the cancellation for the scaling limit. Taking into account these improvements, [69] allows to prove that the higher-order contributions to linear response are summable and vanishingly small as $\eta, \theta \rightarrow 0$, with $0 < |\theta/\eta| \leq C |\log \eta|$.

Finally, we adapt the strategy to study edge transport for $2d$ topological insulators. The complementary question of understanding bulk transport in presence of a bulk spectral gap has been studied in [51], where the validity of a many-body (super-)adiabatic theorem has been established. More recently, the validity of the Kubo formula for many-body system under a *local* gap assumption has been proved in [52]. Here, we focus on the edge transport properties, which are effectively described by a massless quasi-1d system, and we prove the validity of the Kubo formula for the edge current and for the edge density. This expression is the starting point of the analyses of [2, 61], which also considered interacting systems. In particular, the explicit computation of the edge linear response, combined with the subleading estimate for the sum of all the higher-order terms, allows us to prove the quantisation of the edge conductance from quantum dynamics.

1.3 Response theory of interacting systems

The main limitation of the setting described in the previous section is that we considered non-interacting fermionic models. It is of course a physically relevant question to investigate under which conditions the results of [69] can be extended to the case of non-integrable, weakly interacting gapless systems on the lattice. In these cases, the explicit analysis of Euclidean correlation functions performed in [69] is no longer possible. Instead, one has to resort to rigorous renormalisation group methods. In the last years, these methods allowed to rigorously compute transport coefficients for $1d$ and quasi-1d systems, and to put on rigorous grounds several predictions based on formal bosonisation arguments; see e.g. [2, 12, 13, 18, 58, 61]. The extension of [69] is carried out in [70], and will be briefly

reviewed in this section.

The work [70] focuses on one-dimensional fermions, but we believe that, with some extra effort, the analysis could be extended to study edge modes of weakly interacting $2d$ systems. In paper [70], or equivalently in Chapter 4 of this thesis, we consider a system of interacting fermions on a one-dimensional lattice Γ_L of length L , with periodic boundary conditions, $\Lambda_L = \mathbb{Z}/L\mathbb{Z}$. At equilibrium, the Gibbs state of the system is given by

$$\rho_{\beta,\mu,L} = \frac{e^{-\beta(\mathcal{H}-\mu N)}}{\text{Tr}_{\mathcal{H}_L} e^{-\beta(\mathcal{H}-\mu N)}} , \quad (1.3.1)$$

where now $\mathcal{H}$ is a *bona fide* many-body Hamiltonian, *i.e.* it contains a two-body interaction term of density-density type, namely

$$\mathcal{H} = \sum_{x,y \in \Gamma_L} \sum_{\rho,\rho' \in S_M} a_{x,\rho}^* H_{\rho\rho'}(x-y) a_{y,\rho'} + \lambda \sum_{x,y \in \Lambda_L} \sum_{\rho,\rho' \in S_M} a_{x,\rho}^* a_{x,\rho} w(x-y) a_{y,\rho'}^* a_{y,\rho'} ,$$

for $\lambda \in \mathbb{R}$, and $w(\cdot)$ a finite-ranged two-body potential. Just like in [69], we consider the case in which β, L are eventually taken to infinity, and no uniform spectral gap separates μ from the spectrum of $\mathcal{H}$.

The single-particle Hamiltonian H is taken as in [69], with the further requirement that, in the 2-body scattering, the conservation of Fermi momenta only holds with momenta equal in pairs (elastic scattering, see Equation (4.1.11)); this is generically true for spinless fermions in the absence of special symmetries. An example of H that fits the setting is the case of the lattice Laplacian in $1d$, for spinless fermions, for chemical potential μ chosen in the spectrum, away from band edges.

In order to generate a non-trivial charge transport, we employ the adiabatic protocol

$$\mathcal{H}(\eta t) = \mathcal{H} + e^{\eta t} \mathcal{P} , \quad (1.3.2)$$

with

$$\mathcal{P} = \theta \sum_{x \in \Gamma_L} \sum_{\rho \in S_M} \mu(\theta x) a_{x,\rho}^* a_{x,\rho} \quad (1.3.3)$$

for $\mu(\cdot)$ given by the restriction to Γ_L of a smooth, fast-decaying function on $\mathbb{R}$ with compactly supported Fourier transform.

We consider the adiabatic protocol (1.3.2). The main result of [70] is the proof of validity of the Kubo formula for the expectation of the density and of the current density operator, as $\eta \rightarrow 0^+$, taken after $\beta, L \rightarrow \infty$, as in [69]. More precisely, we prove that, denoting by $j_{0,x}$ the density operator and by $j_{1,x}$ the current density operator (see Eqs. (4.1.33), (4.1.34)), for θ/η bounded and for $|\lambda|$ small enough:

$$\text{Tr } j_{\nu,x} \rho(0) - \text{Tr } j_{\nu,x} \rho_{\beta,L} = -i \int_{-\infty}^0 dt e^{\eta t} \langle [j_{\nu,x}, \tau_t(\mathcal{P})] \rangle_{\beta,L} + O(\theta \eta^\gamma) + O\left(\frac{\theta}{\eta^3 \beta}\right) , \quad (1.3.4)$$

where $\gamma > 0$, and where all error terms are bounded uniformly in L . Since the first term is of order θ , Eq. (1.3.4) proves the exactness of linear response as $\beta, L \rightarrow \infty$ followed by $\eta, \theta \rightarrow 0$. It is important to emphasize that $\theta, \eta, |\lambda|$ are chosen small uniformly in β, L ; however, a technical limitation of the proof is that the range of allowed coupling constants λ shrinks to zero as $\theta/\eta \rightarrow +\infty$.

Moreover, the linear response can be explicitly evaluated, in terms of the renormalised Fermi velocities of the quasi-particles; see Theorem 4.1.5 for the explicit expression. For

a general number of Fermi points, to the best of our knowledge these expressions are rigorously derived in [70] for the first time.

The proof follows the strategy of [69], however in a much more difficult context, due to the presence of the many-body interaction. The strategy of [69] was based on an explicit analysis of the Euclidean correlation functions of the system, performed using the quasi-freeness of the state. Here, the correlation functions of the system are not explicit, since the many-body system is interacting. In order to deal with this issue, we use rigorous renormalisation group methods, that allow to express the Euclidean correlation functions of the system at equilibrium in terms of a convergent renormalised expansion. These RG methods have been applied to a wide class of statistical mechanics and condensed matter systems in the last three decades, see [58] for a review. The class of models considered in [70] belongs to the (multichannel) Luttinger liquid universality class: the Luttinger liquid is a $1 + 1$ dimensional QFT that describes interacting massless chiral fermions, and it has special integrability features [62]. The rigorous RG analysis of $1d$ quantum systems belonging to this universality class has been pioneered in [14], for spinless fermions, and in [22] for spinful ones. These works relied on the exact solvability of the Luttinger model, which represented a limitation of the approach. Such restriction has been removed in [16, 18], via a different strategy, based on the combination of Ward identities and Schwinger-Dyson equations, that ultimately allows to control the flow of all RG couplings without relying on the exact solution. The outcome is a construction of a nontrivial RG fixed point, characterised by anomalous scaling exponents. In the applications to condensed matter physics, the method has been used to compute response functions for interacting $1d$ systems, and to prove the validity of the Haldane relations connecting anomalous exponents and transport coefficients, [12, 13]. More recently, these techniques have been used to compute edge transport coefficients for interacting $2d$ topological insulators, and to prove the quantisation of the edge conductance, [2, 61].

The technical core of [70] is a renormalisation group analysis of the m -point density-density and current-density Euclidean correlation functions for the interacting $1d$ system at equilibrium. Similar analyses have been performed *e.g.* in [11], for the Thirring model, and in [38], for the interacting Ising model. With respect to these works, here we obtain an improved estimate for the Euclidean correlations for the special external momenta compatible with our adiabatic protocol (after the complex deformation to imaginary times, the temporal component of all independent incoming momenta of the Euclidean correlations is equal to η), which ultimately allows to prove the summability of the Duhamel expansion without any smallness requirement on the perturbation $\mu(\cdot)$. Furthermore, we extract the scaling limit contribution from the general m -point correlation, and we prove that it is vanishing. This step was done via an explicit computation in [69], and it is here derived via the RG analysis of the scaling limit of the model. A similar cancellation has been derived in the past for the Thirring model or the Luttinger model [11, 46]. Here we extend the cancellation to the multichannel Luttinger liquid universality class. As a technical note, we only need to establish the cancellation for momentum-space correlations with momenta compatible with the adiabatic protocol, which is easier than working in configuration space as in [11, 46]. Finally, the explicit form of the susceptibility and of the conductance is determined using lattice Ward identities, following a strategy similar to [61].

As a future perspective, it would be very interesting to connect the results and the methods used in [70] with the derivation of hydrodynamic models for the non-equilibrium

dynamics of integrable systems, a subject on which a lot of progress has been obtained in recent years starting from [21, 67], see [30, 32] for reviews.

1.4 Organisation of the thesis

This Introduction constitutes the first chapter of the present work of thesis.

In Chapter 2, we review well known facts about many-body lattice models, which are the starting point of the subsequent chapters. In Section 2.1, we introduce the setting for a general model of lattice fermions in d dimensions, in the grand-canonical ensemble. In Section 2.2, we derive the Duhamel expansion for dynamics for time-dependent perturbations, and we show how this can be converted in an expansion in Euclidean cumulants, via the Wick rotation method developed in [48]. Then, in Section 2.3, we prove Wick's rule for Euclidean time-ordered correlation functions of non-interacting Gibbs states, and we provide general bounds for the fermionic two-point function. Finally, in Section 2.4, we provide a brief review of the perturbative expansion of correlation functions of a given interacting Gibbs state in terms of those of the corresponding non-interacting one; furthermore, we show how these correlations admit a Grassmann representation. In relation to this, Appendix A collects a short review of Grassmann algebras and Grassmann Gaussian measures.

Chapter 3 is organised as follows. In Section 3.1, we introduce the class of one-dimensional models we considered in [69], and we define the current and density operators. In Section 3.2, we state our main result for one-dimensional fermions, Theorem 3.2.1, that gives an explicit expression for the full response of the density and of the current operator on large scales. In Section 3.3, we discuss the proof of Theorem 3.2.1. Finally, in Section 3.4, we discuss the extension of Theorem 3.2.1 to the case of edge transport in two-dimensional topological insulators. Our main result here is Theorem 3.4.3, that gives the explicit expression of the large-scale response of the system, and in particular proves the quantisation of the edge conductance.

Finally, Chapter 4 is structured as follows. In Section 4.1 we define model studied in [70], in Fock space, and we state our main result, Theorem 4.1.5. In Section 4.2 we review the Wick rotation of the Duhamel series. In Section 4.3 we discuss the renormalisation group analysis of the model, which takes as starting point the Grassmann integral representation of the theory. Until Section 4.3.6, the analysis is similar to previous applications of rigorous RG methods [2, 61]. Then, in Section 4.3.6 we discuss the RG analysis of the multi-point correlation functions, and their improved bounds, which allow to prove summability of the Duhamel expansion. These bounds rely on the Gallavotti-Nicolò tree expansion, reviewed in Section 4.3.5. In Section 4.4 we define the reference model, and we prove the Ward identity for the m -point density correlations, which allows to prove the vanishing of all correlations with $m \geq 3$. In Section 4.5 we prove our main result, by comparing the renormalised expansions of lattice and reference model, extracting the cancellation of the reference model, and estimating the remainder taking into account the improvement of Section 4.3.6. Also, we compute the susceptibility and the charge conductance, using the explicit form of the scaling limit, and lattice Ward identities to fix the form of the non-universal part. In addition, Appendix B collects the details of the proof of the vanishing of the density cumulants of the reference model of order higher than two.

Chapter 2

Lattice many-body Fermi systems

In this chapter, we introduce some general preliminary notions and provide results about the dynamics and statistical mechanics of lattice fermions, which are well-known, and which will be employed in later chapters. In Section 2.1, we introduce various setups for the analysis of many-body lattice Fermi systems in d dimensions, with a specific focus on the grand-canonical ensemble.

In Section 2.2, we derive the Duhamel expansion for time-dependent perturbations, and we show how this can be converted in an expansion in Euclidean cumulants, via Wick rotation. For non-interacting Gibbs states, such Euclidean correlation functions of local observables are essentially determined by a single function, namely the fermionic two-point correlation, thanks to Wick's Theorem, as shown in Section 2.3. Finally, Section 2.4 contains a brief review of the perturbative expansion of the correlation functions of a given interacting Gibbs state in terms of those of the non-interacting one underlying it; furthermore, we show how these correlations admit a Grassmann representation.

2.1 Lattice fermions

We will describe systems of electrons on a lattice, as models for crystals in the tight-binding approximation. Each lattice site corresponds to an atom or molecule, to which the electron is attracted via electrostatic forces, so that effectively its motion can be interpreted as a hopping between adjacent sites.

2.1.1 Single particle on a lattice

Denoting with d the space dimension, let $\Gamma_L = (\mathbb{Z}/L\mathbb{Z})^d$, and define the following distance on it:

$$\|x - y\|_L := \min_{n \in \mathbb{Z}^d} \|x - y - nL\|, \quad (2.1.1)$$

with $\|\cdot\|$ being the standard norm on $\mathbb{R}^d$. Let $S_M = \{1, \dots, M\}$ be a set of labels parameterising internal degrees of freedom (such as spin, flavour, *etc.*) and possible sub-lattices. Therefore the decorated lattice

$$\Lambda_L := \Gamma_L \times S_M \quad (2.1.2)$$

will describe all the degrees of freedom of a single lattice fermion. We shall denote points in Γ_L with x , labels in S_M with ρ , and points in Λ_L with $\mathbf{x} = (x, \rho)$.

The Hilbert space of a single quantum particle on Λ_L is then given by the vector space

$$\ell^2(\Lambda_L) := \left\{ \psi : \Lambda_L \rightarrow \mathbb{C} \mid \sum_{\mathbf{x} \in \Lambda_L} |\psi(\mathbf{x})|^2 < +\infty \right\}, \quad (2.1.3)$$

endowed with the standard scalar product

$$\langle \psi, \phi \rangle := \sum_{\mathbf{x} \in \Lambda_L} \overline{\psi(\mathbf{x})} \phi(\mathbf{x}).$$

A single-particle *pure state* is then specified by a *line* in such Hilbert space, *i.e.* by a rank 1 projector $|\psi\rangle\langle\psi|$. Observables are represented by self-adjoint operators on $\ell^2(\Lambda_L)$. The expectation value of an observable O on a normalised state ψ is given by $\langle O \rangle_\psi \equiv \langle \psi, O\psi \rangle$.

Given a Hamiltonian H , the dynamics of the state ψ is generated by the Schrödinger equation:

$$i\partial_t \psi_t = H\psi_t, \quad \psi_t|_{t=0} = \psi. \quad (2.1.4)$$

and the solution is $\psi_t = e^{-iHt}\psi$, for all $t \in \mathbb{R}$. We will consider Hamiltonians of the form:

$$(H\psi)(\mathbf{x}) = \sum_{\mathbf{y}} H(\mathbf{x}; \mathbf{y}) \psi(\mathbf{y}), \quad (2.1.5)$$

with $H(\mathbf{x}; \mathbf{y}) \equiv H_{\rho\rho'}(x; y)$ a finite range hopping kernel: there exists $R > 0$ independent of L such that $|H(\mathbf{x}, \mathbf{y})| = 0$ for all $\|\mathbf{x} - \mathbf{y}\| \geq R$.

Remark 2.1.1. For most of the present work, precisely from Chapter 3 onwards, we shall also assume H to be translation-invariant, namely

$$H_{\rho\rho'}(x + z; y + z) = H_{\rho\rho'}(x; y) \quad \forall z \in \Gamma_L, \quad (2.1.6)$$

and we shall therefore write it as $H_{\rho\rho'}(x - y)$.

Example 2.1.2. A typical choice for the Hamiltonian is (minus) the Laplacian on Λ_L :

$$(H\psi)(\mathbf{x}) = (-\Delta\psi)(\mathbf{x}) := - \sum_{\substack{\mathbf{y} \in \Lambda_L \\ \|\mathbf{y} - \mathbf{x}\|_L = 1}} [\psi(\mathbf{y}) - \psi(\mathbf{x})], \quad (2.1.7)$$

The dynamics generated by this operator describes diffusion on the lattice Λ_L , via allowing hopping between neighbouring sites.

A widely employed class of examples is provided by lattice Schrödinger operators: an external potential $V : \Lambda_L \rightarrow \mathbb{R}$ is added to the Laplacian, obtaining

$$(H\psi)_\rho(x) := (-\Delta\psi)_\rho(x) + \sum_{\rho' \in S_M} V_{\rho\rho'}(x) \psi_{\rho'}(x). \quad (2.1.8)$$

For the purpose of describing metals, the potential V is usually chosen so that H has a continuous spectrum part (in the $L \rightarrow \infty$ limit), allowing for diffusive behaviour.

2.1.2 Canonical formalism

Because of the Pauli exclusion principle, a system of N identical fermions will be described by the N -fold wedge product of the single-particle Hilbert space,

$$\ell^2(\Lambda_L)^{\wedge N} := \{ \psi_N \in \ell^2(\Lambda_L)^{\otimes N} \mid \psi_N(\mathbf{x}_\sigma) = \text{sgn}(\sigma) \psi_N(\mathbf{x}) \ \forall \sigma \in S_N, \forall \mathbf{x} \in \Lambda_L^N \}, \quad (2.1.9)$$

where for any $\mathbf{x} = (\mathbf{x}_1, \dots, \mathbf{x}_N) \in \Lambda_L^N$ and any permutation σ , we set

$$\mathbf{x}_\sigma := (\mathbf{x}_{\sigma(1)}, \dots, \mathbf{x}_{\sigma(N)}).$$

The Hamiltonian H_N of such a system will contain the contribution of the energy of each particle plus an interaction term between these: we will have

$$H_N = \sum_{i=1}^N H^{(i)} + \lambda \sum_{i < j}^N w^{(i,j)} \quad (2.1.10)$$

with $H^{(i)} := \mathbb{1}^{\otimes(i-1)} \otimes H \otimes \mathbb{1}^{\otimes(N-i)}$, and w_{ij} being a multiplication operator acting as

$$(w^{(i,j)} \psi_N)(\mathbf{x}_1, \dots, \mathbf{x}_N) = w(\mathbf{x}_i; \mathbf{x}_j) \psi_N(\mathbf{x}_1, \dots, \mathbf{x}_N) \quad (2.1.11)$$

for a two-body interaction potential¹ $w : \Lambda_L^2 \rightarrow \mathbb{R}$ which we take to be bounded and compactly supported (uniformly in L), analogously to the single-particle Hamiltonian. The parameter $\lambda \in \mathbb{R}$ may be thought of as a useful book-keeping parameter in perturbation theory.

The dynamics is governed by the many-body Schrödinger equation

$$i\partial_t \psi_{N,t} = H_N \psi_{N,t}, \quad \psi_{N,0} = \psi_N, \quad (2.1.12)$$

and in general one is not able to compute exactly the spectrum of H_N or the flow e^{-itH_N} . Nevertheless, for suitable initial data ψ_N , one usually expects the dynamics to be well approximated by a non-linear equation, in the limit of large N .

The system is assumed to have reached equilibrium at a temperature β^{-1} , so the initial state is described by a mixed state, called the (canonical) *Gibbs state*,

$$\rho_{\beta,L,N} = \frac{e^{-\beta H_N}}{\text{Tr } e^{-\beta H_N}}, \quad (2.1.13)$$

and one is interested in investigating what the response of the system is when it is slowly driven out of equilibrium via an external perturbation, in the thermodynamic limit (*i.e.* when $N, L \rightarrow \infty$ by keeping the density of particles $\varrho := N/L^d$ fixed).

2.1.3 Grand-canonical formalism

We shall describe our particle system in the grand-canonical formalism. This is not strictly necessary when considering non-interacting systems, since in that case the spectral properties of the many-body Hamiltonian H_N can be derived from the ones of the single-particle Hamiltonian H , but this description will be important for the discussion of the interacting case.

¹At this level, we are not assuming the interaction to be translation-invariant, but all physically reasonable two-body potentials are of such kind. We shall assume the interaction to be translation-invariant in all of the later chapters.

Definition 2.1.3 (Fermionic Fock space). We set the Fock space to be the exterior algebra over the single particle Hilbert space $\ell^2(\Lambda_L)$:

$$\mathcal{F}_L := \bigoplus_{N=0}^{ML^d} \mathcal{F}_L^{(N)} \quad \mathcal{F}_L^{(N)} = \begin{cases} \mathbb{C} & N=0 \\ \ell^2(\Lambda_L)^{\wedge N} & \text{else.} \end{cases} \quad (2.1.14)$$

Its elements are sequences of N -particle wavefunctions $\Psi = (\psi_N)_{N=0}^{ML^d}$, and it inherits a natural inner product, defined by taking the scalar products component-wise, which makes it into a Hilbert space of dimension 4^{ML^d} . The subspace $\mathcal{F}_L^{(N)}$ is called N particle sector, and $\mathcal{F}_L^{(0)}$ is often referred to as vacuum sector.

The grand-canonical Hamiltonian $\mathcal{H}$ is defined as the *second quantisation* of the single-particle Hamiltonian H and the two-body interaction w , *i.e.* it acts on each N particle sector $\mathcal{F}_L^{(N)}$ as the N -body Hamiltonian H_N , that is

$$\mathcal{H} := d\Gamma(H) + \lambda d\Gamma(w) = \bigoplus_{N=0}^{ML^d} H_N \quad (2.1.15)$$

where the second-quantisation map $d\Gamma$ is defined by

$$d\Gamma(H) = \bigoplus_N \sum_{i=1}^N H^{(i)}, \quad d\Gamma(w) = \bigoplus_N \sum_{i < j} w^{(i,j)}. \quad (2.1.16)$$

The dynamics is then governed by the following Schrödinger equation,

$$i\partial_t \Psi_t = \mathcal{H} \Psi_t, \quad \Psi_0 = \Psi, \quad (2.1.17)$$

whose solution is given by the one parameter group of unitaries $(e^{-it\mathcal{H}})_{t \in \mathbb{R}}$.

Definition 2.1.4 (Creation/annihilation operators). Importantly, the Fock space carries a representation of the *Canonical Anticommutation Relations* (CAR) algebra; that is, for any point $\mathbf{x} \in \Lambda_L$, there exist linear operators

$$a_{\mathbf{x}} : \mathcal{F}_L \rightarrow \mathcal{F}_L \quad a_{\mathbf{x}}^* : \mathcal{F}_L \rightarrow \mathcal{F}_L$$

that map each $\mathcal{F}_L^{(N)}$ to $\mathcal{F}_L^{(N-1)}$ and $\mathcal{F}_L^{(N+1)}$, respectively, and that satisfy the canonical anticommutation relations:

$$\{a_{\mathbf{x}}^*, a_{\mathbf{x}'}^*\} = \{a_{\mathbf{x}}, a_{\mathbf{x}'}\} = 0, \quad \{a_{\mathbf{x}}, a_{\mathbf{x}'}^*\} = \delta_{\mathbf{x}, \mathbf{x}'} . \quad (2.1.18)$$

Respectively, a and a^* are called annihilation and creation operators.

Explicitly, their action on a given wavefunction $\Psi = (\psi_N)_N$ is

$$\begin{aligned} (a_{\mathbf{x}}^* \Psi)_N(\mathbf{x}_1, \dots, \mathbf{x}_N) &= (\delta_{\mathbf{x}} \wedge \psi_{N-1})(\mathbf{x}_1, \dots, \mathbf{x}_N), & (N \geq 1), \\ (a_{\mathbf{x}} \Psi)_N(\mathbf{x}_1, \dots, \mathbf{x}_N) &= \sqrt{N+1} \psi_{N+1}(\mathbf{x}, \mathbf{x}_1, \dots, \mathbf{x}_N), & (N \geq 0), \end{aligned} \quad (2.1.19)$$

with $\delta_{\mathbf{x}} \in \ell^2(\Lambda_L)$ being the indicator function of site $\mathbf{x}$, and with the understanding that

$$a_{\mathbf{x}} \Omega = 0 \quad \forall \mathbf{x} \in \Lambda_L,$$

with $\Omega = (1, 0, 0, \dots, 0) \in \mathcal{F}_L^{(0)}$ being the vacuum vector. Therefore, $a_{\mathbf{x}}^*$ (resp. $a_{\mathbf{x}}$) may be thought as adding (resp. removing) a particle at site $\mathbf{x}$.

Lemma 2.1.5. *The following points hold true.*

1. *Creation is the adjoint of annihilation:* $(a_{\mathbf{x}}^*)^* = a_{\mathbf{x}}$ or all $\mathbf{x} \in \Lambda_L$.
2. *Boundedness:* $\|a_{\mathbf{x}}^\sharp\| \leq 1$ or all $\mathbf{x} \in \Lambda_L$, $\sharp = \emptyset, *$;
3. *Cyclicity of Ω :* any vector $\Psi \in \mathcal{F}_L$ can be written as

$$\Psi = P(a^*)\Omega$$

for some polynomial P in the creation operators.

Proof. The first point can be checked explicitly on an orthonormal basis. The second point, instead, follows directly from the last of equation (2.1.18): we have

$$\|a_{\mathbf{x}}\Psi\|^2 = \langle \Psi, a_{\mathbf{x}}^* a_{\mathbf{x}} \Psi \rangle = \|\Psi\|^2 - \langle \Psi, a_{\mathbf{x}} a_{\mathbf{x}}^* \Psi \rangle \leq \|\Psi\|^2, \quad (2.1.20)$$

for all $\Psi \in \mathcal{F}_L$.

Finally, to prove the last claim notice that $\{\delta_{\mathbf{x}}\}_{\mathbf{x} \in \Lambda_L}$ is an orthonormal basis for $\mathcal{F}_L^{(1)}$, and that, for any N , any N -body wavefunction ψ_N can be written as a linear combination of wedge products of N elements of such basis:

$$\psi_N = \sum_{\substack{I \subseteq \Lambda_L \\ |I|=N}} c_I \bigwedge_{\mathbf{x} \in I} \delta_{\mathbf{x}}. \quad (2.1.21)$$

On the other hand, we have

$$\delta_{\mathbf{x}} = a_{\mathbf{x}}^* \Omega, \quad (2.1.22)$$

hence the thesis. $\square$

Hence, any observable $\mathcal{O}$ on Fock space can be written as a linear combination of monomials in the creation and annihilation operators, namely

$$\mathcal{O} = \sum_{X \subseteq \Lambda_L} \mathcal{O}_X \quad \mathcal{O}_X = O(X) \prod_{\mathbf{x} \in X} a_{\mathbf{x}}^{\sharp_{\mathbf{x}}} \quad (2.1.23)$$

for suitable kernels $O(X)$.

Definition 2.1.6 (Local Observables). We will mostly consider *local* observables, *i.e.* operators that satisfy the following properties:

1. they are *finite-ranged*, that is $O(X) = 0$ if $|X| > R$, with $R > 0$ independent of L , and $|O(X)| < S$, for some $S > 0$ independent of L , for all $X \subseteq \Lambda_L$;
2. they are *number/charge preserving*, *i.e.* they commute with the number operator $\mathcal{N}$, defined as

$$\mathcal{N} := d\Gamma(\mathbb{1}_{\ell^2(\Lambda_L)}) = \bigoplus_{N=0}^{ML} N \mathbb{1}_{\mathcal{F}_L^{(N)}} = \sum_{\mathbf{x} \in \Lambda_L} a_{\mathbf{x}}^* a_{\mathbf{x}};$$

notice that this implies that all of the $\mathcal{O}_X$ have the same number² of a and a^* operators.

²To be precise, the condition $[\mathcal{O}, \mathcal{N}] = 0$ is satisfied also by observables with even monomials.

In particular, the Hamiltonian $\mathcal{H}$ is a local, extensive³ observable; written in creation and annihilation operators, it reads:

$$\mathcal{H} = \sum_{\mathbf{x}, \mathbf{y} \in \Lambda_L} H(\mathbf{x}; \mathbf{y}) a_{\mathbf{x}}^* a_{\mathbf{y}} + \frac{\lambda}{2} w(\mathbf{x}; \mathbf{y}) a_{\mathbf{x}}^* a_{\mathbf{y}}^* a_{\mathbf{y}} a_{\mathbf{x}} . \quad (2.1.24)$$

Also, we may define the grand-canonical Gibbs state of the system at inverse temperature β and chemical potential $\mu \in \mathbb{R}$, and its associated quantum expectation, as:

$$\rho_{\beta, \mu, L} = \frac{e^{-\beta(\mathcal{H} - \mu \mathcal{N})}}{\text{Tr } e^{-\beta(\mathcal{H} - \mu \mathcal{N})}} , \quad \langle \mathcal{O} \rangle_{\beta, L} = \text{Tr } \mathcal{O} \rho_{\beta, \mu, L} . \quad (2.1.25)$$

Remark 2.1.7. The chemical potential can be thought of as a function of the inverse temperature β , and of the density ϱ , via the relation

$$\varrho = \frac{\langle \mathcal{N} \rangle_{\beta, L}}{|\Gamma_L|} = \frac{\langle \mathcal{N} \rangle_{\beta, L}}{L^d} . \quad (2.1.26)$$

2.2 Duhamel expansion and Wick rotation

2.2.1 Dynamics and linear response

We will be interested in the response properties of the system, after exposing it to a time-dependent and slowly varying perturbation. We shall consider time-dependent Hamiltonians of the form:

$$\mathcal{H}(\eta t) = \mathcal{H} + e^{\eta t} \mathcal{P} , \quad (2.2.1)$$

for $\eta > 0$ and $t \leq 0$, with $\mathcal{P}$ a finite-range, charge preserving perturbation (as in Definition 2.1.6). In later chapters, for each case, we will make a more specific choice of the perturbation. The time variation of the perturbation is the usual adiabatic switching, with the explicit choice of the exponential switch function, widely used in applications.

Let $t, s \leq 0$, and let $\mathcal{U}(t; s)$ be the two-parameter unitary group generated by $\mathcal{H}(\eta t)$:

$$i \partial_t \mathcal{U}(t; s) = \mathcal{H}(\eta t) \mathcal{U}(t; s), \quad \mathcal{U}(s; s) = \mathbb{1}_{\mathcal{F}_L} . \quad (2.2.2)$$

Let us consider the evolution of the Gibbs state $\rho_{\beta, \mu, L}$,

$$\rho(t) = \lim_{T \rightarrow -\infty} \mathcal{U}(t; T) \rho_{\beta, \mu, L} \mathcal{U}(t; T)^* . \quad (2.2.3)$$

Definition 2.2.1 (Response functions). For $t \leq 0$, the response function $\chi_{\mathcal{O}\mathcal{P}}^{\beta, L}(t)$, associated to a given observable $\mathcal{O}$ and to a given perturbation $\mathcal{P}$, is set to be the variation at time t of the expectation of $\mathcal{O}$:

$$\chi_{\mathcal{O}\mathcal{P}}^{\beta, L}(t) = \text{Tr } \mathcal{O} \rho(t) - \text{Tr } \mathcal{O} \rho_{\beta, \mu, L} . \quad (2.2.4)$$

For selected observables $\mathcal{O}$, we will be interested on the dependence of such response functions on the external perturbation. We will be interested in the following order of limits: first $L \rightarrow \infty$, then $\beta \rightarrow \infty$ and then $\eta \rightarrow 0^+$. For small perturbations, the dynamics of the system can be studied via the Duhamel series around the initial equilibrium state.

³By extensive we mean $\|\mathcal{H}\| \propto L^d$.

Proposition 2.2.2 (Duhamel series). *Let $\tau_t(\mathcal{A})$ be the Heisenberg evolution of the observable $\mathcal{A}$:*

$$\tau_t(\mathcal{A}) = e^{i\mathcal{H}t} \mathcal{A} e^{-i\mathcal{H}t}. \quad (2.2.5)$$

The following identity holds true:

$$\chi_{\mathcal{O}\mathcal{P}}^{\beta,L}(t) = \sum_{n=1}^{\infty} (-i)^n \int_{\Delta_t^n} e^{\eta(s_1+\dots+s_n)} \langle [\dots [\tau_t(\mathcal{O}), \tau_{s_1}(\mathcal{P})], \tau_{s_2}(\mathcal{P})] \dots, \tau_{s_n}(\mathcal{P}) \rangle_{\beta,L} d\underline{s}, \quad (2.2.6)$$

where

$$\Delta_t^n := \{\underline{s} \in \mathbb{R}^n \mid -\infty \leq s_n \leq \dots \leq s_1 \leq t\}.$$

For $\eta > 0$, the sum in the right-hand side is absolutely convergent.

Proof. It is convenient to define an interaction picture time evolution unitary as

$$\mathcal{U}_I(t, s) = e^{it\mathcal{H}} \mathcal{U}(t, s) e^{-is\mathcal{H}}$$

so that $\mathcal{U}_I$ satisfies the equation

$$i\partial_t \mathcal{U}_I(t, s) = e^{\eta t} \tau_t(\mathcal{P}) \mathcal{U}_I(t, s) \quad \mathcal{U}_I(s, s) = \mathbb{1}_{\mathcal{H}_L}.$$

As a consequence, employing the cyclicity of trace and the invariance of the Gibbs state with respect to the autonomous time-evolution τ_t generated by the unperturbed Hamiltonian $\mathcal{H}$, keeping (2.2.3) in mind, equation (2.2.4) reads as

$$\begin{aligned} \text{Tr } \mathcal{O}(\rho(t) - \rho_{\beta,\mu,L}) &= \text{Tr } [\tau_t(\mathcal{O})(\mathcal{U}_I(t, T) \rho_{\beta,L} \mathcal{U}_I^*(t, T) - \rho_{\beta,\mu,L})] \\ &= -i \int_T^t e^{\eta s} \text{Tr } (\tau_t(\mathcal{O}) [\tau_s(\mathcal{P}), \mathcal{U}_I(s, T) \rho_{\beta,\mu,L} \mathcal{U}_I^*(s, T)]) ds \\ &= -i \int_T^t e^{\eta s} \text{Tr } ([\tau_t(\mathcal{O}), \tau_s(\mathcal{P})] \mathcal{U}_I(s, T) \rho_{\beta,\mu,L} \mathcal{U}_I^*(s, T)) ds, \end{aligned} \quad (2.2.7)$$

where, in the second passage, $\mathcal{U}_I(T, T) = \mathbb{1}_{\mathcal{H}_L}$ and the equation for $\mathcal{U}_I$ were used, and in the third one cyclicity of trace was employed once more.

Iterating m times equation (2.2.7), we get:

$$\begin{aligned} \chi_{\mathcal{O}\mathcal{P}}^{\beta,L}(t) &= \sum_{n=1}^m (-i)^n \int_{\Delta_{t,T}^n} e^{\eta(s_1+\dots+s_n)} \langle [\dots [\tau_t(\mathcal{O}), \tau_{s_1}(\mathcal{P})], \tau_{s_2}(\mathcal{P})] \dots, \tau_{s_n}(\mathcal{P}) \rangle_{\beta,L} d\underline{s} \\ &\quad + (-i)^{m+1} \int_{\Delta_{t,T}^{m+1}} e^{\eta(s_1+\dots+s_{m+1})} \text{Tr } \{ [\dots [\tau_t(\mathcal{O}), \tau_{s_1}(\mathcal{P})], \tau_{s_2}(\mathcal{P})] \dots, \tau_{s_{m+1}}(\mathcal{P}) \} \\ &\quad \cdot \mathcal{U}_I(s_{m+1}, T) \rho_{\beta,\mu,L} \mathcal{U}_I^*(s_{m+1}, T) \} d\underline{s}, \end{aligned} \quad (2.2.8)$$

with $\Delta_{t,T}^n := \{\underline{s} \in \mathbb{R}^n \mid T \leq s_n \leq \dots \leq s_1 \leq t\}$. The remainder at the second line of (2.2.8) can be easily estimated by

$$2^{m+1} \|\mathcal{O}\| \|\mathcal{P}\|^{m+1} \int_{\Delta_{t,T}^{m+1}} e^{\eta(s_1+\dots+s_{m+1})} \leq \|\mathcal{O}\| \frac{(2\|\mathcal{P}\|/\eta)^{m+1}}{(m+1)!}$$

where we used that $\Delta_{t,T}^{m+1} \subset \Delta_t^{m+1}$. Hence, for $\eta > 0$, the iteration may be performed uniformly in T very negative, and the resulting series is absolutely convergent. $\square$

The linear response approximation, also referred to as *Kubo formula*, amounts to the truncation to first order of the Duhamel series for the non-autonomous dynamics generated by $\mathcal{H}(\eta t)$. We have:

$$\chi_{\mathcal{O}\mathcal{P}}^{\beta,L}(t) = -i \int_{-\infty}^t ds e^{\eta s} \text{Tr} [\tau_t(\mathcal{O}), \tau_s(\mathcal{P})] \rho_{\beta,\mu,L} + \text{higher orders in } \mathcal{P}. \quad (2.2.9)$$

In many applications, it is tacitly assumed that the linear response approximation is valid; this is very often motivated by the quantitative agreement with experimental observations. From a mathematically rigorous viewpoint, however, proving that indeed linear response dominates the expansion is a nontrivial task, particularly so in gapless cases as the ones considered in Chapters 3-4.

2.2.2 Euclidean correlation functions and Wick rotation.

Euclidean correlation functions. For $t \in \mathbb{R}$, we define the imaginary time evolution of an observable $\mathcal{O}$ as:

$$\gamma_t(\mathcal{O}) := e^{t(\mathcal{H}-\mu\mathcal{N})} \mathcal{O} e^{-t(\mathcal{H}-\mu\mathcal{N})}. \quad (2.2.10)$$

Observe that if $[\mathcal{O}, \mathcal{N}] = 0$, the following identity holds:

$$\gamma_t(\mathcal{O}) = \tau_{-it}(\mathcal{O}). \quad (2.2.11)$$

An important consequence of the definition of the imaginary time dynamics (2.2.10) is the Kubo-Martin-Schwinger (KMS) identity, for all $t, s \in \mathbb{R}$:

$$\langle \gamma_t(\mathcal{A}) \gamma_s(\mathcal{B}) \rangle_{\beta,L} = \langle \gamma_{s+\beta}(\mathcal{B}) \gamma_t(\mathcal{A}) \rangle_{\beta,L}. \quad (2.2.12)$$

In our finite-dimensional setting, the KMS identity is a simple consequence of the definitions, and of the cyclicity of the trace. Nevertheless, it will have far-reaching consequences. In order to discuss them, we need to introduce the notion of time-ordering and of Euclidean correlation functions.

Definition 2.2.3 (Time-ordering). Let $0 \leq t_i < \beta$, for $i = 1, 2, \dots, n$, such that $t_i \neq t_j$ for $i \neq j$. We define the time-ordering of $\gamma_{t_1}(a_{\mathbf{x}_1}^{\#1}), \dots, \gamma_{t_n}(a_{\mathbf{x}_n}^{\#n})$ as:

$$\mathbf{T} \gamma_{t_1}(a_{\mathbf{x}_1}^{\#1}) \cdots \gamma_{t_n}(a_{\mathbf{x}_n}^{\#n}) = \text{sgn}(\pi) \gamma_{t_{\pi(1)}}(a_{\mathbf{x}_{\pi(1)}}^{\#\pi(1)}) \cdots \gamma_{t_{\pi(n)}}(a_{\mathbf{x}_{\pi(n)}}^{\#\pi(n)}), \quad (2.2.13)$$

where π is the permutation needed in order to bring the times in a decreasing order, from the left, with parity⁴ $\text{sgn}(\pi)$. In case two or more times are equal, the definition (2.2.13) is extended via normal ordering.

Remark 2.2.4. The normal ordering $:\cdots:$ of a monomial in a and a^* is obtained by anticommuting its constituents until all creation operators are to the left of the annihilation ones. For example

$$: a_1 a_1^* a_2 : \equiv - : a_1^* a_1 a_2 : = - a_1^* a_1 a_2.$$

Other prescriptions that extend (2.2.13) at equal times are of course possible; it is worth anticipating that in our applications this arbitrariness will play no role, ultimately because it involves a zero measure set of times, and because the algebra of fermionic operators on a finite lattice consists of bounded operators.

⁴A permutation is *even* (resp. *odd*) if it obtained from the identity by the composition of an even (resp. odd) number of simple exchanges.

The above definition extends to all operators on the fermionic Fock space by linearity. In particular, for $\mathcal{O}_1, \dots, \mathcal{O}_n$ even in the number of creation and annihilation operators, we have:

$$\begin{aligned} \mathbf{T}\gamma_{t_1}(\mathcal{O}_1) \cdots \gamma_{t_n}(\mathcal{O}_n) \\ = \sum_{\pi \in S_n} \mathbf{1}_{\{t_{\pi(1)} \geq t_{\pi(2)} \geq \dots \geq t_{\pi(n)}\}} \gamma_{t_{\pi(1)}}(\mathcal{O}_{\pi(1)}) \cdots \gamma_{t_{\pi(n)}}(\mathcal{O}_{\pi(n)}) , \end{aligned} \quad (2.2.14)$$

where $\mathbf{1}_X$ represents the indicator function of the set X . Notice that, for any choice of pairwise different times, only one summand is non zero; nevertheless, this writing is useful when the times are made to vary. Taking the grand-canonical thermal expectation of such time-ordered operators yields what we shall call Euclidean correlations.

Definition 2.2.5 (Euclidean correlation functions). Let $t_i \in [0, \beta)$, for $i = 1, \dots, n$. Given operators $\mathcal{O}_1, \dots, \mathcal{O}_n$, we define the time-ordered Euclidean correlation function as:

$$\langle \mathbf{T}\gamma_{t_1}(\mathcal{O}_1) \cdots \gamma_{t_n}(\mathcal{O}_n) \rangle_{\beta, L} . \quad (2.2.15)$$

From the definition of fermionic time-ordering, and from the KMS identity, it is not difficult to check that:

$$\begin{aligned} \langle \mathbf{T}\gamma_{t_1}(\mathcal{O}_1) \cdots \gamma_{t_n}(\mathcal{O}_n) \rangle_{\beta, L} \\ = (\pm 1) \langle \mathbf{T}\gamma_{t_1}(\mathcal{O}_1) \cdots \gamma_0(\mathcal{O}_k) \cdots \gamma_{t_n}(\mathcal{O}_n) \rangle_{\beta, L} ; \end{aligned} \quad (2.2.16)$$

in the special case in which the operators involve an even number of creation and annihilation operators, which will be particularly relevant for our analysis, the overall sign is $+1$. The property (2.2.16) allows to extend in a periodic (sign $+1$) or antiperiodic (sign -1) way the correlation functions to all times $t_i \in \mathbb{R}$. From now on, when discussing time-ordered correlations we shall always assume that this extension has been taken, unless otherwise specified.

Next, we define the connected time-ordered Euclidean correlation functions, or time-ordered Euclidean cumulants, as:

$$\begin{aligned} \langle \mathbf{T}\gamma_{t_1}(\mathcal{O}_1); \cdots ; \gamma_{t_n}(\mathcal{O}_n) \rangle_{\beta, L} \\ := \frac{\partial^n}{\partial \xi_1 \cdots \partial \xi_n} \log \left\{ 1 + \sum_{I \subseteq \{1, 2, \dots, n\}} \xi(I) \langle \mathbf{T}\mathcal{O}(I) \rangle_{\beta, L} \right\} \Big|_{\xi_i=0} \end{aligned} \quad (2.2.17)$$

where I is a non-empty ordered subset of $\{1, 2, \dots, n\}$, $\xi(I) = \prod_{i \in I} \xi_i$ and $\mathcal{O}(I) = \prod_{i \in I} \gamma_{t_i}(\mathcal{O}_i)$.

Let us list a couple of simple cases:

1. for $n = 1$, this definition reduces to

$$\langle \mathbf{T}\gamma_{t_1}(\mathcal{O}_1) \rangle_{\beta, L} \equiv \langle \gamma_{t_1}(\mathcal{O}_1) \rangle_{\beta, L} = \langle \mathcal{O}_1 \rangle_{\beta, L} ;$$

2. for $n = 2$ one gets

$$\begin{aligned} \langle \mathbf{T}\gamma_{t_1}(\mathcal{O}_1); \gamma_{t_2}(\mathcal{O}_2) \rangle_{\beta, L} &= \langle \mathbf{T}\gamma_{t_1}(\mathcal{O}_1)\gamma_{t_2}(\mathcal{O}_2) \rangle_{\beta, L} - \langle \mathbf{T}\gamma_{t_1}(\mathcal{O}_1) \rangle_{\beta, L} \langle \mathbf{T}\gamma_{t_2}(\mathcal{O}_2) \rangle_{\beta, L} \\ &= \langle \mathbf{T}\gamma_{t_1}(\mathcal{O}_1)\gamma_{t_2}(\mathcal{O}_2) \rangle_{\beta, L} - \langle \mathcal{O}_1 \rangle_{\beta, L} \langle \mathcal{O}_2 \rangle_{\beta, L} . \end{aligned} \quad (2.2.18)$$

More in general, the following relation between correlation functions and connected correlation function holds true:

$$\langle \mathbf{T} \gamma_{t_1}(\mathcal{O}_1) \cdots \gamma_{t_n}(\mathcal{O}_n) \rangle_{\beta, L} = \sum_P \prod_{J \in P} \langle \mathbf{T} \gamma_{t_{j_1}}(\mathcal{O}_{j_1}); \cdots; \gamma_{t_{j_{|J|}}}(\mathcal{O}_{j_{|J|}}) \rangle_{\beta, L},$$

where P is the set of all partitions of $\{1, 2, \dots, n\}$ into ordered subsets, and J is an element of the partition P , $J = \{j_1, \dots, j_{|J|}\}$.

Wick rotation. The next propositions establish the connection between the real time perturbation theory generated by the Duhamel series of Proposition 2.2.2 and the Euclidean correlation functions, for special choices of the adiabatic parameter η .

We first consider the Kubo formula.

Lemma 2.2.6 (Wick rotation for Kubo formula). *Let $\eta_\beta \in \frac{2\pi}{\beta}(\mathbb{N} \setminus \{0\})$. Let $\mathcal{P}$ be such that $[\mathcal{P}, \mathcal{N}] = 0$. Then, the Kubo formula admits the following rewriting:*

$$-i \int_{-\infty}^t ds e^{\eta_\beta s} \langle [\tau_t(\mathcal{O}), \tau_s(\mathcal{P})] \rangle_{\beta, L} = -e^{\eta_\beta t} \int_0^\beta ds \langle \mathbf{T} \gamma_s(\mathcal{P}); \mathcal{O} \rangle_{\beta, L}. \quad (2.2.19)$$

Proof. We start by writing, for $T < t \leq 0$:

$$\begin{aligned} \int_T^t ds e^{\eta_\beta s} \langle [\tau_t(\mathcal{O}), \tau_s(\mathcal{P})] \rangle_{\beta, L} &= e^{\eta_\beta t} \int_{T-t}^0 ds e^{\eta_\beta s} \langle [\mathcal{O}, \tau_s(\mathcal{P})] \rangle_{\beta, L} \\ &= e^{\eta_\beta t} \int_{T-t}^0 ds \left(e^{\eta_\beta s} \langle \mathcal{O} \tau_s(\mathcal{P}) \rangle_{\beta, L} - e^{\eta_\beta s} \langle \tau_s(\mathcal{P}) \mathcal{O} \rangle_{\beta, L} \right) \\ &= e^{\eta_\beta t} \int_{T-t}^0 ds \left(e^{\eta_\beta (s-i\beta)} \langle \tau_{s-i\beta}(\mathcal{P}) \mathcal{O} \rangle_{\beta, L} - e^{\eta_\beta s} \langle \tau_s(\mathcal{P}) \mathcal{O} \rangle_{\beta, L} \right), \end{aligned} \quad (2.2.20)$$

where in the last line we used the KMS condition (2.2.12) and the property $\exp(i\eta_\beta \beta) = 1$. Using the short-hand

$$f(z) := e^{\eta_\beta z} \langle \tau_z(\mathcal{P}) \mathcal{O} \rangle_{\beta, L},$$

we then have

$$-i \int_T^t ds e^{\eta_\beta s} \langle [\tau_t(\mathcal{O}), \tau_s(\mathcal{P})] \rangle_{\beta, L} = ie^{\eta_\beta t} \left[\int_{T-t}^0 ds f(s) - \int_{T-t}^0 ds f(s - i\beta) \right]. \quad (2.2.21)$$

Now, we notice that the function $z \mapsto f(z)$ is *holomorphic* in $z \in \mathbb{C}$. Therefore, by Cauchy theorem we conclude that the integral of f along the oriented boundary Γ of the rectangle

$$0 \rightarrow -i\beta \rightarrow (T - t - i\beta) \rightarrow (T - t) \rightarrow 0 \quad (2.2.22)$$

is equal to zero (see 2.1). The two terms in (2.2.21) correspond respectively to the paths $(T - t) \rightarrow 0$ and $-i\beta \rightarrow (T - t - i\beta)$.

Thus we can write

$$-i \int_T^t ds e^{\eta_\beta s} \langle [\tau_t(\mathcal{O}), \tau_s(\mathcal{P})] \rangle_{\beta, L} = -e^{\eta_\beta t} \int_0^\beta ds f(-is) + e^{\eta_\beta t} \int_{-\beta}^0 ds f(T - t + is), \quad (2.2.23)$$

and we notice that the second term on the right-hand side satisfies

$$\left| e^{\eta_\beta t} \int_{-\beta}^0 ds f(T - t + is) \right| = e^{\eta_\beta T} \left| \int_{-\beta}^0 ds \langle \tau_{T-t+is}(\mathcal{P}) \mathcal{O} \rangle_{\beta, L} \right| \leq \|\mathcal{O}\| \|\mathcal{P}\|_\beta e^{2\beta \|\mathcal{H}\| + \eta_\beta T};$$

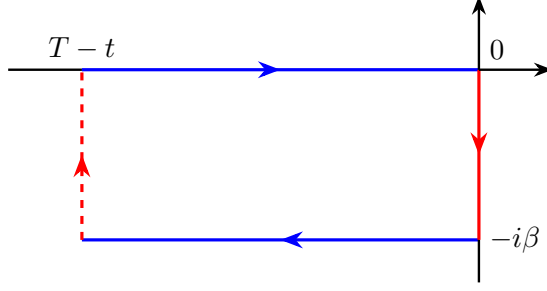


Figure 2.1: Representation of the closed contour $\Gamma \subset \mathbb{C}$ given by (2.2.22). The paths present in (2.2.21) are coloured in blue, and the ones contributing to (2.2.23) are highlighted in red.

therefore, in the $T \rightarrow -\infty$ limit we are led to

$$-i \int_{-\infty}^t ds e^{\eta_\beta s} \langle [\tau_t(\mathcal{O}), \tau_s(\mathcal{P})] \rangle_{\beta, L} = -e^{\eta_\beta t} \int_0^\beta ds e^{-i\eta_\beta s} \langle \mathbf{T} \gamma_s(\mathcal{P}) \mathcal{O} \rangle_{\beta, L}, \quad (2.2.24)$$

where we added $\mathbf{T}$ for free, since $s > 0$. Finally, since $\eta_\beta \neq 0$, we have that

$$e^{\eta_\beta t} \int_0^\beta ds e^{-i\eta_\beta s} \langle \mathcal{P} \rangle_{\beta, L} \langle \mathcal{O} \rangle_{\beta, L} = 0,$$

therefore we may add it to (2.2.24) to reconstruct the cumulant (recall (2.2.18)), obtaining thus the claim. $\square$

Proposition 2.2.7 (Wick rotation). *Let $\eta_\beta \in \frac{2\pi}{\beta}\mathbb{N}$. Let $\mathcal{P}$ be such that $[\mathcal{P}, \mathcal{N}] = 0$. Then, the following identity holds true:*

$$\begin{aligned} & (-i)^n \int_{\Delta_t^n} d\mathbf{s} e^{\eta_\beta(s_1 + \dots + s_n)} \langle [\dots [\tau_t(\mathcal{O}), \tau_{s_1}(\mathcal{P})], \tau_{s_2}(\mathcal{P})] \dots \tau_{s_n}(\mathcal{P})] \rangle_{\beta, L} \\ &= \frac{(-1)^n e^{n\eta_\beta t}}{n!} \int_{\mathbb{T}_\beta^n} d\mathbf{s} e^{-i\eta_\beta(s_1 + \dots + s_n)} \langle \mathbf{T} \gamma_{s_1}(\mathcal{P}); \gamma_{s_2}(\mathcal{P}); \dots; \gamma_{s_n}(\mathcal{P}); \mathcal{O} \rangle_{\beta, L}, \end{aligned} \quad (2.2.25)$$

where $\mathbb{T}_\beta^n$ is the n -dimensional torus of side-lengths β .

Proof. We refer the reader to [48] for the proof of this identity, which holds for general lattice models in d dimensions. The key idea is a complex deformation argument similar to the one presented in the proof of Lemma 2.2.6, but iterated in each time variable. $\square$

Equation (2.2.25) introduces a very useful representation of the Duhamel series, as it converts a question about (adiabatic) dynamics into a problem of (quantum) statistical physics.

Furthermore, the real-time correlation functions appearing on the left-hand side are oscillating functions, due to the unitarity of the time evolution τ , hence the integral is regularised by the damping factors given by the adiabatic switch function, and in order to prove convergence as $\eta \rightarrow 0^+$, a cancellation (due to high oscillations at large times) is needed; on the other hand, the imaginary time evolution γ in the right-hand side is non-unitary, and one expects the oscillations in the left-hand side to translate into *decay* properties of the Euclidean correlation functions.

Therefore, our aim for the next sections will be to derive some general properties of such Euclidean correlations. Notice that, since every observable can be written in terms of a polynomial in a and a^* ,

$$\mathcal{O} \equiv \sum_{X \subset \Lambda_L} \mathcal{O}_X = \sum_{X \subset \Lambda_L} O(X) \prod_{\mathbf{x} \in X} a_{\mathbf{x}}^{\sharp_{\mathbf{x}}}, \quad (2.2.26)$$

everything boils down to the analysis of correlation functions of the kind

$$\langle \mathbf{T} \gamma_{s_1}(a_{\mathbf{x}_1}^{\sharp_1}) \cdots \gamma_{s_n}(a_{\mathbf{x}_n}^{\sharp_n}) \rangle_{\beta, L} \quad (2.2.27)$$

for suitable $n \in \mathbb{N}$. Notice that these correlations do not depend on the observable or perturbation considered; instead, they depend only on the Gibbs state of the system.

2.2.3 On the approximation of the adiabatic parameter

For general choices of $\eta > 0$, Lemma 2.2.6 and Proposition 2.2.7 do not hold directly. However, following [48], we can use an approximation argument to replace the time-dependent Hamiltonian in 2.2.1 with an auxiliary one, given as

$$\mathcal{H}_{\beta, \eta}(\eta_\beta t) = \mathcal{H} + e^{\eta_\beta t} \mathcal{P}, \quad (2.2.28)$$

with $\eta_\beta \in 2\pi\mathbb{Z}/\beta$ being the best approximant of η such that $\eta_\beta \geq \eta$, by employing Lieb-Robinson bounds for non-autonomous dynamics.

Theorem 2.2.8 (Lieb-Robinson bounds for non-autonomous flows). *Let $\mathcal{O}_X$ and $\mathcal{O}_Y$ be two observables supported on sets X and Y , respectively. Let $\{\mathcal{U}(t, s) \mid t, s \in \mathbb{R}\}$ be a strongly-continuous group of non-autonomous unitaries generated by a local, time-dependent Hamiltonian. Then, there exist positive constants C, c, v_{LR} such that, for any $t, s \in \mathbb{R}$, we have*

$$\|[\mathcal{O}_X, \mathcal{U}(t, s)^* \mathcal{O}_Y \mathcal{U}(t, s)]\| \leq C \|X\| \|Y\| e^{v_{LR}|t-s| - c \text{dist}(X, Y)}, \quad (2.2.29)$$

with $\text{dist}(X, Y) := \inf_{x \in X, y \in Y} \|x - y\|_L$ being the distance between X and Y .

Proof. We refer to Proposition 4.6 of [9] or to Theorem 5.1 of [25]. $\square$

Let us denote by $\tilde{\mathcal{U}}$ the non-autonomous flow associated to $\tilde{\mathcal{H}}_{\beta, \eta}(\eta_\beta t)$, similarly to Eq. 2.2.2. We shall also set

$$\tilde{\rho}(t) := \lim_{T \rightarrow -\infty} \tilde{\mathcal{U}}(t, T) \rho_{\beta, \mu, L} \tilde{\mathcal{U}}(t, T)^*.$$

Then, the following Proposition allows to show that the adiabatic dynamics $\mathcal{U}$ is close to $\tilde{\mathcal{U}}$, in the sense of expectations of local observables, for large β .

Proposition 2.2.9 (Approximation by the auxiliary dynamics). *For any local operator $\mathcal{O}$, and for any $t \leq 0$, we have that*

$$|\text{Tr } \mathcal{O} \rho(t) - \text{Tr } \mathcal{O} \tilde{\rho}(t)| \leq C \frac{\|\mathcal{O}\|}{\eta^{d+2} \beta}, \quad (2.2.30)$$

for some $C > 0$ independent of η, β , and L .

Proof. If $\mathcal{O}$ is extensive, we start by writing it in terms of its finitely supported components, namely

$$\mathrm{Tr} \mathcal{O}[\rho(t) - \tilde{\rho}(t)] = \sum_{X \subset \Lambda_L} \mathrm{Tr} \mathcal{O}_X[\rho(t) - \tilde{\rho}(t)] . \quad (2.2.31)$$

Next, we employ the cyclicity of the trace to write

$$\mathrm{Tr} \mathcal{O}_X[\rho(t) - \tilde{\rho}(t)] = \lim_{T \rightarrow -\infty} \mathrm{Tr}[\mathcal{U}(t, T)^* \mathcal{O}_X \mathcal{U}(t, T) - \tilde{\mathcal{U}}(t, T)^* \mathcal{O}_X \tilde{\mathcal{U}}(t, T)] \rho_{\beta, \mu, L} . \quad (2.2.32)$$

Now, we estimate the argument of the limit as:

$$\begin{aligned} & \left| \mathrm{Tr} \left(\mathcal{U}(t, T)^* \mathcal{O}_X \mathcal{U}(t, T) - \tilde{\mathcal{U}}(t, T)^* \mathcal{O}_X \tilde{\mathcal{U}}(t, T) \right) \rho_{\beta, \mu, L} \right| \\ & \leq \| \mathcal{O}_X - \mathcal{U}(t, T) \tilde{\mathcal{U}}(t, T)^* \mathcal{O}_X \tilde{\mathcal{U}}(t, T) \mathcal{U}(t, T)^* \|, \end{aligned} \quad (2.2.33)$$

where we used that $\rho_{\beta, \mu, L} \geq 0$, $\mathrm{Tr} \rho_{\beta, \mu, L} = 1$ and the unitarity of time-evolution. Next, we rewrite the argument of the norm as:

$$\begin{aligned} & \mathcal{O}_X - \mathcal{U}(t, T) \tilde{\mathcal{U}}(t, T)^* \mathcal{O}_X \tilde{\mathcal{U}}(t, T) \mathcal{U}(t, T)^* \\ & = \int_T^t ds \frac{\partial}{\partial s} [\mathcal{U}(t, s) \tilde{\mathcal{U}}(t, s)^* \mathcal{O}_X \tilde{\mathcal{U}}(t, s) \mathcal{U}(t, s)^*] \\ & = -i \int_T^t ds \mathcal{U}(t, s) [-\mathcal{H}(\eta s) + \mathcal{H}_{\beta, \eta}(\eta_\beta s), \tilde{\mathcal{U}}(t, s)^* \mathcal{O}_X \tilde{\mathcal{U}}(t, s)] \mathcal{U}(t, s)^* \\ & = i \int_T^t ds (e^{\eta s} - e^{\eta_\beta s}) \mathcal{U}(t, s) [\mathcal{P}, \tilde{\mathcal{U}}(t, s)^* \mathcal{O}_X \tilde{\mathcal{U}}(t, s)] \mathcal{U}(t, s)^*, \end{aligned} \quad (2.2.34)$$

where in the second passage we used that $\mathcal{U}(t, s)^* = \mathcal{U}(s, t)$, and in the last one we employed the definitions of $\mathcal{H}(\eta s)$ and $\mathcal{H}_{\beta, \eta}(\eta_\beta s)$. Therefore,

$$\| \mathcal{O}_X - \mathcal{U}(t, T) \tilde{\mathcal{U}}(t, T)^* \mathcal{O}_X \tilde{\mathcal{U}}(t, T) \mathcal{U}(t, T)^* \| \leq \int_T^t ds |e^{\eta s} - e^{\eta_\beta s}| \| [\mathcal{P}, \tilde{\mathcal{U}}(t, s)^* \mathcal{O}_X \tilde{\mathcal{U}}(t, s)] \| ; \quad (2.2.35)$$

inserting 2.2.35 in (2.2.33) and combining the outcome with (2.2.32), we are then led to

$$|\mathrm{Tr} \mathcal{O}_X[\rho(t) - \tilde{\rho}(t)]| \leq \lim_{T \rightarrow -\infty} \int_T^t ds |e^{\eta s} - e^{\eta_\beta s}| \| [\mathcal{P}, \tilde{\mathcal{U}}(t, s)^* \mathcal{O}_X \tilde{\mathcal{U}}(t, s)] \| . \quad (2.2.36)$$

Next, we claim that there is a constant C such that

$$\| [\mathcal{P}, \tilde{\mathcal{U}}(t, s)^* \mathcal{O}_X \tilde{\mathcal{U}}(t, s)] \| \leq C(|t - s|^d + 1) . \quad (2.2.37)$$

The proof of claim 2.2.37) is a standard application of the Lieb-Robinson bounds for non-autonomous dynamics presented in 2.2.8. Representing the perturbation $\mathcal{P}$ in terms of its local potentials, we may distinguish two different contributions:

$$\begin{aligned} \| [\mathcal{P}, \tilde{\mathcal{U}}(t, s)^* \mathcal{O}_X \tilde{\mathcal{U}}(t, s)] \| & \leq \sum_{Y \subset \Lambda_L} \| [\mathcal{P}_Y, \tilde{\mathcal{U}}(t, s)^* \mathcal{O}_X \tilde{\mathcal{U}}(t, s)] \| \\ & = \sum_{\substack{Y \subset \Lambda_L \\ \mathrm{dist}(X, Y) \leq D|t-s|}} \| [\mathcal{P}_Y, \tilde{\mathcal{U}}(t, s)^* \mathcal{O}_X \tilde{\mathcal{U}}(t, s)] \| \\ & \quad + \sum_{\substack{Y \subset \Lambda_L \\ \mathrm{dist}(X, Y) > D|t-s|}} \| [\mathcal{P}_Y, \tilde{\mathcal{U}}(t, s)^* \mathcal{O}_X \tilde{\mathcal{U}}(t, s)] \| \end{aligned} \quad (2.2.38)$$

with D large enough, to be chosen below. By the boundedness of the fermionic operators, and by the unitarity of the dynamics, the first term in the right-hand side is estimated as

$$\sum_{\substack{Y \subset \Lambda_L \\ \text{dist}(X,Y) \leq D|t-s|}} \|[\mathcal{P}_Y, \tilde{\mathcal{U}}(t,s)^* \mathcal{O}_X \tilde{\mathcal{U}}(t,s)]\| \leq \tilde{K}|X|(|t-s|^d + 1), \quad (2.2.39)$$

where we used the fact that the sum is restricted to sets Y of diameter bounded uniformly in L , since the perturbation $\mathcal{P}$ is local. For the second term, we use the Lieb-Robinson bound (2.2.29), to obtain

$$\sum_{\substack{Y \subset \Lambda_L \\ \text{dist}(X,Y) > D|t-s|}} \|[\mathcal{P}_Y, \tilde{\mathcal{U}}(t,s)^* \mathcal{O}_X \tilde{\mathcal{U}}(t,s)]\| \leq \sum_{\substack{Y \subset \Lambda_L \\ \text{dist}(X,Y) > D|t-s| \\ \text{diam}(Y) \leq R}} \tilde{C}|Y||X|e^{v_{LR}|t-s| - c \text{dist}(X,Y)}, \quad (2.2.40)$$

where R denotes the maximum diameter of the perturbation components $\mathcal{P}_Y$. Choosing D large enough, we have:

$$\begin{aligned} \sum_{\substack{Y \subset \Lambda_L \\ \text{dist}(X,Y) > D|t-s|}} \|[\mathcal{P}_Y, \tilde{\mathcal{U}}(t,s)^* \mathcal{O}_X \tilde{\mathcal{U}}(t,s)]\| \\ \leq \tilde{C}|X| \sum_{\substack{Y \subset \Lambda_L \\ \text{dist}(X,Y) > D|t-s| \\ \text{diam}(Y) \leq R}} |Y|e^{-\frac{c}{2}\text{dist}(X,Y)} \leq C'|X|. \end{aligned} \quad (2.2.41)$$

This concludes the check of claim (2.2.37). Using the bound (2.2.37) in (2.2.36), we get:

$$\begin{aligned} |\text{Tr } \mathcal{O}_X[\rho(t) - \tilde{\rho}(t)]| &\leq \lim_{T \rightarrow -\infty} \int_T^t ds |e^{\eta s} - e^{\eta \beta s}| \|[\mathcal{P}, \tilde{\mathcal{U}}(t,s)^* \mathcal{O}_X \tilde{\mathcal{U}}(t,s)]\| \\ &\leq C|X| \lim_{T \rightarrow -\infty} \int_T^t ds |e^{\eta s} - e^{\eta \beta s}| [|t-s|^d + 1] \\ &\leq \frac{C'|X|}{\beta} \lim_{T \rightarrow -\infty} \int_T^t ds e^{\eta s} [|s|^{d+1} + 1] \\ &\leq \frac{C''|X|}{\eta^{d+2}\beta}. \end{aligned} \quad (2.2.42)$$

Equation 2.2.42 now provides a bound for the full observable, namely

$$\begin{aligned} |\text{Tr } \mathcal{O}[\rho(t) - \tilde{\rho}(t)]| &= \left| \sum_{X \subset \Lambda_L} \text{Tr } \mathcal{O}_X[\rho(t) - \tilde{\rho}(t)] \right| \leq \sum_{X \subset \Lambda_L} |\text{Tr } \mathcal{O}_X[\rho(t) - \tilde{\rho}(t)]| \\ &\leq \frac{C''}{\eta^{d+2}\beta} \sum_{X \subset \Lambda_L} |X| \\ &\leq \frac{C'''}{\eta^{d+2}\beta} \|\mathcal{O}\|, \end{aligned} \quad (2.2.43)$$

where in the last passage we used the fact that $|X|$ is bounded uniformly in L , since $\mathcal{O}$ is a local observable. This completes the proof. $\square$

We see now the importance of this proposition: a direct implication is that

$$\begin{aligned}\chi_{\mathcal{O}\mathcal{P}}^{\beta,L} &:= \text{Tr } \mathcal{O}\rho(t) - \text{Tr } \mathcal{O}\rho_{\beta,\mu,L} = \tilde{\chi}_{\mathcal{O}\mathcal{P}}^{\beta,L}(t) + \mathcal{E}_{\mathcal{O}\mathcal{P}}^{\beta,L}(t) \\ \tilde{\chi}_{\mathcal{O}\mathcal{P}}^{\beta,L}(t) &:= \text{Tr } \mathcal{O}\tilde{\rho}(t) - \text{Tr } \mathcal{O}\rho_{\beta,\mu,L},\end{aligned}\tag{2.2.44}$$

with the error term being bounded for all $t \leq 0$ by

$$\mathcal{E}_{\mathcal{O}\mathcal{P}}^{\beta,L}(t) \leq C \frac{\|\mathcal{O}\|}{\eta^{d+2}\beta}.\tag{2.2.45}$$

Remark 2.2.10. Notice that the bound above implies that the error term has the same extensivity⁵ properties of $\mathcal{O}$; in particular, if $\mathcal{O}$ is intensive, then (2.2.45) is uniform in L . We also remark that such bound is convenient for our setting, since we are interested in investigating the $\beta \rightarrow \infty$ limit, at fixed $\eta > 0$, of our response function $\chi_{\mathcal{O}\mathcal{P}}^{\beta,L}$.

Thanks to this approximation argument, we can apply the Duhamel expansion of Proposition 2.2.2 to $\tilde{\chi}_{\mathcal{O}\mathcal{P}}^{\beta,L}$, instead of $\chi_{\mathcal{O}\mathcal{P}}^{\beta,L}$. Then, we can apply Lemma 2.2.6 and Proposition 2.2.7 to each term of the Duhamel expansion of the auxiliary dynamics $\tilde{\mathcal{U}}$, obtaining a Euclidean cumulant expansion for $\chi_{\mathcal{O}\mathcal{P}}^{\beta,L}$, up to the error term $\mathcal{E}_{\mathcal{O}\mathcal{P}}^{\beta,L}$.

2.3 The non-interacting Gibbs state

In this section, we will show that, for non-interacting models, all correlation functions as in (2.2.27) can be computed in terms of the fermionic two-point function, via Wick's rule. We will also derive some estimates for such two-point function.

Remark 2.3.1. Since we are dealing with non-interacting systems, in this section we will set $\lambda = 0$, *i.e.* we will not consider the interaction term in $\mathcal{H}$.

2.3.1 The two-point function and Wick's Theorem

Indicating space-time points with

$$\underline{\mathbf{x}} := (x_0, \mathbf{x}) \equiv (\underline{x}, \rho) \in [0, \beta) \times \Lambda_L,$$

we define the fermionic two-point correlation function (or *propagator*) g to be

$$g(\underline{\mathbf{x}}, \underline{\mathbf{y}}) := \langle \mathbf{T} a_{\underline{\mathbf{x}}} a_{\underline{\mathbf{y}}}^* \rangle_{\beta,L}\tag{2.3.1}$$

where $a_{\underline{\mathbf{x}}}^\sharp := \gamma_{x_0}(a_{\underline{\mathbf{x}}}^\sharp)$. Observe that g is translation-invariant in time, *i.e.* it depends on $x_0 - y_0$, and due to the KMS property, we see that

$$\lim_{x_0 \rightarrow \beta} g(\underline{\mathbf{x}}, \underline{\mathbf{y}}) = \langle a_{(\beta, \mathbf{x})} a_{\underline{\mathbf{y}}}^* \rangle_{\beta,L} = -\langle a_{\underline{\mathbf{y}}}^* a_{(0, \mathbf{x})} \rangle_{\beta,L} = -g((0, \mathbf{x}), \underline{\mathbf{y}});$$

hence it makes sense to extend it anti-periodically to a piece-wise smooth function of $x_0, y_0 \in \mathbb{R}$, by setting

$$g((x_0 + n\beta, \mathbf{x}), (y_0 + m\beta, \mathbf{y})) = (-1)^{n-m} g(\underline{\mathbf{x}}, \underline{\mathbf{y}}).\tag{2.3.2}$$

In the following, we shall assume this extension to have been taken, and by an abuse of notation we shall call it g . The following proposition determines it in terms of the single-particle Hamiltonian H .

⁵We will call an observable $\mathcal{O}$ intensive if $\|\mathcal{O}\|$ is bounded uniformly in the volume $|\Lambda_L|$. It will be called extensive if $\|\mathcal{O}\| \propto |\Lambda_L|$.

Proposition 2.3.2 (Structure of the propagator). *Denoting by $\mathbb{M}_\beta := \frac{2\pi}{\beta}(\mathbb{Z} + \frac{1}{2})$ the set of fermionic Matsubara frequencies, and letting $x_0, y_0 \in [0, \beta)$, we have:*

$$g(\underline{\mathbf{x}}, \underline{\mathbf{y}}) = \tilde{g}(\underline{\mathbf{x}}, \underline{\mathbf{y}}) - \mathbf{1}_{x_0=y_0} \frac{\delta_{\underline{\mathbf{x}}, \underline{\mathbf{y}}}}{2}, \quad (2.3.3)$$

where

$$\tilde{g}(\underline{\mathbf{x}}, \underline{\mathbf{y}}) := \lim_{K \rightarrow \infty} \frac{1}{\beta} \sum_{\substack{k_0 \in \mathbb{M}_\beta \\ |k_0| \leq K}} \frac{e^{ik_0(x_0 - y_0)}}{ik_0 + H - \mu} (\underline{\mathbf{x}}; \underline{\mathbf{y}}). \quad (2.3.4)$$

Remark 2.3.3. The *existence* of the limit follows from the presence of the oscillating factor $e^{ik_0(x_0 - y_0)}$. Notice that, for $x_0 - y_0 = 0 \bmod \beta$, the expression in Eq. (2.3.4) also makes sense: one has

$$\begin{aligned} \sum_{\substack{k_0 \in \mathbb{M}_\beta \\ |k_0| \leq K}} \frac{1}{ik_0 + H - \mu} &= \sum_{\substack{k_0 \in \mathbb{M}_\beta \\ 0 < k_0 \leq K}} \left[\frac{1}{-ik_0 + H - \mu} + \frac{1}{ik_0 + H - \mu} \right] \\ &= \sum_{\substack{k_0 \in \mathbb{M}_\beta \\ 0 < k_0 \leq K}} \frac{2(H - \mu)}{k_0^2 + (H - \mu)^2}. \end{aligned} \quad (2.3.5)$$

The existence of the $K \rightarrow \infty$ limit follows from functional calculus for H (notice that $k_0 \neq 0$ for finite β , *i.e.* a non-zero temperature regularises the propagator).

Proof. We start by noticing that g satisfies a differential equation: recalling the definition of time-ordering

$$g(\underline{\mathbf{x}}, \underline{\mathbf{y}}) = \mathbf{1}_{\{x_0 > y_0\}} \langle a_{\underline{\mathbf{x}}} a_{\underline{\mathbf{y}}}^* \rangle_{\beta, L} - \mathbf{1}_{\{y_0 \geq x_0\}} \langle a_{\underline{\mathbf{y}}}^* a_{\underline{\mathbf{x}}} \rangle_{\beta, L}, \quad (2.3.6)$$

and taking the derivative with respect to x_0 , we obtain

$$\begin{aligned} \partial_{x_0} g(\underline{\mathbf{x}}, \underline{\mathbf{y}}) &= \delta(x_0 - y_0) \langle \{a_{\underline{\mathbf{x}}}, a_{\underline{\mathbf{y}}}^*\} \rangle_{\beta, L} + \langle \mathbf{T} \partial_{x_0} (a_{\underline{\mathbf{x}}}) a_{\underline{\mathbf{y}}}^* \rangle_{\beta, L} \\ &= \delta(x_0 - y_0) \delta_{\underline{\mathbf{x}}, \underline{\mathbf{y}}} + \langle \mathbf{T} [\mathcal{H} - \mu \mathcal{N}, a_{\underline{\mathbf{x}}}] a_{\underline{\mathbf{y}}}^* \rangle_{\beta, L} \end{aligned} \quad (2.3.7)$$

where in the second passage we employed the CAR and the definition of γ_{x_0} . Now, an explicit computation yields

$$[\mathcal{H} - \mu \mathcal{N}, a_{\underline{\mathbf{x}}}] = - \sum_{\mathbf{z} \in \Lambda_L} [H(\underline{\mathbf{x}}; \mathbf{z}) - \mu \delta_{\underline{\mathbf{x}}, \mathbf{z}}] a_{(x_0, \mathbf{z})}, \quad (2.3.8)$$

which, plugged in (2.3.7), produces

$$(\partial_{x_0} + H - \mu) g(x_0 - y_0) = \delta(x_0 - y_0); \quad (2.3.9)$$

here we are imagining $g(x_0 - y_0)$ as a $|\Lambda_L| \times |\Lambda_L|$ matrix. Taking the anti-periodic Fourier transform, we can transform this ODE into

$$(ik_0 + H - \mu) \hat{g}(k_0) = 1 \quad k_0 \in \mathbb{M}_\beta. \quad (2.3.10)$$

Since $k_0 \neq 0$ and H is self-adjoint, this linear equation can be solved by inversion, namely

$$\hat{g}(k_0) = \frac{1}{ik_0 + H - \mu}. \quad (2.3.11)$$

For $x_0 \neq y_0$, we know that g is continuous, hence the Fourier series

$$\lim_{K \rightarrow \infty} \frac{1}{\beta} \sum_{\substack{k_0 \in \mathbb{M}_\beta \\ |k_0| \leq K}} e^{ik_0(x_0 - y_0)} \hat{g}(k) \quad (2.3.12)$$

evaluates g at $x_0 - y_0$.

At $x_0 = y_0$, instead, g is not continuous, but it has the following limits:

$$\begin{aligned} \lim_{x_0 \rightarrow y_0^+} g(\underline{\mathbf{x}}, \underline{\mathbf{y}}) &= \langle a_{\underline{\mathbf{x}}} a_{\underline{\mathbf{y}}}^* \rangle_{\beta, L} = g((0, \underline{\mathbf{x}}), (0, \underline{\mathbf{y}})) \\ \lim_{x_0 \rightarrow y_0^-} g(\underline{\mathbf{x}}, \underline{\mathbf{y}}) &= -\langle a_{\underline{\mathbf{y}}}^* a_{\underline{\mathbf{x}}} \rangle_{\beta, L} = g((0, \underline{\mathbf{x}}), (0, \underline{\mathbf{y}})) + \delta_{\underline{\mathbf{x}}, \underline{\mathbf{y}}} ; \end{aligned} \quad (2.3.13)$$

therefore, by Dirichlet's theorem, the Fourier series (2.3.12), evaluated at $x_0 = y_0$, converges to the average of the limits in (2.3.13). This concludes the proof. $\square$

Next, we report Wick's theorem for general correlations.

Theorem 2.3.4 (Wick's rule). *The following identity holds true: for any $n \in \mathbb{N}$, and for any choice of n -tuples $\{\underline{\mathbf{x}}_i\}_{i=1}^n, \{\underline{\mathbf{x}}'_i\}_{i=1}^n \subseteq [0, \beta) \times \Lambda_L$, we have*

$$\langle \mathbf{T} a_{\underline{\mathbf{x}}_1} \cdots a_{\underline{\mathbf{x}}_n} a_{\underline{\mathbf{x}}'_1}^* \cdots a_{\underline{\mathbf{x}}'_n}^* \rangle_{\beta, L} = \sum_{\pi \in S_n} \text{sgn}(\pi) \prod_{i=1}^n g(\underline{\mathbf{x}}_i, \underline{\mathbf{x}}'_{\pi(i)}) , \quad (2.3.14)$$

where S_n is the symmetric group over n elements, and $\text{sgn}(\pi)$ is the parity of permutation π .

Proof. For notational simplicity, let us call $\tilde{H} := H - \mu$, and

$$G_{2n}(\underline{\mathbf{X}}, \underline{\mathbf{X}}') := \langle \mathbf{T} a_{\underline{\mathbf{x}}_1} \cdots a_{\underline{\mathbf{x}}_n} a_{\underline{\mathbf{x}}'_1}^* \cdots a_{\underline{\mathbf{x}}'_n}^* \rangle_{\beta, L} .$$

In the proof of Proposition 2.3.2 we saw that g is the unique antiperiodic solution of the equation

$$(\partial_{x_0} + \tilde{H})g(\underline{\mathbf{x}}, \underline{\mathbf{x}}') = \delta^2(\underline{\mathbf{x}}, \underline{\mathbf{x}}') , \quad (2.3.15)$$

with $\delta^2(\underline{\mathbf{x}}, \underline{\mathbf{x}}') \equiv \delta(x_0 - x'_0)\delta_{\underline{\mathbf{x}}, \underline{\mathbf{x}}'}$. By linearity, one may check that the equation

$$(\partial_{x_0} + \tilde{H})G(\underline{\mathbf{x}}, \underline{\mathbf{X}}') = \sum_{i=1}^n c_i \delta^2(\underline{\mathbf{x}}, \underline{\mathbf{x}}'_i) . \quad (2.3.16)$$

has $\sum_i c_i g(\underline{\mathbf{x}}; \underline{\mathbf{x}}'_i)$ as the unique antiperiodic solution. We claim that G_{2n} , seen as a function of $\underline{\mathbf{x}}_1$, satisfies (2.3.16) for a suitable choice of $\{c_i\}_i$. Recall that:

$$\mathbf{T} a_{\underline{\mathbf{x}}_1}^{\#1} \cdots a_{\underline{\mathbf{x}}_m}^{\#m} = \sum_{\pi \in S_m} \text{sgn}(\pi) \Theta(x_{0, \pi(1)} - x_{0, \pi(2)}) \cdots \Theta(x_{0, \pi(m-1)} - x_{0, \pi(m)}) a_{\underline{\mathbf{x}}_{\pi(1)}}^{\#_{\pi(1)}} \cdots a_{\underline{\mathbf{x}}_{\pi(m)}}^{\#_{\pi(m)}} , \quad (2.3.17)$$

with Θ being the Heaviside theta function. The operator $(\partial_{x_{0,1}} + \tilde{H})$, when acting on the Θ functions, produces anticommutators between $a_{\underline{\mathbf{x}}_1}$ and one of the $a_{\underline{\mathbf{x}}'_i}^*$, $i = 1, \dots, n$. One has:

$$\begin{aligned} (\partial_{x_{0,1}} + \tilde{H})G_{2n}(\underline{\mathbf{X}}, \underline{\mathbf{X}}') &= \sum_{i=1}^n \delta^2(\underline{\mathbf{x}}_1 - \underline{\mathbf{x}}'_i) (-1)^{n-1+i} \langle \mathbf{T} a_{\underline{\mathbf{x}}_2} \cdots a_{\underline{\mathbf{x}}_1}^* \cdots a_{\underline{\mathbf{x}}'_{i-1}}^* a_{\underline{\mathbf{x}}'_{i+1}}^* \cdots a_{\underline{\mathbf{x}}'_n}^* \rangle_{\beta, L} \\ &\equiv \sum_{i=1}^n \delta^2(\underline{\mathbf{x}}_1 - \underline{\mathbf{x}}'_i) (-1)^{n-1+i} G_{2n-2}(\underline{\mathbf{x}}_2, \dots, \underline{\mathbf{x}}_n, \underline{\mathbf{x}}'_1, \dots, \widehat{\underline{\mathbf{x}}'_i}, \dots, \underline{\mathbf{x}}'_n) . \end{aligned} \quad (2.3.18)$$

where the wide hat indicates omission. Therefore, by (2.3.16) we obtain:

$$G_{2n}(\underline{\mathbf{X}}, \underline{\mathbf{X}}') = \sum_{i=1}^n (-1)^{n-1+i} g(\underline{\mathbf{x}}_1; \underline{\mathbf{x}}'_i) G_{2n-2}(\underline{\mathbf{X}} \setminus \{\underline{\mathbf{x}}_1\}, \underline{\mathbf{X}}' \setminus \{\underline{\mathbf{x}}'_i\}) . \quad (2.3.19)$$

We can now repeat the argument for G_{2n-2} , and so on, until exhaustion of the $\underline{\mathbf{X}}$ variables. This proves the claim. $\square$

Remark 2.3.5. In the Wick-rotated Duhamel expansion, we will need to estimate n -point correlation functions, with n arbitrarily large, and later on, when discussing perturbation theory for interacting fermions, we will need the same. The Wick rule hints at the fact that these correlations might grow ‘factorially’ in the number of points, due to the number of addends in equation (2.3.14): this could be the source of a combinatorial problem for the Duhamel and the perturbative expansions, that could imply the failure for both.

The solution of this combinatorial problem will ultimately rely on the determinant structure of (2.3.14): indeed, we may rewrite it as

$$\langle \mathbf{T} a_{\underline{\mathbf{x}}_1} \cdots a_{\underline{\mathbf{x}}_n} a_{\underline{\mathbf{x}}'_1}^* \cdots a_{\underline{\mathbf{x}}'_n}^* \rangle_{\beta, L} = \det \mathfrak{G}_{2n}(\underline{\mathbf{X}}, \underline{\mathbf{X}}') , \quad \mathfrak{G}_{2n}(\underline{\mathbf{X}}, \underline{\mathbf{X}}')_{i,j} := g(\underline{\mathbf{x}}_i, \underline{\mathbf{x}}'_j) .$$

Determinants can be estimated in a much more efficient way than just counting the number of entries of the matrix.

2.3.2 Estimates for the two-point function

Since Proposition 2.3.2 relates the form of the two-point function to the single-particle Hamiltonian H , we expect to be able to determine decay properties for g by looking at the spectral feature of H .

Proposition 2.3.6 (Explicit form of the propagator). *Let $x_0, y_0 \in [0, \beta)$. Then:*

$$g(\underline{\mathbf{x}}; \underline{\mathbf{y}}) = \begin{cases} \frac{e^{-(x_0-y_0)(H-\mu)}}{1 + e^{-\beta(H-\mu)}}(\underline{\mathbf{x}}; \underline{\mathbf{y}}) & x_0 > y_0 \\ \frac{e^{-(x_0-y_0)(H-\mu)}}{1 + e^{\beta(H-\mu)}}(\underline{\mathbf{x}}; \underline{\mathbf{y}}) & y_0 \geq x_0 . \end{cases} \quad (2.3.20)$$

Proof. Suppose that $x_0 > y_0$, and call $\tau := x_0 - y_0 > 0$. We have then the following representation:

$$\frac{e^{-\tau(H-\mu)}}{1 + e^{-\beta(H-\mu)}} = \int_{\tilde{\Gamma}} \frac{dz}{2\pi i} \frac{e^{-\tau z}}{1 + e^{-\beta z}} \frac{1}{z - (H - \mu)} \quad (2.3.21)$$

where $\tilde{\Gamma}$ is a rectangle encircling the spectrum of $H - \mu$, crossing the real axis at $z = \pm\gamma$, with $\gamma > \|H - \mu\|$, and crossing the imaginary axis at $z = \pm\epsilon$, with $\epsilon < \pi/\beta$. Employing the fact that the integrand has additional poles at for $e^{-\beta z} = -1$, i.e. for $z \in i\mathbb{M}_\beta$, we are led to

$$\frac{e^{-\tau(H-\mu)}}{1 + e^{-\beta(H-\mu)}} = \frac{1}{\beta} \sum_{\substack{k_0 \in \mathbb{M}_\beta \\ |k_0| \leq K}} \frac{e^{ik_0\tau}}{ik_0 + H - \mu} + \int_{\Gamma} \frac{dz}{2\pi i} \frac{e^{-\tau z}}{1 + e^{-\beta z}} \frac{1}{z - (H - \mu)} ; \quad (2.3.22)$$

here, the closed contour $\Gamma \subset \mathbb{C}$ (see Fig. 2.2) is a rectangle encircling the spectrum of $H - \mu$, such that it intersects the imaginary axis at $z = \pm iK$, the real axis at $z = \pm\gamma$, and it stays away from the poles of the integrand.

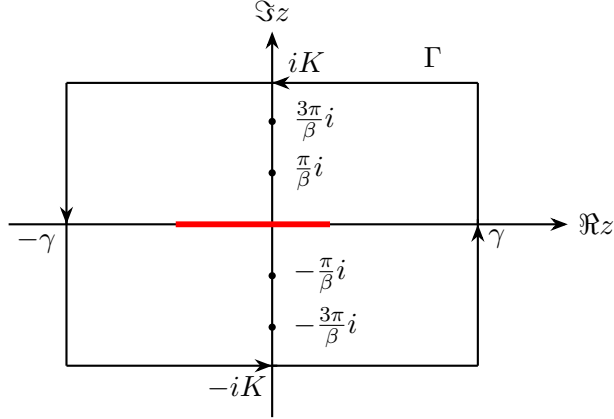


Figure 2.2: Schematic representation of the contour $\Gamma \subset \mathbb{C}$, with some Matsubara frequencies drawn. Since L is finite, the spectrum of $H - \mu$ is discrete: we highlight in red an interval containing it.

Recalling Proposition 2.3.2, we then need to show that the last term vanishes as $K \rightarrow \infty$.

We shall take first the $\gamma \rightarrow \infty$ limit, and then the $K \rightarrow \infty$ limit. Let Γ_3, Γ_4 be the parts of the contour Γ with $\Re z = \gamma, -\gamma$, respectively. These contributions vanish thanks to the presence of the factors $e^{-\tau\gamma}$, for Γ_3 , and of $e^{(\tau-\beta)\gamma}$ for Γ_4 (recall that $0 < \tau < \beta$).

Let now Γ_1, Γ_2 be the parts of the contour Γ with $\Im z = \pm iK$. The z integrals converge uniformly in γ . In fact, for $\Re z \geq 0$ one makes use of the factor $e^{-\tau\Re z}$, while for $\Re z < 0$ one makes use of the factor $e^{(\beta-\tau)\Re z}$. Then, the bound

$$\|(\pm iK - (H - \mu))^{-1}\| \leq K^{-1} \quad (2.3.23)$$

is used to show that the contribution of Γ_1, Γ_2 vanishes as $K \rightarrow \infty$.

The case for $\tau < 0$ is analogous.

Finally, the claim at equal times, *i.e.* for $\tau = 0$, follows from the definition of the extension of g to all imaginary times in $\mathbb{R}$. $\square$

Proposition 2.3.7 (Space-time decay). *There exist positive constants C_β, c_β such that, for all $\underline{x}, \underline{y}$,*

$$|g(\underline{x}; \underline{y})| \leq C_\beta e^{-c_\beta \|\underline{x} - \underline{y}\|_{\beta, L}}. \quad (2.3.24)$$

where $\|\underline{x} - \underline{y}\|_{\beta, L}^2 := |x_0 - y_0|_\beta^2 + \|x - y\|_L^2$ is the space-time distance.

Moreover, suppose that μ lies in a gap of $\sigma(H)$, *i.e.* there exists an interval $I \subset \mathbb{R}$, independent of L , such that $\mu \in I$ and $I \cap \sigma(H) = \emptyset$. Then, the above constants can be chosen to be independent of β :

$$|g(\underline{x}; \underline{y})| \leq C e^{-c \|\underline{x} - \underline{y}\|_{\beta, L}}. \quad (2.3.25)$$

Proof. Let us call $\tau := x_0 - y_0$ and let us suppose $\tau > 0$. Then, by (2.3.20) we obtain:

$$g(\underline{x}; \underline{y}) = \frac{e^{-\tau(H-\mu)}}{1 + e^{-\beta(H-\mu)}}(\underline{x}; \underline{y}) = \int_{\Gamma} \frac{dz}{2\pi i} \frac{e^{-\tau(z-\mu)}}{1 + e^{-\beta(z-\mu)}} \frac{1}{z - H}(\underline{x}; \underline{y}), \quad (2.3.26)$$

where the path $\Gamma \subset \mathbb{C}$ is chosen so that it encircles the spectrum $\sigma(H)$, at a distance $\pi/(2\beta)$. In this way, the only poles contained in Γ are those associated to the resolvent:

the zeroes of $1 - e^{-\beta(z-\mu)}$ are $z = \mu + i\frac{2\pi}{\beta}(n + \frac{1}{2})$, $n \in \mathbb{Z}$; hence, recalling the norm on the circle

$$|\tau|_\beta := \min\{|\tau|, |\beta - \tau|\},$$

we may estimate

$$\left| \frac{e^{-\tau(z-\mu)}}{1 + e^{-\beta(z-\mu)}} \right| \leq K_\beta e^{-c_\beta |\tau|_\beta}, \quad (2.3.27)$$

for suitable β -dependent positive constants K_β , c_β . Additionally, the Combes-Thomas estimate (see *e.g.* [1]) for the resolvent in (2.3.24) yields exponential decay in space:

$$\left| (z - H)^{-1}(\mathbf{x}; \mathbf{y}) \right| \leq K'_\beta e^{-c'_\beta \|x-y\|}. \quad (2.3.28)$$

The combination of (2.3.27)-(2.3.28) proves (2.3.24).

Now, suppose that μ lies in a spectral gap of H , then

$$\sigma(H) = \sigma_-(H) \cup \sigma_+(H),$$

σ_- being the part of the spectrum below μ , and σ_+ being the part above. Let Γ_-, Γ_+ be two disjoint contours encircling $\sigma_-(H)$ and $\sigma_+(H)$, respectively. We choose them so that the distance separating them from the spectrum is of the order of the spectral gap, *e.g.* $|I|/4$.

We then write

$$\frac{e^{-\tau(H-\mu)}}{1 + e^{-\beta(H-\mu)}} = \int_{\Gamma_+} \frac{dz}{2\pi i} \frac{e^{-\tau(z-\mu)}}{1 + e^{-\beta(z-\mu)}} \frac{1}{z - H} + \int_{\Gamma_-} \frac{dz}{2\pi i} \frac{e^{-\tau(z-\mu)}}{1 + e^{-\beta(z-\mu)}} \frac{1}{z - H}. \quad (2.3.29)$$

In both integrals, the resolvent kernel $(z - H)^{-1}(\mathbf{x}; \mathbf{y})$ decays exponentially in $\|x - y\|_L$, uniformly in z as a consequence of the Combes-Thomas estimate. Also, there exist β -independent constants such that

$$\begin{aligned} \left| \frac{e^{-\tau(z-\mu)}}{1 + e^{-\beta(z-\mu)}} \right| &\leq C e^{-\tau(\Re z - \mu)} \leq K e^{-c\tau}, & \text{for } z \in \Gamma_+, \\ \left| \frac{e^{-\tau(z-\mu)}}{1 + e^{-\beta(z-\mu)}} \right| &\leq C e^{(\tau-\beta)|\Re z - \mu|} \leq K e^{-c(\beta-\tau)}, & \text{for } z \in \Gamma_-. \end{aligned}$$

Recalling again the definition of the norm $|\cdot|_\beta$ on $\mathbb{T}_\beta$, we are led to the claim. One may proceed analogously for $\tau := x_0 - y_0 < 0$. $\square$

Remark 2.3.8. Notice, that this result implies that, for $\beta < +\infty$, the non-interacting fermionic two-point g decays exponentially, with c_β playing the role of a thermal correlation length; in particular, it follows that g is in $L^1(\mathbb{T}_\beta \times \Lambda_L)$ *uniformly* in the lattice volume $|\Gamma_L| = L^d$. However, for β large, the constants C_β and c_β behave linearly in β , therefore

$$\|g\|_{L^1(\mathbb{T}_\beta \times \Lambda_L)} \leq C_\beta c_\beta^{d+1} \propto \beta^{d+2},$$

hence this bound is not good enough at low temperatures. Nevertheless, for non-interacting *gapped* systems the second inequality implies a stronger result, namely

$$\|g\|_{L^1(\mathbb{T}_\beta \times \Lambda_L)} \leq C c^{d+1} \equiv \tilde{C},$$

for all β, L .

2.4 The interacting Gibbs state

In this section, we will set λ to be non-zero, and we will write the grand-canonical Hamiltonian as the sum of its non-interacting part $\mathcal{H}_0$ plus the quartic interaction $\mathcal{V}$:

$$\mathcal{H} = \mathcal{H}_0 + \lambda \mathcal{V} ; \quad (2.4.1)$$

furthermore, we shall denote by $\gamma_t(\mathcal{O})$ the imaginary-time evolution associated to $\mathcal{H}$, and by $\mathcal{O}_t$ the corresponding imaginary-time evolution generated by $\mathcal{H}_0$, namely

$$\mathcal{O}_t := e^{t(\mathcal{H}_0 - \mu \mathcal{N})} \mathcal{O} e^{-t(\mathcal{H}_0 - \mu \mathcal{N})} . \quad (2.4.2)$$

In what follows, we shall set up a perturbative expansion for the Euclidean correlations appearing in the Duhamel expansion, by employing Wick's rule, we shall highlight some critical issues with the same, and we shall hint at their solution.

2.4.1 Perturbation theory

The starting point of perturbation theory is an interpolation of the interacting Gibbs state with respect to the non-interacting one.

Lemma 2.4.1 (Expansion for the Gibbs state). *For any finite β, L , and for any time $t \in (0, \beta]$, the map*

$$\mathbb{C} \rightarrow \mathcal{B}(\mathcal{F}_L) \quad \lambda \mapsto e^{t\mathcal{H}_0} e^{-t\mathcal{H}}$$

is an entire function, and it admits the following expansion:

$$e^{t\mathcal{H}_0} e^{-t\mathcal{H}} = \mathbb{1} + \sum_{k \geq 1} \frac{(-\lambda)^k}{k!} \int_{[0,t]^k} d\mathbf{s} \, \mathbf{T} \mathcal{V}_{s_1} \cdots \mathcal{V}_{s_k} . \quad (2.4.3)$$

Proof. The starting point is the following identity:

$$\begin{aligned} e^{t\mathcal{H}_0} e^{-t\mathcal{H}} &= \mathbb{1} + \int_0^t ds \frac{d}{ds} e^{s\mathcal{H}_0} e^{-s\mathcal{H}} \\ &= \mathbb{1} + \int_0^t ds e^{s\mathcal{H}_0} (\mathcal{H}_0 - \mathcal{H}) e^{-s\mathcal{H}} \\ &= \mathbb{1} - \lambda \int_0^t ds \mathcal{V}_s e^{s\mathcal{H}_0} e^{-s\mathcal{H}} . \end{aligned}$$

where in the last passage we employed the definition of $\mathcal{V}_s$. Iterating the expression above, one obtains

$$\begin{aligned} e^{t\mathcal{H}_0} e^{-t\mathcal{H}} &= \mathbb{1} + \sum_{k \geq 1} (-\lambda)^k \int_0^t ds_1 \int_0^{s_1} ds_2 \cdots \int_0^{s_{k-1}} ds_k \mathcal{V}_{s_1} \mathcal{V}_{s_2} \cdots \mathcal{V}_{s_k} \\ &= \mathbb{1} + \sum_{k \geq 1} \frac{(-\lambda)^k}{k!} \int_{[0,t]^k} ds_1 \cdots ds_k \mathbf{T} \mathcal{V}_{s_1} \cdots \mathcal{V}_{s_k} . \end{aligned} \quad (2.4.4)$$

Notice that, for finite L , the left-hand side is an entire function of λ . Equivalently, the series on the right-hand side is absolutely convergent for any λ ; to see this, notice first that, for all $\Psi \in \mathcal{F}_L$,

$$\begin{aligned} \|\mathcal{V}_s \Psi\|^2 &= \langle e^{-s\mathcal{H}_0} \Psi, \mathcal{V}_s^* \mathcal{V}_s e^{-s\mathcal{H}_0} \Psi \rangle \\ &\leq \|e^{-s\mathcal{H}_0} \Psi\|^2 \|\mathcal{V}\|^2 \\ &\leq e^{2s\|\mathcal{H}_0\|} \|\Psi\|^2 \|\mathcal{V}\|^2 , \end{aligned}$$

hence $\|\mathcal{V}_s\| \leq \exp(s\|\mathcal{H}_0\|)\|\mathcal{V}\|$. Also, thanks to the boundedness of the fermionic operators, and since $\mathcal{H}_0$ and $\mathcal{V}$ have short-range kernels H and w , we get:

$$\|\mathcal{H}_0\| \leq \sum_{\mathbf{x}, \mathbf{y}} |H(\mathbf{x}, \mathbf{y})| \leq C|\Lambda_L|, \quad \|\mathcal{V}\| \leq \sum_{\mathbf{x}, \mathbf{y}} |w(\mathbf{x}, \mathbf{y})| \leq C|\Lambda_L|,$$

Therefore, we can estimate the k -th order in the right-hand side of (2.4.4) as:

$$|\lambda|^k \int_0^t ds_1 \int_0^{s_1} ds_2 \cdots \int_0^{s_{k-1}} ds_k e^{C|\Lambda_L|(s_1+\cdots+s_k)} (C|\Lambda_L|)^k \leq \frac{|\lambda|^k}{k!} (C\beta|\Lambda_L|e^{C\beta|\Lambda_L|})^k. \quad (2.4.5)$$

where we used $0 < t \leq \beta$ to obtain the right-hand side. This proves the convergence of the expansion for any $\lambda \in \mathbb{C}$, at finite β, L . $\square$

Definition 2.4.2 (Partition function). We define the grand-canonical partition functions of the free and the interacting models to be

$$\mathcal{Z}_{\beta,L}^0 = \text{Tr} e^{-\beta(\mathcal{H}_0 - \mu N)} \quad \mathcal{Z}_{\beta,L}^\lambda := \text{Tr} e^{-\beta(\mathcal{H} - \mu N)}.$$

We will call their ratio the relative partition function:

$$\zeta_{\beta,L}^\lambda := \frac{\mathcal{Z}_{\beta,L}^\lambda}{\mathcal{Z}_{\beta,L}^0}.$$

Since $\mathcal{Z}_{\beta,L}^0 > 0$, this ratio is well-defined and, according to the previous lemma, we see that $\lambda \mapsto \zeta_{\beta,L}^\lambda$ is an entire function.

Employing Lemma 2.4.1, we now obtain an expansion for the Euclidean correlation functions of the interacting model in terms of the correlations of the non-interacting one.

Proposition 2.4.3 (Interpolation of correlations). *The following points are true.*

1. For any $\lambda \in \mathbb{C}$, the relative partition function is given by the following expansion:

$$\zeta_{\beta,L}^\lambda = 1 + \sum_{k \geq 1} \frac{(-\lambda)^k}{k!} \int_{\mathbb{T}_\beta^k} d\mathbf{s} \langle \mathbf{T} \mathcal{V}_{s_1} \cdots \mathcal{V}_{s_k} \rangle_{\beta,L}^0. \quad (2.4.6)$$

where $\langle \cdot \rangle_{\beta,L}^0$ denotes the expectation with respect to the non-interacting Gibbs state.

2. There exists $\bar{\lambda}_{\beta,L} > 0$ such that, for any $|\lambda| < \bar{\lambda}_{\beta,L}$, for any collection of charge-preserving observables $\{\mathcal{O}_i\}_{i=1}^n$, and for any choice of times $\{t_i\}_i$, we have:

$$\langle \mathbf{T} \gamma_{t_1}(\mathcal{O}_1) \cdots \gamma_{t_n}(\mathcal{O}_n) \rangle_{\beta,L} = \frac{1}{\zeta_{\beta,L}^\lambda} \sum_{k \geq 0} \frac{(-\lambda)^k}{k!} \int_{\mathbb{T}_\beta^k} d\mathbf{s} \langle \mathbf{T} \mathcal{V}_{s_1} \cdots \mathcal{V}_{s_k} \mathcal{O}_{1,t_1} \cdots \mathcal{O}_{n,t_n} \rangle_{\beta,L}^0. \quad (2.4.7)$$

Proof. Since both $\mathcal{H}_0$ and $\mathcal{H}$ are number-preserving, by Lemma 2.4.1 and by linearity of the expectation $\langle \cdot \rangle_{\beta,L}^0$, it is easy to check that the denominator is given by

$$\begin{aligned} \mathcal{Z}_{\beta,L}^\lambda &= \text{Tr}[e^{-\beta\mathcal{H}} e^{\beta\mu N}] = \mathcal{Z}_{\beta,L}^0 \langle e^{\beta\mathcal{H}_0} e^{-\beta\mathcal{H}} \rangle_{\beta,L}^0 \\ &= \mathcal{Z}_{\beta,L}^0 \left[1 + \sum_{k \geq 1} \frac{(-\lambda)^k}{k!} \int_{[0,\beta]^k} d\mathbf{s} \langle \mathbf{T} \mathcal{V}_{s_1} \cdots \mathcal{V}_{s_k} \rangle_{\beta,L}^0 \right]. \end{aligned}$$

This proves that $\zeta_{\beta,L}^\lambda$ is given by equation (2.4.6). While $\zeta_{\beta,L}^\lambda$ is non-zero for real λ , it might vanish for some λ complex; nevertheless, the convergent expansion (2.4.6) implies that there exists a β, L -dependent positive constant $\bar{\lambda}_{\beta,L}$ such that

$$\zeta_{\beta,L}^\lambda \neq 0 \quad \forall |\lambda| < \bar{\lambda}_{\beta,L} . \quad (2.4.8)$$

Now, we turn to the proof of (2.4.7). Without loss of generality, we may assume the times to be such that $t_1 \geq t_2 \geq \dots \geq t_n$. Then, by definition of expectation, we have

$$\langle \mathbf{T} \gamma_{t_1}(\mathcal{O}_1) \cdots \gamma_{t_n}(\mathcal{O}_n) \rangle_{\beta,L} = \frac{1}{\zeta_{\beta,L}^\lambda \mathcal{Z}_{\beta,L}^0} \text{Tr}[e^{-\beta(\mathcal{H}-\mu\mathcal{N})} \gamma_{t_1}(\mathcal{O}_1) \cdots \gamma_{t_n}(\mathcal{O}_n)] \quad (2.4.9)$$

which, thanks to (2.4.8), is well defined for $|\lambda| < \bar{\lambda}_{\beta,L}$. Let us define

$$\mathcal{J}(t, t') := e^{t\mathcal{H}_0} e^{-(t-t')\mathcal{H}} e^{-t'\mathcal{H}_0} .$$

It satisfies the Cauchy problem

$$\partial_t \mathcal{J}(t, t') = -\lambda \mathcal{V}_t \mathcal{J}(t, t') , \quad \mathcal{J}(t', t') = \mathbb{1} , \quad (2.4.10)$$

hence proceeding as in Lemma 2.4.1 one obtains

$$\mathcal{J}(t, t') = \sum_{k \geq 0} \frac{(-\lambda)^k}{k!} \int_{(t', t)^k} d\mathbf{s} \, \mathbf{T} \mathcal{V}_{s_1} \cdots \mathcal{V}_{s_k} , \quad (2.4.11)$$

i.e. $\mathcal{J}(t, t')$ is given only by operators evolved at times in the interval (t', t) . Combining this property with the group composition law

$$\mathcal{J}(t, t') \mathcal{J}(t', t'') = \mathcal{J}(t, t'') ,$$

we see that

$$\mathbf{T} \mathcal{J}(t, t') \mathcal{O}_{\bar{t}} = \mathcal{J}(t, \bar{t}) \mathcal{O}_{\bar{t}} \mathcal{J}(\bar{t}, t') \quad (2.4.12)$$

for any $\bar{t} \in (t', t)$. Using (2.4.12), the argument of the trace in (2.4.9) can be written as

$$\begin{aligned} e^{-\beta(\mathcal{H}-\mu\mathcal{N})} \gamma_{t_1}(\mathcal{O}_1) \cdots \gamma_{t_n}(\mathcal{O}_n) &= e^{-\beta(\mathcal{H}_0-\mu\mathcal{N})} \mathcal{J}(\beta, t_1) \mathcal{O}_{1,t_1} \mathcal{J}(t_1, t_2) \cdots \mathcal{J}(t_{n-1}, t_n) \mathcal{O}_{n,t_n} \mathcal{J}(t_n, 0) \\ &= e^{-\beta(\mathcal{H}_0-\mu\mathcal{N})} \mathbf{T} \mathcal{J}(\beta, 0) \mathcal{O}_{1,t_1} \cdots \mathcal{O}_{n,t_n} , \end{aligned}$$

so that, by (2.4.11),

$$\begin{aligned} e^{-\beta(\mathcal{H}-\mu\mathcal{N})} \gamma_{t_1}(\mathcal{O}_1) \cdots \gamma_{t_n}(\mathcal{O}_n) &= e^{-\beta(\mathcal{H}_0-\mu\mathcal{N})} \mathbf{T} \mathcal{J}(\beta, 0) \mathcal{O}_{1,t_1} \cdots \mathcal{O}_{n,t_n} \\ &= \sum_{k \geq 0} \frac{(-\lambda)^k}{k!} \int_{\mathbb{T}_\beta^k} d\mathbf{s} \, e^{-\beta(\mathcal{H}_0-\mu\mathcal{N})} \mathbf{T} \mathcal{V}_{s_1} \cdots \mathcal{V}_{s_k} \mathcal{O}_{1,t_1} \cdots \mathcal{O}_{n,t_n} . \end{aligned}$$

Inserting it in (2.4.9), we are led to equation (2.4.7). $\square$

Remark 2.4.4 (On the analyticity of correlations). Since (2.4.7) expresses the interacting correlation functions as the ratio of an entire function and the relative partition function, their domain of analyticity certainly contains the set where $\zeta_{\beta,L} \neq 0$. In the proof, we showed that this is true for a disk of radius $\bar{\lambda}_{\beta,L}$. In many cases, such as the one considered in Chapter 4, a Renormalisation Group analysis allows to prove that the relative free energy, defined as

$$f_{\beta,L}(\lambda) := -\frac{1}{\beta|\Lambda_L|} \log \zeta_{\beta,L}^\lambda ,$$

is analytic in a disk of radius $\bar{\lambda}$ independent of β, L . This implies that the correlation functions are analytic for $|\lambda| < \bar{\lambda}$, uniformly in $\beta, L \rightarrow \infty$.

Remark 2.4.5. The formula (2.4.7) is very useful, as it allows us to employ Wick's rule, stated for the non-interacting Gibbs state in Theorem 2.3.4, on the right-hand side of the same.

Since in the end we are also interested in integrating over the imaginary times $\{t_i\}_i$, we obtain

$$\int_{\mathbb{T}_\beta^n} d\underline{t} \langle \mathbf{T} \gamma_{t_1}(\mathcal{O}_1) \cdots \gamma_{t_n}(\mathcal{O}_n) \rangle_{\beta,L} = \frac{1}{\zeta_{\beta,L}^\lambda} \sum_{k \geq 0} \frac{(-\lambda)^k}{k!} \int_{\mathbb{T}_\beta^{k+n}} d\underline{s} \langle \mathbf{T} \mathcal{V}_{s_1} \cdots \mathcal{O}_{n,s_{k+n}} \rangle_{\beta,L}^0. \quad (2.4.13)$$

2.4.2 Diagrammatic representation of perturbation theory

The discussion in the previous section did not use any of the features of the model, in addition to the CAR and the short range of H , w . In order to improve the above estimate on the analyticity of the correlations functions, we will try to exploit information on the decay of the two-point function. To do so, it is convenient to use the Wick rule to reduce the right-hand side of (2.4.13) to suitable combinations of products of propagators g .

Without loss of generality, let us assume that $\mathcal{O}_i$ is a homogeneous normal-ordered⁶ polynomial of degree $2m_i$, that is

$$\mathcal{O}_i = \sum_{\substack{\mathbf{X}=\{\mathbf{x}_j\} \\ \mathbf{Y}=\{\mathbf{y}_j\}}} O_i(\mathbf{X}, \mathbf{Y}) a_{\mathbf{x}_1}^* \cdots a_{\mathbf{x}_{m_i}}^* a_{\mathbf{y}_1} \cdots a_{\mathbf{y}_{m_i}}.$$

Consider $\int_{\mathbb{T}_\beta^{k+n}} d\underline{s} \langle \mathbf{T} \mathcal{V}_{s_1} \cdots \mathcal{O}_{n,s_{k+n}} \rangle_{\beta,L}^0$; we may rewrite it as:

$$\begin{aligned} \int_{\mathbb{T}_\beta^{k+n}} d\underline{s} \langle \mathbf{T} \mathcal{V}_{s_1} \cdots \mathcal{O}_{n,s_{k+n}} \rangle_{\beta,L}^0 &= \int \prod_{i=1}^k d\underline{\mathbf{x}}_i d\underline{\mathbf{y}}_i \frac{w(\underline{\mathbf{x}}_i; \underline{\mathbf{y}}_i)}{2} \prod_{i'=1}^n d\underline{\mathbf{x}}_{i'} d\underline{\mathbf{y}}_{i'} O_{i'}(\underline{\mathbf{x}}_{i'}; \underline{\mathbf{y}}_{i'}) \\ &\quad \cdot \langle \mathbf{T} a_{\underline{\mathbf{x}}_1}^* a_{\underline{\mathbf{y}}_1}^* a_{\underline{\mathbf{y}}_1} a_{\underline{\mathbf{x}}_1} \cdots a_{\underline{\mathbf{x}}_n}^* a_{\underline{\mathbf{y}}_n} \rangle_{\beta,L}^0. \end{aligned} \quad (2.4.14)$$

where we employed the short-hands

$$\int d\underline{\mathbf{x}} \equiv \int_0^\beta dx_0 \sum_{\mathbf{x} \in \Lambda_L}, \quad \int d\underline{\mathbf{x}}_i \equiv \prod_{j=1}^{m_i} \int d\underline{\mathbf{x}}_{i,j}, \quad a_{\underline{\mathbf{x}}_i}^\# = \prod_{j=1}^{m_i} a_{\underline{\mathbf{x}}_{i,j}}^\#,$$

and we set

$$\begin{aligned} w(\underline{\mathbf{x}}; \underline{\mathbf{y}}) &:= \delta(x_0 - y_0) w(\mathbf{x}; \mathbf{y}), \\ O(\underline{\mathbf{X}}; \underline{\mathbf{Y}}) &:= \prod_{j=2}^m \delta(x_{1,0} - x_{j,0}) \prod_{j=1}^m \delta(x_{1,0} - y_{j,0}) O(\mathbf{X}; \mathbf{Y}). \end{aligned} \quad (2.4.15)$$

Applying the Wick rule, one gets the following graphical expansion for the summands in the numerator of (2.4.13),

$$(-\lambda)^k \int_{\mathbb{T}_\beta^{k+n}} d\underline{s} \langle \mathbf{T} \mathcal{V}_{s_1} \cdots \mathcal{O}_{n,s_{k+n}} \rangle_{\beta,L}^0 = \sum_{G \in \mathcal{G}_{k;n}} \text{Val}(G), \quad (2.4.16)$$

where $\mathcal{G}_{k;n}$ is the set of all labelled graphs with oriented edges G , often called *Feynman diagrams*, that can be obtained via the following rules:

⁶Any non normal-ordered homogeneous polynomial can be reduced to a (non-homogeneous) normal-ordered one of the same degree.

1. G has $k + n$ vertices, k of which we call of λ -type, and n of which we call of $\mathcal{O}$ -type;
 - (a) each vertex of λ -type is associated to an interaction term $-\lambda w(\underline{\mathbf{x}}; \underline{\mathbf{y}}) a_{\underline{\mathbf{x}}}^* a_{\underline{\mathbf{y}}}^* a_{\underline{\mathbf{y}}} a_{\underline{\mathbf{x}}}$, so we decorate it with the label $-\lambda w(\underline{\mathbf{x}}; \underline{\mathbf{y}}) a_{\underline{\mathbf{x}}}^*$;
 - (b) each vertex of $\mathcal{O}$ -type is associated to an observable term $O(\underline{\mathbf{X}}; \underline{\mathbf{Y}}) a_{\underline{\mathbf{X}}}^* a_{\underline{\mathbf{Y}}}$, so we associate to it the label $O(\underline{\mathbf{X}}; \underline{\mathbf{Y}})$;
2. calling $m = \sum_{i=1}^n m_i$, G has $4k + 2m$ oriented half-edges, distributed and labelled as follows:
 - (a) each λ -vertex has four half-edges; two of which are oriented as exiting the vertex, and two as entering it; the half-edges are labelled by $\underline{\mathbf{x}}$ or $\underline{\mathbf{y}}$ in a way that no half-edge with the same orientation has the same label;
 - (b) each $\mathcal{O}_i$ -vertex has m_i outgoing half-edges, labelled by $\{\underline{\mathbf{x}}_{i,j}\}_{j=1}^{m_i}$ and m_i ingoing ones, labelled by $\{\underline{\mathbf{y}}_{i,j}\}_{j=1}^{m_i}$;
3. the set of edges $E(G)$ is obtained by contracting all the half-edges, compatibly with their orientations; therefore, G has a total of $2k + m$ oriented edges; each oriented edge $\ell = (\underline{\mathbf{z}}', \underline{\mathbf{z}}) \in E(G)$ is decorated with the propagator $g_\ell \equiv g(\underline{\mathbf{z}}, \underline{\mathbf{z}}')$.

Given a Feynman diagram G (see Figure 2.3 for an example), its value is then defined as

$$\text{Val}(G) = \int \prod_{i=1}^k [d\underline{\mathbf{x}}_i d\underline{\mathbf{y}}_i (-\lambda w(\underline{\mathbf{x}}_i; \underline{\mathbf{y}}_i))] \prod_{i'=1}^n [d\underline{\mathbf{X}}_{i'} d\underline{\mathbf{Y}}_{i'} O_{i'}(\underline{\mathbf{X}}_{i'}; \underline{\mathbf{Y}}_{i'})] \text{sgn}(G) \left[\prod_{\ell \in E(G)} g_\ell \right], \quad (2.4.17)$$

where $\text{sgn}(G)$ is the sign of the permutation arising from the Wick rule (see Equation (2.3.14)).

Remark 2.4.6. Notice that Val is multiplicative: if a Feynman graph G has r connected components $G_1, \dots, G_r$, then

$$\text{Val}(G) = \prod_{i=1}^r \text{Val}(G_i). \quad (2.4.18)$$

Applying the same reasoning for the denominator of (2.4.13), we therefore obtain

$$\int_{\mathbb{T}_\beta^n} d\underline{\mathbf{t}} \langle \mathbf{T} \gamma_{t_1}(\mathcal{O}_1) \cdots \gamma_{t_n}(\mathcal{O}_n) \rangle_{\beta, L} = \frac{\sum_{k \geq 0} \frac{1}{k!} \sum_{G \in \mathcal{G}_{k;n}} \text{Val}(G)}{\sum_{k \geq 0} \frac{1}{k!} \sum_{G \in \mathcal{G}_k} \text{Val}(G)}, \quad (2.4.19)$$

where $\mathcal{G}_k \equiv \mathcal{G}_{k;0}$ is called the set of *vacuum* Feynman diagrams of order k .

On the other hand, we wish to find a diagrammatic expansion for the interacting cumulants, since those are the ones appearing in the (Wick-rotated) Duhamel expansion (2.2.25). In order to do so, the following combinatorial identity is useful.

Proposition 2.4.7 (Linked cluster theorem). *For any $|\lambda| < \bar{\lambda}_{\beta, L}$, it follows that*

$$\sum_{k \geq 0} \frac{1}{k!} \sum_{G \in \mathcal{G}_k} \text{Val}(G) = \exp \left[\sum_{k \geq 1} \frac{1}{k!} \sum_{G \in \mathcal{G}_k^c} \text{Val}(G) \right], \quad (2.4.20)$$

where $\mathcal{G}_k^c \subset \mathcal{G}_k$ is the subset of connected Feynman diagrams.

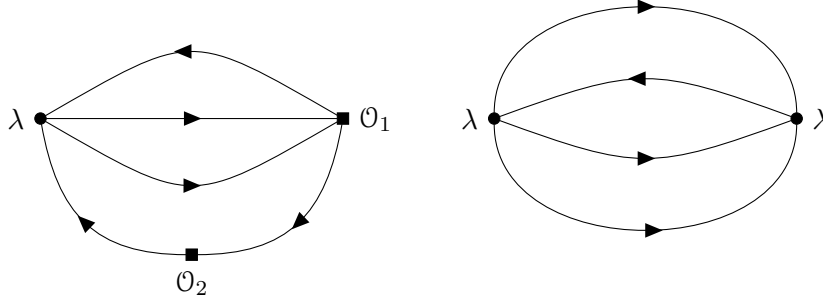


Figure 2.3: Example of Feynman diagram presenting two connected components: the first one has one λ -vertex, an $\mathcal{O}_1$ -vertex with $m_1 = 2$, and an $\mathcal{O}_2$ -vertex with $m_2 = 1$; the second component has two interaction vertices, and therefore it is an example of vacuum diagram.

Proof. Consider a graph $G \in \mathcal{G}_k$, with value $\text{Val}(G)$. Let $\{G_i\}_{i=1}^r$ be its connected components, with G_i having k_i vertices, with $\sum_{i=1}^r k_i = k$ and $k_i \geq 1$. Then, by equation (2.4.18), we may write

$$\sum_{G \in \mathcal{G}_k} \text{Val}(G) = \sum_{r=0}^k \frac{1}{r!} \sum_{\substack{k_1, \dots, k_r \\ \sum_i k_i = k}} \sum_{\substack{* \\ i=1}}^r \prod \sum_{G_i \in \mathcal{G}_{k_i}^c} \text{Val}(G_i), \quad (2.4.21)$$

where $\sum^*$ is the sum over the partitions of the vertices $\{1, \dots, k\}$ into subsets containing $k_1, \dots, k_r$ objects, with the constraint $\sum_{i=1}^r k_i = k$. Then, setting

$$f(k_i) := \sum_{G \in \mathcal{G}_{k_i}^c} \text{Val}(G)$$

equation (2.4.21) transforms into

$$\sum_{G \in \mathcal{G}_k} \text{Val}(G) = \sum_{r=0}^k \frac{1}{r!} \sum_{\substack{k_1, \dots, k_r \\ \sum_i k_i = k}} \binom{k}{k_1 \dots k_r} \prod_i f(k_i). \quad (2.4.22)$$

The right-hand side of the claim then is given by

$$\sum_{k \geq 0} \sum_{G \in \mathcal{G}_k} \text{Val}(G) = \sum_{k \geq 0} \frac{1}{k!} \sum_{r=0}^k \frac{1}{r!} \sum_{\substack{k_1, \dots, k_r \\ \sum_i k_i = k}} \frac{k!}{k_1! \dots k_r!} \prod_{i=1}^r f(k_i). \quad (2.4.23)$$

Finally, we rearrange the sums (notice that the sum over k_i start from 1: otherwise, the i -th connected component would be empty),

$$\sum_{k \geq 0} \sum_{r=0}^k \sum_{\substack{k_1, \dots, k_r \\ \sum_i k_i = k}} (\dots) = \sum_{r \geq 0} \sum_{k_1, \dots, k_r \geq 1} (\dots),$$

to obtain

$$\sum_{k \geq 0} \sum_{G \in \mathcal{G}_k} \text{Val}(G) = \sum_{r \geq 0} \frac{1}{r!} \left[\sum_{k \geq 1} \frac{f(k)}{k!} \right]^r,$$

which is the claim. $\square$

Remark 2.4.8. Actually, one can generalise Proposition 2.4.7 to the case in which Feynman diagrams have n different species of vertices, namely

$$\sum_K \frac{1}{K!} \sum_{G \in \mathcal{G}_K} \text{Val}(G) = \exp \left[\sum_{K \neq (0, \dots, 0)} \frac{1}{K!} \sum_{G \in \mathcal{G}_K^c} \text{Val}(G) \right] \quad (2.4.24)$$

where now the left-hand side summation is over multi-indices $K = (k_1, \dots, k_n)$ of a fixed length n , and $K! := k_1! \cdots k_n!$.

Theorem 2.4.9 (Diagrammatic expansion for cumulants). *For $|\lambda| < \bar{\lambda}_{\beta, L}$, we have:*

$$\int_{\mathbb{T}_\beta^n} d\mathbf{t} \langle \mathbf{T}\gamma_{t_1}(\mathcal{O}_1); \dots; \gamma_{t_n}(\mathcal{O}_n) \rangle_{\beta, L} = \sum_{k \geq 0} \frac{1}{k!} \sum_{G \in \mathcal{G}_{k; n}^c} \text{Val}(G), \quad (2.4.25)$$

where $\mathcal{G}_{k; n}^c$ is the set of connected Feynman graphs.

Proof. Recalling the definition (2.2.17) of cumulant, one can verify that, equivalently

$$\begin{aligned} & \langle \mathbf{T}\gamma_{t_1}(\mathcal{O}_1); \dots; \gamma_{t_n}(\mathcal{O}_n) \rangle_{\beta, L} \\ &= \frac{\partial^n}{\partial \xi_1 \cdots \partial \xi_n} \log \left[\sum_{m_1, \dots, m_n \geq 0} \frac{\xi_1^{m_1} \cdots \xi_n^{m_n}}{m_1! \cdots m_n!} \langle \mathbf{T}\gamma_{t_1}(\mathcal{O}_1^{m_1}) \cdots \gamma_{t_n}(\mathcal{O}_n^{m_n}) \rangle_{\beta, L} \right] \Big|_{\xi=0}. \end{aligned} \quad (2.4.26)$$

We may express the simple correlations appearing in the argument of the logarithm in terms of Feynman diagrams, so that such argument becomes

$$\frac{1}{\zeta_{\beta, L}^\lambda} \sum_{m_1, \dots, m_n} \frac{\xi_1^{m_1} \cdots \xi_n^{m_n}}{m_1! \cdots m_n!} \frac{1}{k!} \sum_{G \in \mathcal{G}_{k; m_1, \dots, m_n}} \text{Val}(G(\mathbf{t})), \quad (2.4.27)$$

where $\mathcal{G}_{k, m_1, \dots, m_n}$ is the set of Feynman diagrams with k vertices of type λ and m_i vertices of type $\mathcal{O}_i$, and $\text{Val}(G(\mathbf{t}))$ is the value of G obtained by not integrating over the times of the observables. Now, by choosing $K = (k, m_1, \dots, m_n)$ as multi-index, we can apply (2.4.24) to (2.4.27), so that (2.4.26) becomes

$$\begin{aligned} & \langle \mathbf{T}\gamma_{t_1}(\mathcal{O}_1); \dots; \gamma_{t_n}(\mathcal{O}_n) \rangle_{\beta, L} \\ &= \frac{\partial^n}{\partial \xi_1 \cdots \partial \xi_n} \log \left\{ \frac{1}{\zeta_{\beta, L}^\lambda} \exp \left[\sum_{\substack{k, m_1, \dots, m_n \\ k + \sum_i m_i \neq 0}} \frac{\xi_1^{m_1} \cdots \xi_n^{m_n}}{m_1! \cdots m_n!} \frac{1}{k!} \sum_{G \in \mathcal{G}_{k, m_1, \dots, m_n}^c} \text{Val}(G(\mathbf{t})) \right] \right\} \Big|_{\xi=0} \\ &= \frac{\partial^n}{\partial \xi_1 \cdots \partial \xi_n} \left[\sum_{\substack{k, m_1, \dots, m_n \\ k + \sum_i m_i \neq 0}} \frac{\xi_1^{m_1} \cdots \xi_n^{m_n}}{m_1! \cdots m_n!} \frac{1}{k!} \sum_{G \in \mathcal{G}_{k, m_1, \dots, m_n}^c} \text{Val}(G(\mathbf{t})) \right] \Big|_{\xi=0}. \end{aligned} \quad (2.4.28)$$

where in the second passage we employed the crucial fact that the relative partition function $\zeta_{\beta, L}^\lambda$ is not zero for $|\lambda| < \bar{\lambda}_{\beta, L}$, hence its logarithm is a finite constant that is annihilated by the derivatives. Now, the only summand that survives is the one of degree 1 in each ξ_i , so that

$$\langle \mathbf{T}\gamma_{t_1}(\mathcal{O}_1); \dots; \gamma_{t_n}(\mathcal{O}_n) \rangle_{\beta, L} = \sum_{k \geq 0} \frac{1}{k!} \sum_{G \in \mathcal{G}_{k; n}^c} \text{Val}(G(\mathbf{t})). \quad (2.4.29)$$

where we used the identification $\mathcal{G}_{k, 1, \dots, 1}^c \equiv \mathcal{G}_{k; n}^c$. Integrating over the Euclidean times $\mathbf{t} = (t_i)_{i=1}^n$, we are led to (2.4.25). $\square$

Remark 2.4.10. A straightforward implication is a diagrammatic expansion for the n -th order of the (Wick rotated) Duhamel expansion, Eq. (2.2.25). In particular, we have

$$\int_{\mathbb{T}_\beta^n} d\underline{s} e^{-i\eta_\beta(s_1+\dots+s_n)} \langle \mathbf{T} \gamma_{s_1}(\mathcal{P}); \gamma_{s_2}(\mathcal{P}); \dots; \gamma_{s_n}(\mathcal{P}); \mathcal{O} \rangle_{\beta,L} = \sum_{k \geq 0} \frac{1}{k!} \sum_{G \in \mathcal{G}_{k,n-1,1}^c} \text{Val}(G) \quad (2.4.30)$$

where $\mathcal{G}_{k,n-1,1}^c$ is the set of Feynman graphs with k interaction vertices, $n-1$ $\mathcal{P}$ -vertices, and one vertex of type $\mathcal{O}$, with the following differences in the evaluation of $\text{Val}(G)$:

1. there is no time integration for the $\mathcal{O}$ -type vertex;
2. supposing $\mathcal{P}$ to be a homogeneous polynomial of degree $2m$, each $\mathcal{P}$ -vertex is associated with a space-time kernel that carries an extra phase:

$$P(\underline{\mathbf{X}}; \underline{\mathbf{Y}}) = e^{-i\eta_\beta x_{1,0}} \prod_{j=2}^m [\delta(x_{1,0} - x_{j',0})] \prod_{j=1}^m [\delta(x_{1,0} - y_{j,0})] P(\underline{\mathbf{X}}; \underline{\mathbf{Y}}) .$$

If we try to provide term-by-term bounds for the diagrammatic expansion (2.4.25), which are uniform in β and L , we are faced immediately with two problems.

Infrared divergences. Given a connected Feynman diagram $G \in \mathcal{G}_{k,n}^c$, *i.e.* involving observables $\mathcal{O}_i$ with $2m_i$ half-edges and k interactions $\mathcal{V}$, we know that its number of edges is $2k + m$, with $m = \sum_{i=1}^n m_i$. At finite β , we also know that Proposition 2.3.7 implies $g \in L^1(\mathbb{T}_\beta \times \Lambda_L) \cap L^\infty(\mathbb{T}_\beta \times \Lambda_L)$.

Therefore, an estimate for its value can be obtained by selecting a tree⁷ sub-graph T_G spanning all $k + n$ vertices, using the propagators g associated to the lines in T_G to perform the integration over the space-time coordinates, and by bounding in L^∞ all the remaining propagators. Since the cardinality of T is $k + n - 1$, and since the observables $\mathcal{O}_i$ and the interaction $\mathcal{V}$ are finite-ranged, we obtain, for some $C_m > 0$,

$$|\text{Val}(G)| \leq \beta |\Lambda_L| C_m^{k+n} |\lambda|^k \|g\|_1^{k+n-1} \|g\|_\infty^{k+m-n+1} \quad (2.4.31)$$

where the factor $\beta |\Lambda_L|$ is due to the integration of the first coordinate along the tree T .

For models in which the single-particle Hamiltonian has a spectral gap uniform in the system size, (2.4.31) allows one to estimate the value of a single diagram uniformly in β, L , while also ensuring smallness in the interaction strength λ .

Instead, for models in which the single-particle Hamiltonian has no such gap, even if one might hope to obtain bounds for $\|g\|_\infty$ which are uniform in β, L , $\|g\|_1$ remains highly divergent in the inverse temperature.

Combinatorial problem. Even if we assume the single-diagram bound (2.4.31) to be uniform in β, L , the insertion of (2.4.31) in (2.4.25) yields

$$\left| \int_{\mathbb{T}_\beta^n} d\underline{t} \langle \mathbf{T} \gamma_{t_1}(\mathcal{O}_1); \dots; \gamma_{t_n}(\mathcal{O}_n) \rangle_{\beta,L} \right| \leq \sum_{k \geq 0} \frac{\tilde{C}^{k+n} |\lambda|^k}{k!} |\mathcal{G}_{k,n}^c| ; \quad (2.4.32)$$

⁷By tree we mean a connected, acyclic graph.

so we are left with the problem of counting the number of connected Feynman graphs of a given order k , that is the cardinality of $\mathcal{G}_{k;n}^c$. It can be shown that such cardinality grows as $(2k + m)!$; this gives rise to a combinatorial problem, since

$$\frac{(2k + m)!}{k!} \sim c^{k+m} m! \frac{(2k)!}{k!}.$$

This estimate would not allow us to obtain a radius of convergence for perturbation theory which is uniform as $\beta, L \rightarrow \infty$.

While the first problem has a deeper physical meaning, the second one is simply due to us not considering crucial cancellations between different Feynman diagrams of the same order: indeed, we obtained the diagrammatic expansion by applying Wick's rule, but we never employed the determinantal structure of the latter (see Remark 2.3.5) in (2.4.32).

This combinatorial issue can be solved employing the celebrated *Brydges-Battle-Federbush formula* (see Chapter 4), which provides an expansion for cumulants that keeps track of the aforementioned determinantal structure; such an expansion, combined with Gram-Hadamard bounds for determinants, yields bounds that lose only a $k!$ instead of a $(2k)!$ factor. This crucial improvement will ultimately allow us to solve the combinatorial problem described above.

2.4.3 Functional integral representation of perturbation theory

A convenient way to deal with the two issues shown at the end of the previous section is to suitably translate the perturbative expansion of the second-quantised lattice model into a Grassmann functional integral. For a quick review of Grassmann algebras and Grassmann integration, we refer to Appendix A.

It turns out that the free energy and the correlation functions of the initial lattice model can be rewritten in terms of Grassmann variables, due to the fact the non-interacting theories verify the same fermionic Wick rule. The advantage of the Grassmann formulation with respect to the initial second-quantised model is that Gaussian fields satisfy the addition principle that will allow us to set up a multiscale analysis for the interacting theory, rewriting the full non-interacting two-point function as a sum of single-scale propagators, which will be integrated in an iterative way.

Let $\chi : \mathbb{R} \rightarrow [0, 1]$ be a smooth, even cutoff function such that:

$$\chi(t) = \begin{cases} 1 & |t| \leq 1 \\ 0 & |t| \geq \frac{3}{2} \end{cases} \quad (2.4.33)$$

and, for $N_0 \in \mathbb{N}$, set $\chi_{N_0}(\cdot) = \chi(2^{-N_0} \cdot)$. Let us define $\mathbb{M}_{\beta, N_0}$ to be the set of regularised Matsubara frequencies:

$$\mathbb{M}_{\beta, N_0} = \{k_0 \in \mathbb{M}_\beta \mid \chi_{N_0}(k_0) > 0\}. \quad (2.4.34)$$

Due to the properties of χ , the set $\mathbb{M}_{\beta, N_0}$ is finite. Then, we consider the Grassmann $\mathbb{C}$ -algebra $\text{Gr}(\mathbb{M}_{\beta, N_0} \times \Lambda_L)$ over the set $\mathbb{M}_{\beta, N_0} \times \Lambda_L$, of generators (also known as Grassmann fields) $\{\hat{\Psi}_{k_0, \mathbf{x}}^-, \hat{\Psi}_{k_0, \mathbf{x}}^+\}_{(k_0, \mathbf{x}) \in \mathbb{M}_{\beta, N_0} \times \Lambda_L}$.

Defining the frequency-space propagator

$$\hat{g}(k_0; \mathbf{x}, \mathbf{y}) = \frac{1}{ik_0 + H - \mu}(\mathbf{x}; \mathbf{y}), \quad (2.4.35)$$

the regularised Euclidean propagator reads

$$\begin{aligned} g^{(\leq N_0)}(\underline{\mathbf{x}}, \underline{\mathbf{y}}) &:= \frac{1}{\beta} \sum_{k_0 \in \mathbb{M}_{\beta, N_0}} e^{ik_0(x_0 - y_0)} \hat{g}^{(\leq N_0)}(k_0; \underline{\mathbf{x}}, \underline{\mathbf{y}}) \\ \hat{g}^{(\leq N_0)}(k_0; \underline{\mathbf{x}}, \underline{\mathbf{y}}) &:= \chi_{N_0}(k_0) \hat{g}(k_0; \underline{\mathbf{x}}, \underline{\mathbf{y}}) . \end{aligned} \quad (2.4.36)$$

This expression can be thought of as smoothened version of the Fourier partial sums which defined $\tilde{g}$ in (2.3.4); in particular, we still have that

$$\lim_{N_0 \rightarrow \infty} g^{(\leq N_0)}(\underline{\mathbf{x}}, \underline{\mathbf{y}}) = \tilde{g}(\underline{\mathbf{x}}, \underline{\mathbf{y}}) = g(\underline{\mathbf{x}}, \underline{\mathbf{y}}) + \frac{1}{2} \mathbf{1}_{x_0=y_0} \delta_{\underline{\mathbf{x}}, \underline{\mathbf{y}}} . \quad (2.4.37)$$

We may now define the Grassmann Gaussian measure P_{N_0} with covariance $g^{(\leq N_0)}$ as

$$\begin{aligned} \int P_{N_0}(d\hat{\Psi})(\cdot) &= \frac{1}{\mathcal{Z}_{\beta, L, N_0}^0} \int d\hat{\Psi} \exp\left(-\frac{1}{\beta} \sum_{k_0, \underline{\mathbf{x}}, \underline{\mathbf{y}}} \hat{\Psi}_{k_0, \underline{\mathbf{x}}}^+ (\hat{g}^{(\leq N_0)})^{-1}(k_0; \underline{\mathbf{x}}, \underline{\mathbf{y}}) \hat{\Psi}_{k_0, \underline{\mathbf{y}}}^-\right)(\cdot) \\ \mathcal{Z}_{\beta, L, N_0}^0 &= \frac{1}{\prod_{k_0} \beta \det[\hat{g}^{(\leq N_0)}(k_0)]} , \end{aligned} \quad (2.4.38)$$

and it is straightforward to check that, defining the time-domain fields

$$\Psi_{\underline{\mathbf{x}}}^{\pm} = \frac{1}{\beta} \sum_{k_0 \in \mathbb{M}_{\beta, N_0}} e^{\mp i k_0 x_0} \hat{\Psi}_{k_0, \underline{\mathbf{x}}}^{\pm} , \quad (2.4.39)$$

we have

$$\int P_{N_0}(d\Psi) \Psi_{\underline{\mathbf{x}}}^- \Psi_{\underline{\mathbf{y}}}^+ = g^{(\leq N_0)}(\underline{\mathbf{x}}, \underline{\mathbf{y}}) . \quad (2.4.40)$$

We define the Grassmann counterpart of the many-body interaction as

$$V(\Psi) = \frac{\lambda}{2} \int d\underline{\mathbf{x}} d\underline{\mathbf{x}}' w(\underline{\mathbf{x}}; \underline{\mathbf{x}}') \Psi_{\underline{\mathbf{x}}}^+ \Psi_{\underline{\mathbf{x}}'}^+ \Psi_{\underline{\mathbf{x}}}^- \Psi_{\underline{\mathbf{x}}'}^- , \quad (2.4.41)$$

where, as before,

$$\int d\underline{\mathbf{x}} = \int_0^\beta dx_0 \sum_{\underline{\mathbf{x}} \in \Lambda_L} , \quad \text{and} \quad w(\underline{\mathbf{x}}; \underline{\mathbf{x}}') = \delta(x_0 - x'_0) w(\underline{\mathbf{x}}; \underline{\mathbf{x}}') .$$

Then, (2.4.40) and the Wick rule for Gaussian Grassmann measures (see *e.g.* [36] for a review) imply that

$$\int P_{N_0}[d\Psi] e^{-V(\Psi)} = \sum_{k \geq 0} \frac{1}{k!} \sum_{G \in \mathcal{G}_k} \text{Val}_{N_0}(G) \quad (2.4.42)$$

where Val_{N_0} is the value of the graph G computed by employing $g^{(\leq N_0)}$ as a propagator. Notice that the sum on the right-hand side is finite, due to the fact that the Grassmann algebra is finitely generated; therefore, the expectation of e^{-V} is a polynomial (of degree $\propto 2^{N_0}$) in λ .

Hence, by Eqs. (2.4.37), the $N_0 \rightarrow \infty$ limit of (2.4.42) recovers the relative partition function of the Fock space model, that is,

$$\zeta_{\beta, L}^\lambda = \lim_{N_0 \rightarrow \infty} \int P_{N_0}[d\Psi] e^{-V(\Psi)} , \quad (2.4.43)$$

since both sides of the equality are determined by the same Feynman diagram expansion.

Analogously, one may obtain a Grassmann representation of Euclidean cumulants by generalising (2.4.42) to a Grassmann generating functional,

$$\mathcal{W}_{\beta,L,N_0}[\xi] = \log \int P_{N_0}[d\Psi] e^{-V(\Psi) + \sum_{i=1}^n \xi_i \int d\mathbf{X} d\mathbf{Y} O_i(\mathbf{X}; \mathbf{Y}) \Psi_{\mathbf{X}}^+ \Psi_{\mathbf{Y}}^-},$$

with the kernels $O_i(\mathbf{X}, \mathbf{Y})$ given as in (2.4.15), so that

$$\int_{\mathbb{T}_{\beta}^n} d\mathbf{t} \langle \mathbf{T} \gamma_{t_1}(\mathcal{O}_1); \dots; \gamma_{t_n}(\mathcal{O}_n) \rangle_{\beta,L} = \lim_{N_0 \rightarrow \infty} \frac{\partial^n \mathcal{W}_{\beta,L,N_0}[\xi]}{\partial \xi_1 \dots \partial \xi_n} \Big|_{\xi=0}. \quad (2.4.44)$$

This is the starting point of the Wilsonian Renormalisation Group (RG) analysis of the model, as the right-hand sides of (2.4.43)-(2.4.44) are amenable to a multiscale expansion, which in the Grassmann framework amounts to a decomposition of the Gaussian measure P_{N_0} in (roughly) a product of Gaussian measures $\prod_{h=h_{\beta}}^{N_0} P_{(h)}$.

In Chapter 4, we shall present in more detail the RG analysis for a wide class of one-dimensional, translation-invariant, gapless systems.

Remark 2.4.11. It should be noted that, in this entire chapter, we never assumed the single-particle Hamiltonian H and the two-body interaction w to be spatially translation-invariant, as it was not a necessary condition for the proof of the general statements we discussed. Nevertheless, most of the systems that we will consider in Chapters 3 and 4 will present such symmetry. In particular, for interacting systems, this will be needed in order to define the multiscale expansion via Bloch-Fourier analysis (and in order to provide suitable L^1 - L^∞ bounds for the single-scale propagators).

Chapter 3

Response theory for free fermions

In this chapter, based on [69], we study the validity of the Kubo formula for the density and current response for a class of non-interacting models of one-dimensional metals. In particular, Section 3.2 contains the main result for such models, Theorem 3.2.1, which gives an explicit expression for the full response of the density and of the current operator on large scales. In Section 3.3, we discuss the proof of Theorem 3.2.1.

Secondly, in Section 3.4, the extension of Theorem 3.2.1 to the case of edge transport in two-dimensional topological insulators is discussed. In this case, Theorem 3.4.3 establishes the validity of the Kubo formula and gives the explicit expression of the large-scale response of the system; in particular, it proves the quantisation of the edge conductance starting from the microscopic Schrödinger dynamics. The same section also contains the proof of this result, obtained by building on the proof of Theorem 3.2.1.

3.1 Lattice fermions

3.1.1 Hamiltonian and Gibbs state

Let $L \in 2\mathbb{N} + 1$. We consider fermions on a one-dimensional lattice

$$\Gamma_L = \left[-\left\lfloor \frac{L}{2} \right\rfloor, \left\lfloor \frac{L}{2} \right\rfloor \right] \cap \mathbb{Z}$$

endowed with periodic boundary conditions. We shall allow for internal degrees of freedom, *e.g.* the spin, and we shall denote by $S_M = \{1, \dots, M\}$ the set of their labels. We shall use the notation $\Lambda_L = \Gamma_L \times S_M$, which we can view as a decorated 1d lattice. Given $x \in \Gamma_L$ and $\rho \in S_M$, we shall denote by $\mathbf{x} = (x, \rho)$ the corresponding point in Λ_L . We will use the following distance on Γ_L :

$$|x - y|_L^2 = \min_{n \in \mathbb{Z}} |x - y + nL|^2 . \quad (3.1.1)$$

Let H be a single-particle Hamiltonian on Λ_L . That is, H is a self-adjoint matrix, with entries given by $H(\mathbf{x}; \mathbf{y}) \equiv H_{\rho\rho'}(x; y)$. We shall suppose that the Hamiltonian is finite-ranged. Also, we shall assume that the Hamiltonian H is the periodisation of an Hamiltonian H^∞ defined over $\ell^2(\mathbb{Z} \times S_M)$:

$$H_{\rho\rho'}(x; y) = \sum_{n \in \mathbb{Z}} H_{\rho\rho'}^\infty(x + nL; y) . \quad (3.1.2)$$

Furthermore, we assume that H^∞ , and hence H , are translation-invariant:

$$H_{\rho\rho'}^\infty(x; y) = H_{\rho\rho'}^\infty(x + z; y + z) \quad \text{for all } z \in \mathbb{Z}. \quad (3.1.3)$$

A natural example is the lattice Laplacian, for spinless fermions ($M = 1$):

$$(H^\infty\psi)(x) = t(\psi(x-1) + \psi(x+1) - 2\psi(x)) , \quad (3.1.4)$$

with $t \in \mathbb{R}$ the hopping parameter.

Translation-invariance allows to conveniently express the Hamiltonian in momentum space. To this end, let us introduce the set of allowed quasi-momenta:

$$B_L = \left\{ k = \frac{2\pi}{L}n \mid 0 \leq n \leq L-1 \right\} . \quad (3.1.5)$$

The set B_L is the (discretized) Brillouin zone. Given a function f on Γ_L , we define its Bloch transform as:

$$\hat{f}(k) = \sum_{x \in \Gamma_L} e^{-ikx} f(x) \quad \text{for all } k \in B_L. \quad (3.1.6)$$

This relation can be inverted as:

$$f(x) = \frac{1}{L} \sum_{k \in B_L} e^{ikx} \hat{f}(k) . \quad (3.1.7)$$

We will always consider functions that satisfy periodic boundary conditions on B_L . Thus, it is natural to introduce the following norm in momentum space:

$$|k|_{\mathbb{T}} = \min_{n \in \mathbb{Z}} |k + 2\pi n| \quad \text{for all } k \in B_L. \quad (3.1.8)$$

Next, we define the Bloch Hamiltonian $\hat{H}(k)$ as the matrix with entries:

$$\hat{H}_{\rho\rho'}(k) = \sum_{x \in \Gamma_L} e^{-ikx} H_{\rho\rho'}(x; 0) . \quad (3.1.9)$$

That is, $\hat{H}(k)$ is an $M \times M$ Hermitian matrix. By (3.1.3), it is given by the restriction of $\hat{H}^\infty(k)$, defined on the circle $\mathbb{T}$ of length 2π , to B_L . We shall make the following assumptions on the spectrum of the Bloch Hamiltonian and on the chemical potential $\mu \in \mathbb{R}$.

Assumption 3.1.1 (Low energy spectrum.). There exists $\Delta > 0$ such that the following is true.

- (i) There exist $N \in \mathbb{N}$, disjoint sets $I_\omega \subset \mathbb{T}$ labelled by $\omega = 1, \dots, N$, and strictly monotone smooth functions $e_\omega : I_\omega \rightarrow \mathbb{R}$ such that:

$$\sigma(H^\infty) \cap (\mu - \Delta, \mu + \Delta) = \bigcup_{\omega=1}^N \text{Ran}(e_\omega) . \quad (3.1.10)$$

- (ii) We introduce the ω -Fermi point $k_F^\omega \in I_\omega$ and the ω -Fermi velocity v_ω as

$$e_\omega(k_F^\omega) = \mu , \quad v_\omega = \partial_k e_\omega(k_F^\omega) . \quad (3.1.11)$$

Notice that $v_\omega \neq 0$, by the strict monotonicity of e_ω .

(iii) For any $\omega = 1, \dots, N$, and $k \in I_\omega$, $e_\omega(k)$ is a non-degenerate eigenvalue of $\hat{H}^\infty(k)$.

Remark 3.1.2. (i) The simplest example of Hamiltonian that fits the setting is the Laplacian on $\mathbb{Z}$, whose energy band is:

$$\varepsilon(k) = 2t(\cos(k) - 1) , \quad (3.1.12)$$

and where $N = 2$.

- (ii) In general, it is a well-known fact that, for short-ranged one-dimensional lattice models, the number of Fermi points N must be even, with net chirality equal to zero: $\sum_{\omega=1}^N v_\omega/|v_\omega| = 0$.
- (iii) The above setting is generic and covers a large class of translation-invariant, $1d$ quantum systems. We are, however, ruling out Fermi points with zero velocity, and degenerate Fermi points. We believe that the latter restriction could be avoided, by a technical but straightforward extension of the analysis carried out in the present chapter. Instead, we think that the presence of Fermi points with zero velocity would have a dramatic effect on our analysis, and we do not know whether our results would extend to that case.

We will describe the many-particle system in a grand-canonical setting, in the fermionic Fock space. Let us briefly review how the general definitions of Chapter 2 apply to this specific case. The fermionic Fock space associated with the model on a finite lattice is:

$$\mathcal{F}_L = \mathbb{C} \oplus \bigoplus_{n \geq 1} \ell_a^2(\Lambda_L^n) , \quad (3.1.13)$$

with $\ell_a^2(\Lambda_L^n)$ the set of antisymmetric, square summable functions on Λ_L . We shall use the notation $\mathbf{x} = (x, \rho)$ for points in Λ_L , with $x \in \Gamma_L$ and $\rho \in S_M$, and we shall denote by $a_{\mathbf{x}}^*, a_{\mathbf{x}}$ the usual fermionic creation and annihilation operators, satisfying the canonical anticommutation relations:

$$\{a_{\mathbf{x}}, a_{\mathbf{y}}^*\} = \delta_{\mathbf{x}, \mathbf{y}} , \quad \{a_{\mathbf{x}}, a_{\mathbf{y}}\} = \{a_{\mathbf{x}}^*, a_{\mathbf{y}}^*\} = 0 . \quad (3.1.14)$$

In terms of these objects, we shall lift the Hamiltonian H to the Fock space as:

$$\mathcal{H} = \sum_{\mathbf{x}, \mathbf{y} \in \Lambda_L} a_{\mathbf{x}}^* H(\mathbf{x}; \mathbf{y}) a_{\mathbf{y}} . \quad (3.1.15)$$

In the following, it will also be convenient to rewrite the Hamiltonian in momentum space. For any $k \in B_L$, we define the Bloch transform of the fermionic operators as:

$$\hat{a}_{(k, \rho)} = \sum_{x \in \Gamma_L} a_{(x, \rho)} e^{-ikx} , \quad \hat{a}_{(k, \rho)}^* = \sum_{x \in \Gamma_L} a_{(x, \rho)}^* e^{ikx} , \quad (3.1.16)$$

and we shall also set $\hat{a}_{\mathbf{k}}^\sharp \equiv \hat{a}_{(k, \rho)}^\sharp$ with $\mathbf{k} = (k, \rho)$. Eqs. (3.1.16) can be inverted as, for $x \in \Gamma_L$:

$$a_{(x, \rho)} = \frac{1}{L} \sum_{k \in B_L} e^{ikx} \hat{a}_{(k, \rho)} , \quad a_{(x, \rho)}^* = \frac{1}{L} \sum_{k \in B_L} e^{-ikx} \hat{a}_{(k, \rho)}^* . \quad (3.1.17)$$

In terms of the momentum-space operators, the Hamiltonian can be rewritten as:

$$\mathcal{H} = \frac{1}{L} \sum_{k \in B_L} \sum_{\rho, \rho' \in S_M} \hat{a}_{(k, \rho)}^* \hat{H}_{\rho \rho'}(k) \hat{a}_{(k, \rho')} . \quad (3.1.18)$$

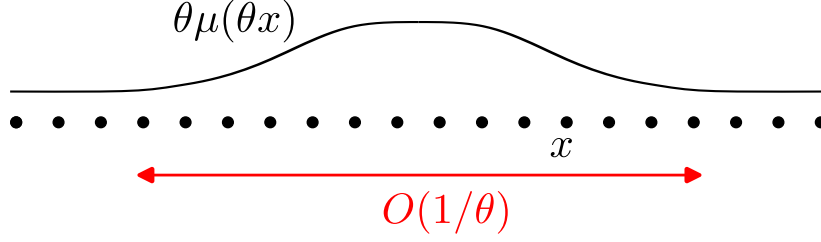


Figure 3.1: Qualitative representation of the rescaled local potential entering the definition of $\mathcal{P}$, Eq. (3.1.22).

Finally, the Gibbs state of the system at inverse temperature $\beta > 0$ is:

$$\langle \mathcal{O} \rangle_{\beta, L} = \text{Tr } \mathcal{O} \rho_{\beta, \mu, L}, \quad \rho_{\beta, \mu, L} = \frac{e^{-\beta(\mathcal{H} - \mu \mathcal{N})}}{\text{Tr } e^{-\beta(\mathcal{H} - \mu \mathcal{N})}}, \quad (3.1.19)$$

with $\mu \in \mathbb{R}$ the chemical potential and $\mathcal{N}$ the number operator. We shall assume that Assumption 3.1.1 holds, for our choice of the chemical potential μ .

3.1.2 Dynamics and linear response

Perturbing the system. We will be interested in the response properties of the system, after exposing it to a time-dependent and slowly varying perturbation. We shall consider time-dependent Hamiltonians of the form:

$$\mathcal{H}(\eta t) = \mathcal{H} + e^{\eta t} \mathcal{P}, \quad (3.1.20)$$

for $\eta > 0$ and $t \leq 0$. We shall consider finite-ranged perturbations:

$$\mathcal{P} = \sum_{X \subseteq \Lambda_L} \Phi_X \quad (3.1.21)$$

with $\Phi_X = 0$ if $|X| > R$, with $R > 0$ independent of L .

In this Chapter, we shall consider perturbations of the following type:

$$\mathcal{P} = \theta \sum_{\mathbf{x} \in \Lambda_L} \mu(\theta x) a_{\mathbf{x}}^* a_{\mathbf{x}}; \quad (3.1.22)$$

that is, we shall consider the second-quantisation of a time-dependent local potential (see Figure 3.1), slowly varying in space, on a length scale $1/\theta$, for $0 < \theta \ll 1$. Concerning the function $\mu(\theta x)$, we assume it to be the periodisation of a smooth, compactly supported function $\mu_{\infty}(\theta x)$ on $\mathbb{R}$:

$$\mu(\theta x) = \sum_{a \in \mathbb{Z}} \mu_{\infty}(\theta(x + aL)). \quad (3.1.23)$$

In particular, we can rewrite the function $\mu(\theta x)$ in terms of the Fourier transform of μ_{∞} as:

$$\mu(\theta x) = \frac{1}{L} \sum_{p \in B_L} e^{ipx} \hat{\mu}_{\theta}(p), \quad \hat{\mu}_{\theta}(p) = \sum_{n \in \mathbb{Z}} \hat{\mu}_{\infty}((p + 2\pi n)/\theta). \quad (3.1.24)$$

Considering $t, s \leq 0$, and letting $\mathcal{U}(t; s)$ be the unitary group generated by $\mathcal{H}(\eta t)$, namely

$$i\partial_t \mathcal{U}(t; s) = \mathcal{H}(\eta t) \mathcal{U}(t; s), \quad \mathcal{U}(s; s) = \mathbb{1}, \quad (3.1.25)$$

let us consider the evolution of the Gibbs state $\rho_{\beta,\mu,L}$,

$$\rho(t) = \lim_{T \rightarrow -\infty} \mathcal{U}(t; T) \rho_{\beta,\mu,L} \mathcal{U}(t; T)^*. \quad (3.1.26)$$

We recall that that we are interested in the variation of physical observables,

$$\text{Tr } \mathcal{O} \rho(t) - \text{Tr } \mathcal{O} \rho_{\beta,\mu,L}, \quad (3.1.27)$$

and on the dependence on the external perturbation. Recall also that we are interested in the following order of limits: first $L \rightarrow \infty$, then $\beta \rightarrow \infty$ and then $\eta \rightarrow 0^+$. For small perturbations, the dynamics of the system can be studied via the Duhamel series (obtained in 2.2.2) around the initial equilibrium state.

Density and current operators. Let:

$$n_{\mathbf{x}} = a_{\mathbf{x}}^* a_{\mathbf{x}}, \quad n_x = \sum_{\rho \in S_M} n_{(x,\rho)}. \quad (3.1.28)$$

In order to introduce the current operator, let us consider the time-variation of the density operator:

$$\begin{aligned} \partial_t \tau_t(n_{\mathbf{x}}) &= i \tau_t([\mathcal{H}, n_{\mathbf{x}}]) \\ &= \sum_{\mathbf{y} \in \Lambda_L} \tau_t(i a_{\mathbf{y}}^* H(\mathbf{y}; \mathbf{x}) a_{\mathbf{x}} - i a_{\mathbf{x}}^* H(\mathbf{x}; \mathbf{y}) a_{\mathbf{y}}) \\ &\equiv \sum_{\mathbf{y} \in \Lambda_L} \tau_t(j_{\mathbf{y}, \mathbf{x}}), \end{aligned} \quad (3.1.29)$$

where $j_{\mathbf{y}, \mathbf{x}} = i a_{\mathbf{y}}^* H(\mathbf{y}; \mathbf{x}) a_{\mathbf{x}} - i a_{\mathbf{x}}^* H(\mathbf{x}; \mathbf{y}) a_{\mathbf{y}}$ is the bond current between $\mathbf{x}$ and $\mathbf{y}$. Let us derive the conservation law for n_x . To this end, in (3.1.29) we sum over the internal label ρ in $\mathbf{x} = (x, \rho)$, and we introduce the discrete derivative d_x as $(d_x f)(x) = f(x) - f(x-1)$ to obtain the continuity equation

$$\partial_t \tau_t(n_x) = -d_x \tau_t(j_x), \quad (3.1.30)$$

where we introduced the lattice current j_x , determined by

$$d_x j_x := \sum_{y \in \Gamma_L} \sum_{\rho, \rho' \in S_M} j_{(x,\rho), (y,\rho')}. \quad (3.1.31)$$

The simple identity (3.1.30) will have far reaching consequences in our analysis. Let $j_{\nu, x}$, $\nu = 0, 1$, be defined as $j_{0,x} = n_x$, $j_{1,x} = j_x$, then we may characterise them in momentum-space as:

$$j_{\nu, x} = \frac{1}{L} \sum_{p \in B_L} e^{ipx} \hat{j}_{\nu, p} \quad \hat{j}_{\nu, p} = \frac{1}{L} \sum_{k \in B_L} (a_{k-p}^* \hat{J}_{\nu}(k, p) a_k), \quad (3.1.32)$$

where we used the shorthand notation $(a^*, J a) = \sum_{\rho, \rho'} a_{\rho}^* J_{\rho \rho'} a_{\rho'}$. We have

$$\hat{J}_0(k, p) = 1, \quad \hat{J}_1(k, p) = i \frac{\hat{H}(k) - \hat{H}(k-p)}{1 - e^{-ip}}. \quad (3.1.33)$$

Remark 3.1.3. If the Hamiltonian H has range 1, then $j_{1,x}$ takes a particularly simple form, given by the bond current between sites x and $x+1$:

$$j_{1,x} \equiv j_x = \sum_{\rho, \rho' \in S_M} j_{(x,\rho), (x+1, \rho')} . \quad (3.1.34)$$

Finally, we shall define a smeared version of the density and of the current operators as

$$j_\nu(\mu_\theta) = \sum_{x \in \Gamma_L} j_{\nu,x} \theta_\mu(\theta x) . \quad (3.1.35)$$

3.2 Large scale response of 1d systems

Let us define the full response as:

$$\chi_\nu^{\beta,L}(x; \eta, \theta) := \frac{1}{\theta} \left(\text{Tr } j_{\nu,x} \rho(0) - \text{Tr } j_{\nu,x} \rho_{\beta,\mu,L} \right) \quad \nu = 0, 1. \quad (3.2.1)$$

The quantity $\chi_0^{\beta,L}$ is called the susceptibility, while the quantity $\chi_1^{\beta,L}$ is called the conductance. The next theorem gives precise information about these objects.

Theorem 3.2.1 (Response of one-dimensional systems). *Let $\theta = a\eta$. There exist constants $w, \gamma, \eta_0 > 0$ such that, for all $0 \leq \eta < \eta_0$ and $a \leq w |\log \eta|$, and for β, L large enough:*

$$\chi_\nu^{\beta,L}(x; \eta, \theta) = - \sum_{\omega=1}^N \frac{v_\omega^\nu}{2\pi |v_\omega|} \int_{\mathbb{R}} \hat{\mu}_\infty(q) e^{iq\theta x} \frac{v_\omega q}{-i/a + v_\omega q} \frac{dq}{2\pi} + O(\eta^\gamma) + O\left(\frac{1}{\eta^3 \beta}\right) \quad (3.2.2)$$

where $v_\omega^\nu = \delta_{\nu,0} + v_\omega \delta_{\nu,1}$.

Corollary 3.2.2. *In particular, we can isolate three regimes.*

1. Suppose that $a \rightarrow 0$ as $\eta \rightarrow 0^+$. Then,

$$\chi_\nu(x, \eta, \theta) = O(a) + O(\eta^\gamma) + O\left(\frac{1}{\eta^3 \beta}\right) . \quad (3.2.3)$$

2. Suppose that a is constant in η . Then:

$$\begin{aligned} & \chi_\nu^{\beta,L}(x; \eta, \theta) \\ &= - \sum_{\omega=1}^N \frac{v_\omega^\nu}{2\pi |v_\omega|} \int_{\mathbb{R}} \mu_\infty(\theta x - y) \left[\delta(y) - \frac{1}{a|v_\omega|} e^{-y/av_\omega} \Theta(y/v_\omega) \right] dy + O(\eta^\gamma) + O\left(\frac{1}{\eta^3 \beta}\right) , \end{aligned} \quad (3.2.4)$$

where Θ is the Heaviside step function.

3. Finally, suppose that $a \rightarrow \infty$ as $\eta \rightarrow 0^+$. Then:

$$\chi_\nu^{\beta,L}(x; \eta, \theta) = -\mu_\infty(\theta x) \sum_{\omega=1}^N \frac{v_\omega^\nu}{2\pi |v_\omega|} + O(\eta^\gamma) + O\left(\frac{1}{\eta^3 \beta}\right) . \quad (3.2.5)$$

In particular,

$$\lim_{\eta \rightarrow 0^+} \lim_{\beta \rightarrow \infty} \lim_{L \rightarrow \infty} \chi_1^{\beta,L}(x; \eta, \theta) = -\mu_\infty(0) \sum_{\omega=1}^N \frac{\text{sgn}(v_\omega)}{2\pi} = 0 \quad (3.2.6)$$

since the number of left chiral fermionic modes equals the number of right chiral modes, for short-ranged one-dimensional systems.

- Remark 3.2.3.** (i) The above theorem justifies the validity of linear response for non-interacting one dimensional systems in the absence of a spectral gap, in the zero temperature limit and in the thermodynamic limit. Remarkably, the leading contribution to transport only depends on the Fermi velocities v_ω ; all the higher-order contributions in the perturbation are taken into account by the $O(\eta^\gamma)$ corrections.
- (ii) The parameter $a = \theta/\eta$ defines the relative magnitude between the variations in time and in space of the perturbation. The regime $a \ll 1$ corresponds to the situation in which the system is effectively exposed to the perturbation for a shorter time, while $a \gg 1$ corresponds to a longer time. The choice $a = 1$ defines a natural transition between the two regimes. It coincides with the Euler scaling, which, as mentioned in the introduction, is the starting point for hydrodynamic effective models. This framework can also be used to derive a linear response theory; see *e.g.* [28] and references therein.
- (iii) As the proof will show, due to the absence of a spectral gap all terms in the Duhamel expansion of the full response (3.2.1) are apparently of the same order in θ , and in particular of the same order of the linear response. If so, this would not allow us to prove that the linear term gives the dominant contribution to (3.2.1), since we are not making any assumption on the size of $\mu(\theta x)$. The validity of linear response arises thanks to a crucial cancellation taking place at higher orders in the Duhamel expansion. This cancellation allows to prove that all terms after the linear response, up to a large θ -dependent order $n(\theta)$, are suppressed by an extra factor η^γ with $\gamma > 0$, and hence they vanish as $\eta \rightarrow 0$; this is ultimately due to a cancellation taking place in the scaling limit of the model, often called ‘loop cancellation’ in quantum field theory, related to bosonisation [34]. Concerning the orders n larger than $n(\theta)$, we will prove that they give a small contribution, thanks to improved estimates on the Duhamel series, which allow one to prove its summability.

3.3 Proof of Theorem 3.2.1

3.3.1 Auxiliary dynamics

Let $\eta_\beta \in (2\pi/\beta)\mathbb{N}$, such that $\eta_\beta \geq \eta$ and $\eta_\beta - \eta = \min_{\xi \in \frac{2\pi}{\beta}\mathbb{N}}(\xi - \eta)$. We define:

$$\mathcal{H}_{\beta,\eta}(t) = \mathcal{H} + \theta e^{\eta_\beta t} \sum_{\mathbf{x} \in \Lambda_L} \mu_\alpha(\theta x) a_{\mathbf{x}}^* a_{\mathbf{x}} , \quad (3.3.1)$$

where $\mu_\alpha(\theta x)$ is obtained from $\mu(\theta x)$ after cutting off large momenta, in a smooth way:

$$\mu_\alpha(\theta x) = \frac{1}{L} \sum_{p \in B_L} e^{ipx} \hat{\mu}_{\alpha,\theta}(p) , \quad \hat{\mu}_{\alpha,\theta}(p) = \hat{\mu}_\theta(p) \chi(\theta^{\alpha-1}|p|_{\mathbb{T}}) , \quad (3.3.2)$$

with $\alpha \in (0, 1)$ and $\chi(\cdot) : \mathbb{R}^+ \rightarrow [0, 1]$ a smooth cutoff function such that:

$$\chi(t) = \begin{cases} 1 & t \leq 1 \\ 0 & t \geq 2. \end{cases} \quad (3.3.3)$$

With respect to the original perturbation, in the definition of the function $\mu_\alpha(\theta x)$ we are cutting off all momenta p of norm greater than $2\theta^{1-\alpha}$. By the smoothness of the function $\mu(\cdot)$, this regularisation will have a minor effect.

Let us denote by $\tilde{\mathcal{U}}(t; s)$ the two-parameter unitary group generated by $\mathcal{H}_{\beta, \eta}(t)$,

$$i\partial_t \tilde{\mathcal{U}}(t; s) = \mathcal{H}_{\beta, \eta}(t) \tilde{\mathcal{U}}(t; s), \quad \tilde{\mathcal{U}}(s; s) = \mathbb{1}. \quad (3.3.4)$$

The next result allows to control the error introduced by replacing the dynamics generated by $\mathcal{H}(\eta t)$ with the dynamics generated by $\mathcal{H}_{\beta, \eta}(t)$.

Proposition 3.3.1 (Approximation by the auxiliary dynamics). *Under the same assumptions of Theorem 3.2.1, it follows that, for any $m \in \mathbb{N}$:*

$$\left\| \tilde{\mathcal{U}}(t; -\infty)^* \mathcal{O}_X \tilde{\mathcal{U}}(t; -\infty) - \mathcal{U}(t; -\infty)^* \mathcal{O}_X \mathcal{U}(t; -\infty) \right\| \leq C_m \frac{\theta^{1+\alpha(m-1)}}{\eta^2} + \frac{C\theta}{\eta^3 \beta}. \quad (3.3.5)$$

Remark 3.3.2. Let us briefly comment on the reason for introducing the auxiliary dynamics. Let $\mathcal{A}$ be such that $[\mathcal{A}, \mathcal{N}] = 0$, and let $\tilde{\tau}_t(\mathcal{A}) = e^{\eta t} \tau_t(\mathcal{A})$, which we can view as a (non-unitary) regularisation of the original dynamics, damped in the past. Then, if $\eta \in \frac{2\pi}{\beta} \mathbb{N}$, we observe that $\tilde{\tau}_t(\cdot)$ satisfies the KMS identity:

$$\langle \tilde{\tau}_t(\mathcal{A}) \mathcal{B} \rangle_{\beta, L} = \langle \mathcal{B} \tilde{\tau}_{t+i\beta}(\mathcal{A}) \rangle_{\beta, L}, \quad (3.3.6)$$

where we used the trivial but crucial identity $e^{\eta t} = e^{\eta(t+i\beta)}$. The KMS identity for the regularised dynamics (3.3.6) plays a key role in the mapping of the real time Duhamel series into an imaginary time expansion, which can be efficiently studied [48] (see Proposition 2.2.7). Due to the fact that the difference between any $\eta > 0$ and its best approximation $\eta_\beta \in \frac{2\pi}{\beta} \mathbb{N}$ vanishes as $\beta \rightarrow \infty$, this approach is suitable to study zero or low temperature systems (lower than some η -dependent value).

Concerning the momentum-space regularisation of the perturbation, this will be used to rule out the scattering of different Fermi points, at orders in the expansion that are less than some θ -dependent (high) order.

Proof of Proposition 3.3.1. The proof of this proposition is a standard argument, based on Lieb-Robinson bounds. It is a slight generalisation of the analogous result in Proposition 2.2.9. We reproduce it for completeness. We start by writing:

$$\begin{aligned} & \left\| \tilde{\mathcal{U}}(t; -\infty)^* \mathcal{O}_X \tilde{\mathcal{U}}(t; -\infty) - \mathcal{U}(t; -\infty)^* \mathcal{O}_X \mathcal{U}(t; -\infty) \right\| \\ &= \left\| \mathcal{O}_X - \mathcal{U}_I(t; -\infty)^* \mathcal{O}_X \mathcal{U}_I(t; -\infty) \right\|, \end{aligned} \quad (3.3.7)$$

where $\mathcal{U}_I(t; s) = \mathcal{U}(t; s) \tilde{\mathcal{U}}(t; s)^*$. Then, since:

$$i\partial_s \mathcal{U}_I(t; s) = -\mathcal{U}_I(t; s) \tilde{\mathcal{U}}(t; s) (\mathcal{H}(\eta s) - \mathcal{H}_{\eta, \beta}(s)) \tilde{\mathcal{U}}(t; s)^* \quad (3.3.8)$$

we have:

$$\begin{aligned} & \mathcal{O}_X - \mathcal{U}_I(t; -\infty)^* \mathcal{O}_X \mathcal{U}_I(t; -\infty) \\ &= \int_{-\infty}^t ds \frac{d}{ds} \mathcal{U}_I(t; s)^* \mathcal{O}_X \mathcal{U}_I(t; s) \\ &= i \int_{-\infty}^t ds \mathcal{U}_I(t; s)^* \left[\mathcal{O}_X, \tilde{\mathcal{U}}(t; s) (\mathcal{H}(\eta s) - \mathcal{H}_{\eta, \beta}(s)) \tilde{\mathcal{U}}(t; s)^* \right] \mathcal{U}_I(t; s), \end{aligned} \quad (3.3.9)$$

which implies:

$$\begin{aligned}
& \left\| \mathcal{O}_X - \mathcal{U}_I(t; -\infty)^* \mathcal{O}_X \mathcal{U}_I(t; -\infty) \right\| \\
& \leq \int_{-\infty}^t ds \left\| \left[\tilde{\mathcal{U}}(t; s)^* \mathcal{O}_X \tilde{\mathcal{U}}(t; s), (\mathcal{H}(\eta s) - \mathcal{H}_{\eta, \beta}(s)) \right] \right\| \\
& \leq \int_{-\infty}^t ds \left(\left\| \left[\tilde{\mathcal{U}}(t; s)^* \mathcal{O}_X \tilde{\mathcal{U}}(t; s), \mathcal{Q}_1(s) \right] \right\| + \left\| \left[\tilde{\mathcal{U}}(t; s)^* \mathcal{O}_X \tilde{\mathcal{U}}(t; s), \mathcal{Q}_2(s) \right] \right\| \right),
\end{aligned} \tag{3.3.10}$$

where:

$$\begin{aligned}
\mathcal{Q}_1(s) &= \theta(e^{\eta s} - e^{\eta \beta s}) \sum_{\mathbf{x} \in \Lambda_L} \mu(\theta x) a_{\mathbf{x}}^* a_{\mathbf{x}} \\
\mathcal{Q}_2(s) &= \theta e^{\eta \beta s} \sum_{\mathbf{x} \in \Lambda_L} (\mu(\theta x) - \mu_{\alpha}(\theta x)) a_{\mathbf{x}}^* a_{\mathbf{x}}.
\end{aligned} \tag{3.3.11}$$

Consider the second term. We have:

$$\begin{aligned}
& \left\| \left[\tilde{\mathcal{U}}(t; s)^* \mathcal{O}_X \tilde{\mathcal{U}}(t; s), \mathcal{Q}_2(s) \right] \right\| \\
& \leq \theta e^{\eta \beta s} \sum_{\mathbf{x} \in \Lambda_L} |\mu(\theta x) - \mu_{\alpha}(\theta x)| \left\| \left[\tilde{\mathcal{U}}(t; s)^* \mathcal{O}_X \tilde{\mathcal{U}}(t; s), n_{\mathbf{x}} \right] \right\| \\
& \leq \theta e^{\eta \beta s} \|\mu - \mu_{\alpha}\|_{\infty} \sum_{\mathbf{x} \in \Lambda_L} \left\| \left[\tilde{\mathcal{U}}(t; s)^* \mathcal{O}_X \tilde{\mathcal{U}}(t; s), n_{\mathbf{x}} \right] \right\|.
\end{aligned} \tag{3.3.12}$$

By the Lieb-Robinson bound, we can further estimate:

$$\left\| \left[\tilde{\mathcal{U}}(t; s)^* \mathcal{O}_X \tilde{\mathcal{U}}(t; s), \mathcal{Q}_2(s) \right] \right\| \leq C_X \theta e^{\eta \beta s} (1 + |t - s|) \|\mu - \mu_{\alpha}\|_{\infty}. \tag{3.3.13}$$

The norm can be bounded as:

$$\|\mu - \mu_{\alpha}\|_{\infty} \leq \frac{1}{L} \sum_{p \in B_L} |\mu_{\theta}(p)| [1 - \chi(\theta^{\alpha-1} |p|_{\mathbb{T}})]. \tag{3.3.14}$$

Next, by the smoothness of $\mu(x)$, we have $(1 + |q|_{\mathbb{T}}^m) |\hat{\mu}(q)| \leq C_m$, which easily gives:

$$\frac{1}{L} \sum_{p \in B_L} |\mu_{\theta}(p)| [1 - \chi(\theta^{\alpha-1} |p|_{\mathbb{T}})] \leq K_m \theta^{\alpha(m-1)}, \quad \forall m \in \mathbb{N}. \tag{3.3.15}$$

Plugging this bound in (3.3.13), we obtain:

$$\left\| \left[\tilde{\mathcal{U}}(t; s)^* \mathcal{O}_X \tilde{\mathcal{U}}(t; s), \mathcal{Q}_2(s) \right] \right\| \leq C_X K_m \theta e^{\eta \beta s} (1 + |t - s|) \theta^{\alpha(m-1)}, \tag{3.3.16}$$

which allows to estimate the corresponding contribution to (3.3.10) as:

$$\int_{-\infty}^t ds \left\| \left[\tilde{\mathcal{U}}(t; s)^* \mathcal{O}_X \tilde{\mathcal{U}}(t; s), \mathcal{Q}_2(s) \right] \right\| \leq C_m \frac{\theta^{1+\alpha(m-1)}}{\eta^2}. \tag{3.3.17}$$

Consider now the contribution due to $\mathcal{Q}_1(s)$ in the right-hand side of (3.3.10). This term can be estimated as in Proposition 2.2.9, we omit the details. The result is:

$$\int_{-\infty}^t ds \left\| \left[\tilde{\mathcal{U}}(t; s)^* \mathcal{O}_X \tilde{\mathcal{U}}(t; s), \mathcal{Q}_1(s) \right] \right\| \leq \frac{C\theta}{\eta^3 \beta}. \tag{3.3.18}$$

Combining (3.3.7), (3.3.10), (3.3.17), (3.3.18), the final result (3.3.5) follows. $\square$

Therefore, by Proposition 3.3.1 we have:

$$\mathrm{Tr} j_{\nu,x} \rho(t) = \mathrm{Tr} j_{\nu,x} \tilde{\rho}(t) + \mathcal{E}_{\nu}^{\beta,L}(x, t; \eta, \theta), \quad (3.3.19)$$

where $\tilde{\rho}(t) = \tilde{\mathcal{U}}(t; -\infty) \rho_{\beta,\mu,L} \tilde{\mathcal{U}}(t; -\infty)^*$ and, by Proposition 3.3.1:

$$|\mathcal{E}_{\nu}^{\beta,L}(x, t; \eta, \theta)| \leq C_m \frac{\theta^{1+\alpha(m-1)}}{\eta^2} + \frac{C\theta}{\eta^3 \beta}. \quad (3.3.20)$$

We require this error term is vanishing once divided by $\theta = a\eta$, recall (3.2.1). To this end, it is sufficient to consider:

$$a = o\left(\eta^{\frac{2}{\alpha(m-1)}-1}\right). \quad (3.3.21)$$

In order to leave room for $a \rightarrow \infty$ as $\eta \rightarrow 0^+$, we will choose $\alpha \in (0, 1)$ and we will pick $m \in \mathbb{N}$ so that $\alpha(m-1) > 2$. Later, we will have to introduce stronger constraints on a .

3.3.2 Duhamel series of the auxiliary dynamics

Consider the main term in the right-hand side of (3.3.19). By Proposition 2.2.7:

$$\begin{aligned} & \mathrm{Tr} j_{\nu,x} \tilde{\rho}(t) - \mathrm{Tr} j_{\nu,x} \rho_{\beta,\mu,L} \\ &= \sum_{n \geq 1} \frac{(-1)^n e^{n\eta\beta t}}{n!} \int_{[0,\beta]^n} d\underline{s} e^{-i\eta\beta(s_1+\dots+s_n)} \langle \mathbf{T} \gamma_{s_1}(\tilde{\mathcal{P}}); \dots; \gamma_{s_n}(\tilde{\mathcal{P}}); j_{\nu,x} \rangle_{\beta,L}. \end{aligned} \quad (3.3.22)$$

It is convenient to rewrite the n -th order contribution to the expansion in Fourier space. Recalling the definition (3.1.35) and the form of the perturbation (3.1.22),

$$\begin{aligned} & \int_{[0,\beta]^n} d\underline{s} e^{-i\eta\beta(s_1+\dots+s_n)} \langle \mathbf{T} \gamma_{s_1}(\tilde{\mathcal{P}}); \dots; \gamma_{s_n}(\tilde{\mathcal{P}}); j_{\nu,x} \rangle_{\beta,L} \\ &= \int_{[0,\beta]^n} d\underline{s} e^{-i\eta\beta(s_1+\dots+s_n)} \langle \mathbf{T} \gamma_{s_1}(j_0(\mu_{\alpha,\theta})); \dots; \gamma_{s_n}(j_0(\mu_{\alpha,\theta})); j_{\nu,x} \rangle_{\beta,L}. \end{aligned} \quad (3.3.23)$$

Next, using that:

$$j_0(\mu_{\alpha,\theta}) = \frac{\theta}{L} \sum_{p \in B_L} \hat{\mu}_{\alpha,\theta}(-p) \hat{j}_{\nu,p}, \quad \hat{j}_{\nu,p} = \sum_{\mathbf{x} \in \Lambda_L} e^{-ipx} \hat{j}_{\nu,\mathbf{x}}, \quad (3.3.24)$$

and writing, for $\underline{p} = (p_0, p_1)$ with $p_1 = p$ and $p_0 = \eta\beta$,

$$\hat{n}_{\underline{p}} := \int_0^\beta ds e^{-i\eta\beta s} \gamma_s(\hat{n}_p), \quad (3.3.25)$$

we can rewrite the right-hand side of (3.3.23) as:

$$\begin{aligned} & \int_{[0,\beta]^n} d\underline{s} e^{-i\eta\beta(s_1+\dots+s_n)} \langle \mathbf{T} \gamma_{s_1}(j_0(\mu_{\alpha,\theta})); \dots; \gamma_{s_n}(j_0(\mu_{\alpha,\theta})); j_{\nu,x} \rangle_{\beta,L} \\ &= \frac{\theta^n}{L^n} \sum_{\{p_i\} \in B_L^n} \left[\prod_{i=1}^n \hat{\mu}_{\alpha,\theta}(-p_i) \right] e^{ip_{n+1}x} \frac{1}{\beta L} \langle \mathbf{T} \hat{n}_{\underline{p}_1}; \dots; \hat{n}_{\underline{p}_n}; \hat{j}_{\nu,\underline{p}_{n+1}} \rangle_{\beta,L}, \end{aligned} \quad (3.3.26)$$

where $\underline{p}_{n+1} = -\underline{p}_1 - \dots - \underline{p}_n$. In order to derive (3.3.26), we used the space-time translation invariance of the Gibbs state. Next, by Wick's rule:

$$\begin{aligned} & \frac{1}{\beta L} \langle \mathbf{T} \hat{n}_{\underline{p}_1} ; \dots ; \hat{n}_{\underline{p}_n} ; \hat{j}_{\nu, \underline{p}_{n+1}} \rangle_{\beta, L} \\ &= -\frac{1}{\beta L} \sum_{\pi \in S_n} \sum_{\underline{k} \in \mathbb{M}_\beta \times B_L} \text{Tr} \left[\hat{J}_\nu(k, p_{n+1}) \prod_{i=1}^{n+1} g\left(\underline{k} + \sum_{j < i} \underline{p}_{\pi(j)}\right) \right] \end{aligned} \quad (3.3.27)$$

where S_n is the set of permutations π of $\{1, \dots, n\}$; $\mathbb{M}_\beta = (2\pi/\beta)(\mathbb{Z} + 1/2)$ is the set of the fermionic Matsubara frequencies, and $g(\underline{k})$ is given by

$$g(\underline{k}) = \frac{1}{ik_0 + \hat{H}(k) - \mu}. \quad (3.3.28)$$

In the following, it will be convenient to decompose the fermionic propagator in a singular plus a regular part. Let $\chi(\cdot)$ be a smooth cutoff function as in (3.3.3). We write:

$$\begin{aligned} g(\underline{k}) &= g(k) \chi(|\hat{H}(k) - \mu|/\Delta) + g(k)(1 - \chi(|\hat{H}(k) - \mu|/\Delta)) \\ &= g_a(\underline{k}) + g_b(\underline{k}); \end{aligned} \quad (3.3.29)$$

the propagator g_b satisfies the estimate:

$$\|d_{k_0}^{n_0} d_{k_1}^{n_1} g_b(\underline{k})\| \leq \frac{C_{n_0, n_1}}{1 + |k_0|^{n_0+1}} \quad \text{uniformly in } k. \quad (3.3.30)$$

Consider now the first term in (3.3.29). It is:

$$g_a(\underline{k}) = \sum_{\omega=1}^N \frac{\chi(|e_\omega(k) - \mu|/\Delta)}{ik_0 + e_\omega(k) - \mu} P_\omega(k), \quad (3.3.31)$$

with $P_\omega(k)$ a rank-one projector,

$$P_\omega(k) := |\xi_\omega(k)\rangle \langle \xi_\omega(k)| \quad \hat{H}(k) \xi_\omega(k) = e_\omega(k) \xi_\omega(k). \quad (3.3.32)$$

We can further rewrite $g_a(\underline{k})$, as follows. By assumption, the Fermi point k_F^ω is not a critical point for e_ω , that is $v_\omega = e'_\omega(k_F^\omega) \neq 0$. Thus, setting $\underline{k} := (k_0, k_1)$ and $\underline{k}_F^\omega := (0, k_F^\omega)$,

$$\sum_{\omega=1}^N \frac{\chi(|e_\omega(k) - \mu|/\Delta)}{ik_0 + e_\omega(k) - \mu} P_\omega(k) = \sum_{\omega=1}^N \frac{\chi_\delta^\omega(\underline{k})}{D_\omega(\underline{k})} P_\omega(k) + \sum_{\omega=1}^N \left(\frac{\chi(|e_\omega(k) - \mu|/\Delta)}{ik_0 + e_\omega(k) - \mu} - \frac{\chi_\delta^\omega(\underline{k})}{D_\omega(\underline{k})} \right) P_\omega(k) \quad (3.3.33)$$

where: $\chi_\delta^\omega(\underline{k}) := \chi(\|\underline{k} - \underline{k}_F^\omega\|_\omega / \delta)$, $\|q\|_\omega^2 := q_0^2 + v_\omega^2 |q|_\mathbb{T}^2$ and $D_\omega(\underline{k}_F^\omega + \underline{q}) := iq_0 + v_\omega q$. The parameter δ is chosen so that $\delta \ll \inf_{\omega \neq \omega'} |k_F^\omega - k_F^{\omega'}|_\mathbb{T}$ and

$$|e_\omega(k) - \mu| \leq \Delta \quad \forall k \in \mathcal{B}_{2\delta}^\omega(k_F^\omega) := \{k \in [0, 2\pi] : |k - k_F^\omega|_\mathbb{T} < 2\delta/|v_\omega|\} \quad (3.3.34)$$

for all ω . Therefore, we can rewrite:

$$\begin{aligned} g(\underline{k}) &= g_s(\underline{k}) + g_r(\underline{k}) \quad \text{with} \\ g_s(\underline{k}) &:= \sum_{\omega=1}^N \frac{\chi_\delta^\omega(\underline{k})}{D_\omega(\underline{k})} P_\omega(k) \\ g_r(\underline{k}) &:= g_b(\underline{k}) + \sum_{\omega=1}^N \left(\frac{\chi(|e_\omega(k) - \mu|/\Delta)}{ik_0 + e_\omega(k) - \mu} - \frac{\chi_\delta^\omega(\underline{k})}{D_\omega(\underline{k})} \right) P_\omega(k). \end{aligned} \quad (3.3.35)$$

It is not difficult to see that:

$$\|g_r(k)\| \leq \frac{C}{1+|k_0|}, \quad \|d_{k_a} g_r(k)\| \leq \sum_{\omega=1}^N \frac{C}{1+|k_0|} \frac{1}{\|\underline{k} - \underline{k}_F^\omega\|}. \quad (3.3.36)$$

Using the above splitting we can decompose the left-hand side of (3.3.27) as:

$$\frac{1}{\beta L} \langle \mathbf{T} \hat{n}_{\underline{p}_1}; \dots; \hat{n}_{\underline{p}_n}; \hat{j}_{\nu, \underline{p}_{n+1}} \rangle_{\beta, L} = \tilde{S}_{n; \nu}^{\beta, L}(\underline{p}_1, \dots, \underline{p}_n) + \tilde{R}_{n; \nu}^{\beta, L}(\underline{p}_1, \dots, \underline{p}_n) \quad (3.3.37)$$

where $\tilde{S}_{n; \nu}^{\beta, L}(\underline{p}_1, \dots, \underline{p}_n)$ collects the contribution due only to the singular propagators g_s ,

$$\tilde{S}_{n; \nu}^{\beta, L}(\underline{p}_1, \dots, \underline{p}_n) := -\frac{1}{\beta L} \sum_{\underline{k} \in \mathbb{M}_\beta \times B_L} \sum_{\pi \in S_n} \text{Tr} \left[\hat{J}_\nu(k, p_{n+1}) \prod_{i=1}^{n+1} g_s\left(\underline{k} + \sum_{j < i} \underline{p}_{\pi(j)}\right) \right], \quad (3.3.38)$$

while $\tilde{R}_{n; \nu}^{\beta, L}(\underline{p}_1, \dots, \underline{p}_n)$ contains at least one bounded propagator g_r . The next lemma allows to rewrite $\tilde{S}_{n; \nu}^{\beta, L}$ in a more explicit way, up to subleading terms. In the following, we shall use the notation:

$$\int_{\beta, L} \frac{d\underline{k}}{(2\pi)^2} [\dots] := \frac{1}{\beta L} \sum_{\underline{k} \in \mathbb{M}_\beta \times B_L} [\dots]; \quad (3.3.39)$$

in fact, as $\beta, L \rightarrow \infty$, the sum converges to an integral over $\underline{k} \in \mathbb{R} \times \mathbb{T}$.

Lemma 3.3.3 (Structure of the singular part). *Let $v_\omega^0 = 1$ and $v_\omega^1 = v_\omega$. Then for any $p_1, \dots, p_n$ in the support of $\hat{\mu}_{\alpha, \theta}$,*

$$\tilde{S}_{n; \nu}^{\beta, L}(\underline{p}_1, \dots, \underline{p}_n) = S_{n; \nu}^{\beta, L}(\underline{p}_1, \dots, \underline{p}_n) + T_{n; \nu}^{\beta, L}(\underline{p}_1, \dots, \underline{p}_n) \quad (3.3.40)$$

where, for $g_\omega(\underline{k}) = \chi_\delta^\omega(\underline{k})/D_\omega(\underline{k})$,

$$\begin{aligned} S_{n; \nu}^{\beta, L}(\underline{p}_1, \dots, \underline{p}_n) &= - \sum_{\omega=1}^N v_\omega^\nu \int_{\beta, L} \frac{d\underline{k}}{(2\pi)^2} \sum_{\pi \in S_n} \prod_{i=1}^{n+1} g_\omega\left(\underline{k} + \sum_{j < i} \underline{p}_{\pi(j)}\right) \\ T_{n; \nu}^{\beta, L}(\underline{p}_1, \dots, \underline{p}_n) &= - \sum_{\omega=1}^N \int_{\beta, L} \frac{d\underline{k}}{(2\pi)^2} \sum_{\pi \in S_n} f_{n, \omega}^\nu(k; p_{\pi(1)}, \dots, p_{\pi(n)}) \prod_{i=1}^{n+1} g_\omega\left(\underline{k} + \sum_{j < i} \underline{p}_{\pi(j)}\right), \end{aligned} \quad (3.3.41)$$

where:

$$|f_{n, \omega}^\nu(k; p_1, \dots, p_n)| \leq C_\delta \left(\delta_{\nu, 1} |k - k_F^\omega|_{\mathbb{T}} + n \sum_{i=1}^n |p_i|_{\mathbb{T}} \right). \quad (3.3.42)$$

Proof. We start with the following remark. Since all external momenta $\{p_j\}$ are in the support of $\hat{\mu}_{\alpha, \theta}$, we have that $|p_j|_{\mathbb{T}} \leq \theta^{1-\alpha}$. Furthermore, g_ω and $g_{\omega'}$ have disjoint support for $\omega \neq \omega'$, recall the discussion after (3.3.34). Thus, for θ small enough, and for all $i = 1, \dots, n$:

$$g_\omega(\underline{k} + \sum_{j < i} \underline{p}_j) g_{\omega'}(\underline{k} + \sum_{j < i+1} \underline{p}_j) = 0 \quad \text{if } \omega' \neq \omega. \quad (3.3.43)$$

Hence, $\tilde{S}_{n; \nu}^{\beta, L}$ reduces to

$$\tilde{S}_{n; \nu}^{\beta, L}(\underline{p}_1, \dots, \underline{p}_n) = - \sum_{\omega=1}^N \sum_{\pi \in S_n} \int_{\beta, L} \frac{d\underline{k}}{(2\pi)^2} \left[\prod_{i=1}^{n+1} g_\omega\left(\underline{k} + \sum_{j < i} \underline{p}_{\pi(j)}\right) \right] t_\omega^\nu(k; p_{\pi(1)}, \dots, p_{\pi(n)}) \quad (3.3.44)$$

with

$$t_\omega^\nu(k; p_{\pi(1)}, \dots, p_{\pi(n)}) := \text{Tr} \left[\hat{J}_\nu(k, p_{n+1}) \prod_{i=1}^{n+1} P_\omega \left(k + \sum_{j < i} p_{\pi(j)} \right) \right]. \quad (3.3.45)$$

Noticing that $t_\omega^0(k; 0, \dots, 0) \equiv 1$ and $t_\omega^1(k; 0, \dots, 0) = e'_\omega(k)$, we Taylor expand $t_\omega^\nu(k; \cdot)$ around $(0, \dots, 0)$, and e'_ω around k_F^ω , obtaining

$$t_\omega^\nu(k; p_1, \dots, p_n) = v_\omega^\nu + f_{n,\omega}^\nu(k; p_1, \dots, p_n); \quad (3.3.46)$$

the bound (3.3.42) easily follows from the smoothness of $\hat{J}_\nu(k, p)$ and of $P_\omega(k)$. $\square$

Remark 3.3.4. This decomposition allows to isolate, at order n , the most singular contribution from the n -th order term in the Duhamel expansion. Indeed, a simple rescaling argument shows that each summand appearing in $\theta^{n-1} S_{n;\nu}^{\beta,L}$ is $O((\theta/\eta)^{n-1})$, whereas the summands in $\theta^{n-1} T_{n;\nu}^{\beta,L}$ and $\theta^{n-1} \tilde{R}_{n;\nu}^{\beta,L}$ are $O(\theta(\theta/\eta)^{n-1})$. We will see that the smallness of $\theta^{n-1} S_{n;\nu}^{\beta,L}$ will be guaranteed by a crucial cancellation.

Let $R_{n;\nu}^{\beta,L} := \tilde{R}_{n;\nu}^{\beta,L} + T_{n;\nu}^{\beta,L}$. We obtained the following representation for the Duhamel expansion of the response functions, recall (3.3.22)-(3.3.26),

$$\begin{aligned} \chi_\nu^{\beta,L}(x; \eta, \theta) = & - \sum_{n=1}^{\infty} \frac{(-\theta)^{n-1}}{n!} \frac{1}{L^n} \sum_{\{p_i\} \in B_L^n} \left[\prod_{j=1}^n \hat{\mu}_{\alpha,\theta}(-p_j) \right] e^{ip_{n+1}x} \\ & \cdot [S_{n;\nu}^{\beta,L}(\underline{p}_1, \dots, \underline{p}_n) + R_{n;\nu}^{\beta,L}(\underline{p}_1, \dots, \underline{p}_n)] + E_\nu^{\beta,L}(x; \eta, \theta), \end{aligned} \quad (3.3.47)$$

where: $\underline{p}_i := (\eta_\beta, p_i)$ and $\underline{p}_{n+1} := -\sum_{i=1}^n \underline{p}_i$; the term $S_{n;\nu}^{\beta,L}$ is given by the first of (3.3.41); the term $R_{n;\nu}^{\beta,L}(\underline{p}_1, \dots, \underline{p}_n)$ will be proven to be subleading as $\theta \rightarrow 0$; and, recall (3.3.20):

$$|E_\nu^{\beta,L}(x; \eta, \theta)| \leq \frac{C\theta^{\alpha(m-1)}}{\eta^2} + \frac{C}{\eta^3\beta}. \quad (3.3.48)$$

Our next task will be to evaluate explicitly the linear response, and to bound the higher-order terms, starting from the identity (3.3.47).

3.3.3 Evaluation of the linear response

In this subsection we will evaluate the main term in the expression for the response function, Eq. (3.2.2), as $\beta, L \rightarrow \infty$. The analysis is based on lattice conservation laws, regularity of correlations, and explicit computation of the scaling limit contribution. It has been already used to determine the linear response of 1d and quasi-1d systems, [2, 19, 20, 60, 61], in a setting that also allow to include many-body interactions. For completeness, we shall reproduce all the steps here, for general 1d non-interacting systems.

Let $S_{n;\nu} = \lim_{\beta, L \rightarrow \infty} S_{n;\nu}^{\beta,L}$ and $R_{n;\nu} = \lim_{\beta, L \rightarrow \infty} R_{n;\nu}^{\beta,L}$. In this limit, the $n = 1$ term in (3.3.47) reads, setting $\underline{p} = \theta \underline{q}$, with $\underline{q} = (a^{-1}, q)$,

$$\chi_\nu^{\text{lin}}(x; \eta, \theta) = - \int_{\mathbb{T}_{\theta^{-1}}} \frac{dq}{2\pi} \hat{\mu}_\alpha(-q) e^{-i\theta q x} [S_{1;\nu}(\theta \underline{q}) + R_{1;\nu}(\theta \underline{q})], \quad (3.3.49)$$

with $\hat{\mu}_\alpha(q) = \hat{\mu}(q)\chi(\theta^\alpha|q|)$, and $\mathbb{T}_{\theta^{-1}}$ is the torus $\mathbb{T}$ rescaled by a factor $1/\theta$. Let us start by discussing the $R_{1;\nu}$ term.

Lemma 3.3.5 (Continuity of the remainder). *For $\alpha \in (0, 1)$, there exists $C_\alpha > 0$ such that:*

$$|R_{1;\nu}(\underline{0})| \leq C, \quad |R_{1;\nu}(\underline{p}) - R_{1;\nu}(\underline{0})| \leq C_\alpha \|\underline{p}\|^\alpha. \quad (3.3.50)$$

Proof. We have:

$$\tilde{R}_{1;\nu}(\underline{p}) = \int_{\mathbb{R} \times \mathbb{T}} \frac{d\underline{k}}{(2\pi)^2} \text{Tr} \left(\hat{J}_\nu(k, -p) (g_r(\underline{k}) g_s(\underline{k} + \underline{p}) + g_s(\underline{k}) g_r(\underline{k} + \underline{p}) + g_r(\underline{k}) g_r(\underline{k} + \underline{p})) \right). \quad (3.3.51)$$

By the estimate (3.3.36), $\tilde{R}_{1;\nu}(\underline{0})$ is finite. Furthermore, by the smoothness of $\hat{J}_\nu(k, p)$ and using the bound, with $0 < \alpha < 1$,

$$\|g_r(\underline{k} + \underline{p}) - g_r(\underline{k})\| \leq C \sum_{\omega} \frac{1}{1 + |\underline{k}_0|^\alpha} \frac{\|\underline{p}\|^\alpha}{\|\underline{k} - \underline{k}_F^\omega\|^\alpha} \quad (3.3.52)$$

which follows from the second of (3.3.36), we easily get:

$$|\tilde{R}_{1;\nu}(\underline{p}) - \tilde{R}_{1;\nu}(\underline{0})| \leq C_\alpha \|\underline{p}\|^\alpha. \quad (3.3.53)$$

Consider now the term $T_{1;\nu}$. From the estimate (3.3.42), and recalling that $g_\omega(\underline{k}) = \chi_\delta^\omega(\underline{k})/D_\omega(\underline{k})$, we have:

$$\begin{aligned} |T_{1;\nu}(\underline{p})| &\leq C \sum_{\omega=1}^N \int d\underline{k} |f_{1,\omega}^\nu(k; p)| |g_\omega(\underline{k} + \underline{p})| |g_\omega(\underline{k})| \\ &\leq K \sum_{\omega=1}^N \int d\underline{k} (|\underline{k} - \underline{k}_F^\omega|_{\mathbb{T}} + |p|_{\mathbb{T}}) |g_\omega(\underline{k} + \underline{p})| |g_\omega(\underline{k})| \\ &\leq \tilde{K} (1 + \|\underline{p}\| |\log \|\underline{p}\||). \end{aligned} \quad (3.3.54)$$

Similarly,

$$\begin{aligned} |T_{1;\nu}(\underline{p}) - T_{1;\nu}(\underline{0})| &\leq K \sum_{\omega=1}^N \int d\underline{k} |\underline{k} - \underline{k}_F^\omega|_{\mathbb{T}} |g_\omega(\underline{k} + \underline{p}) - g_\omega(\underline{k})| |g_\omega(\underline{k})| \\ &\quad + K \sum_{\omega=1}^N \int d\underline{k} |p|_{\mathbb{T}} |g_\omega(\underline{k} + \underline{p})| |g_\omega(\underline{k})|; \end{aligned} \quad (3.3.55)$$

the second term is bounded proportionally to $\|\underline{p}\| |\log \|\underline{p}\||$. Consider the first term. Using that, for $0 < \alpha < 1$:

$$|g_\omega(\underline{k} + \underline{p}) - g_\omega(\underline{k})| \leq C \left(\frac{\chi_\omega(\underline{k})}{\|\underline{k} - \underline{k}_F^\omega\|^{1-\alpha}} + \frac{\chi_\omega(\underline{k} + \underline{p})}{\|\underline{k} - \underline{k}_F^\omega + \underline{p}\|^{1-\alpha}} \right) \frac{\|\underline{p}\|^\alpha}{\|\underline{k} - \underline{k}_F^\omega + \underline{p}\|^\alpha \|\underline{k} - \underline{k}_F^\omega\|^\alpha} \quad (3.3.56)$$

we obtain:

$$\begin{aligned} &\int d\underline{k} |\underline{k} - \underline{k}_F^\omega| |g_\omega(\underline{k} + \underline{p}) - g_\omega(\underline{k})| |g_\omega(\underline{k})| \\ &\leq C \int d\underline{k} \left(\frac{\chi_\omega(\underline{k})}{\|\underline{k} - \underline{k}_F^\omega\|^{1-\alpha}} + \frac{\chi_\omega(\underline{k} + \underline{p})}{\|\underline{k} - \underline{k}_F^\omega + \underline{p}\|^{1-\alpha}} \right) \frac{\|\underline{p}\|^\alpha}{\|\underline{k} - \underline{k}_F^\omega + \underline{p}\|^\alpha \|\underline{k} - \underline{k}_F^\omega\|^\alpha} \\ &\leq \tilde{C}_\alpha \|\underline{p}\|^\alpha. \end{aligned} \quad (3.3.57)$$

Thus, we found:

$$\left| T_{1;\nu}(\underline{p}) - T_{1;\nu}(0) \right| \leq C_\alpha \|\underline{p}\|^\alpha . \quad (3.3.58)$$

This concludes the proof of (3.3.50). $\square$

Next, we shall consider the singular term $S_{1;\nu}$ in (3.3.49), whose explicit form is

$$\begin{aligned} S_{1;\nu}(\theta \underline{q}) &= - \sum_{\omega=1}^N v_\omega^\nu \int_{\mathbb{R} \times \mathbb{T}} g_\omega(\underline{k}) g_\omega(\underline{k} + \theta \underline{q}) \frac{d\underline{k}}{(2\pi)^2} \\ &\equiv \sum_{\omega=1}^N v_\omega^\nu \mathfrak{B}_\delta^\omega(\theta \underline{q}) , \end{aligned} \quad (3.3.59)$$

where $\mathfrak{B}_\delta^\omega$ is the relativistic bubble diagram:

$$\mathfrak{B}_\delta^\omega(\theta \underline{q}) := - \int_{\mathbb{R}^2} \frac{\chi_\delta^\omega(\underline{k}) \chi_\delta^\omega(\underline{k} + \theta \underline{q})}{D_\omega(\underline{k}) D_\omega(\underline{k} + \theta \underline{q})} \frac{d\underline{k}}{(2\pi)^2} . \quad (3.3.60)$$

The next proposition allows to compute it. The result is well-known, and we reproduce it here for completeness.

Proposition 3.3.6 (The relativistic bubble diagram). *Let $\alpha \in (0, 1)$. For any $\underline{q} = (q_0, \underline{q}) \neq \underline{0}$ such that $\|\underline{q}\| \leq \theta^{-\alpha}$, we have:*

$$\mathfrak{B}_\delta^\omega(\theta \underline{q}) = \frac{1}{4\pi|v_\omega|} \frac{-iq_0 + v_\omega q}{iq_0 + v_\omega q} + O(\theta^{1-\alpha}) . \quad (3.3.61)$$

Proof. First of all, by performing the change of coordinates $\underline{k} \rightarrow \theta \underline{k}$ we obtain

$$\mathfrak{B}_\delta^\omega(\theta \underline{q}) = \mathfrak{B}_{\delta/\theta}^\omega(\underline{q}) . \quad (3.3.62)$$

Then, we rewrite:

$$\begin{aligned} \mathfrak{B}_{\delta/\theta}^\omega(\underline{q}) &= - \int_{\mathbb{R}^2} \frac{\chi_{\delta/\theta}^\omega(\underline{k}) \chi_{\delta/\theta}^\omega(\underline{k} + \underline{q})}{D_\omega(\underline{k}) D_\omega(\underline{k} + \underline{q})} \frac{d\underline{k}}{(2\pi)^2} \\ &= - \frac{1}{D_\omega(\underline{q})} \int_{\mathbb{R}^2} \chi_{\delta/\theta}^\omega(\underline{k}) \chi_{\delta/\theta}^\omega(\underline{k} + \underline{q}) \left(\frac{1}{D_\omega(\underline{k})} - \frac{1}{D_\omega(\underline{k} + \underline{q})} \right) \frac{d\underline{k}}{(2\pi)^2} \\ &= - \frac{1}{D_\omega(\underline{q})} \int_{\mathbb{R}^2} \frac{\chi_{\delta/\theta}^\omega(\underline{k}) \chi_{\delta/\theta}^\omega(\underline{k} + \underline{q}) - \chi_{\delta/\theta}^\omega(\underline{k} - \underline{q}) \chi_{\delta/\theta}^\omega(\underline{k})}{D_\omega(\underline{k})} \frac{d\underline{k}}{(2\pi)^2} . \end{aligned} \quad (3.3.63)$$

Observing that:

$$\chi_{\delta/\theta}^\omega(\underline{k} + \underline{q}) - \chi_{\delta/\theta}^\omega(\underline{k} - \underline{q}) = 2\underline{q} \cdot \nabla_{\underline{k}} \chi_{\delta/\theta}^\omega(\underline{k}) + r_\omega(\underline{k}, \underline{q}) \quad (3.3.64)$$

with $|r_\omega(\underline{k}, \underline{q})| \leq C_\delta \theta^2 \|\underline{q}\|^2$, we see that, for $\|\underline{q}\| \leq \theta^{-\alpha}$, using that $r_\omega(\underline{k}, \underline{q})$ is supported for $\|\underline{k}\| \sim (\delta/\theta)$:

$$\left| \frac{1}{D_\omega(\underline{q})} \int_{\mathbb{R}^2} \frac{\chi_{\delta/\theta}^\omega(\underline{k}) r_\omega(\underline{k}, \underline{q})}{D_\omega(\underline{k})} \frac{d\underline{k}}{(2\pi)^2} \right| \leq C_\delta \theta^{-\alpha+1} \quad (3.3.65)$$

which vanishes for $\theta \rightarrow 0$. Consider now the main term,

$$\begin{aligned} & -\frac{2}{D_\omega(\underline{q})} \int_{\mathbb{R}^2} \frac{\chi_{\delta/\theta}^\omega(\underline{k}) \underline{q} \cdot \nabla_{\underline{k}} \chi_{\delta/\theta}^\omega(\underline{k})}{D_\omega(\underline{k})} \frac{d\underline{k}}{(2\pi)^2} \\ & = -\frac{2q_0}{D_\omega(\underline{q})} \int_{\mathbb{R}^2} \frac{\chi_{\delta/\theta}^\omega(\underline{k}) \partial_0 \chi_{\delta/\theta}^\omega(\underline{k})}{D_\omega(\underline{k})} \frac{d\underline{k}}{(2\pi)^2} - \frac{2q_1}{D_\omega(\underline{q})} \int_{\mathbb{R}^2} \frac{\chi_{\delta/\theta}^\omega(\underline{k}) \partial_1 \chi_{\delta/\theta}^\omega(\underline{k})}{D_\omega(\underline{k})} \frac{d\underline{k}}{(2\pi)^2}. \end{aligned} \quad (3.3.66)$$

After a rescaling and a change of variable in the k_1 variable, we get;

$$\begin{aligned} & -\frac{2}{D_\omega(\underline{q})} \int_{\mathbb{R}^2} \frac{\chi_{\delta/\theta}^\omega(\underline{k}) \underline{q} \cdot \nabla_{\underline{k}} \chi_{\delta/\theta}^\omega(\underline{k})}{D_\omega(\underline{k})} \frac{d\underline{k}}{(2\pi)^2} \\ & = -\frac{2q_0}{D_\omega(\underline{q})} \frac{1}{|v_\omega|} \int_{\mathbb{R}^2} \frac{\chi(\underline{k}) \partial_0 \chi(\underline{k})}{ik_0 + k_1} \frac{d\underline{k}}{(2\pi)^2} - \frac{2q_1}{D_\omega(\underline{q})} \frac{v_\omega}{|v_\omega|} \int_{\mathbb{R}^2} \frac{\chi(\underline{k}) \partial_1 \chi(\underline{k})}{ik_0 + k_1} \frac{d\underline{k}}{(2\pi)^2}, \end{aligned} \quad (3.3.67)$$

where $\chi(\underline{k}) \equiv \chi(\|\underline{k}\|)$, see Eq. (3.3.3). Then, one observes that this expression can be further rewritten as:

$$\begin{aligned} & -\frac{2}{D_\omega(\underline{q})} \int_{\mathbb{R}^2} \frac{\chi_{\delta/\theta}^\omega(\underline{k}) \underline{q} \cdot \nabla_{\underline{k}} \chi_{\delta/\theta}^\omega(\underline{k})}{D_\omega(\underline{k})} \frac{d\underline{k}}{(2\pi)^2} \\ & = -\frac{2(q_0 + iv_\omega q_1)}{D_\omega(\underline{q})|v_\omega|} \int_{\mathbb{R}^2} \frac{\chi(\underline{k}) \partial_0 \chi(\underline{k})}{ik_0 + k_1} \frac{d\underline{k}}{(2\pi)^2} \\ & = -\frac{2(q_0 + iv_\omega q_1)}{D_\omega(\underline{q})|v_\omega|} \int_{\mathbb{R}^2} \frac{k_0}{\|\underline{k}\|} \frac{\chi(\|\underline{k}\|) \chi'(\|\underline{k}\|)}{ik_0 + k_1} \frac{d\underline{k}}{(2\pi)^2} \\ & = -\frac{q_0 + iv_\omega q_1}{D_\omega(\underline{q})|v_\omega|} \int_{\mathbb{R}^2} \frac{k_0}{\|\underline{k}\|} \frac{(\chi^2(\|\underline{k}\|))'}{ik_0 + k_1} \frac{d\underline{k}}{(2\pi)^2}. \end{aligned} \quad (3.3.68)$$

The last integral can be computed switching to polar coordinates, and one finds:

$$-\frac{2}{D_\omega(\underline{q})} \int_{\mathbb{R}^2} \frac{\chi_{\delta/\theta}^\omega(\underline{k}) \underline{q} \cdot \nabla_{\underline{k}} \chi_{\delta/\theta}^\omega(\underline{k})}{D_\omega(\underline{k})} \frac{d\underline{k}}{(2\pi)^2} = \frac{q_0 + iv_\omega q_1}{D_\omega(\underline{q})|v_\omega|} \frac{-i}{4\pi} \quad (3.3.69)$$

which reproduces the main term in (3.3.61). $\square$

The evaluation of the relativistic bubble diagram, combined with lattice conservation laws, allow to compute (3.3.49).

Proposition 3.3.7 (Evaluation of the linear response). *We have:*

$$\chi_\nu^{\text{lin}}(x; \eta, \theta) = -\sum_{\omega=1}^N \chi_{\nu, \omega} \int_{\mathbb{R}} \hat{\mu}_\infty(q) e^{iq\theta x} \frac{v_\omega q}{-i/a + v_\omega q} \frac{dq}{2\pi} + O(\theta^{1-\alpha}), \quad (3.3.70)$$

with $\chi_{\nu, \omega} = v_\omega^\nu / 2\pi |v_\omega|$. Let $\theta = a\eta$. We can distinguish three regimes.

(i) Suppose that $a \rightarrow 0$ as $\eta \rightarrow 0^+$. Then $\chi_\nu^{\text{lin}}(x; \eta, \theta) = O(a)$.

(ii) Suppose that a is constant in η . Then:

$$\chi_\nu^{\text{lin}}(x; \eta, \theta) = -\sum_{\omega=1}^N \chi_{\nu, \omega} \int_{\mathbb{R}} \mu_\infty(\theta x - y) \left[\delta(y) - \frac{1}{a|v_\omega|} e^{-y/av_\omega} \Theta(y/v_\omega) \right] dy + O(\eta^{1-\alpha}). \quad (3.3.71)$$

(iii) Suppose that $a \rightarrow \infty$ as $\eta \rightarrow 0^+$, so that $a\eta \rightarrow 0$. Then:

$$\chi_\nu^{\text{lin}}(x; \eta, \theta) = -\mu_\infty(\theta x) \sum_{\omega=1}^N \chi_{\nu, \omega} + O(\theta^{1-\alpha}). \quad (3.3.72)$$

Proof. By (3.3.59), we rewrite the linear response (3.3.49) as:

$$\begin{aligned} \chi_\nu^{\text{lin}}(x, \eta, \theta) &= - \int_{\mathbb{T}_{\theta^{-1}}} \frac{dq}{2\pi} \hat{\mu}_\alpha(-q) e^{-i\theta qx} [S_{1;\nu}(\theta \underline{q}) + R_{1;\nu}(\theta \underline{q})] \\ &= - \int_{\mathbb{R}} \frac{dq}{2\pi} \hat{\mu}_\infty(-q) e^{-i\theta qx} \left[\sum_{\omega=1}^N v_\omega^\nu \mathfrak{B}_\delta^\omega(\theta \underline{q}) + R_{1;\nu}(\underline{0}) \right] + E_{\nu;1}(x; \eta, \theta), \end{aligned} \quad (3.3.73)$$

where, by (3.3.50), and by the regularity properties of the function $\mu_\infty(x)$:

$$|E_{\nu;1}(x; \eta, \theta)| \leq K\theta. \quad (3.3.74)$$

Next, by (3.3.61), (3.3.62), $\mathfrak{B}_\delta^\omega(\theta \underline{q}) = \mathfrak{B}_\infty^\omega(\underline{q}) + O(\theta^{1-\alpha})$, with:

$$\mathfrak{B}_\infty^\omega(\underline{q}) = \frac{1}{4\pi|v_\omega|} \frac{-iq_0 + v_\omega q}{iq_0 + v_\omega q}; \quad (3.3.75)$$

thus, recalling that $\underline{q} = (a^{-1}, q)$ and performing the change of variables $q \rightarrow -q$,

$$\begin{aligned} \chi_\nu^{\text{lin}}(x; \eta, \theta) &= - \int_{\mathbb{R}} \frac{dq}{2\pi} \hat{\mu}_\infty(q) e^{i\theta qx} \left[\sum_{\omega=1}^N v_\omega^\nu \frac{1}{4\pi|v_\omega|} \frac{i/a + v_\omega q}{-i/a + v_\omega q} + R_{1;\nu}(\underline{0}) \right] + \sum_{i=1,2} E_{\nu;i}(x; \eta, \theta) \end{aligned} \quad (3.3.76)$$

where:

$$|E_{\nu;2}(x; \eta, \theta)| \leq C\theta^{1-\alpha}. \quad (3.3.77)$$

Let us now determine $R_{1;\nu}(\underline{0})$. We shall compute it by exploiting lattice conservation laws. Thanks to the continuity equation (3.1.30), the following equality holds:

$$i\partial_{x_0} \langle \mathbf{T} n_{\underline{x}}; j_{\nu, \underline{y}} \rangle_{\beta, L} + \text{d}_x \langle \mathbf{T} j_{\underline{x}}; j_{\nu, \underline{y}} \rangle_{\beta, L} = i\delta(x_0 - y_0) \langle [n_{\underline{x}}, j_{\nu, \underline{y}}] \rangle_{\beta, L}, \quad (3.3.78)$$

where $\mathcal{O}_{\underline{x}} := \gamma_{x_0}(\mathcal{O}_{x_1})$. The contact term on the right-hand side arises due to the definition of time-ordering. Taking the Fourier transform of both left-hand side and right-hand side, we obtain:

$$p_0 \frac{1}{\beta L} \langle \mathbf{T} \hat{n}_{\underline{p}}; \hat{j}_{\nu, -\underline{p}} \rangle_{\beta, L} + (1 - e^{-ip}) \frac{1}{\beta L} \langle \mathbf{T} \hat{j}_{\underline{p}}; \hat{j}_{\nu, -\underline{p}} \rangle_{\beta, L} = i \sum_{x \in \mathbb{Z}} e^{-ipx} \langle [n_x, j_{\nu, 0}] \rangle_{\beta, L}. \quad (3.3.79)$$

Let $\underline{p} = (p_0, 0)$. Using that $[\sum_{x \in \mathbb{Z}} n_x, j_{\nu, 0}] = 0$, we obtain $\langle \mathbf{T} \hat{n}_{\underline{p}}; \hat{j}_{\nu, -\underline{p}} \rangle_{\beta, L} = 0$. Hence, recalling that $(1/\beta L) \langle \mathbf{T} \hat{n}_{\underline{p}}; \hat{j}_{\nu, -\underline{p}} \rangle_{\beta, L} = S_{1;\nu}^{\beta, L}(\underline{p}) + R_{1;\nu}^{\beta, L}(\underline{p})$, we obtain, in the $\beta, L \rightarrow \infty$ limit, and by the continuity of $R_{1;\nu}(\underline{p})$ at $\underline{p} = \underline{0}$:

$$S_{1;\nu}((p_0, 0)) + R_{1;\nu}((p_0, 0)) = 0 \quad \implies \quad R_{1;\nu}(\underline{0}) = - \lim_{p_0 \rightarrow 0} S_{1;\nu}((p_0, 0)) = \sum_{\omega=1}^N \frac{v_\omega^\nu}{4\pi|v_\omega|}. \quad (3.3.80)$$

Plugging this identity in (3.3.76), we obtain:

$$\chi_\nu^{\text{lin}}(x; \eta, \theta) = - \sum_{\omega=1}^N \frac{v_\omega^\nu}{2\pi|v_\omega|} \int_{\mathbb{R}} \frac{dq}{2\pi} \hat{\mu}_\infty(q) e^{iq\theta x} \frac{v_\omega q}{-i/a + v_\omega q} + \sum_{i=1,2} E_{\nu;i}(x; \eta, \theta) \quad (3.3.81)$$

which proves (3.3.70). Let us further analyse the integral. From (3.3.81), it is clear that if $a \rightarrow 0$, the integral is vanishing. More generally, we can rewrite (3.3.81) as:

$$\chi_\nu^{\text{lin}}(x; \eta, \theta) = - \sum_{\omega=1}^N \chi_{\nu,\omega} \left[\mu_\infty(\theta x) + \frac{i}{a} \int_{\mathbb{R}} \frac{dq}{2\pi} \frac{\hat{\mu}_\infty(q) e^{iq\theta x}}{-i/a + v_\omega q} \right] + \sum_{i=1,2} E_{\nu;i}(x; \eta, \theta). \quad (3.3.82)$$

To compute the integral, we employ the good decay properties of $\hat{\mu}_\infty(q)$ to rewrite it as:

$$\begin{aligned} \int_{\mathbb{R}} \frac{\hat{\mu}_\infty(q) e^{iq\theta x}}{-i/a + v_\omega q} \frac{dq}{2\pi} &= \lim_{Q \rightarrow \infty} \int_{|q| \leq Q} \frac{\hat{\mu}_\infty(q) e^{iq\theta x}}{-i/a + v_\omega q} \frac{dq}{2\pi} \\ &= \lim_{Q \rightarrow \infty} \int dy \mu_\infty(y) \int_{|q| \leq Q} \frac{e^{-iqy} e^{iq\theta x}}{-i/a + v_\omega q} \frac{dq}{2\pi} \\ &= \lim_{Q \rightarrow \infty} \int dz \mu_\infty(z + \theta x) \int_{|q| \leq Q} \frac{e^{-iqz}}{-i/a + v_\omega q} \frac{dq}{2\pi}. \end{aligned} \quad (3.3.83)$$

Let $z > 0$. By Cauchy theorem for holomorphic functions:

$$\begin{aligned} \int_{|q| \leq Q} \frac{e^{-iqz}}{-i/a + v_\omega q} \frac{dq}{2\pi} &= \frac{1}{2\pi v_\omega} \int_{|q| \leq Q} \frac{e^{-iqz}}{-i/(av_\omega) + q} dq \\ &= -\frac{1}{2\pi v_\omega} (2\pi i) e^{z/(av_\omega)} \mathbf{1}(v_\omega < 0) + g_Q^+(z) \end{aligned} \quad (3.3.84)$$

where $g_Q^+(z)$ is bounded uniformly in z and $\lim_{Q \rightarrow \infty} g_Q^+(z) = 0$. Similarly, if $z < 0$:

$$\int_{|q| \leq Q} \frac{e^{-iqz}}{-i/a + v_\omega q} \frac{dq}{2\pi} = \frac{1}{2\pi v_\omega} (2\pi i) e^{z/(av_\omega)} \mathbf{1}(v_\omega > 0) + g_Q^-(z) \quad (3.3.85)$$

with $g_Q^-(z)$ is bounded uniformly in z and $\lim_{Q \rightarrow \infty} g_Q^-(z) = 0$. All in all, for $z \neq 0$:

$$\lim_{Q \rightarrow \infty} \int_{|q| \leq Q} \frac{e^{-iqz}}{-i/a + v_\omega q} \frac{dq}{2\pi} = \frac{i}{|v_\omega|} e^{z/(av_\omega)} \mathbf{1}((z/v_\omega) < 0). \quad (3.3.86)$$

Plugging this in (3.3.83), we obtain, performing a change of variable $z \rightarrow -z$:

$$\int_{\mathbb{R}} \frac{\hat{\mu}_\infty(q) e^{iq\theta x}}{-i/a + v_\omega q} \frac{dq}{2\pi} = \frac{i}{|v_\omega|} \int dz \mu_\infty(\theta x - z) e^{-z/(av_\omega)} \theta(z/v_\omega) \quad (3.3.87)$$

with $\theta(\cdot)$ the Heaviside step function, equal to 1 for positive argument and zero otherwise. Inserting this formula in (3.3.82), we get:

$$\begin{aligned} \chi_\nu^{\text{lin}}(x; \eta, \theta) &= - \sum_{\omega=1}^N \chi_{\nu,\omega} \left[\mu_\infty(\theta x) - \frac{1}{a|v_\omega|} \int dz \mu_\infty(\theta x - z) e^{-z/(av_\omega)} \theta(z/v_\omega) \right] \\ &\quad + \sum_{i=1,2} E_{\nu;i}(x; \eta, \theta) \end{aligned} \quad (3.3.88)$$

which proves (3.3.71). Finally, the last claim (3.3.72) follows after taking the $a \rightarrow \infty$ limit, and using that μ is integrable. This concludes the proof of Proposition 3.3.7. $\square$

3.3.4 Estimates for the higher-order corrections

Let:

$$\mathfrak{B}_{n+1}^\omega(\underline{p}_1, \dots, \underline{p}_n) := - \int_{\mathbb{R}^2} \prod_{i=1}^{n+1} g_\omega\left(\underline{k} + \sum_{j<i} \underline{p}_j\right) \frac{d\underline{k}}{(2\pi)^2}, \quad (3.3.89)$$

with the understanding that $\underline{p}_{n+1} = -\sum_{j=1}^n \underline{p}_j$. Thus, from (3.3.41), in the $\beta, L \rightarrow \infty$ limit:

$$S_{n;\nu}(\underline{p}_1, \dots, \underline{p}_n) = \sum_{\omega=1}^N v_\omega^\nu \sum_{\pi \in S_n} \mathfrak{B}_{n+1}^\omega(\underline{p}_{\pi(1)}, \dots, \underline{p}_{\pi(n)}). \quad (3.3.90)$$

Our main goal will be to show that there are non-trivial cancellations in the sum over permutations. These cancellations are ultimately implied by the identity:

$$g_\omega(\underline{k})g_\omega(\underline{k} + \underline{p}) = \frac{1}{D_\omega(\underline{p})} \left(g_\omega(\underline{k}) - g_\omega(\underline{k} + \underline{p}) \right) + F_\delta^\omega(\underline{k}; \underline{p}) \quad (3.3.91)$$

where:

$$F_\delta^\omega(\underline{k}; \underline{p}) := \frac{\chi_\delta^\omega(\underline{k})[\chi_\delta^\omega(\underline{k} + \underline{p}) - 1]}{D_\omega(\underline{k})D_\omega(\underline{k} + \underline{p})} + \frac{\chi_\delta^\omega(\underline{k} + \underline{p}) - \chi_\delta^\omega(\underline{k})}{D_\omega(\underline{p})D_\omega(\underline{k} + \underline{p})}. \quad (3.3.92)$$

This identity is a tree-level Ward identity, for the vertex function. It can be viewed as generated by the invariance of the relativistic 1 + 1 dimensional model under chiral gauge transformations. The term $F_\delta^\omega(\underline{k}; \underline{p})$ in the right-hand side introduces an anomaly in the usual Ward identity, due to the presence of the momentum cutoff. The starting point is the following proposition. A similar cancellation, for chiral fermions in the absence of UV cutoff, has been discussed in [34].

Proposition 3.3.8 (Loop cancellation.). *For any $\omega = 1, \dots, N$, the following holds:*

$$\begin{aligned} & \sum_{\pi \in S_n} \mathfrak{B}_{n+1}^\omega(\underline{p}_{\pi(1)}, \dots, \underline{p}_{\pi(n)}) \\ &= -\frac{1}{n+1} \sum_{\pi \in S_{n+1}} \int F_\delta^\omega(\underline{k}; \underline{p}_{\pi(1)}) \prod_{i=3}^{n+1} g_\omega\left(\underline{k} + \sum_{j<i} \underline{p}_{\pi(j)}\right) \frac{d\underline{k}}{(2\pi)^2}. \end{aligned} \quad (3.3.93)$$

Proof. To begin, we define:

$$\mathfrak{B}_{n+1}^\omega(\underline{p}_1, \dots, \underline{p}_n, \underline{p}_{n+1}) := \mathfrak{B}_{n+1}^\omega(\underline{p}_1, \dots, \underline{p}_n). \quad (3.3.94)$$

Observe that $\mathfrak{B}_{n+1}^\omega(\underline{p}_1, \dots, \underline{p}_n, \underline{p}_{n+1})$ is invariant under the abelian subgroup of maximal cycles σ of the permutations of $\{1, 2, \dots, n+1\}$,

$$\mathfrak{B}_{n+1}^\omega(\underline{p}_1, \dots, \underline{p}_{n+1}) = \mathfrak{B}_{n+1}^\omega(\underline{p}_{\sigma(1)}, \dots, \underline{p}_{\sigma(n)}, \underline{p}_{\sigma(n+1)}). \quad (3.3.95)$$

In fact, let us consider the integrand in (3.3.89):

$$g_\omega(\underline{k})g_\omega(\underline{k} + \underline{p}_1)g_\omega(\underline{k} + \underline{p}_1 + \underline{p}_2) \cdots g_\omega(\underline{k} + \underline{p}_1 + \dots + \underline{p}_n). \quad (3.3.96)$$

Using that $\underline{p}_{n+1} = -\underline{p}_1 - \dots - \underline{p}_n$ we can rewrite it as:

$$g_\omega(\underline{k} + \underline{p}_1 + \dots + \underline{p}_{n+1})g_\omega(\underline{k} + \underline{p}_1)g_\omega(\underline{k} + \underline{p}_1 + \underline{p}_2) \cdots g_\omega(\underline{k} + \underline{p}_1 + \dots + \underline{p}_n). \quad (3.3.97)$$

Now, perform the change of variables $\underline{k} + \underline{p}_1 \rightarrow \underline{k}$ in the integral in the definition (3.3.89). We get:

$$g_\omega(\underline{k} + \underline{p}_2 + \dots + \underline{p}_{n+1}) g_\omega(\underline{k}) g_\omega(\underline{k} + \underline{p}_2) \cdots g_\omega(\underline{k} + \underline{p}_2 + \dots + \underline{p}_n), \quad (3.3.98)$$

which we can also rewrite as, after a rearrangement of the factors:

$$g_\omega(\underline{k} + \underline{p}_1 + \dots + \underline{p}_{n+1}) g_\omega(\underline{k} + \underline{p}_2) \cdots g_\omega(\underline{k} + \underline{p}_2 + \dots + \underline{p}_n) g_\omega(\underline{k} + \underline{p}_2 + \dots + \underline{p}_{n+1}) \quad (3.3.99)$$

which is precisely what one obtains from (3.3.97) replacing the ordered sequence of external momenta $\{\underline{p}_1, \underline{p}_2, \dots, \underline{p}_n, \underline{p}_{n+1}\}$ with $\{\underline{p}_2, \underline{p}_3, \dots, \underline{p}_{n+1}, \underline{p}_1\}$. This proves (3.3.95).

Next, we rewrite the left-hand side of (3.3.93) as

$$\sum_{\pi \in S_n} \mathfrak{B}_{n+1}^\omega(\underline{p}_{\pi(1)}, \dots, \underline{p}_{\pi(n)}, \underline{p}_{n+1}) = \sum_{\substack{\pi \in S_n \\ \pi(n+1)=n+1}} \mathfrak{B}_{n+1}^\omega(\underline{p}_{\pi(1)}, \dots, \underline{p}_{\pi(n+1)}); \quad (3.3.100)$$

using the aforementioned cyclicity property, we can also rewrite:

$$\sum_{\substack{\pi \in S_n \\ \pi(n+1)=n+1}} \mathfrak{B}_{n+1}^\omega(\underline{p}_{\pi(1)}, \dots, \underline{p}_{\pi(n+1)}) = \frac{1}{n+1} \sum_{i=1}^{n+1} \sum_{\substack{\pi \in S_{n+1} \\ \pi(i)=n+1}} \mathfrak{B}_{n+1}^\omega(\underline{p}_{\pi(1)}, \dots, \underline{p}_{\pi(n+1)}), \quad (3.3.101)$$

where we used the invariance under maximally cyclic permutations. Thus,

$$\sum_{\pi \in S_n} \mathfrak{B}_{n+1}^\omega(\underline{p}_{\pi(1)}, \dots, \underline{p}_{\pi(n)}, \underline{p}_{n+1}) = \frac{1}{n+1} \sum_{\pi \in S_{n+1}} \mathfrak{B}_{n+1}^\omega(\underline{p}_{\pi(1)}, \dots, \underline{p}_{\pi(n+1)}). \quad (3.3.102)$$

We further rewrite:

$$\sum_{\pi \in S_n} \mathfrak{B}_{n+1}^\omega(\underline{p}_{\pi(1)}, \dots, \underline{p}_{\pi(n)}, \underline{p}_{n+1}) = \frac{1}{n+1} \sum_{i=1}^{n+1} \sum_{\substack{\pi \in S_{n+1} \\ \pi(1)=i}} \mathfrak{B}_{n+1}^\omega(\underline{p}_i, \dots, \underline{p}_{\pi(n+1)}); \quad (3.3.103)$$

applying the Ward identity (3.3.91) to $g_\omega(\underline{k} + \underline{p}_i) g_\omega(\underline{k})$ we get:

$$\begin{aligned} \mathfrak{B}_{n+1}^\omega(\underline{p}_i, \dots, \underline{p}_{\pi(n+1)}) &= \frac{\mathfrak{B}_n^\omega(\underline{p}_{\pi(2)} + \underline{p}_i, \dots, \underline{p}_{\pi(n+1)}) - \mathfrak{B}_n^\omega(\underline{p}_{\pi(2)}, \dots, \underline{p}_{\pi(n+1)} + \underline{p}_i)}{D_\omega(\underline{p}_i)} \\ &\quad - \int \frac{d\underline{k}}{(2\pi)^2} F_\delta^\omega(\underline{k}; \underline{p}_i) \prod_{m=3}^{n+1} g_\omega\left(\underline{k} + \sum_{j < m} \underline{p}_{\pi(j)}\right). \end{aligned} \quad (3.3.104)$$

Let us plug the first term in the right-hand side of (3.3.104) in the rightmost sum in

(3.3.103). We get:

$$\begin{aligned}
& \sum_{\substack{\pi \in S_{n+1} \\ \pi(1)=i}} \mathfrak{B}_{n+1}^\omega(\underline{p}_i, \dots, \underline{p}_{\pi(n+1)}) \\
&= \frac{1}{D_\omega(\underline{p}_i)} \sum_{\substack{\pi \in S_{n+1} \\ \pi(1)=i}} \left(\mathfrak{B}_n^\omega(\underline{p}_{\pi(2)} + \underline{p}_i, \dots, \underline{p}_{\pi(n+1)}) - \mathfrak{B}_n^\omega(\underline{p}_{\pi(2)}, \dots, \underline{p}_{\pi(n+1)} + \underline{p}_i) \right) \\
&= \frac{1}{D_\omega(\underline{p}_i)} \sum_{\substack{\pi \in S_{n+1} \\ \pi(1)=i}} \left(\mathfrak{B}_n^\omega(\underline{p}_{\pi(2)} + \underline{p}_i, \underline{p}_{\pi(3)}, \dots, \underline{p}_{\pi(n+1)}) - \mathfrak{B}_n^\omega(\underline{p}_{\pi(n+1)} + \underline{p}_i, \underline{p}_{\pi(2)}, \dots, \underline{p}_{\pi(n)}) \right) \\
&= 0,
\end{aligned} \tag{3.3.105}$$

where the last identity follows from the fact that given a permutation π such that $\pi(1) = i$, we can define the new permutation $\tilde{\pi}$ such that $\tilde{\pi}(1) = i$ and $\tilde{\pi}(2) = \pi(n+1)$, $\tilde{\pi}(3) = \pi(2)$, $\dots$, $\tilde{\pi}(n+1) = \pi(n)$, a relation that defines a one-to-one map between permutations. Hence, (3.3.93) follows. $\square$

The cancellation highlighted in the previous proposition implies an improved estimate for the n -th order contribution to the full response.

Proposition 3.3.9 (Loop estimates: relativistic contributions). *Let $p_1, \dots, p_n$ such that $\hat{\mu}_{\alpha, \theta}(-p_i) \neq 0$. Suppose that $2 \leq n < (1/4)\theta^{\alpha-1}$. Then, for β, L large enough:*

$$\frac{\theta^{n-1}}{n!} |S_{n;\nu}^{\beta, L}(\underline{p}_1, \dots, \underline{p}_n)| \leq C^m \theta^{n-1}. \tag{3.3.106}$$

Suppose instead that $n \geq (1/4)\theta^{\alpha-1}$, then:

$$\frac{\theta^{n-1}}{n!} |S_{n;\nu}^{\beta, L}(\underline{p}_1, \dots, \underline{p}_n)| \leq C^m \frac{a^{n-1}}{n!} |\log \eta|. \tag{3.3.107}$$

Proof. Let us prove (3.3.106). Let $2 \leq n < (1/4)\theta^{\alpha-1}$. It will be enough to prove the bound (3.3.106) for the $\beta, L \rightarrow \infty$ limit. We start from the following rewriting, consequence of (3.3.8) and of (3.3.90):

$$\begin{aligned}
S_{n;\nu}(\underline{p}_1, \dots, \underline{p}_n) &= \sum_{\omega=1}^N v_\omega^\nu \sum_{\pi \in S_n} \mathfrak{B}_{n+1}^\omega(\underline{p}_{\pi(1)}, \dots, \underline{p}_{\pi(n)}, \underline{p}_{n+1}) \\
&= -\frac{1}{n+1} \sum_{\omega=1}^N v_\omega^\nu \sum_{\pi \in S_{n+1}} \int \frac{d\underline{k}}{(2\pi)^2} F_\delta^\omega(\underline{k}; \underline{p}_{\pi(1)}) \prod_{i=3}^{n+1} g_\omega\left(\underline{k} + \sum_{j<i} \underline{p}_{\pi(j)}\right).
\end{aligned} \tag{3.3.108}$$

The proof of (3.3.106) is based on the support properties of the function F_δ^ω . For all $\|\underline{p}\|_\omega < (1/4)\delta$, we write:

$$\chi_\delta^\omega(\underline{k} + \underline{p}) = \chi_\delta^\omega(\underline{k}) + \frac{1}{\delta} \int_0^1 \chi'(\|\underline{k} - \underline{k}_F^\omega + t\underline{p}\|_\omega/\delta) \frac{\langle \underline{k} - \underline{k}_F^\omega + t\underline{p}, \underline{p} \rangle_\omega}{\|\underline{k} - \underline{k}_F^\omega + t\underline{p}\|_\omega} dt \tag{3.3.109}$$

where $\langle \underline{a}, \underline{b} \rangle_\omega := a_0 b_0 + v_\omega^2 \underline{a} \underline{b}$ is the scalar product associated to the norm $\|\cdot\|_\omega$. Thus, from the definition (3.3.92), we have:

$$\begin{aligned} & |F_\delta^\omega(\underline{k}; \underline{p})| \\ & \leq \frac{\chi_\delta^\omega(\underline{k})[1 - \chi_\delta^\omega(\underline{k})]}{\|\underline{k} - \underline{k}_F^\omega\|_\omega \|\underline{k} - \underline{k}_F^\omega + \underline{p}\|_\omega} \\ & \quad + \int_0^1 \frac{|\chi'(\|\underline{k} - \underline{k}_F^\omega + t\underline{p}\|_\omega/\delta)|}{\delta \|\underline{k} - \underline{k}_F^\omega + \underline{p}\|_\omega} \left[\frac{\|\underline{p}\|_\omega}{\|\underline{k} - \underline{k}_F^\omega\|_\omega} \chi_\delta^\omega(\underline{k}) + 1 \right] dt. \end{aligned} \quad (3.3.110)$$

Since $\text{supp}(\chi') \subseteq [1, 2]$, we can estimate, for some $C_\lambda > 0$:

$$|F_\delta^\omega(\underline{k}; \underline{p})| \leq \begin{cases} C_\lambda/\delta^2 & \|\underline{k} - \underline{k}_F^\omega\|_\omega \in [(3/4)\delta, (9/4)\delta] \\ 0 & \text{otherwise.} \end{cases} \quad (3.3.111)$$

Let us use this bound to estimate (3.3.108). Observe that, since $\underline{p}_i$ is such that $\hat{\mu}_{\alpha, \theta}(p_i) \neq 0$, we have $|p_i| \leq \theta^{1-\alpha}$; hence we can assume that $\|\underline{p}_i\| \leq \theta^{1-\alpha} + \theta/a \leq (1/4)\delta$ for θ small enough, and we can use the bound (3.3.111). In particular, this estimate allows to show that, if the order n is not too large, all the propagators are bounded uniformly in their arguments. In fact, from (3.3.111), and for $n < (1/4)\theta^{\alpha-1}$:

$$\begin{aligned} \|\underline{k} - \underline{k}_F^\omega + \sum_{j \leq i} \underline{p}_{\pi(j)}\|_\omega & \geq \|\underline{k} - \underline{k}_F^\omega\|_\omega - \|\sum_{j \leq i} \underline{p}_{\pi(j)}\|_\omega \\ & \geq \|\underline{k} - \underline{k}_F^\omega\|_\omega - n(\theta^{1-\alpha} + a\theta) \geq \delta/2. \end{aligned} \quad (3.3.112)$$

Thus, integration over $\underline{k}$ gives:

$$\int \frac{d\underline{k}}{(2\pi)^2} |F_\delta^\omega(\underline{k}; \underline{p}_{\pi(1)})| \prod_{i=3}^{n+1} \left| \frac{\chi_\delta^\omega(\underline{k} + \sum_{j < i} \underline{p}_{\pi(j)})}{D_\omega(\underline{k} + \sum_{j < i} \underline{p}_{\pi(j)})} \right| \leq (K/\delta)^{n-1}; \quad (3.3.113)$$

this bound, combined with (3.3.108), implies:

$$|S_{n; \nu}(\underline{p}_1, \dots, \underline{p}_n)| \leq C^n n!, \quad (3.3.114)$$

which proves Eq. (3.3.106). Since $S_{n; \nu}^{\beta, L}$ at nonzero external momenta converges to a limit for $\beta, L \rightarrow \infty$, an analogous estimate (3.3.114) holds for β, L large enough and $n \leq (1/4)\theta^{\alpha-1}$.

Next, let us prove (3.3.107). Suppose that $n \geq (1/4)\theta^{\alpha-1}$. For these large values of n , we cannot use the previous argument, because the condition (3.3.112) might fail. Here we shall use that, thanks to the fact that the temporal part of the external momenta is $p_{i,0} = \eta$ for all $i = 1, \dots, n$, there is no accumulation of singularities in the loop integral. Starting from (3.3.41), we estimate:

$$|S_{n; \nu}^{\beta, L}(\underline{p}_1, \dots, \underline{p}_n)| \leq C \sum_{\omega=1}^N \int_{\beta, L} \frac{d\underline{k}}{(2\pi)^2} \sum_{\pi \in S_n} \prod_{i=1}^{n+1} \left| g_\omega\left(\underline{k} + \sum_{j < i} \underline{p}_{\pi(j)}\right) \right| \quad (3.3.115)$$

and:

$$\int_{\beta, L} \frac{d\underline{k}}{(2\pi)^2} \prod_{i=1}^{n+1} \left| g_\omega\left(\underline{k} + \sum_{j < i} \underline{p}_j\right) \right| \leq \int_{\beta, L} \frac{d\underline{k}}{(2\pi)^2} \prod_{i=1}^{n+1} \frac{\chi_\delta^\omega(\underline{k} + \sum_{j < i} \underline{p}_j)}{D_\omega(\underline{k} + \sum_{j < i} \underline{p}_j)}. \quad (3.3.116)$$

By the support properties of the cutoff function, the domain of summation over $\underline{k}$ is contained in $B_{2\delta}^\omega = \{\underline{k} \mid \|\underline{k} - \underline{k}_F^\omega\|_\omega \leq 2\delta\}$; in particular, $|k_0| \leq 2\delta$. We further split the Matsubara frequencies as:

$$\{k_0 \in \mathbb{M}_\beta \mid |k_0| \leq 2\delta\} = \mathcal{A} \cup \mathcal{A}^c, \quad (3.3.117)$$

with, for $\bar{n} = \min\{n, \lfloor 2\delta/\eta \rfloor\}$:

$$\begin{aligned} \mathcal{A} &:= \bigcup_{j=1}^{\bar{n}} \mathcal{A}_j \\ \mathcal{A}_j &:= \left\{ k_0 \in \mathbb{M}_\beta \mid k_0 \in \eta \left[-j - \frac{1}{2}, -j + \frac{1}{2} \right] \right\}. \end{aligned} \quad (3.3.118)$$

Observe that, for $k_0 \in \mathcal{A}_j$ and $i \neq j$:

$$|D_\omega(\underline{k} + \sum_{l \leq i} \underline{p}_l)| \geq |k_0 + \eta i| \geq \eta[|i - j| - 1/2] \geq \eta|i - j|/2. \quad (3.3.119)$$

Therefore, for $k_0 \in \mathcal{A}_j$:

$$\begin{aligned} \prod_{i=0}^n \frac{1}{|D_\omega(\underline{k} + \sum_{l \leq i} \underline{p}_l)|} &\leq \frac{(2/\eta)^n}{\|\underline{k} - \underline{k}_F^\omega + \sum_{l \leq j} \underline{p}_l\|_\omega} \prod_{\substack{i=0 \\ i \neq j}}^n \frac{1}{|i - j|} \\ &= \frac{1}{n!} \binom{n}{j} \frac{(2/\eta)^n}{\|\underline{k} - \underline{k}_F^\omega + \sum_{l \leq j} \underline{p}_l\|_\omega}. \end{aligned} \quad (3.3.120)$$

Hence:

$$\begin{aligned} &\int_{\beta, L} \mathbf{1}(k_0 \in \mathcal{A}_j) \prod_{i=0}^n \frac{\chi_\delta^\omega(\underline{k} + \sum_{j \leq i} \underline{p}_j)}{|D_\omega(\underline{k} + \sum_{j \leq i} \underline{p}_j)|} \frac{d\underline{k}}{(2\pi)^2} \\ &\leq \frac{(2/\eta)^n}{n!} \sum_{j=0}^{\bar{n}} \binom{n}{j} \int_{\beta, L} \mathbf{1}(k_0 \in \mathcal{A}_j) \frac{\chi_\delta^\omega(\underline{k} - \underline{k}_F^\omega + \sum_{l \leq j} \underline{p}_l)}{\|\underline{k} - \underline{k}_F^\omega + \sum_{l \leq j} \underline{p}_l\|_\omega} d\underline{k} \\ &\leq \frac{C^n \eta^{1-n}}{n!} |\log \eta| \sum_{j=1}^{\bar{n}} \binom{n}{j} \\ &\leq \frac{(2C)^n \eta^{1-n}}{n!} |\log \eta|, \end{aligned} \quad (3.3.121)$$

since $\bar{n} \leq n$. Consider now the contribution of the region $\mathcal{A}^c$. Here, we shall use that:

$$|D_\omega(\underline{k} + \sum_{l \leq i} \underline{q}_l)| \geq |k_0 + \eta i| \geq \begin{cases} (i+1)\eta/2 & \text{if } k_0 \geq 1/2 \\ (n-i+1)\eta/2 & \text{if } k_0 \leq -n-1/2, \end{cases} \quad (3.3.122)$$

where the second case is needed only if $\bar{n} = n$, i.e. $n < \lfloor 2\delta/\eta \rfloor$. In both cases, we can extract the inverse of a factorial, as before. We have:

$$\begin{aligned} &\int_{\beta, L} \mathbf{1}(k_0 \in \mathcal{A}^c) \prod_{i=0}^n \frac{\chi_\delta^\omega(\underline{k} + \sum_{j \leq i} \underline{p}_j)}{|D_\omega(\underline{k} + \sum_{j \leq i} \underline{p}_j)|} d\underline{k} \\ &\leq \frac{(C/\eta)^{n-1}}{(n-1)!} \int_{\beta, L} \mathbf{1}(k_0 \in \mathcal{A}^c) \frac{\chi_\delta^\omega(\underline{k}) \chi_\delta^\omega(\underline{k} + \underline{p}_1)}{|D_\omega(\underline{k}) D_\omega(\underline{k} + \underline{p}_1)|} d\underline{k} \\ &\leq \frac{(K/\eta)^{n-1}}{(n-1)!} |\log \eta|. \end{aligned} \quad (3.3.123)$$

Putting together the above estimates we obtained:

$$\int_{\beta,L} \frac{d\mathbf{k}}{(2\pi)^2} \prod_{i=1}^{n+1} \left| g_{\omega} \left(\mathbf{k} + \sum_{j<i} \mathbf{p}_j \right) \right| \leq \frac{C^n}{n!} \frac{|\log \eta|}{\eta^{n-1}} ; \quad (3.3.124)$$

this implies, from (3.3.115):

$$|S_{n;\nu}^{\beta,L}(\mathbf{p}_1, \dots, \mathbf{p}_n)| \leq C^n \frac{|\log \eta|}{\eta^{n-1}} , \quad (3.3.125)$$

which proves (3.3.107). This concludes the proof of Proposition 3.3.9. $\square$

We shall now estimate the contribution of the terms $R_{n;\nu}^{\beta,L}(\mathbf{p}_1, \dots, \mathbf{p}_n)$ in (3.3.47). To this end, let us recall the splitting:

$$R_{n;\nu}^{\beta,L}(\mathbf{p}_1, \dots, \mathbf{p}_n) = \tilde{R}_{n;\nu}^{\beta,L}(\mathbf{p}_1, \dots, \mathbf{p}_n) + T_{n;\nu}^{\beta,L}(\mathbf{p}_1, \dots, \mathbf{p}_n) , \quad (3.3.126)$$

with:

$$\tilde{R}_{n;\nu}^{\beta,L}(\mathbf{p}_1, \dots, \mathbf{p}_n) = - \sum_{\pi \in S_n} \sum_{\substack{f \in \{r,s\}^{n+1} \\ f \neq s}} \int_{\beta,L} \text{Tr} \left[\hat{J}_{\nu}(k, p_{n+1}) \prod_{i=1}^{n+1} g_{f(i)} \left(\mathbf{k} + \sum_{j<i} \mathbf{p}_{\pi(j)} \right) \right] \frac{d\mathbf{k}}{(2\pi)^2} , \quad (3.3.127)$$

and:

$$T_{n;\nu}^{\beta,L}(\mathbf{p}_1, \dots, \mathbf{p}_n) = - \sum_{\omega=1}^N \sum_{\pi \in S_n} \int_{\beta,L} f_{n,\omega}^{\nu}(k; p_{\pi(1)}, \dots, p_{\pi(n)}) \prod_{i=1}^{n+1} g_{\omega} \left(\mathbf{k} + \sum_{j<i} \mathbf{p}_{\pi(j)} \right) \frac{d\mathbf{k}}{(2\pi)^2} , \quad (3.3.128)$$

where:

$$|f_{n,\omega}^{\nu}(k; p_1, \dots, p_n)| \leq C_{\delta} \left(\left| k - k_F^{\omega} |_{\mathbb{T}} \delta_{\nu,1} + n \sum_{i=1}^n |p_i|_{\mathbb{T}} \right| \right) . \quad (3.3.129)$$

Proposition 3.3.10 (Loop estimates: remainder terms). *There exists $C > 0$ such that for all $n \geq 2$:*

$$\frac{\theta^{n-1}}{n!} |R_{n;\nu}^{\beta,L}(\mathbf{p}_1, \dots, \mathbf{p}_n)| \leq C^n a^{n-1} |\log \eta| \sum_{j=1}^{n-1} \frac{\eta^j + \delta_{j,1} \sum_{i=1}^n |p_i|_{\mathbb{T}}}{(n-j)!} \quad (3.3.130)$$

for all $\mathbf{p}_i = (\eta_{\beta}, p_i)$, $i = 1, \dots, n$, such that $\hat{\mu}_{\alpha,\theta}(-p_i) \neq 0$.

Proof. With reference to (3.3.126), let us start by proving an estimate for $\tilde{R}_{n;\nu}^{\beta,L}(\mathbf{p}_1, \dots, \mathbf{p}_n)$. To this end, it is convenient to introduce:

$$\tilde{R}_{\nu;n,j}^{\beta,L}(\mathbf{p}_1, \dots, \mathbf{p}_n) := - \sum_{\pi \in S_n} \sum_{\substack{f \in \{r,s\}^{n+1} \\ |f^{-1}(r)|=j}} \int_{\beta,L} \text{Tr} \left[\hat{J}_{\nu}(k, p_{n+1}) \prod_{i=1}^{n+1} g_{f(i)} \left(\mathbf{k} + \sum_{l<i} \mathbf{p}_{\pi(l)} \right) \right] \frac{d\mathbf{k}}{(2\pi)^2} ; \quad (3.3.131)$$

that is, $\tilde{R}_{\nu;n,j}^{\beta,L}$ collects the contribution coming from the loops with exactly j bounded propagators g_r . Then, we can estimate:

$$|\tilde{R}_{\nu;n,j}^{\beta,L}(\mathbf{p}_1, \dots, \mathbf{p}_n)| \leq C^j \sum_{\pi \in S_n} \sum_{\substack{I \subseteq \{1, \dots, n+1\} \\ |I|=j}} \sum_{\Omega \in \{1, \dots, N\}^{I^c}} \int_{\beta,L} \prod_{i \in I^c} \left| g_{\Omega(i)} \left(\mathbf{k} + \sum_{l<i} \mathbf{p}_{\pi(l)} \right) \right| d\mathbf{k}, \quad (3.3.132)$$

where the last sum runs over chiralities $\Omega(i)$ with $i \in I^c$, and the constant C^j takes into account the estimate for the bounded propagators. For the moment, we suppose that $j \leq n-1$, so that in the integral we have at least two relativistic propagators. Let us label the elements of I^c as $I^c = \{m_0, m_1, \dots, m_{n-j}\}$ in increasing order. Performing the change of variables $\underline{k} + \sum_{i \leq m_0} \underline{p}_i \rightarrow \underline{k}$, we can estimate:

$$\int_{\beta, L} \prod_{i \in I^c} \left| g_{\Omega(i)} \left(\underline{k} + \sum_{l < i} \underline{p}_{\pi(l)} \right) \right| d\underline{k} \leq \int_{\beta, L} \prod_{i=0}^{n-j} \frac{\chi_{\delta}^{\Omega(m_i)}(\underline{k} + \sum_{l > m_0}^{m_i} \underline{p}_l)}{|D_{\Omega(m_i)}(\underline{k} + \sum_{l > m_0}^{m_i} \underline{p}_l)|} d\underline{k}. \quad (3.3.133)$$

Next, we proceed as after (3.3.116). We write:

$$\begin{aligned} & \int_{\beta, L} \prod_{i=0}^{n-j} \frac{\chi_{\delta}^{\Omega(m_i)}(\underline{k} + \sum_{l > m_0}^{m_i} \underline{p}_l)}{|D_{\Omega(m_i)}(\underline{k} + \sum_{l > m_0}^{m_i} \underline{p}_l)|} d\underline{k} \\ &= \int_{\beta, L} \mathbf{1}(k_0 \in \mathcal{A}^c) \prod_{i=0}^{n-j} \frac{\chi_{\delta}^{\Omega(m_i)}(\underline{k} + \sum_{l > m_0}^{m_i} \underline{p}_l)}{|D_{\Omega(m_i)}(\underline{k} + \sum_{l > m_0}^{m_i} \underline{p}_l)|} d\underline{k} \\ &+ \sum_{\ell=1}^{\bar{n}} \int_{\beta, L} \mathbf{1}(k_0 \in \mathcal{A}_{\ell}) \prod_{i=0}^{n-j} \frac{\chi_{\delta}^{\Omega(m_i)}(\underline{k} + \sum_{l > m_0}^{m_i} \underline{p}_l)}{|D_{\Omega(m_i)}(\underline{k} + \sum_{l > m_0}^{m_i} \underline{p}_l)|} d\underline{k}. \end{aligned} \quad (3.3.134)$$

Consider the argument of the sum with $\ell = m_s - m_0$ for $m_s \in I^c$. Proceeding as in (3.3.121):

$$\begin{aligned} & \int_{\beta, L} \mathbf{1}(k_0 \in \mathcal{A}_{\ell}) \prod_{i=0}^{n-j} \frac{\chi_{\delta}^{\Omega(m_i)}(\underline{k} + \sum_{l > m_0}^{m_i} \underline{p}_l)}{|D_{\Omega(m_i)}(\underline{k} + \sum_{l > m_0}^{m_i} \underline{p}_l)|} d\underline{k} \\ & \leq (C/\eta)^{n-j} (\eta |\log \eta|) \prod_{\substack{i=0 \\ |m_i - m_s| > 0}}^{n-j} \frac{1}{|m_i - m_s|}. \end{aligned} \quad (3.3.135)$$

Using that $|m_i - m_s| \geq |i - s|$, we have:

$$\begin{aligned} & \sum_{s=0}^{n-j} \int_{\beta, L} \mathbf{1}(k_0 \in \mathcal{A}_{m_s - m_0}) \prod_{i=0}^{n-j} \frac{\chi_{\delta}^{\Omega(m_i)}(\underline{k} + \sum_{l > m_0}^{m_i} \underline{p}_l)}{|D_{\Omega(m_i)}(\underline{k} + \sum_{l > m_0}^{m_i} \underline{p}_l)|} d\underline{k} \\ & \leq (K/\eta)^{n-j} (\eta |\log \eta|) \frac{1}{(n-j)!} \sum_{s=0}^{n-j} \binom{n-j}{s} \\ & \leq (C/\eta)^{n-j} (\eta |\log \eta|) \frac{1}{(n-j)!}. \end{aligned} \quad (3.3.136)$$

Instead, the contribution of all ℓ such that $\ell + m_0 \notin I^c$ is:

$$\sum_{\ell: \ell + m_0 \notin I^c} \int_{\beta, L} \mathbf{1}(k_0 \in \mathcal{A}_{\ell}) \prod_{i=0}^{n-j} \frac{\chi_{\delta}^{\Omega(m_i)}(\underline{k} + \sum_{l > m_0}^{m_i} \underline{p}_l)}{|D_{\Omega(m_i)}(\underline{k} + \sum_{l > m_0}^{m_i} \underline{p}_l)|} d\underline{k} \leq (C/\eta)^{n-j} \frac{\eta}{(n-j)!}. \quad (3.3.137)$$

Consider now the contribution due to the region $\mathcal{A}^c$. Also here, none of the propagators in the product is singular. Therefore, proceeding as in (3.3.123):

$$\int_{\beta, L} \mathbf{1}(k_0 \in \mathcal{A}^c) \prod_{i=0}^{n-j} \frac{\chi_{\delta}^{\Omega(m_i)}(\underline{k} + \sum_{l > m_0}^{m_i} \underline{p}_l)}{|D_{\Omega(m_i)}(\underline{k} + \sum_{l > m_0}^{m_i} \underline{p}_l)|} d\underline{k} \leq (C/\eta)^{n-j+1} \frac{1}{(n-j-1)!} |\log \eta|. \quad (3.3.138)$$

All in all, from (3.3.133), (3.3.136), (3.3.137), (3.3.138) we obtained (with a different C):

$$\int_{\beta,L} \prod_{i=0}^{n-j} \frac{\chi_{\delta}^{\Omega(m_i)}(\underline{k} + \sum_{l>m_0}^{m_i} \underline{p}_l)}{|D_{\Omega(m_i)}(\underline{k} + \sum_{l>m_0}^{m_i} \underline{p}_l)|} d\underline{k} \leq (C/\eta)^{n-j} (\eta |\log \eta|) \frac{1}{(n-j)!} . \quad (3.3.139)$$

Plugging this bound in (3.3.132), we get:

$$|\tilde{R}_{\nu;n,j}^{\beta,L}(\underline{p}_1, \dots, \underline{p}_n)| \leq C^n n! \eta^{j-n} (\eta |\log \eta|) \frac{1}{(n-j)!} . \quad (3.3.140)$$

Suppose that that $j = n$ or $j = n + 1$. If $j = n + 1$, we use two propagators g_r to sum over $\underline{k}$. The final estimate is:

$$|\tilde{R}_{\nu;n,n+1}^{\beta,L}(\underline{p}_1, \dots, \underline{p}_n)| \leq C^n n! . \quad (3.3.141)$$

If $j = n$, we sum over $\underline{k}$ with the relativistic propagator and with one propagator g_r . Also in this case, the final estimate is:

$$|\tilde{R}_{\nu;n,n}^{\beta,L}(\underline{p}_1, \dots, \underline{p}_n)| \leq C^n n! . \quad (3.3.142)$$

Therefore, we obtained:

$$\frac{\theta^{n-1}}{n!} \sum_j |\tilde{R}_{\nu;n,j}^{\beta,L}(\underline{p}_1, \dots, \underline{p}_n)| \leq C^n \theta^{n-1} + C^n a^{n-1} \eta |\log \eta| \sum_{j=1}^{n-1} \frac{\eta^{j-1}}{(n-j)!} . \quad (3.3.143)$$

Finally, consider the error term (3.3.128). The estimate for this term is performed as in the $j = 0$ version of the argument above, keeping into account a factor $(|k - k_F^{\omega}|_{\mathbb{T}} + n \sum_{i=1}^n |p_i|_{\mathbb{T}})$ arising from the estimate (3.3.129). Since the combination $|k - k_F^{\omega}|_{g_{\omega}(\underline{k})}$ is bounded, we have:

$$\frac{\theta^{n-1}}{n!} |T_{n;\nu}^{\beta,L}(\underline{p}_1, \dots, \underline{p}_n)| \leq \frac{C^n a^{n-1}}{n!} |\log \eta| (\eta + \sum_i |p_i|_{\mathbb{T}}) , \quad (3.3.144)$$

we omit the details. Putting together (3.3.143) and (3.3.144), Eq. (3.3.130) follows. $\square$

We are now ready to prove Theorem 3.2.1.

Proof of Theorem 3.2.1. Recall the expression (3.3.47) for the full response of the system. We are interested in proving an estimate for the sum of all contributions with $n \geq 2$, which is subleading with respect to the linear response as $\eta \rightarrow 0^+$. Let $n(\theta) = (1/4)\theta^{\alpha-1}$. We have, from Proposition 3.3.9 and Proposition 3.3.10, for β, L large enough:

$$\begin{aligned} & \sum_{n=2}^{\infty} \frac{\theta^{n-1}}{n!} \frac{1}{L^n} \sum_{\{p_i\} \in B_L^n} \left[\prod_{j=1}^n |\hat{\mu}_{\alpha,\theta}(-p_j)| \right] \left(|S_{n;\nu}^{\beta,L}(\underline{p}_1, \dots, \underline{p}_n)| + |R_{n;\nu}^{\beta,L}(\underline{p}_1, \dots, \underline{p}_n)| \right) \\ & \leq \sum_{n=2}^{n(\theta)} C^n \|\hat{\mu}_{\alpha,\theta}\|_1^n \theta^{n-1} + \sum_{n>n(\theta)} C^n \|\hat{\mu}_{\alpha,\theta}\|_1^n \frac{a^{n-1}}{n!} |\log \eta| \\ & \quad + \sum_{n \geq 2} C^n a^{n-1} (\eta + \theta) |\log \eta| \sum_{j=1}^{n-1} \frac{\eta^{j-1}}{(n-j)!} . \end{aligned} \quad (3.3.145)$$

The first term in the right-hand side comes from (3.3.106); the second term from (3.3.107); the last term from (3.3.130). In particular, the factor θ in the third term comes from $\|\hat{\mu}_{\alpha,\theta}(p_i)|p_i|_{\mathbb{T}}\|_1 \leq C\theta$. We estimate the last sum as:

$$\begin{aligned}
& (\eta + \theta) |\log \eta| \sum_{n \geq 2} C^n a^{n-1} \sum_{j=1}^{n-1} \frac{\eta^{j-1}}{(n-j)!} \\
& \leq \tilde{C}(\eta + \theta) |\log \eta| \sum_{n \geq 0} (Ka)^n \sum_{j=0}^n \frac{\eta^j}{(n-j)!} \\
& \leq \tilde{C}(\eta + \theta) |\log \eta| \sum_{j \geq 0} (Ka\eta)^j \sum_{n \geq j} \frac{(Ka)^{n-j}}{(n-j)!} \\
& = \tilde{C}(\eta + \theta) |\log \eta| \frac{e^{Ka}}{1 - Ka\eta} .
\end{aligned} \tag{3.3.146}$$

We choose $a \leq a(\eta)$, so that right-hand side of (3.3.146) vanishes as $\eta \rightarrow 0$. This holds true if $a(\eta) = w |\log \eta|$ with $w > 0$ small enough. Therefore, putting together (3.3.145), (3.3.146), (3.3.47) we get, for $\gamma > 0$, and β, L large enough:

$$|\chi_{\nu}^{\beta,L}(x; \eta, \theta) - \chi_{\nu}^{\text{lin}}(x; \eta, \theta)| \leq C\eta^{\gamma} \tag{3.3.147}$$

with $\chi_{\nu}^{\text{lin}}(x; \eta, \theta)$ given by (3.3.70). This concludes the proof of Theorem 3.2.1. $\square$

3.4 Large scale response of edge modes of 2d systems

In this section we will adapt the previous analysis to study the response of edge currents and edge densities at the boundary of 2d topological insulators.

3.4.1 Lattice fermions on the cylinder

Hamiltonian and Gibbs state. Give $L \in 2\mathbb{N} + 1$, we consider fermions on the two-dimensional lattice

$$\Gamma_L = \left\{ x \in \mathbb{Z}^2 \mid -\left\lfloor \frac{L}{2} \right\rfloor \leq x_1 \leq \left\lfloor \frac{L}{2} \right\rfloor, 0 \leq x_2 \leq L-1 \right\}, \tag{3.4.1}$$

endowed with periodic boundary conditions in the x_1 coordinate and Dirichlet boundary condition in the x_2 coordinate:

$$f(x_1, x_2) = f(x_1 + L, x_2) \quad \text{and} \quad f(x_1, 0) = f(x_1, L-1) = 0 \quad \text{for all } x \in \Gamma_L. \tag{3.4.2}$$

We shall introduce the following distance on Γ_L :

$$|x - y|_L^2 = \min_{n \in \mathbb{Z}} |x_1 - y_1 + nL|^2 + |x_2 - y_2|^2. \tag{3.4.3}$$

As for the one-dimensional case, we will allow for the presence of internal degrees of freedom; the resulting decorated lattice is $\Lambda_L = \Gamma_L \times S_M$, and we denote by $\mathbf{x} = (x, \sigma)$ with $x \in \Gamma_L$ and $\sigma \in S_M$ the points on Λ_L .

We shall suppose that the Hamiltonian H on $\ell^2(\Lambda_L)$ is the periodisation of a Hamiltonian H^∞ on $\ell^2((\mathbb{Z} \times [0, L-1] \cap \mathbb{Z}) \times S_M)$:

$$H((x, \rho); (y, \rho')) = \sum_{a \in \mathbb{N}} H^\infty((x + aL, \rho); (y, \rho')) . \quad (3.4.4)$$

We assume that the Hamiltonian H^∞ is finite-ranged and translation-invariant. Without loss of generality we can assume that the range is $\sqrt{2}$, up to increasing the number of internal degrees of freedom. As for the one-dimensional case, translation-invariance implies that the Hamiltonian H can be fibered in momentum space. Let $k \in B_L$, with B_L as in (3.1.5). We define the Bloch Hamiltonian as:

$$\hat{H}_{\rho\rho'}(k; x_2, y_2) = \sum_x e^{-ikx} H_{\rho\rho'}((x, x_2); (0, y_2)) . \quad (3.4.5)$$

The operator $\hat{H}(k)$ is self-adjoint on $\ell^2([0, L-1] \cap \mathbb{Z} \times S_M)$. By (3.1.3), it is given by the restriction to B_L of the Bloch Hamiltonian $\hat{H}^\infty(k)$ associated with H^∞ , which is defined for $k \in \mathbb{T}$. We shall make the following assumptions on the spectrum of the Bloch Hamiltonian and on the chemical potential μ .

Assumption 3.4.1 (Low-energy spectrum). There exists $\Delta > 0$ such that the following is true.

- (i) There exists $N \in \mathbb{N}$, disjoint sets $I_\omega \subset \mathbb{T}$ labelled by $\omega = 1, \dots, N$, and strictly monotone, smooth functions $e_\omega : I_\omega \rightarrow \mathbb{R}$, such that

$$\sigma(H^\infty) \cap (\mu - \Delta, \mu + \Delta) = \bigcup_{\omega=1}^N \text{Ran}(e_\omega) . \quad (3.4.6)$$

- (ii) We introduce the ω -Fermi point $k_F^\omega \in I_\omega$ and the ω -Fermi velocity v_ω as

$$e_\omega(k_F^\omega) = \mu , \quad v_\omega = \partial_k e_\omega(k_F^\omega) . \quad (3.4.7)$$

Notice that $v_\omega \neq 0$, by the strict monotonicity of e_ω .

- (iii) For any $\omega = 1, \dots, N$, and $k \in I_\omega$, $e_\omega(k)$ is a non-degenerate eigenvalue of $\hat{H}^\infty(k)$. Let $\xi^\omega(k)$ be the corresponding eigenfunction,

$$\hat{H}^\infty(k) \xi^\omega(k) = e_\omega(k) \xi^\omega(k) . \quad (3.4.8)$$

We assume that $\xi^\omega(k)$ is exponentially localised near one of the boundaries of the cylinder:

$$|\partial_k^n \xi_{\omega;\rho}(k; x_2)| \leq C_n e^{-cx_2} \quad \text{or} \quad |\partial_k^n \xi_{\omega;\rho}(k; x_2)| \leq C_n e^{-c(L-x_2)} , \quad (3.4.9)$$

for all $n \in \mathbb{N}$.

Remark 3.4.2. For short, we shall write $|\partial_k^n \xi_{\omega;\rho}(k; x_2)| \leq C_n e^{-c|x_2|_\omega}$, with the notation:

$$|x_2|_\omega = x_2 \quad \text{or} \quad |x_2|_\omega = L - x_2 , \quad \text{for all } x_2. \quad (3.4.10)$$

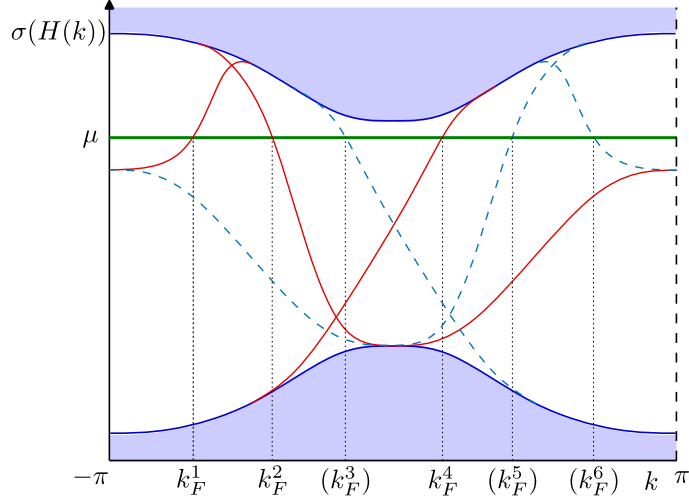


Figure 3.2: Typical form for the spectrum of H . The purple region corresponds to the “bulk spectrum”. The red curves are the edge modes localised at $x_2 = 0$, while the dotted curves are the edge modes localised at $x_2 = L - 1$.

The second-quantised grand-canonical Hamiltonian is

$$\mathcal{H} = \sum_{\mathbf{x}, \mathbf{y} \in \Lambda_L} a_{\mathbf{x}}^* H(\mathbf{x}; \mathbf{y}) a_{\mathbf{y}} \quad (3.4.11)$$

with $a_{\mathbf{x}}$ and $a_{\mathbf{x}}^*$ the usual fermionic creation/annihilation operators, compatible with the boundary conditions on Γ_L . Similarly to (3.1.16), (3.1.17) we define, for all $k \in B_L$:

$$\hat{a}_{(k, x_2, \rho)} = \sum_{x_1 \in \Gamma_L} a_{(x, \rho)} e^{-ikx_1}, \quad \hat{a}_{(k, x_2, \rho)}^* = \sum_{x_1 \in \Gamma_L} a_{(x, \rho)}^* e^{ikx_1}, \quad (3.4.12)$$

which can be inverted as:

$$a_{(x, \rho)} = \frac{1}{L} \sum_{k \in B_L} e^{ikx_1} \hat{a}_{(k, x_2, \rho)}, \quad a_{(x, \rho)}^* = \frac{1}{L} \sum_{k \in B_L} e^{-ikx_1} \hat{a}_{(k, x_2, \rho)}^*. \quad (3.4.13)$$

In terms of the momentum-space operators, the Hamiltonian reads:

$$\mathcal{H} = \frac{1}{L} \sum_{k \in B_L} \sum_{x_2, y_2=1}^L \sum_{\rho, \rho' \in S_M} \hat{a}_{(k, x_2, \rho)}^* \hat{H}_{\rho\rho'}(k; x_2, y_2) \hat{a}_{(k, y_2, \rho')}. \quad (3.4.14)$$

The grand-canonical Gibbs state of the system at inverse temperature $\beta > 0$ is:

$$\langle \mathcal{O} \rangle_{\beta, L} = \text{Tr } \mathcal{O} \rho_{\beta, \mu, L}, \quad \rho_{\beta, \mu, L} = \frac{e^{-\beta(\mathcal{H} - \mu N)}}{\text{Tr } e^{-\beta(\mathcal{H} - \mu N)}}, \quad (3.4.15)$$

with $\mu \in \mathbb{R}$ the chemical potential and N the number operator. We will suppose that μ is chosen so that Assumption 3.4.1 holds.

Perturbing the system. Similarly to the one-dimensional case, we expose our system to a time-dependent and slowly varying perturbation, by introducing the Hamiltonian:

$$\mathcal{H}(\eta t) = \mathcal{H} + e^{\eta t} \mathcal{P} \quad \eta > 0, t \leq 0, \quad (3.4.16)$$

where the perturbation $\mathcal{P}$ is given by

$$\mathcal{P} = \theta \sum_{\mathbf{x} \in \Lambda_L} \mu(\theta x) a_{\mathbf{x}}^* a_{\mathbf{x}} ; \quad (3.4.17)$$

given $\mu_\infty \in \mathcal{C}_c^\infty(\mathbb{R} \times \mathbb{R}^+)$ supported near the boundary $x_2 = 0$, the function $\mu(\theta x)$ is its periodisation:

$$\mu(\theta x) = \sum_{n \in \mathbb{Z}} \mu_\infty(\theta(x + nLe_1)) . \quad (3.4.18)$$

We can further write:

$$\begin{aligned} \mu(\theta x) &= \frac{1}{L} \sum_{p \in B_L} e^{ipx_1} \hat{\mu}_\theta(p, x_2) , \\ \hat{\mu}_\theta(p, x_2) &= \sum_{m \in \mathbb{Z}} \frac{1}{\theta} \hat{\mu}_\infty((p + 2\pi m)/\theta, \theta x_2) \quad \text{for all } p \in B_L. \end{aligned} \quad (3.4.19)$$

Let $\mathcal{U}(t; s)$ be the two-parameter unitary group generated by $\mathcal{H}(\eta t)$:

$$i\partial_t \mathcal{U}(t; s) = \mathcal{H}(\eta t) \mathcal{U}(t; s) , \quad \mathcal{U}(s; s) = \mathbb{1} . \quad (3.4.20)$$

The evolution of the Gibbs state $\rho_{\beta, \mu, L}$ is given as:

$$\rho(t) = \lim_{T \rightarrow +\infty} \mathcal{U}(t; -T) \rho_{\beta, \mu, L} \mathcal{U}(t; -T)^* . \quad (3.4.21)$$

As in the one-dimensional case, we will be interested in the variation of physical observables,

$$\text{Tr } \mathcal{O} \rho(t) - \text{Tr } \mathcal{O} \rho_{\beta, \mu, L} , \quad (3.4.22)$$

for $\mathcal{O}$ given by the density and by the current operator, defined below.

Density and current operators. Similarly to Section 3.1, we set:

$$n_{\mathbf{x}} = a_{\mathbf{x}}^* a_{\mathbf{x}} , \quad n_x = \sum_{\rho \in S_M} n_{(x, \rho)} . \quad (3.4.23)$$

Proceeding as after Eq. (3.1.29), we obtain the lattice continuity equation:

$$\partial_t \tau_t(n_x) = -\text{div}_x \tau_t(\vec{j}_x) \equiv -d_1 j_{1,x} - d_2 j_{2,x} , \quad (3.4.24)$$

where $d_i f_x = f(x) - f(x - e_i)$ and with the current densities:

$$\begin{aligned} j_{1,x} &= j_{(x, x+e_1)} + \frac{1}{2} \left[j_{(x, x+e_1-e_2)} + j_{(x, x+e_1+e_2)} + j_{(x-e_2, x+e_1)} + j_{(x+e_2, x+e_1)} \right] \\ j_{2,x} &= j_{(x, x+e_2)} + \frac{1}{2} \left[j_{(x, x+e_2-e_1)} + j_{(x, x+e_2+e_1)} + j_{(x-e_1, x+e_2)} + j_{(x+e_1, x+e_2)} \right] . \end{aligned} \quad (3.4.25)$$

Setting $j_{0,x} = n_x$, we define the smeared current and density operators as:

$$j_\nu(\mu_\theta) = \sum_{x \in \Gamma_L} j_{\nu,x} \theta \mu(\theta x) . \quad (3.4.26)$$

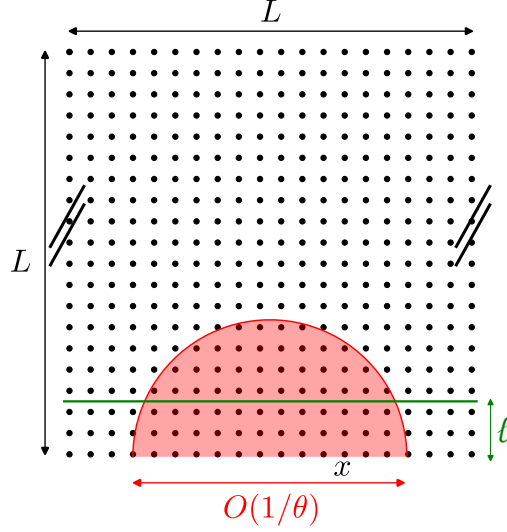


Figure 3.3: Representation of the lattice Γ_L : the red semi-circle represents the typical support of the perturbation $\theta\mu(\theta x)$, whereas $1 \ll \ell \ll 1/\theta$ is the length of the fiducial line entering the definition of the edge 2-current $j_{\nu,x}^\ell$.

Also, we introduce the current and the density operators associated with a strip of length ℓ in proximity of the $x_2 = 0$ boundary as:

$$j_{\nu,x_1}^\ell := \sum_{x_2=0}^{\ell-1} j_{\nu,(x_1,x_2)} . \quad (3.4.27)$$

We shall call j_{0,x_1}^ℓ the edge density and j_{1,x_1}^ℓ the edge current, corresponding to the point x_1 at the edge $x_2 = 0$. Finally, in what follows it will also be convenient to consider the total current across the fiducial line at x_1 , $j_{\nu,x_1}^L \equiv j_{\nu,x_1}$. In Fourier space, for $\nu = 0, 1$:

$$\begin{aligned} \hat{j}_{\nu,p} &= \sum_{x_1} e^{-ipx_1} j_{\nu,x_1} \\ &= \frac{1}{L} \sum_{k \in B_L} \sum_{y,y'=0}^{L-1} (a_{(k-p,y)}^*, \hat{J}_\nu(k,p;y,y') a_{(k,y')}) \end{aligned} \quad (3.4.28)$$

where the brackets denote summation over the internal degrees of freedom, as in (3.1.32), and:

$$\hat{J}_0(k,p) = 1 \quad \hat{J}_1(k,p) = i \frac{\hat{H}(k) - \hat{H}(k-p)}{1 - e^{-ip}} . \quad (3.4.29)$$

In terms of these operators, the current (3.4.27) can be expressed in momentum space as:

$$\hat{j}_{\nu,p}^\ell = \frac{1}{L} \sum_{k \in B_L} \sum_{y,y'=0}^{L-1} (a_{(k-p,y)}^*, \hat{J}_\nu^\ell(k,p;y,y') a_{(k,y')}) \quad (3.4.30)$$

where $\hat{J}_\nu^\ell(k,p;y,y')$ is a bounded and smooth kernel such that:

$$\begin{aligned} \hat{J}_\nu^\ell(k,p;y,y') &= 0 & \text{if } y > \ell \text{ or } y' > \ell, \\ \hat{J}_\nu^\ell(k,p;y,y') &= \hat{J}_\nu(k,p;y,y') & \text{if } y < \ell - 1 \text{ and } y' < \ell - 1. \end{aligned} \quad (3.4.31)$$

3.4.2 Validity of edge linear response

In analogy with the one-dimensional case, let us define:

$$\chi_\nu^{\beta,L,\ell}(x; \eta, \theta) := \frac{1}{\theta} \left(\text{Tr } j_{\nu,x}^\ell \rho(0) - \text{Tr } j_{\nu,x}^\ell \rho_{\beta,\mu,L} \right) \quad \nu = 0, 1. \quad (3.4.32)$$

The quantity $\chi_\nu^{\beta,L,\ell}$ describes the edge response of the system, after introducing a time-dependent perturbation in proximity of the edge, as in (3.4.17). We are interested in studying this quantity in the following order of limits: first $L \rightarrow \infty$; then $\beta \rightarrow \infty$; then $\eta, \theta \rightarrow 0$; and then $\ell \rightarrow \infty$.

Theorem 3.4.3 (Edge response of two-dimensional systems). *Let $\theta = a\eta$. There exist constants $w, \gamma, \eta_0 > 0$ such that, for all $0 \leq \eta < \eta_0$ and $a \leq w |\log \eta|$, for β, L large enough:*

$$\begin{aligned} \chi_\nu^{\beta,L,\ell}(x; \eta, \theta) \\ = - \sum_{\omega}^* \frac{v_\omega^\nu}{2\pi|v_\omega|} \int_{\mathbb{R}} \hat{\mu}_\infty(q, 0) e^{iq\theta x} \frac{v_\omega q}{-i/a + v_\omega q} \frac{dq}{2\pi} + O(\ell\eta^\gamma) + O(e^{-c\ell}) + O\left(\frac{\ell}{\eta^4\beta}\right) \end{aligned} \quad (3.4.33)$$

where the asterisk denotes summation over the edge modes localised near the lower edge, recall (3.4.9), and where $v_\omega^\nu = \delta_{\nu,0} + v_\omega \delta_{\nu,1}$.

Corollary 3.4.4. *In particular, we can isolate three regimes.*

1. Suppose that $a \rightarrow 0$ as $\eta \rightarrow 0^+$. Then,

$$\chi_\nu^{\beta,L,\ell}(x; \eta, \theta) = O(a) + O(\ell\eta^\gamma) + O(e^{-c\ell}) + O\left(\frac{\ell}{\eta^4\beta}\right). \quad (3.4.34)$$

2. Suppose that a is constant in η . Then, calling Θ the Heaviside step function:

$$\begin{aligned} \chi_\nu^{\beta,L,\ell}(x; \eta, \theta) = - \sum_{\omega}^* \frac{v_\omega^\nu}{2\pi|v_\omega|} \int_{\mathbb{R}} \mu_\infty(\theta x - y, 0) \left[\delta(y) - \frac{1}{a|v_\omega|} e^{-y/av_\omega} \Theta(y/v_\omega) \right] dy \\ + O(\ell\eta^\gamma) + O(e^{-c\ell}) + O\left(\frac{\ell}{\eta^4\beta}\right). \end{aligned} \quad (3.4.35)$$

3. Finally, suppose that $a \rightarrow \infty$ as $\eta \rightarrow 0^+$. Then:

$$\chi_\nu^{\beta,L,\ell}(x; \eta, \theta) = -\mu_\infty(\theta x, 0) \sum_{\omega}^* \frac{v_\omega^\nu}{2\pi|v_\omega|} + O(1/a) + O(\ell\eta^\gamma) + O(e^{-c\ell}) + O\left(\frac{\ell}{\eta^4\beta}\right); \quad (3.4.36)$$

in particular,

$$\lim_{\ell \rightarrow \infty} \lim_{\eta \rightarrow 0^+} \lim_{\beta \rightarrow \infty} \lim_{L \rightarrow \infty} \chi_1^{\beta,L,\ell}(x; \eta, \theta) = -\mu_\infty(0, 0) \sum_{\omega}^* \frac{\text{sgn}(v_\omega)}{2\pi}. \quad (3.4.37)$$

Remark 3.4.5. (i) Theorem 3.4.3 and Corollary 3.4.4 extend Theorem 3.2.1 and Corollary 3.2.2 to the case of edge currents of $2d$ quantum systems. In contrast to the $1d$ case, the edge conductance (3.4.37) might be non-zero. Whenever that is the case, by the bulk-edge duality the system is in a non-trivial topological phase.

(ii) The proof of Theorem 3.4.3 is an adaptation of the proof of Theorem 3.2.1, discussed below. It relies on the emergent one-dimensional nature of the edge modes, and on the loop cancellation for chiral relativistic fermions, Proposition 3.3.8.

3.4.3 Proof of Theorem 3.4.3

Auxiliary dynamics. Let:

$$\mathcal{H}_{\beta,\eta}(t) = \mathcal{H} + \theta e^{\eta\beta t} \sum_{\mathbf{x} \in \Lambda_L} \mu_\alpha(\theta x) a_{\mathbf{x}}^* a_{\mathbf{x}} , \quad (3.4.38)$$

with:

$$\mu_\alpha(\theta x) := \frac{1}{L} \sum_{p \in B_L} e^{ipx} \hat{\mu}_{\alpha,\theta}(p, x_2) , \quad \hat{\mu}_{\alpha,\theta}(p, x_2) := \hat{\mu}_\theta(p, x_2) \chi(\theta^{\alpha-1} |p|_{\mathbb{T}}) , \quad (3.4.39)$$

where $\alpha \in (0, 1)$. As in the 1d case, we are cutting off all momenta p of norm greater than $2\theta^{1-\alpha}$. Let $\tilde{\mathcal{U}}(t; s)$ the two-parameter unitary group generated by $\mathcal{H}_{\beta,\eta}(t)$. Then, the following holds.

Proposition 3.4.6 (Approximation by the auxiliary dynamics.). *Under the same assumptions of Theorem 3.4.3, it follows that, for any $m \in \mathbb{N}$:*

$$\left\| \tilde{\mathcal{U}}(t; -\infty)^* j_{\nu,x}^\ell \tilde{\mathcal{U}}(t; -\infty) - \mathcal{U}(t; -\infty)^* j_{\nu,x}^\ell \mathcal{U}(t; -\infty) \right\| \leq C_m \frac{\theta^{1+\alpha(m-1)} \ell}{\eta^3} + \frac{C\theta\ell}{\eta^4\beta} . \quad (3.4.40)$$

Proof. The proof is obtained by straightforwardly adapting the Proof of Proposition 3.3.1 to this two-dimensional setting, with the following differences. The first is that the Lieb-Robinson ball has volume proportional to $|t - s|^2$ (instead of $|t - s|$), and this produces errors proportional to an η^{-4} bound (instead of η^{-3}); the second is the presence of the factor ℓ in the estimate, due to $\|j_\nu^\ell\| \leq C\ell$. $\square$

Thanks to Proposition 3.4.6 we can write

$$\text{Tr } j_{\nu,x}^\ell \rho(t) = \text{Tr } j_{\nu,x}^\ell \tilde{\rho}(t) + \mathcal{E}_\nu^{\beta,L,\ell}(x, t; \eta, \theta) , \quad (3.4.41)$$

where $\tilde{\rho}(t) = \tilde{\mathcal{U}}(t; -\infty) \rho_{\beta,\mu,L} \tilde{\mathcal{U}}(t; -\infty)^*$ and the error term $\mathcal{E}_\nu^{\beta,L,\ell}(x, t; \eta, \theta)$ takes into account the approximation by the auxiliary dynamics:

$$|\mathcal{E}_\nu^{\beta,L,\ell}(x, t; \eta, \theta)| \leq C_m \frac{\theta^{1+\alpha(m-1)} \ell}{\eta^3} + \frac{C\theta\ell}{\eta^4\beta} . \quad (3.4.42)$$

We require that this error term, once divided by $\theta = a\eta$ (recall the definition (3.4.32)), vanishes as $\eta \rightarrow 0^+$. This forces to choose the parameter a such that:

$$a = o\left(\eta^{\frac{3}{\alpha(m-1)}-1}\right) . \quad (3.4.43)$$

To allow for a divergent a as $\eta \rightarrow 0^+$, we choose $m \in \mathbb{N}$ such that $\alpha(m-1) > 3$. Later, we will have to introduce stronger constraints on a .

Analysis of the Duhamel series. By Proposition 2.2.7, we can represent the Duhamel series as:

$$\begin{aligned} & \text{Tr } j_{\nu,x}^\ell \tilde{\rho}(0) - \text{Tr } j_{\nu,x}^\ell \rho_{\beta,\mu,L} \\ &= \sum_{n=1}^{\infty} \frac{(-1)^n}{n!} \int_{[0,\beta]^n} d\mathbf{s} e^{-i\eta\beta(s_1+\dots+s_n)} \langle \mathbf{T} \gamma_{s_1}(j_0(\mu_{\alpha,\theta})); \dots; \gamma_{s_n}(j_0(\mu_{\alpha,\theta})); j_{\nu,x}^\ell \rangle_{\beta,L} . \end{aligned} \quad (3.4.44)$$

Proceeding as in (3.3.24)-(3.3.25), one obtains

$$\begin{aligned} \text{Tr } j_{\nu,x}^\ell \tilde{\rho}(0) - \text{Tr } j_{\nu,x}^\ell \rho_{\beta,\mu,L} &= \sum_{n=1}^{\infty} \frac{(-\theta)^n}{n! L^n} \sum_{\{p_i\} \in B_L^n} \sum_{\{y_i\} \in \{1,\dots,L\}^n} \left[\prod_{i=1}^n \hat{\mu}_{\alpha,\theta}(-p_i, y_i) \right] e^{-ip_{n+1}x} \\ &\quad \cdot \frac{1}{\beta L} \langle \mathbf{T} \hat{n}_{\underline{p}_1, y_1} ; \dots ; \hat{n}_{\underline{p}_n, y_n} ; \hat{j}_{\nu, \underline{p}_{n+1}}^\ell \rangle_{\beta, L} , \end{aligned} \quad (3.4.45)$$

with the usual conventions $\underline{p}_i = (\eta_\beta, p_i)$, $\underline{p}_{n+1} = -\sum_{i=1}^n \underline{p}_i$. The proof of Theorem 3.4.3 is based on a detailed analysis of the cumulant expansion in (3.4.45). By Wick's rule:

$$\begin{aligned} &\frac{1}{\beta L} \langle \mathbf{T} \hat{n}_{\underline{p}_1, y_1} ; \dots ; \hat{n}_{\underline{p}_n, y_n} ; \hat{j}_{\nu, \underline{p}_{n+1}}^\ell \rangle_{\beta, L} \\ &= - \sum_{\pi \in S_n} \int_{\beta, L} \frac{d\underline{k}}{(2\pi)^2} \sum_{z, z'} \sum_{\{y_i\}} \text{Tr} \left[\hat{j}_\nu^\ell(k, p_{n+1}; z, z') \prod_{i=1}^{n+1} g\left(\underline{k} + \sum_{j < i} \underline{p}_{\pi(j)}; y_{\pi(i-1)}, y_{\pi(i)}\right) \right] \end{aligned} \quad (3.4.46)$$

where the trace is over the internal degrees of freedom, we set $y_{\pi(0)} \equiv z'$ and $y_{\pi(n+1)} \equiv z$, and where the propagator g is given by

$$g(\underline{k}) = \frac{1}{ik_0 + \hat{H}(\underline{k}) - \mu} . \quad (3.4.47)$$

As done after (3.3.28), we decompose the propagator in a singular plus a more regular part:

$$g(\underline{k}) = g_a(\underline{k}) + g_b(\underline{k}) , \quad (3.4.48)$$

where:

$$g_b(\underline{k}) = g(\underline{k})[1 - \chi(|\hat{H}(\underline{k}) - \mu|/\Delta)] \quad (3.4.49)$$

with μ, Δ as in Assumption 3.4.1. The propagator $g_b(\underline{k})$ satisfies, by a Combes-Thomas argument (see *e.g.* [26, Appendix A] for more details):

$$\left| d_{k_0}^{n_0} d_{k_1}^{n_1} g_{b;\rho,\rho'}(\underline{k}; y, z) \right| \leq \frac{C_{n_0, n_1}}{1 + |k_0|} e^{-c|x_2 - y_2|} . \quad (3.4.50)$$

Instead, the propagator $g_a(\underline{k})$ is given by:

$$g_a(\underline{k}) = \sum_{\omega=1}^N \frac{\chi(|e_\omega(\underline{k}) - \mu|/\Delta)}{ik_0 + e_\omega(\underline{k}) - \mu} P_\omega(\underline{k}) \quad (3.4.51)$$

where e_ω is the dispersion relation of the ω edge mode, and P_ω is the rank-1 projector onto the edge mode ξ_ω . Next, proceeding as after (3.3.33), we extract from $g_a(\underline{k})$ its relativistic part, up to remainder term which is more regular. All in all, we have:

$$\begin{aligned} g(\underline{k}) &= g_s(\underline{k}) + g_r(\underline{k}) \\ g_s(\underline{k}) &= \sum_{\omega=1}^N \frac{\chi_\delta^\omega(\underline{k})}{D_\omega(\underline{k})} P_\omega(\underline{k}) , \quad \left\| d_{k_\alpha}^n g_r(\underline{k}; y, z) \right\| \leq \sum_{\omega=1}^N \frac{C_n}{1 + |k_0|} \frac{e^{-c|y-z|}}{\|\underline{k} - \underline{k}_F^\omega\|^n} , \end{aligned} \quad (3.4.52)$$

with $\chi_\delta^\omega(\underline{k})$ and $D_\omega(\underline{k})$ as in (3.3.33). We omit the details.

Proceeding as in (3.3.37), we rewrite the correlation function in (3.4.46) as:

$$\begin{aligned} & \frac{1}{\beta L} \langle \mathbf{T} \hat{n}_{\underline{p}_1, y_1} ; \dots ; \hat{n}_{\underline{p}_n, y_n} ; \hat{j}_{\nu, \underline{p}_{n+1}}^\ell \rangle_{\beta, L} \\ &= \tilde{S}_{n; \nu}^{\beta, L, \ell}(\underline{p}_1, y_1, \dots, \underline{p}_n, y_n) + \tilde{R}_{n; \nu}^{\beta, L, \ell}(\underline{p}_1, y_1, \dots, \underline{p}_n, y_n), \end{aligned} \quad (3.4.53)$$

where:

$$\begin{aligned} & \tilde{S}_{n; \nu}^{\beta, L, \ell}(\underline{p}_1, y_1, \dots, \underline{p}_n, y_n) \\ &= - \sum_{\pi \in S_n} \int_{\beta, L} \frac{d\mathbf{k}}{(2\pi)^2} \sum_{z, z'} \text{Tr} \left[\hat{J}_\nu^\ell(k, p_{n+1}; z, z') \prod_{i=1}^{n+1} g_s \left(\mathbf{k} + \sum_{j < i} \underline{p}_{\pi(j)}; y_{\pi(i-1)}, y_{\pi(i)} \right) \right], \end{aligned} \quad (3.4.54)$$

and where the error term $\tilde{R}_{n; \nu}^{\beta, L, \ell}$ contains at least one propagator g_r . The next lemma is the analogue of Lemma 3.3.3.

Lemma 3.4.7 (Structure of the singular part). *For any $p_1, \dots, p_n$ in the support of $\hat{\mu}_{\alpha, \theta}$,*

$$\tilde{S}_{n; \nu}^{\beta, L, \ell}(\underline{p}_1, y_1, \dots, \underline{p}_n, y_n) = S_{n; \nu}^{\beta, L, \ell}(\underline{p}_1, y_1, \dots, \underline{p}_n, y_n) + T_{n; \nu}^{\beta, L, \ell}(\underline{p}_1, y_1, \dots, \underline{p}_n, y_n) \quad (3.4.55)$$

where: for $g_\omega(\underline{k}) = \chi_\delta^\omega(\underline{k})/D_\omega(\underline{k})$,

$$\begin{aligned} S_{n; \nu}^{\beta, L, \ell}(\underline{p}_1, y_1, \dots, \underline{p}_n, y_n) &= - \sum_{\omega=1}^N v_\omega^{\nu, \ell} \prod_{i=1}^n \|\xi_\omega(k_F^\omega; y_i)\|^2 \int_{\beta, L} \frac{d\mathbf{k}}{(2\pi)^2} \sum_{\pi \in S_n} \prod_{i=1}^{n+1} g_\omega \left(\mathbf{k} + \sum_{j < i} \underline{p}_{\pi(j)} \right) \\ T_{n; \nu}^{\beta, L, \ell}(\underline{p}_1, y_1, \dots, \underline{p}_n, y_n) &= - \sum_{\omega=1}^N \int_{\beta, L} \frac{d\mathbf{k}}{(2\pi)^2} \sum_{\pi \in S_n} f_{n, \omega}^{\nu, \ell}(k; p_{\pi(1)}, y_{\pi(1)}, \dots, p_{\pi(n)}, y_{\pi(n)}) \prod_{i=1}^{n+1} g_\omega \left(\mathbf{k} + \sum_{j < i} \underline{p}_{\pi(j)} \right), \end{aligned} \quad (3.4.56)$$

with $v_\omega^{\nu, \ell}$ such that:

$$\begin{aligned} |v_\omega^{\nu, \ell} - \delta_{\nu, 0} - \delta_{\nu, 1} v_\omega| &\leq C e^{-c\ell} && \text{if the edge mode } \omega \text{ is localised at } x_2 = 0 \\ |v_\omega^{\nu, \ell}| &\leq C e^{-c(L-\ell)} && \text{if the edge mode } \omega \text{ is localised at } x_2 = L - 1, \end{aligned} \quad (3.4.57)$$

and $\|\xi_\omega(k_F^\omega; y_i)\|^2 = \sum_{\rho=1}^M |\xi_{\omega; \rho}(k_F^\omega; y_i)|^2$. Furthermore,

$$|f_{n, \omega}^{\nu, \ell}(k; p_1, y_1, \dots, p_n, y_n)| \leq C \left(|k - k_F^\omega|_{\mathbb{T}} \delta_{\nu, 1} + n \sum_{i=1}^n |p_i|_{\mathbb{T}} \right) e^{-c \sum_{i=1}^n |y_i|_\omega}. \quad (3.4.58)$$

Proof. Let us write $g_s(\mathbf{k}) = \sum_\omega g_\omega(\mathbf{k}) P_\omega(k)$ in (3.4.54). As in the one-dimensional case, for $\eta, |p_i|$ small enough, in the loop integral no jump between different chiralities is allowed. Thus, we have:

$$\begin{aligned} & \tilde{S}_{n; \nu}^{\beta, L, \ell}(\underline{p}_1, y_1, \dots, \underline{p}_n, y_n) \\ &= - \sum_{\pi \in S_n} \sum_{\omega=1}^N \int_{\beta, L} \frac{d\mathbf{k}}{(2\pi)^2} \sum_{z, z'} \text{Tr} \left[\hat{J}_\nu^\ell(k, p_{n+1}; z, z') \prod_{i=1}^{n+1} P_\omega \left(k + \sum_{j < i} p_{\pi(j)}; y_{\pi(i-1)}, y_{\pi(i)} \right) \right] \\ & \quad \cdot \prod_{i=1}^{n+1} g_\omega \left(\mathbf{k} + \sum_{j < i} \underline{p}_{\pi(j)} \right). \end{aligned} \quad (3.4.59)$$

Consider the trace in (3.4.59). We rewrite it as:

$$\begin{aligned} & \sum_{z, z'} \text{Tr} \left[\hat{J}_\nu^\ell(k, p_{n+1}; z, z') \prod_{i=1}^{n+1} P_\omega \left(k + \sum_{j < i} p_{\pi(j)}; y_{\pi(i-1)}, y_{\pi(i)} \right) \right] \\ &= \sum_{z, z'} \text{Tr} \left[\hat{J}_\nu^\ell(k, 0; z, z') \prod_{i=1}^{n+1} P_\omega \left(k; y_{\pi(i-1)}, y_{\pi(i)} \right) \right] + \mathcal{E}_{n; \omega; 1}^\nu(k, p_1, \dots, p_n; y_1, \dots, y_n); \end{aligned} \quad (3.4.60)$$

using the decay estimates of the edge modes (3.4.9) and the smoothness of the kernel of $\hat{J}_\nu(k, p)$, the error term can be estimated as:

$$|\mathcal{E}_{n; \omega; 1}^\nu(k, p_1, \dots, p_n; y_1, \dots, y_n)| \leq C n e^{-c \sum_{i=1}^n |y_i|_\omega} \sum_i |p_i|_\mathbb{T}. \quad (3.4.61)$$

Consider now the main term in (3.4.60). Let:

$$\sum_{z, z'} \sum_{\rho, \rho'} \overline{\xi_{\omega, \rho}(k; z)} \hat{J}_{\nu; \rho, \rho'}^\ell(k, 0; z, z') \xi_{\omega, \rho'}(k; z') =: v_\omega^{\nu, \ell}(k); \quad (3.4.62)$$

if ω labels an edge mode localised at $x_2 = L - 1$, then $|v_\omega^{\nu, \ell}(k)| \leq C e^{-c(L-\ell)}$ by the exponential decay of the edge modes. Instead, if ω labels an edge mode localised at $x_2 = 0$,

$$v_\omega^{0, L}(k) = 1, \quad v_\omega^{1, L}(k) = \partial_k e_\omega(k), \quad (3.4.63)$$

where the last identity follows from the Feynman-Hellmann argument. Also, by the decay and the regularity of the edge modes,

$$\left| v_\omega^{\nu, \ell}(k) - v_\omega^{\nu, \ell}(k) \right| \leq C e^{-c|\ell|_\omega}, \quad \left| v_\omega^{\nu, \ell}(k) - v_\omega^{\nu, \ell}(k_F^\omega) \right| \leq C |k - k_F^\omega|_\mathbb{T}. \quad (3.4.64)$$

We then write:

$$\sum_{z, z'} \text{Tr} \left[\hat{J}_\nu^\ell(k, 0; z, z') \prod_{i=1}^{n+1} P_\omega \left(k; y_{\pi(i)}, y_{\pi(i-1)} \right) \right] = v_\omega^{\nu, \ell}(k) \prod_{i=1}^n \|\xi_\omega(k; y_i)\|^2. \quad (3.4.65)$$

Combining (3.4.59)-(3.4.65), we obtain the leading term in (3.4.55), using once more the regularity properties (3.4.9) of the edge modes. The bound for the error term (3.4.58) follows putting together all the estimates for the error terms accumulated. We omit the details. $\square$

Let:

$$R_{n; \nu}^{\beta, L, \ell}(\underline{p}_1, y_1, \dots, \underline{p}_n, y_n) = T_{n; \nu}^{\beta, L, \ell}(\underline{p}_1, y_1, \dots, \underline{p}_n, y_n) + \tilde{R}_{n; \nu}^{\beta, L, \ell}(\underline{p}_1, y_1, \dots, \underline{p}_n, y_n). \quad (3.4.66)$$

Recalling the definition (3.4.32) of the response $\chi_\nu^{\beta, L, \ell}(x; \eta, \theta)$, and the expansion (3.4.45), we have:

$$\begin{aligned} \chi_\nu^{\beta, L, \ell}(x; \eta, \theta) &= - \sum_{n=1}^{\infty} \frac{(-\theta)^n}{n! L^n} \sum_{\{p_i\} \in B_L^n} \sum_{\{y_i\} \in \{1, \dots, L\}^n} \left[\prod_{i=1}^n \hat{\mu}_{\alpha, \theta}(-p_i, y_i) \right] e^{i p_{n+1} x} \\ &\quad \cdot [S_{n; \nu}^{\beta, L, \ell}(\underline{p}_1, y_1, \dots, \underline{p}_n, y_n) + R_{n; \nu}^{\beta, L, \ell}(\underline{p}_1, y_1, \dots, \underline{p}_n, y_n)] + \frac{1}{\theta} \mathcal{E}_\nu^{\beta, L, \ell}(x; \eta, \theta), \end{aligned} \quad (3.4.67)$$

with the error $\mathcal{E}_\nu^{\beta, L, \ell}$ satisfying (3.4.42).

Evaluation of the linear response. Let us consider the $\beta, L \rightarrow \infty$ contribution to the linear response,

$$\chi_\nu^{\ell, \text{lin}}(x; \eta, \theta) = - \int_{\mathbb{T}} \frac{dp}{2\pi} \sum_{y=0}^{\infty} \hat{\mu}_{\alpha, \theta}(-p, y) e^{-ipx} [S_{1; \nu}^\ell(\underline{p}, y) + R_{1; \nu}^\ell(\underline{p}, y)] . \quad (3.4.68)$$

The next proposition is the analogue of Proposition 3.3.7.

Proposition 3.4.8 (Evaluation of the linear response). *We have:*

$$\chi_\nu^{\ell, \text{lin}}(x, \eta, \theta) = - \sum_{\omega}^* \chi_{\nu, \omega} \int_{\mathbb{R}} \frac{dq}{2\pi} \hat{\mu}_{\infty}(q, 0) e^{iq\theta x} \frac{v_{\omega} q}{-i/a + v_{\omega} q} + O(\ell\theta^\alpha) + O(e^{-c\ell}) , \quad (3.4.69)$$

with $\chi_{\nu, \omega} = v_{\omega}' / (2\pi |v_{\omega}|)$, and where the sum runs only over the edge modes localised at $x_2 = 0$.

Proof. By the smoothness of the test function, recall (3.4.19):

$$|\hat{\mu}_{\alpha, \theta}(p, y) - \hat{\mu}_{\alpha, \theta}(p, 0)| \leq \frac{C_r y}{1 + |p/\theta|_{\mathbb{T}_{\theta-1}}^r} \quad \text{for all } r \in \mathbb{N}. \quad (3.4.70)$$

Using the exponential decay of the edge modes and the estimates on the propagator, see discussion after (3.4.48), we have:

$$\chi_\nu^{\ell, \text{lin}}(x; \eta, \theta) = - \int_{\mathbb{T}} \frac{dp}{2\pi} \sum_{y=1}^{\infty} \hat{\mu}_{\alpha, \theta}(-p, 0) e^{-ipx} [S_{1; \nu}^\ell(\underline{p}, y) + R_{1; \nu}^\ell(\underline{p}, y)] + O(\ell\theta) . \quad (3.4.71)$$

The same estimates also allow to show, for $\alpha > 0$:

$$\sum_{y=0}^{\infty} |R_{1; \nu}^\ell(\underline{p}, y) - R_{1; \nu}^\ell(\underline{0}, y)| \leq C\ell \|\underline{p}\|^\alpha + Ce^{-c\ell} . \quad (3.4.72)$$

Consider the contribution of $S_{1; \nu}^\ell$ to the linear response. Recalling the equation (3.3.60) defining the relativistic bubble diagram $\mathfrak{B}_\delta^\omega$, we have:

$$\begin{aligned} \sum_{y=0}^{\infty} S_{1; \nu}^\ell(\underline{p}, y) &= - \sum_{y=0}^{\infty} \sum_{\omega}^* v_{\omega}^{\nu, \ell} \|\xi_{\omega}(k_F^\omega; y)\|^2 \int \frac{dk}{(2\pi)^2} g_{\omega}(\underline{k}) g_{\omega}(\underline{k} + \underline{p}) \\ &= \sum_{\omega}^* v_{\omega}^{\nu, \ell} \mathfrak{B}_\delta^\omega(\underline{p}) , \end{aligned} \quad (3.4.73)$$

where the sum is restricted to the edge modes localised at $y = 0$. Therefore:

$$\begin{aligned} \chi_\nu^{\ell, \text{lin}}(x; \eta, \theta) &= - \int_{\mathbb{T}} \frac{dp}{2\pi} \sum_{y=0}^{\infty} \hat{\mu}_{\alpha, \theta}(-p, 0) e^{-ipx} [S_{1; \nu}^\ell(\underline{p}, y) + R_{1; \nu}^\ell(\underline{p}, y)] + O(\ell\theta) \\ &= - \int_{\mathbb{T}} \frac{dp}{2\pi} \hat{\mu}_{\alpha, \theta}(-p, 0) e^{-ipx} \left[\sum_{\omega}^* v_{\omega}^{\nu, \ell} \mathfrak{B}_\delta^\omega(\underline{p}) + \sum_{y=0}^{\infty} R_{1; \nu}^\ell(\underline{0}, y) \right] \\ &\quad + O(\ell\theta^\alpha) + O(e^{-c\ell}) , \end{aligned} \quad (3.4.74)$$

where the error terms take into account (3.4.72) and the replacement of $\hat{\mu}_{\alpha,\theta}(p, 0)$ with $\hat{\mu}_\theta(p, 0)$. Next, we fix the second term in the integral using the lattice continuity equation. Let us introduce the shorthand $\mathcal{O}_{\underline{x}, x_2} = \gamma_{x_0}(\mathcal{O}_{(x_1, x_2)})$; then, Eq. (3.4.24) implies:

$$\begin{aligned} i\partial_{x_0}\langle \mathbf{T}n_{\underline{x}, x_2}; j_{\nu, \underline{x}}^\ell \rangle_{\beta, L} + d_{x_1}\langle \mathbf{T}j_{1, \underline{x}, x_2}; j_{\nu, \underline{x}}^\ell \rangle_{\beta, L} + d_{x_2}\langle \mathbf{T}j_{2, \underline{x}, x_2}; j_{\nu, \underline{x}}^\ell \rangle_{\beta, L} \\ = i\delta(x_0 - x'_0)\langle [n_{(x, x_2)}, j_{\nu, x'}^\ell] \rangle_{\beta, L}. \end{aligned} \quad (3.4.75)$$

Taking the Fourier transform, summing over x_2 , and using the Dirichlet boundary conditions:

$$p_0 \frac{1}{\beta L} \langle \mathbf{T}\hat{n}_{\underline{p}}^L; \hat{j}_{\nu, -\underline{p}}^\ell \rangle_{\beta, L} + (1 - e^{-ip}) \frac{1}{\beta L} \langle \mathbf{T}\hat{j}_{1, \underline{p}}^L; \hat{j}_{\nu, -\underline{p}}^\ell \rangle_{\beta, L} = i \sum_{x \in \Gamma_L} e^{ipx_1} \langle [n_x, j_{\nu, 0}^\ell] \rangle_L. \quad (3.4.76)$$

Choosing $\underline{p} = (\eta_\beta, 0) \neq \underline{0}$, the right-hand side vanishes because $\sum_{x \in \Gamma_L} n_x = \mathcal{N}$ is the number operator, which commutes with $j_{\nu, 0}^\ell$. Thus:

$$\sum_{x_2=0}^{L-1} \langle \mathbf{T}\hat{n}_{(\eta_\beta, 0), x_2}; \hat{j}_{\nu, (-\eta_\beta, 0)}^\ell \rangle_{\beta, L} = 0 \implies \sum_{y=0}^{L-1} R_{\nu; 1}^{\beta, L, \ell}((\eta_\beta, 0), y) = - \sum_{y=0}^{L-1} S_{\nu; 1}^{\beta, L, \ell}((\eta_\beta, 0), y). \quad (3.4.77)$$

Taking the limit $\beta, L \rightarrow \infty$, and using the continuity in $\underline{p}$ of $R_{\nu; 1}^\ell(\underline{p}, y)$, we can determine the second term in the integral in (3.4.74) in terms of (3.4.73). We ultimately get:

$$\chi_\nu^{\ell, \text{lin}}(x, \eta, \theta) = - \sum_{\omega}^* \frac{v_\omega^{\nu, \ell}}{2\pi|v_\omega|} \int_{\mathbb{T}} \frac{dp}{(2\pi)} \hat{\mu}_\theta(p, 0) e^{ipx} \frac{v_\omega p}{-i\eta + v_\omega p} + O(\ell\theta^\alpha) + O(e^{-c\ell}). \quad (3.4.78)$$

The final claim (3.4.69) follows recalling the definition of $\hat{\mu}_\theta(p, 0)$ in terms of $\hat{\mu}_\infty(p/\theta, 0)$, Eq. (3.4.19), using the fast decay of $\hat{\mu}_\infty(p)$, performing the change of variables $p/\theta = q$, and recalling that $|v_\omega^{\nu, \ell} - v_\omega^\nu| \leq Ce^{-c\ell}$. $\square$

We are now ready to prove the main result of this section, Theorem 3.4.3.

Proof of Theorem 3.4.3. The starting point is the identity (3.4.67). The explicit form of the linear response term, Eq. (3.4.35), is obtained proceeding as in Proposition 3.3.7. Let us now consider the higher-order terms,

$$\begin{aligned} - \sum_{n=2}^{\infty} \frac{(-\theta)^{n-1}}{n!L^n} \sum_{\{p_i\} \in B_L^n} \sum_{\{y_i\} \in \{0, \dots, L-1\}^n} \left[\prod_{i=1}^n \hat{\mu}_{\alpha, \theta}(-p_i, y_i) \right] e^{ip_{n+1}x} \\ \cdot [S_{n; \nu}^{\beta, L, \ell}(\underline{p}_1, y_1, \dots, \underline{p}_n, y_n) + R_{n; \nu}^{\beta, L, \ell}(\underline{p}_1, y_1, \dots, \underline{p}_n, y_n)]. \end{aligned} \quad (3.4.79)$$

We claim that, for some $\gamma > 0$:

$$|(3.4.79)| \leq C\ell\eta^\gamma. \quad (3.4.80)$$

This estimate, combined with the evaluation of the linear response, Proposition 3.4.8, and with the bound (3.4.42) for the error terms produced by the comparison with the auxiliary dynamics, implies the claim of Theorem 3.4.3, Eq. (3.4.42).

Let us prove (3.4.80). Consider the contribution associated with $S_{n;\nu}^{\beta,L,\ell}$. By (3.4.56),

$$\begin{aligned} & - \sum_{n=2}^{\infty} \frac{(-\theta)^{n-1}}{n!L^n} \sum_{\{p_i\} \in B_L^n} \sum_{\{y_i\} \in \{0, \dots, L-1\}^n} \left[\prod_{i=1}^n \hat{\mu}_{\alpha, \theta}(-p_i, y_i) \right] e^{ip_{n+1}x} S_{n;\nu}^{\beta,L,\ell}(\underline{p}_1, y_1, \dots, \underline{p}_n, y_n) \\ & = - \sum_{n=2}^{\infty} \frac{(-\theta)^{n-1}}{n!L^n} \sum_{\{p_i\} \in B_L^n} \sum_{\omega=1}^N \left[\prod_{i=1}^n \tilde{\mu}_{\alpha, \theta; \omega}(-p_i) \right] e^{ip_{n+1}x} S_{n;\nu; \omega}^{\beta,L,\ell}(\underline{p}_1, \dots, \underline{p}_n) \end{aligned} \quad (3.4.81)$$

where:

$$\tilde{\mu}_{\alpha, \theta; \omega}(p_i) = \sum_{y_i=0}^{L-1} \hat{\mu}_{\alpha, \theta}(p_i, y_i) \|\xi_{\omega}(k_F^{\omega}; y_i)\|^2, \quad (3.4.82)$$

and:

$$S_{n;\nu; \omega}^{\beta,L,\ell}(\underline{p}_1, \dots, \underline{p}_n) = -v_{\omega}^{\nu, \ell} \int_{\beta, L} \frac{dk}{(2\pi)^2} \sum_{\pi \in S_n} \prod_{i=1}^{n+1} g_{\omega}\left(\underline{k} + \sum_{j < i} \underline{p}_{\pi(j)}\right). \quad (3.4.83)$$

The right-hand side of eq. (3.4.81) is the same type of contribution that has been studied for one-dimensional systems. By Proposition 3.3.9, and proceeding as in the proof of Theorem 3.2.1, we have, for $\gamma > 0$:

$$\sum_{n=2}^{\infty} \frac{\theta^{n-1}}{n!L^n} \sum_{\{p_i\} \in B_L^n} \sum_{\omega=1}^N \left[\prod_{i=1}^n |\tilde{\mu}_{\alpha, \theta; \omega}(-p_i)| \right] |S_{n;\nu; \omega}^{\beta,L,\ell}(\underline{p}_1, \dots, \underline{p}_n)| \leq C\eta^{\gamma}. \quad (3.4.84)$$

Let us now discuss the error terms $R_{n;\nu}^{\beta,L,\ell}$ in (3.4.67). Consider the term $T_{n;\nu}^{\beta,L,\ell}$ in (3.4.66). We have:

$$\begin{aligned} & \sum_{n=2}^{\infty} \frac{\theta^{n-1}}{n!L^n} \sum_{\{p_i\} \in B_L^n} \sum_{\{y_i\} \in \{0, \dots, L-1\}^n} \left[\prod_{i=1}^n |\hat{\mu}_{\alpha, \theta}(-p_i, y_i)| \right] |T_{n;\nu}^{\beta,L,\ell}(\underline{p}_1, \dots, \underline{p}_n)| \\ & \leq \sum_{n=2}^{\infty} \frac{\theta^{n-1}}{n!L^n} \sum_{\{p_i\} \in B_L^n} \sum_{\{y_i\} \in \{0, \dots, L-1\}^n} \left[\prod_{i=1}^n |\hat{\mu}_{\alpha, \theta}(-p_i, y_i)| \right] \\ & \quad \cdot \sum_{\omega=1}^N \int_{\beta, L} \frac{dk}{(2\pi)^2} \sum_{\pi \in S_n} |f_{n, \omega}^{\nu, \ell}(k; p_{\pi(1)}, y_{\pi(1)}, \dots, p_{\pi(n)}, y_{\pi(n)})| \prod_{i=1}^{n+1} \left| g_{\omega}\left(\underline{k} + \sum_{j < i} \underline{p}_{\pi(j)}\right) \right|; \end{aligned} \quad (3.4.85)$$

using (3.4.58):

$$\begin{aligned} & \sum_{\{y_i\} \in \{0, \dots, L-1\}^n} \left[\prod_{i=1}^n |\hat{\mu}_{\alpha, \theta}(-p_i, y_i)| \right] |f_{n, \omega}^{\nu, \ell}(k; p_1, y_1, \dots, p_n, y_n)| \\ & \leq \sum_{\{y_i\} \in \{0, \dots, L-1\}^n} \left[\prod_{i=1}^n |\hat{\mu}_{\alpha, \theta}(-p_i, y_i)| \right] C^n \left(|k - k_F^{\omega}|_{\mathbb{T}} + \sum_{i=1}^n |p_i|_{\mathbb{T}} \right) e^{-c \sum_{i=1}^n |y_i|_{\omega}} \quad (3.4.86) \\ & \equiv C^n \left(|k - k_F^{\omega}|_{\mathbb{T}} + \sum_{i=1}^n |p_i|_{\mathbb{T}} \right) \left[\prod_{i=1}^n |\tilde{\mu}_{\alpha, \theta; \omega}(-p_i)| \right]. \end{aligned}$$

Plugging (3.4.86) into (3.4.85), we obtain an expression completely analogous to the one-dimensional case. Proceeding as for (3.3.144):

$$\begin{aligned} (3.4.85) &\leq \sum_{n \geq 2} \frac{C^n a^{n-1}}{n!} |\log \eta| (\eta + n\theta) \\ &\leq (1 + a) e^{Ca} |\log \eta| \eta. \end{aligned} \quad (3.4.87)$$

To conclude, let us estimate the terms $\tilde{R}_{n;\nu}^{\beta,L,\ell}$ in (3.4.66). Recall the representation:

$$\begin{aligned} &\tilde{R}_{n;\nu}^{\beta,L,\ell}(\underline{p}_1, y_1, \dots, \underline{p}_n, y_n) \\ &= \sum_{\pi \in S_n} \sum_{\substack{f \in \{r,s\}^{n+1} \\ f \neq s}} \sum_{z, z'} \int_{\beta, L} \frac{dk}{(2\pi)^2} \text{Tr} \left[\hat{J}_\nu^\ell(k, p_{n+1}; z, z') \prod_{i=1}^{n+1} g_{f(i)} \left(\underline{k} + \sum_{l < i} \underline{p}_{\pi(l)}; y_{\pi(i-1)}, y_{\pi(i)} \right) \right]. \end{aligned} \quad (3.4.88)$$

Let $\tilde{R}_{n;\nu,j}^{\beta,L,\ell}$ be the contribution to $\tilde{R}_{n;\nu}^{\beta,L,\ell}$ with j propagators labelled by r . Let

$$\tilde{g}_r(\underline{k}; y_i, y_j) = (1 + |k_0|) g_r(\underline{k}; y_i, y_j) \quad (3.4.89)$$

and recall:

$$g_s(\underline{k}; y_i, y_j) = \sum_{\omega} g_{\omega}(\underline{k}) P_{\omega}(\underline{k}; y_i, y_j). \quad (3.4.90)$$

Let $m_0, m_1, \dots, m_{n-j}$ be the labels such that $f(m_k) = s$, in increasing order. Denoting by I the set of labels associated with the g_r propagators, we can rewrite the trace in (3.4.88) as:

$$\begin{aligned} &\text{Tr} \left[\hat{J}_\nu^\ell(k, p_{n+1}; z, z') \prod_{i=1}^{n+1} g_{f(i)} \left(\underline{k} + \sum_{l < i} \underline{p}_{\pi(l)}; y_{\pi(i-1)}, y_{\pi(i)} \right) \right] \\ &= \sum_{\omega_{m_0}, \dots, \omega_{m_{n-j}}} g_{\omega_{m_0}} \left(\underline{k} + \sum_{l < m_0} \underline{p}_{\pi(l)} \right) \cdots g_{\omega_{m_{n-j}}} \left(\underline{k} + \sum_{l < m_{n-j}} \underline{p}_{\pi(l)} \right) \prod_{q \in I} \frac{1}{|k_0 + q\eta| + 1} \\ &\quad \cdot \text{Tr} \left[\hat{J}_\nu^\ell(k, p_{n+1}; z, z') \prod_{i=1}^{n+1} A_{f(i)} \left(\underline{k} + \sum_{l < i} \underline{p}_{\pi(l)}; y_{\pi(i-1)}, y_{\pi(i)} \right) \right], \end{aligned} \quad (3.4.91)$$

where:

$$A_{f(i)}(\underline{k}) = \begin{cases} P_{\omega_i}(k) & \text{if } i \in I^c \\ \tilde{g}_r(\underline{k}) & \text{if } i \in I. \end{cases} \quad (3.4.92)$$

Using the exponential decay of the edge modes and the exponential decay in $|y_i - y_j|$ of $\tilde{g}_r(\underline{k}; y_i, y_j)$, we easily get:

$$\begin{aligned} &\sum_{z, z'} \sum_{\{y_i\} \in \{0, \dots, L-1\}^n} \left[\prod_{i=1}^n |\hat{\mu}_{\alpha, \theta}(-p_i, y_i)| \right] \\ &\quad \cdot \left| \text{Tr} \left[\hat{J}_\nu^\ell(k, p_{n+1}; z, z') \prod_{i=1}^{n+1} A_{f(i)} \left(\underline{k} + \sum_{l < i} \underline{p}_{\pi(l)}; y_{\pi(i-1)}, y_{\pi(i)} \right) \right] \right| \\ &\leq C^n \ell \left[\prod_{i=1}^n |\nu_{\alpha, \theta}(p_i)| \right] \end{aligned} \quad (3.4.93)$$

for:

$$\nu_{\alpha,\theta}(p) := \frac{1}{\theta} \frac{\chi(\theta^{\alpha-1}|p|)}{1 + |p/\theta|_{\mathbb{T}_{\theta-1}}^2} . \quad (3.4.94)$$

Thus, the contribution associated with (3.4.91) to (3.4.88) summed over the $\{y_i\}$ variables is bounded as:

$$\begin{aligned} & C^n \ell \sum_{\pi \in S_n} \int_{\beta, L} \frac{dk}{(2\pi)^2} \sum_{\{\omega_{m_i}\}} \left| g_{\omega_{m_0}} \left(\underline{k} + \sum_{l < m_0} \underline{p}_{\pi(l)} \right) \cdots g_{\omega_{m_{n-j}}} \left(\underline{k} + \sum_{l < m_{n-j}} \underline{p}_{\pi(l)} \right) \right| \\ & \cdot \prod_{q \in I} \left[\frac{1}{|k_0 + q\eta| + 1} \right] \cdot \left[\prod_{i=1}^n |\nu_{\alpha,\theta}(p_i)| \right] ; \end{aligned} \quad (3.4.95)$$

from now on, the discussion proceeds exactly as in the one-dimensional case, see the discussion after (3.3.131). In fact, the functions in the product over $q \in I$ in Eq. (3.4.95) play the role of the bounded propagators in the one-dimensional case. Therefore, recalling (3.3.145), choosing $a \leq w |\log \eta|$ for w small enough, and for some $\gamma > 0$, the final result is:

$$\sum_{n=2}^{\infty} \frac{\theta^{n-1}}{n! L^n} \sum_{\{p_i\} \in B_L^n} \sum_{\{y_i\} \in \{0, \dots, L-1\}^n} \left[\prod_{i=1}^n |\hat{\mu}_{\alpha,\theta}(p_i, y_i)| \right] |\tilde{R}_{\nu;n}^{\beta, L, \ell}(\underline{p}_1, y_1, \dots, \underline{p}_n, y_n)| \leq C \ell \eta^\gamma . \quad (3.4.96)$$

Eqs. (3.4.87), (3.4.96) prove (3.4.80), and conclude the proof of Theorem 3.4.3. $\square$

Chapter 4

Response theory for interacting fermions

This Chapter 4, based on [70], is devoted to the generalisation of Theorem 3.2.1 to the case of (weakly) interacting one-dimensional systems. In Section 4.1, we start by reviewing basic facts about one-dimensional lattice fermions, we define the class of models that we are able to consider, and we state our main result, Theorem 4.1.5, which justifies the Kubo approximation of dynamical response. The theorem provides an explicit expression for the response of the density and current observables after introducing a weak and slowly varying external perturbation. Remarkably, we find that the addition of the interaction modifies such response by renormalising the parameters of the underlying free theory (namely, the value of the Fermi velocities), and by mixing the contribution of different Fermi points.

In Section 4.2 we briefly review the Wick rotation of the Duhamel series to obtain the expansion in Euclidean cumulants of density type. In Section 4.3, we discuss the renormalisation group analysis of the model, which takes as a starting point the Grassmann integral representation of the theory, presented at the end of Chapter 2. In particular, in Section 4.3.6 we discuss the RG analysis of the multi-point correlation functions, and their improved bounds, which allow us to prove summability of the Duhamel expansion. These bounds rely on the Gallavotti-Nicolò tree expansion, reviewed in Section 4.3.5.

In Section 4.4 we define the reference model, and we prove the Ward identity for the m -point density correlations, which allows us to prove the vanishing of all correlations with $m \geq 3$. In Section 4.5 we prove our main result, by comparing the renormalised expansions of the lattice and reference model, extracting the cancellation of the reference model, and estimating the remainder taking into account the improvement of Section 4.3.6. Also, we compute the susceptibility and the charge conductance, using the explicit form of the scaling limit, and lattice Ward identities to fix the form of the non-universal part.

4.1 Interacting lattice fermions

4.1.1 Hamiltonian and Gibbs state

Let $L \in 2\mathbb{N} + 1$. We consider fermions on a one-dimensional lattice

$$\Gamma_L = \left[-\left\lfloor \frac{L}{2} \right\rfloor, \left\lfloor \frac{L}{2} \right\rfloor \right] \cap \mathbb{Z}$$

endowed with periodic boundary conditions. We shall allow for internal degrees of freedom, *e.g.* the spin, and we shall denote by $S_M = \{1, \dots, M\}$ the set of their labels. We shall use the notation $\Lambda_L = \Gamma_L \times S_M$, which we can view as a decorated $1d$ lattice. Given $x \in \Gamma_L$ and $\rho \in S_M$, we shall denote by $\mathbf{x} = (x, \rho)$ the corresponding point in Λ_L . Further, we will make use of the following distance on Γ_L :

$$|x - y|_L = \min_{n \in \mathbb{Z}} |x - y + nL|. \quad (4.1.1)$$

Let H be a single-particle Hamiltonian on Λ_L . That is, H is a self-adjoint matrix, with entries given by $H(\mathbf{x}; \mathbf{y}) \equiv H_{\rho\rho'}(x; y)$. We shall suppose that the Hamiltonian is finite-ranged, and we shall assume that the Hamiltonian H is the periodisation of a Hamiltonian H^∞ defined over $\ell^2(\mathbb{Z} \times S_M)$:

$$H_{\rho\rho'}(x; y) = \sum_{n \in \mathbb{Z}} H_{\rho\rho'}^\infty(x + nL; y) \quad \text{for } x, y \in \Gamma_L. \quad (4.1.2)$$

Furthermore, we will assume H^∞ , and hence H , to be translation-invariant:

$$H_{\rho\rho'}^\infty(x; y) = H_{\rho\rho'}^\infty(x + z; y + z) \quad \text{for all } z \in \mathbb{Z}. \quad (4.1.3)$$

Translation invariance allows us to conveniently express the Hamiltonian H in momentum space. To this end, recall that Brillouin zone is defined as:

$$B_L = \left\{ k = \frac{2\pi}{L}n \mid 0 \leq n \leq L - 1 \right\}. \quad (4.1.4)$$

and that the Bloch transform of a function f is given by:

$$\hat{f}(k) = \sum_{x \in \Gamma_L} e^{-ikx} f(x) \quad \text{for all } k \in B_L. \quad (4.1.5)$$

This relation can be inverted as:

$$f(x) = \frac{1}{L} \sum_{k \in B_L} e^{ikx} \hat{f}(k) \quad \text{for all } x \in \Gamma_L. \quad (4.1.6)$$

As a function on $\frac{2\pi}{L}\mathbb{Z}$, the function $\hat{f}(k)$ is periodic with period 2π . Thus, it is natural to introduce the following norm for quasi-momenta:

$$|k|_{\mathbb{T}} = \min_{n \in \mathbb{Z}} |k + 2\pi n| \quad \text{for all } k \in B_L. \quad (4.1.7)$$

Next, we define the Bloch Hamiltonian $\hat{H}(k)$ as the $M \times M$ Hermitian matrix with entries:

$$\hat{H}_{\rho\rho'}(k) = \sum_{x \in \Gamma_L} e^{-ikx} H_{\rho\rho'}(x; 0). \quad (4.1.8)$$

Observe that, due to equation (4.1.3), $\hat{H}(k)$ is the restriction to B_L of $\hat{H}^\infty(k)$, defined on the circle $\mathbb{T}$ of length 2π .

We shall make the following assumptions on the spectrum of the Bloch Hamiltonian and on the chemical potential $\mu \in \mathbb{R}$.

Assumption 4.1.1 (Low energy spectrum.). There exists $\Delta > 0$ such that the following points are true.

- (i) There exist $N_f \in \mathbb{N}$ disjoint sets $I_\omega \subset \mathbb{T}$, labelled by $\omega = 1, \dots, N_f$, and strictly monotone smooth functions $e_\omega : I_\omega \rightarrow \mathbb{R}$ such that:

$$\sigma(H^\infty) \cap (\mu - \Delta, \mu + \Delta) = \bigcup_{\omega=1}^{N_f} \text{Ran}(e_\omega) . \quad (4.1.9)$$

The functions e_ω are the eigenvalue branches of the Bloch Hamiltonian $\hat{H}^\infty$, crossing the Fermi level μ .

- (ii) We introduce the ω -Fermi point $k_F^\omega \in I_\omega$ and the ω -Fermi velocity v_ω as the real numbers satisfying:

$$e_\omega(k_F^\omega) = \mu , \quad v_\omega = \partial_k e_\omega(k_F^\omega) . \quad (4.1.10)$$

Notice that $v_\omega \neq 0$, by the strict monotonicity of e_ω . The set of Fermi points $\{k_F^\omega\}_{\omega=1, \dots, N_f}$ will be referred to as the (discrete) *Fermi surface* of the system.

- (iii) For any $\omega = 1, \dots, N_f$, and $k \in I_\omega$, $e_\omega(k)$ is a non-degenerate eigenvalue of $\hat{H}^\infty(k)$. In particular, the Fermi points k_F^ω , $\omega = 1, \dots, N_f$ are all different.
- (iv) For any quadruple $(\omega_1, \omega_2, \omega_3, \omega_4)$, we have that

$$k_F^{\omega_1} - k_F^{\omega_2} = k_F^{\omega_3} - k_F^{\omega_4} \quad \text{mod } 2\pi \quad (4.1.11)$$

only if $\omega_1 = \omega_2$ and $\omega_3 = \omega_4$, or if $\omega_1 = \omega_3$ and $\omega_2 = \omega_4$.

Remark 4.1.2. (i) A paradigmatic example satisfying the assumptions is (minus) the Laplacian on $\mathbb{Z}$, for spinless fermions. The spectrum is described by:

$$\varepsilon(k) = 2t(1 - \cos(k)) , \quad (4.1.12)$$

with $t > 0$ the hopping parameter, so that $N_f = 2$ for any $\mu \in (0, 4t)$.

- (ii) In general, it is a well-known fact that short-range one-dimensional lattice models necessarily present an even number N_f of Fermi points, with net chirality equal to zero: $\sum_{\omega=1}^{N_f} v_\omega / |v_\omega| = 0$.
- (iii) Eq. (4.1.11) can be viewed as an elastic scattering assumption for the Fermi momenta of the system. It is generically true for spinless fermions, in the absence of special symmetry properties of the spectrum. Observe that this assumption was not needed in the non-interacting case, Chapter 3. The reason for its introduction is to prevent the appearance of dangerous marginal terms in the renormalisation group analysis of the interacting model.

Next, we will introduce the class of many-body interacting systems we will consider in this chapter, in the grand-canonical setting. To this end, it is convenient to lift the many particle systems to the fermionic Fock space $\mathcal{F}_L$,

$$\mathcal{F}_L = \mathbb{C} \oplus \bigoplus_{n \geq 1} \ell^2(\Lambda_L)^{\wedge n} , \quad (4.1.13)$$

where $\wedge$ is the antisymmetric tensor product. We shall use the notation $\mathbf{x} = (x, \rho)$ for points in Λ_L , with $x \in \Gamma_L$ and $\rho \in S_M$, and we shall denote by $a_{\mathbf{x}}^*, a_{\mathbf{x}}$ the usual fermionic creation and annihilation operators, satisfying the canonical anticommutation relations:

$$\{a_{\mathbf{x}}, a_{\mathbf{y}}^*\} = \delta_{\mathbf{x}, \mathbf{y}} , \quad \{a_{\mathbf{x}}, a_{\mathbf{y}}\} = \{a_{\mathbf{x}}^*, a_{\mathbf{y}}^*\} = 0 . \quad (4.1.14)$$

Later, it will also be convenient to switch to Fourier space. For any $k \in B_L$, we define the Bloch transform of the fermionic operators as:

$$\hat{a}_{(k,\rho)} = \sum_{x \in \Gamma_L} a_{(x,\rho)} e^{-ikx}, \quad \hat{a}_{(k,\rho)}^* = \sum_{x \in \Gamma_L} a_{(x,\rho)}^* e^{ikx}. \quad (4.1.15)$$

Equations (4.1.15) can be inverted as, for $(x, \rho) \in \Lambda_L$:

$$a_{(x,\rho)} = \frac{1}{L} \sum_{k \in B_L} e^{ikx} \hat{a}_{(k,\rho)}, \quad a_{(x,\rho)}^* = \frac{1}{L} \sum_{k \in B_L} e^{-ikx} \hat{a}_{(k,\rho)}^*. \quad (4.1.16)$$

In terms of the creation and annihilation operators, we define the many-body Hamiltonian $\mathcal{H}$ acting on $\mathcal{F}_L$ as:

$$\mathcal{H} = \sum_{\mathbf{x}, \mathbf{y} \in \Lambda_L} a_{\mathbf{x}}^* H(\mathbf{x}; \mathbf{y}) a_{\mathbf{y}} + \lambda \sum_{\mathbf{x}, \mathbf{y} \in \Lambda_L} \left(n_{\mathbf{x}} - \frac{1}{2} \right) w(\mathbf{x}; \mathbf{y}) \left(n_{\mathbf{y}} - \frac{1}{2} \right), \quad (4.1.17)$$

where: $\lambda \in \mathbb{R}$ is the coupling constant, $n_{\mathbf{x}} = a_{\mathbf{x}}^* a_{\mathbf{x}}$ is the density operator; and $w(\mathbf{x}; \mathbf{y})$ is the two-body potential. As for the single-particle Hamiltonian, we will suppose that w is translation-invariant, compatible with the periodic boundary conditions, and finite-ranged. Furthermore, for simplicity, we will assume that $w(\mathbf{x}; \mathbf{y}) \equiv w(x; y)$. That is,

$$w(\mathbf{x}; \mathbf{y}) = \frac{1}{L} \sum_{k \in B_L} e^{ik(x-y)} \hat{w}(k), \quad (4.1.18)$$

where $\hat{w}(k)$ is the restriction to B_L of a real-analytic function on $\mathbb{T}$.

Remark 4.1.3. The factors $-1/2$ in (4.1.17) amount to a shift of the many-body Hamiltonian by a constant term and by a term proportional to the number operator, which can be reabsorbed in the definition of the chemical potential of the interacting Gibbs state. These factors are introduced to simplify the Grassmann representation of the Gibbs state, discussed later.

Finally, the Gibbs state of the system at inverse temperature $\beta > 0$ is:

$$\langle \mathcal{O} \rangle_{\beta, L} = \text{Tr } \mathcal{O} \rho_{\beta, \mu, L}, \quad \rho_{\beta, L} = \frac{e^{-\beta(\mathcal{H} - \mu \mathcal{N})}}{\text{Tr } e^{-\beta(\mathcal{H} - \mu \mathcal{N})}}, \quad (4.1.19)$$

with $\mu \in \mathbb{R}$ the chemical potential and $\mathcal{N}$ the number operator. We shall assume Assumption 4.1.1 to hold for our choice of the chemical potential μ . Notice that the Gibbs state depends on the parameters λ, μ as well, but in order to simplify the notation, we have chosen to write explicitly only the dependence on the variables β, L .

4.1.2 Dynamics and linear response

Perturbing the system. We will be interested in the response properties of the system, after exposing it to a time-dependent and slowly varying perturbation. We shall consider time-dependent Hamiltonians of the form:

$$\mathcal{H}(\eta t) = \mathcal{H} + e^{\eta t} \mathcal{P}, \quad (4.1.20)$$

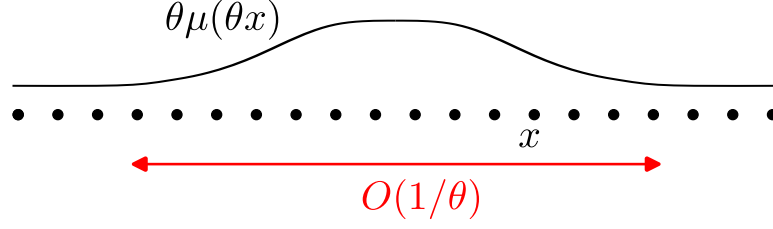


Figure 4.1: Qualitative representation of the spatial lattice Γ_L and typical form of the rescaled local potential entering the definition of $\mathcal{P}$, equation (4.1.21).

for $\eta > 0$ and $t \leq 0$. We will consider external perturbations that couple to the density of the system:

$$\mathcal{P} = \theta \sum_{\mathbf{x} \in \Lambda_L} \mu(\theta x) a_{\mathbf{x}}^* a_{\mathbf{x}}, \quad (4.1.21)$$

where: $\mu(\cdot)$ (see Figure 4.1) is the restriction to Γ_L of the periodisation of a function $\mu_{\infty}(x)$ in $\mathbb{R}$ such that $\hat{\mu}_{\infty}(p)$ has compact support:

$$\mu(x) = \sum_{n \in \mathbb{Z}} \mu_{\infty}(x + nL), \quad \text{for } x \in \Gamma_L. \quad (4.1.22)$$

Thus, $\hat{\mu}(p)$ is $\hat{\mu}_{\infty}(p)$ restricted to $p \in (2\pi/L)\mathbb{Z}$. The compact support restriction could be replaced by fast decay in momentum space, but we prefer to work in a slightly more restrictive setting to avoid extra technicalities. Observe that the parameter $\theta > 0$ in (4.1.21) defines simultaneously the strength and the space-variation of the perturbation, and will be suitably chosen later on.

Let $t, s \leq 0$, and let $\mathcal{U}(t; s)$ be the unitary group generated by $\mathcal{H}(\eta t)$:

$$i\partial_t \mathcal{U}(t; s) = \mathcal{H}(\eta t) \mathcal{U}(t; s), \quad \mathcal{U}(s; s) = \mathbb{1}. \quad (4.1.23)$$

Let us consider the evolution of the Gibbs state $\rho_{\beta, L}$,

$$\rho(t) = \lim_{T \rightarrow -\infty} \mathcal{U}(t; T) \rho_{\beta, L} \mathcal{U}(t; T)^*. \quad (4.1.24)$$

We will be interested in the variation of quantum expectations of physical observables under this perturbation protocol, namely quantities of the form

$$\text{Tr } \mathcal{O} \rho(t) - \text{Tr } \mathcal{O} \rho_{\beta, L}. \quad (4.1.25)$$

We shall take interest in the undertaking of the following order of limits: first $L \rightarrow \infty$, then $\beta \rightarrow \infty$ and then $\eta, \theta \rightarrow 0^+$. For small perturbations, the dynamics of the system can be studied via the Duhamel series (obtained in Proposition 2.2.2 for lattice systems in any dimension and general local observables), which describes the expansion around the unperturbed evolution in the interaction picture.

Indeed, we recall that, letting $\tau_t(\mathcal{A})$ be the Heisenberg evolution of the observable $\mathcal{A}$ with respect to the unperturbed Hamiltonian $\mathcal{H}$,

$$\tau_t(\mathcal{A}) = e^{i\mathcal{H}t} \mathcal{A} e^{-i\mathcal{H}t}, \quad (4.1.26)$$

we have:

$$\begin{aligned} \text{Tr } \mathcal{O} \rho(t) - \text{Tr } \mathcal{O} \rho_{\beta, L} &= \sum_{n=1}^{\infty} (-i)^n \int_{-\infty \leq s_n \leq \dots \leq s_1 \leq t} d\underline{s} e^{\eta(s_1 + \dots + s_n)} \\ &\quad \cdot \langle [\dots [\tau_t(\mathcal{O}), \tau_{s_1}(\mathcal{P})], \tau_{s_2}(\mathcal{P})] \dots \tau_{s_n}(\mathcal{P})] \rangle_{\beta, L}. \end{aligned} \quad (4.1.27)$$

As we showed in Section 2.2, for $\eta > 0$ and for finite L , the expansion in the right-hand side is absolutely convergent. The linear response approximation amounts to the truncation of the above expansion to first order in $\mathcal{P}$:

$$\mathrm{Tr} \, \mathcal{O} \rho(t) - \mathrm{Tr} \, \mathcal{O} \rho_{\beta,L} = -i \int_{-\infty}^t ds e^{\eta s} \mathrm{Tr} [\tau_t(\mathcal{O}), \tau_s(\mathcal{P})] \rho_{\beta,\mu,L} + \text{h.o.t.} \quad (4.1.28)$$

Proving that the error term is small uniformly in β and L is in general a very difficult problem, in particular for gapless interacting models, as in the present chapter.

Density and current operators. We will focus on the dynamics of the expectation values of density and current operators, defined as follows. Let:

$$n_x = \sum_{\rho \in S_M} n_{(x,\rho)} \equiv \sum_{\rho \in S_M} a_{(x,\rho)}^* a_{(x,\rho)} \quad (4.1.29)$$

be the (total) density at $x \in \Gamma_L$. In order to introduce the current operator, let us consider the time-variation of the density operator:

$$\begin{aligned} \partial_t \tau_t(n_{\mathbf{x}}) &= i \tau_t([\mathcal{H}, n_{\mathbf{x}}]) \\ &= \sum_{\mathbf{y} \in \Lambda_L} \tau_t(i a_{\mathbf{y}}^* H(\mathbf{y}; \mathbf{x}) a_{\mathbf{x}} - i a_{\mathbf{x}}^* H(\mathbf{x}; \mathbf{y}) a_{\mathbf{y}}) \\ &\equiv \sum_{\mathbf{y} \in \Lambda_L} \tau_t(j_{\mathbf{y}, \mathbf{x}}) , \end{aligned} \quad (4.1.30)$$

where $j_{\mathbf{y}, \mathbf{x}} = i a_{\mathbf{y}}^* H(\mathbf{y}; \mathbf{x}) a_{\mathbf{x}} - i a_{\mathbf{x}}^* H(\mathbf{x}; \mathbf{y}) a_{\mathbf{y}}$ is the bond current between $\mathbf{x}$ and $\mathbf{y}$. Let us derive the conservation law for n_x . To this end, in (4.1.30) we sum over the internal label ρ in $\mathbf{x} = (x, \rho)$, and we introduce the discrete derivative d_x as $(d_x f)(x) = f(x) - f(x-1)$ to obtain the continuity equation

$$\partial_t \tau_t(n_x) = -d_x \tau_t(j_x) , \quad (4.1.31)$$

where we introduced the lattice current j_x :

$$d_x j_x := \sum_{y \in \Gamma_L} \sum_{\rho, \rho' \in S_M} j_{(x,\rho), (y,\rho')} \quad (4.1.32)$$

Eq. (4.1.31) will have far reaching consequences in our analysis.

It is convenient to rewrite density and current operators in Fourier space. Let $j_{\nu,x}$, $\nu = 0, 1$, be defined as $j_{0,x} = n_x$, $j_{1,x} = j_x$. In Fourier space:

$$j_{\nu,x} = \frac{1}{L} \sum_{p \in B_L} e^{ipx} \hat{j}_{\nu,p} , \quad \hat{j}_{\nu,p} = \frac{1}{L} \sum_{k \in B_L} (a_{k-p}^* \hat{J}_{\nu}(k, p) a_k) , \quad (4.1.33)$$

where we used the shorthand notation $(a^*, J a) = \sum_{\rho, \rho'} a_{\rho}^* J_{\rho \rho'} a_{\rho'}$. An explicit computation gives:

$$\hat{J}_0(k, p) = 1 , \quad \hat{J}_1(k, p) = i \frac{\hat{H}(k) - \hat{H}(k-p)}{1 - e^{-ip}} . \quad (4.1.34)$$

Remark 4.1.4. If H has range 1, then $j_{1,x}$ takes a particularly simple form:

$$j_{1,x} \equiv j_x = \sum_{\rho, \rho' \in S_M} j_{(x,\rho), (x+1,\rho')} \quad (4.1.35)$$

Finally, we define a smeared version of the density and of the current operators as:

$$j_{\nu}(\mu_{\theta}) = \sum_{x \in \Gamma_L} j_{\nu,x} \theta_{\mu}(\theta x) . \quad (4.1.36)$$

4.1.3 Main result

For $x \in \Gamma_L$, let us define the full response of the density and of the current operators as:

$$\chi_\nu^{\beta,L}(x; \eta, \theta) := \frac{1}{\theta} \left(\text{Tr } j_{\nu,x} \rho(0) - \text{Tr } j_{\nu,x} \rho_{\beta,L} \right) \quad \nu = 0, 1. \quad (4.1.37)$$

The quantity $\chi_0^{\beta,L}$ is the susceptibility, while the quantity $\chi_1^{\beta,L}$ is the conductance. The next theorem determines the structure of these response functions, as $\beta, L \rightarrow \infty$, and for θ, η, λ small enough. In the following, we shall denote by η_β the best approximation of η in $\frac{2\pi}{\beta}\mathbb{N}$, such that $\eta_\beta \geq \eta$.

Theorem 4.1.5 (Main result). *Let $\mathcal{H}$ be of the form (4.1.17), and assume that H satisfies Assumption 4.1.1. Let $\mathcal{H}(\eta t)$ be as in (4.1.20)-(4.1.22). Let $a(\eta) := \theta/\eta = O(1)$ as $\eta \rightarrow 0$.*

Validity of linear response. *There exist constants $\gamma, \eta_0, \bar{\lambda} > 0$ such that, for all $0 \leq \eta < \eta_0$ and for all $|\lambda| < \bar{\lambda}$, linear response is valid for β, L large enough, i.e.*

$$\chi_\nu^{\beta,L}(x; \eta, \theta) = \chi_\nu^{\beta,L,\text{lin}}(x; \eta, \theta) + O(\eta^\gamma) + O\left(\frac{1}{\eta^3\beta}\right),$$

with $\chi_\nu^{\beta,L,\text{lin}}$ is the first order term in the Duhamel expansion (4.1.27), after replacing η with η_β .

Structure of linear response. *The linear response can be written as follows:*

$$\begin{aligned} \chi_\nu^{\beta,L,\text{lin}}(x; \eta, \theta) &= \chi_\nu^{\text{lin}}(x; \eta, \theta) + O(\beta^{-1}) + O(L^{-1}) \\ \chi_\nu^{\text{lin}}(x; \theta, \eta) &= - \sum_{\omega, \omega'=1}^{N_f} \int_{\mathbb{R}} \hat{\mu}_\infty(q) e^{i\theta qx} \mathcal{K}_{\omega\omega'}^\nu(q; \lambda) \frac{dq}{2\pi} + O(\eta^{\tilde{\gamma}}), \end{aligned} \quad (4.1.38)$$

for some $\tilde{\gamma} > 0$. The matrix-valued response function $\mathcal{K}^\nu$ is given as follows. There exist an analytic function $v : (-\bar{\lambda}, \bar{\lambda}) \rightarrow \text{GL}(N_f)$, called the matrix of renormalised Fermi velocities, such that

$$v_{\omega\omega'}(\lambda) = \delta_{\omega\omega'} v_\omega(\lambda), \quad v_\omega(\lambda) = v_\omega + O(\lambda),$$

and an analytic function $\Lambda : (-\bar{\lambda}, \bar{\lambda}) \rightarrow \text{GL}(N_f)$ with

$$\Lambda^T(\lambda) = \Lambda(\lambda), \quad \Lambda_{\omega\omega'}(\lambda) = \begin{cases} 0 & \omega = \omega' \\ \lambda + O(\lambda^2) & \omega \neq \omega' \end{cases},$$

such that the matrix $\mathcal{K}^\nu$ is given by

$$\mathcal{K}^\nu(q; \lambda) = \tilde{\mathcal{K}}(q; \lambda) \frac{v^\nu(\lambda)}{2\pi|v(\lambda)|} \frac{v(\lambda)q}{-i/a + v(\lambda)q} \mathfrak{K}^\nu(\lambda),$$

with $v^\nu := \delta_{\nu,0}\mathbb{1} + \delta_{\nu,1}v$, and $\tilde{\mathcal{K}}$ and $\mathfrak{K}^\nu$ defined as

$$\begin{aligned} \tilde{\mathcal{K}}(q) &= \left(1 - \frac{1}{4\pi|v|}\Lambda\right) \left(1 + \frac{1}{4\pi|v|} \frac{i/a + vq}{-i/a + vq} \Lambda\right)^{-1}, \\ \mathfrak{K}^0 &\equiv \mathbb{1}_{N_f}, \quad \mathfrak{K}^1 := \text{sgn } v \frac{1 + (4\pi|v|)^{-1}\Lambda}{1 - (4\pi|v|)^{-1}\Lambda} \text{sgn } v. \end{aligned}$$

Remark 4.1.6. (i) The above theorem justifies the validity of linear response for interacting one-dimensional systems in the absence of a spectral gap. At zero temperature and as $\eta \rightarrow 0^+$ the only contribution surviving to the response of the system is the one associated to Kubo formula, which can be explicitly evaluated.

- (ii) A technical limitation of the proof is that the range of allowed λ is not uniform in $a = \theta/\eta$. In particular, our current estimates imply that $\bar{\lambda}$ shrinks to zero as $a \rightarrow \infty$.
- (iii) Suppose that $a \rightarrow 0$. Then, the above result shows that the response of the system at zero temperature is trivial as $\eta \rightarrow 0^+$, since:

$$\chi_\nu^{\text{lin}}(x, \eta, \theta) = O(a) . \quad (4.1.39)$$

- (iii) Considering non-interacting systems, *i.e.* for $\lambda = 0$, we notice that both $\tilde{\mathcal{K}}$ and $\mathfrak{K}^\nu$ become the identity matrix, and the Fermi velocities reduce to the free ones, so that

$$\mathcal{K}^\nu(q; 0) = \frac{v^\nu(0)}{2\pi|v(0)|} \frac{v(0)q}{-i/a + v(0)q} ,$$

which is in agreement with the result obtained in Theorem 3.2.1 of the previous chapter. Thus, the many-body interaction modifies the response in two concurrent ways: by dressing the velocities of the infrared modes labelled by the Fermi points, and by mixing the contribution of each mode via $\tilde{\mathcal{K}}$ and $\mathfrak{K}^\nu$.

- (iv) As mentioned above, for non-interacting systems a similar result was obtained in Chapter 3, which in turn is based on [69]. The analysis of [69] was based on a careful analysis of the real-time Duhamel expansion after Wick rotation. Here, such direct analysis is not possible, due to the fact that the state is not quasi-free. We will instead resort to rigorous renormalisation group methods, which will allow to prove essentially optimal estimates on the imaginary-time correlators.
- (v) At the price of a straightforward approximation argument, carried out in the non-interacting setting in Chapter 3, the result could be extended to the case in which $\hat{\mu}_\infty(p)$ decays fast enough, rather than assuming $\hat{\mu}_\infty(p)$ to be compactly supported in p . We prefer to avoid this extra technical step, which we believe would not add anything substantial to the main result of this chapter.

Remark 4.1.7 (The case of two chiralities). In the case of two opposite Fermi points with counterpropagating velocities the expression in (4.1.38) drastically simplifies. This case is relevant for instance when H is the nearest-neighbour lattice Laplacian. We will show how the general expression (4.1.38) reproduces known rigorous results, see *e.g.* [13, 16]. For simplicity, we shall evaluate it for $a = 1$, that is $\theta = \eta$; the expression for $a \neq 1$ will be simply given by substituting $q \rightarrow aq$ in the response for $a = 1$.

The dispersion relation is $\varepsilon(k) = 2(1 - \cos k)$, and for any choice of $\mu \in (0, 4)$ we have $N_f = 2$ distinct free Fermi points $k_F^+ = -k_F^-$ with opposite free Fermi velocities $v_\pm = \pm v_0$, with $v_0 = 2 \sin k_F^+ > 0$. Since spatial $\mathbb{Z}_2$ -parity is a symmetry of the model, in this case the renormalised parameters needed are just two, namely $v_\star(\lambda) = v_0 + O(\lambda)$, $\lambda_\star(\lambda) = \lambda + O(\lambda^2)$, so that the matrices v and Λ introduced in Theorem 4.1.5 become

$$v = v_\star \sigma_3 \quad \Lambda = \lambda_\star \sigma_1$$

with σ_1, σ_3 the first and the third Pauli matrices. For simplicity, let us set

$$b := \frac{\lambda_\star}{4\pi v_\star} \quad \mathfrak{b}_\pm(q) := \frac{i \pm v_\star q}{-i \pm v_\star q} \quad B(q) := b \begin{pmatrix} 0 & \mathfrak{b}_+(q) \\ \mathfrak{b}_-(q) & 0 \end{pmatrix}.$$

Then, the response matrices are given by

$$\mathcal{K}^0(q) = \frac{1}{2\pi v_\star} (1 - b\sigma_1) \frac{1}{1 + B(q)} \frac{v_\star q \sigma_3}{-i + v_\star q \sigma_3}, \quad \mathcal{K}^1(q) = v_\star \mathcal{K}^0(q) \frac{1 + b\sigma_1}{1 - b\sigma_1} \sigma_3. \quad (4.1.40)$$

Let us compute $\mathcal{K}^0$ first. By using the properties of σ_3 , and employing the fact that $\det[1 + B(q)] = 1 - b^2 \mathfrak{b}_+(q) \mathfrak{b}_-(q) \equiv 1 - b^2$, we have

$$\frac{1}{1 + B(q)} = \frac{1 - B(q)}{1 - b^2} \quad \frac{v_\star q \sigma_3}{-i + v_\star q \sigma_3} = \frac{v_\star q}{1 + v_\star^2 q^2} (i\sigma_3 + v_\star q),$$

so that

$$\begin{aligned} \frac{1}{1 + B(q)} \frac{v_\star q \sigma_3}{-i + v_\star q \sigma_3} &= \frac{1}{1 - b^2} \frac{v q}{1 + v_\star^2 q^2} \begin{pmatrix} 1 & -b\mathfrak{b}_+(q) \\ -b\mathfrak{b}_-(q) & 1 \end{pmatrix} \begin{pmatrix} i + v_\star q & 0 \\ 0 & -i + v_\star q \end{pmatrix} \\ &= \frac{1}{1 - b^2} \frac{v_\star q}{1 + v_\star^2 q^2} \begin{pmatrix} i + v_\star q & -b(i + v_\star q) \\ b(i - v_\star q) & -i + v_\star q \end{pmatrix}. \end{aligned} \quad (4.1.41)$$

Therefore, inserting (4.1.41) into (4.1.40), the susceptibility matrix takes the form

$$\begin{aligned} \mathcal{K}^0(q) &= \frac{1}{2\pi v_\star} \frac{v_\star q}{1 + v_\star^2 q^2} \frac{1}{1 - b^2} \begin{pmatrix} 1 & -b \\ -b & 1 \end{pmatrix} \begin{pmatrix} i + v_\star q & -b(i + v_\star q) \\ b(i - v_\star q) & -i + v_\star q \end{pmatrix} \\ &= \frac{1}{2\pi v_\star} \frac{v_\star q}{1 + v_\star^2 q^2} \frac{1}{1 - b^2} \begin{pmatrix} (1 - b^2)i + (1 + b^2)vq & -2bv_\star q \\ -2bv_\star q & (b^2 - 1)i + (1 + b^2)v_\star q \end{pmatrix}; \end{aligned}$$

finally, by summing over all the entries of $\mathcal{K}_0$ we obtain the following simple form for the density response function:

$$\sum_{\omega, \omega' = \pm} \mathcal{K}_{\omega\omega'}^0(q) = \frac{1}{\pi v_\star} \frac{4\pi v_\star - \lambda_\star}{4\pi v_\star + \lambda_\star} \frac{v_\star^2 q^2}{1 + v_\star^2 q^2}. \quad (4.1.42)$$

This expression for the susceptibility has been already derived rigorously, from a linear response ansatz, see [13], Theorem 1.1, Eq. (1.15). To compute the current response, we proceed as follows. We write

$$\sum_{\omega, \omega' = \pm} \mathcal{K}_{\omega\omega'}^1(q) \equiv (\vec{1}, \mathcal{K}^1(q) \vec{1}) = v_\star \left(\vec{1}, \mathcal{K}^0(q) \frac{1 + b\sigma_1}{1 - b\sigma_1} \sigma_3 \vec{1} \right) \quad (4.1.43)$$

with $\vec{1}$ the vector with all components equal to 1; now, we have the following identity:

$$\frac{1 + b\sigma_1}{1 - b\sigma_1} \sigma_3 \vec{1} = \sigma_3 \frac{1 - b\sigma_1}{1 + b\sigma_1} \vec{1} = \sigma_3 \frac{(1 - b\sigma_1)^2}{1 - b^2} \vec{1} = \frac{(1 - b)^2}{1 - b^2} \sigma_3 \vec{1},$$

where in the first step we used $\{\sigma_1, \sigma_3\} = 0$, and in the last one we exploited $\sigma_1 \vec{1} = \vec{1}$. This identity allows to write

$$(\vec{1}, \mathcal{K}^1(q) \vec{1}) = v_\star \frac{1 - b}{1 + b} (\vec{1}, \mathcal{K}^0(q) \sigma_3 \vec{1}) \quad (4.1.44)$$

which, up to the prefactor, corresponds to summing the components of the first column of $\mathcal{K}^0$ and subtracting the components of the second one. Therefore:

$$(\vec{1}, \mathcal{K}^1(q) \vec{1}) = \frac{1}{\pi} \frac{4\pi v_\star - \lambda_\star}{4\pi v_\star + \lambda_\star} \frac{iv_\star q}{1 + v_\star^2 q^2} . \quad (4.1.45)$$

Note that, owing to the oddness of the correlator (4.1.45), $\chi_1^{\text{lin}}(x) \equiv 0$ whenever μ_∞ is even, as expected.

4.2 Duhamel expansion and Wick rotation

4.2.1 Auxiliary dynamics

Recall that η_β is the best approximation of η in $\frac{2\pi}{\beta}\mathbb{N}$, such that $\eta_\beta \geq \eta$. We define the auxiliary Hamiltonian by replacing the adiabatic parameter η with η_β :

$$\mathcal{H}_{\beta,\eta}(t) = \mathcal{H} + e^{\eta_\beta t} \theta \sum_{\mathbf{x} \in \Lambda_L} \mu(\theta x) a_{\mathbf{x}}^* a_{\mathbf{x}} . \quad (4.2.1)$$

Let us denote by $\tilde{\mathcal{U}}(t; s)$ the two-parameter unitary group generated by $\mathcal{H}_{\beta,\eta}(t)$,

$$i\partial_t \tilde{\mathcal{U}}(t; s) = \mathcal{H}_{\beta,\eta}(t) \tilde{\mathcal{U}}(t; s) , \quad \tilde{\mathcal{U}}(s; s) = \mathbb{1} . \quad (4.2.2)$$

The next result allows to control the error introduced by replacing the dynamics generated by $\mathcal{H}(\eta t)$ with the dynamics generated by $\mathcal{H}_{\beta,\eta}(t)$.

Proposition 4.2.1 (Approximation by the auxiliary dynamics). *Under the same assumptions of Theorem 4.1.5, it follows that:*

$$\left\| \tilde{\mathcal{U}}(t; -\infty)^* \mathcal{O}_X \tilde{\mathcal{U}}(t; -\infty) - \mathcal{U}(t; -\infty)^* \mathcal{O}_X \mathcal{U}(t; -\infty) \right\| \leq C \frac{\theta}{\eta^3 \beta} \quad (4.2.3)$$

for some $C > 0$ independent of θ, η, β .

Proof of Proposition 4.2.1. The proof is given in Proposition 2.2.9. Compare also to Proposition 3.3.1. $\square$

4.2.2 Duhamel expansion

Setting $\tilde{\rho}(t) = \tilde{\mathcal{U}}(t; -\infty) \rho_{\beta,\mu,L} \tilde{\mathcal{U}}(t; -\infty)^*$, by Proposition 4.2.1 we may write

$$\text{Tr } j_{\nu,x} \rho(t) = \text{Tr } j_{\nu,x} \tilde{\rho}(t) + \mathcal{E}_{\nu}^{\beta,L}(x, t; \eta, \theta) , \quad (4.2.4)$$

with the error term $\mathcal{E}_{\nu}^{\beta,L}$ satisfying the bound

$$|\mathcal{E}_{\nu}^{\beta,L}(x, t; \eta, \theta)| \leq C \frac{\theta}{\eta^3 \beta} . \quad (4.2.5)$$

Consider the main term in the right-hand side of (4.2.4). By Proposition 2.2.7:

$$\begin{aligned} & \text{Tr } j_{\nu,x} \tilde{\rho}(t) - \text{Tr } j_{\nu,x} \rho_{\beta,\mu,L} \\ &= \sum_{n \geq 1} \frac{(-1)^n e^{n\eta_\beta t}}{n!} \int_{[0,\beta)^n} d\mathbf{s} e^{-i\eta_\beta(s_1 + \dots + s_n)} \langle \mathbf{T} \gamma_{s_1}(\mathcal{P}); \dots ; \gamma_{s_n}(\mathcal{P}); j_{\nu,x} \rangle_{\beta,L} . \end{aligned} \quad (4.2.6)$$

It is convenient to rewrite the n -th order contribution to the expansion in Fourier space. Recalling the definition (4.1.36) and the form of the perturbation (4.1.21),

$$\begin{aligned} & \int_{[0,\beta]^n} d\underline{s} e^{-i\eta_\beta(s_1+\dots+s_n)} \langle \mathbf{T}\gamma_{s_1}(\mathcal{P}); \dots; \gamma_{s_n}(\mathcal{P}); j_{\nu,x} \rangle_{\beta,L} \\ &= \int_{[0,\beta]^n} d\underline{s} e^{-i\eta_\beta(s_1+\dots+s_n)} \langle \mathbf{T}\gamma_{s_1}(j_0(\mu_\theta)); \dots; \gamma_{s_n}(j_0(\mu_\theta)); j_{\nu,x} \rangle_{\beta,L}, \end{aligned} \quad (4.2.7)$$

using that

$$j_0(\mu_\theta) = \frac{\theta}{L} \sum_{p \in B_L} \hat{\mu}_\theta(-p) \hat{j}_{\nu,p}, \quad \hat{j}_{\nu,p} = \sum_{\mathbf{x} \in \Lambda_L} e^{-ipx} \hat{j}_{\nu,\mathbf{x}}, \quad (4.2.8)$$

with $\hat{\mu}_\theta(p) := \hat{\mu}(p/\theta)/\theta$, and writing, for $\underline{p} = (p_0, p_1)$ with $p_1 = p$ and $p_0 = \eta_\beta$,

$$\hat{n}_{\underline{p}} := \int_0^\beta ds e^{-i\eta_\beta s} \gamma_s(\hat{n}_p), \quad (4.2.9)$$

we can rewrite the right-hand side of (4.2.7) as:

$$\begin{aligned} & \int_{[0,\beta]^n} d\underline{s} e^{-i\eta_\beta(s_1+\dots+s_n)} \langle \mathbf{T}\gamma_{s_1}(j_0(\mu_\theta)); \dots; \gamma_{s_n}(j_0(\mu_\theta)); j_{\nu,x} \rangle_{\beta,L} \\ &= \frac{\theta^n}{L^n} \sum_{\{p_i\} \in B_L^n} \left[\prod_{i=1}^n \hat{\mu}_\theta(-p_i) \right] e^{ip_{n+1}x} \frac{1}{\beta L} \langle \mathbf{T} \hat{n}_{\underline{p}_1}; \dots; \hat{n}_{\underline{p}_n}; \hat{j}_{\nu,\underline{p}_{n+1}} \rangle_{\beta,L}, \end{aligned} \quad (4.2.10)$$

where $\underline{p}_{n+1} = -\underline{p}_1 - \dots - \underline{p}_n$. In order to derive (4.2.10), we used the space-time translation invariance of the Gibbs state.

Thus, we are led to the following representation for the Duhamel expansion of the response functions: recalling (4.2.6)-(4.2.10), we have

$$\begin{aligned} \chi_\nu^{\beta,L}(x; \eta, \theta) &= - \sum_{m=2}^{\infty} \frac{(-\theta)^{m-2}}{(m-1)!} \frac{1}{L^{m-1}} \sum_{\{p_i\} \in B_L^{m-1}} \left[\prod_{j=1}^{m-1} \hat{\mu}_\theta(-p_j) \right] e^{ip_m x} \\ &\quad \cdot \frac{1}{\beta L} \langle \mathbf{T} \hat{n}_{\underline{p}_1}; \dots; \hat{n}_{\underline{p}_{m-1}}; \hat{j}_{\nu,\underline{p}_m} \rangle_{\beta,L} + E_\nu^{\beta,L}(x; \eta, \theta), \end{aligned} \quad (4.2.11)$$

where: $\underline{p}_i := (\eta_\beta, p_i)$ and $\underline{p}_m := -\sum_{i=1}^{m-1} \underline{p}_i$; the error term $E_\nu^{\beta,L}$ satisfies, by (4.2.5):

$$|E_\nu^{\beta,L}(x; \eta, \theta)| = \frac{|\mathcal{E}_\nu^{\beta,L}(x, 0; \eta, \theta)|}{\theta} \leq \frac{C}{\eta^3 \beta}. \quad (4.2.12)$$

Starting from identity (4.2.11), the remaining part of the proof (and of the chapter) will be devoted to the proof of suitable bounds for the Euclidean density-current correlations appearing in the higher-order ($m \geq 3$) terms, and to the explicit evaluation of the linear ($m = 2$) term.

Remark 4.2.2. Observe that we have written (4.2.11) in terms of the number of points of the correlation functions (m), instead of the Duhamel order ($n = m - 1$).

4.3 RG analysis of the lattice model

4.3.1 Grassmann integral representation of the model

In this section, we will introduce a Grassmann integral formulation that will allow us to rewrite the density Euclidean m -point functions as derivatives of a suitable generating functional, which can be analysed via multiscale methods.

Let $\chi : \mathbb{R} \rightarrow [0, 1]$ be a smooth, even cutoff function such that:

$$\chi(t) = \begin{cases} 1 & |t| \leq 1 \\ 0 & |t| \geq \frac{3}{2} \end{cases} \quad (4.3.1)$$

and, for $N_0 \in \mathbb{N}$ set $\chi_{N_0}(\cdot) = \chi(2^{-N_0}\cdot)$. We define the set of space-time momenta $\mathbb{D}_{\beta,L,N_0}$ as:

$$\mathbb{D}_{\beta,L,N_0} = \left\{ \underline{k} = (k_0, \underline{k}) \in \frac{2\pi}{\beta} \left(\mathbb{Z} + \frac{1}{2} \right) \times B_L \mid \chi_{N_0}(k_0) > 0 \right\}. \quad (4.3.2)$$

Observe that the set $\mathbb{D}_{\beta,L,N_0}$ is finite. Then, consider the Grassmann $\mathbb{C}$ -algebra generated by the set of anti-commuting symbols $\{\hat{\Psi}_{\underline{k},\rho}^{\pm} \mid \underline{k} \in \mathbb{D}_{\beta,L,N_0}, \rho \in S_M\}$. On such Grassmann algebra we define a Gaussian Grassmann measure $P_{N_0}[d\Psi]$ as follows. First, we set the Berezin integral $d\Psi$ to be the linear functional which is non-zero only on monomials of maximal degree, and such that its evaluation on such top degree monomials is:

$$\int d\Psi \prod_{\underline{k} \in \mathbb{D}_{\beta,L,N_0}} \prod_{\rho \in S_M} \hat{\Psi}_{\underline{k},\rho}^- \hat{\Psi}_{\underline{k},\rho}^+ = 1. \quad (4.3.3)$$

Let us define the free propagator as:

$$\hat{g}_{\rho\rho'}^{(\leq N_0)}(\underline{k}) = \left(\frac{\chi_{N_0}(k_0)}{ik_0 + \hat{H}(\underline{k}) - \mu} \right)_{\rho\rho'}. \quad (4.3.4)$$

We define the associated Grassmann Gaussian integration $P_{N_0}[d\Psi]$ as:

$$\int P_{N_0}[d\Psi](\cdot) = \frac{1}{\mathcal{Z}_{\beta,L,N_0}^0} \int d\Psi \exp \left\{ -\frac{1}{\beta L} \sum_{\underline{k} \in \mathbb{D}_{\beta,L,N_0}} (\hat{\Psi}_{\underline{k}}^+, \hat{g}^{(\leq N_0)}(\underline{k})^{-1} \hat{\Psi}_{\underline{k}}^-) \right\} (\cdot) \quad (4.3.5)$$

with the short-hand $(\phi, A\psi) = \sum_{\rho,\rho'} \phi_{\rho} A_{\rho\rho'} \psi_{\rho'}$ and normalisation

$$\mathcal{Z}_{\beta,L,N_0}^0 = \prod_{\underline{k} \in \mathbb{D}_{\beta,L,N_0}} [\beta L] \det[\hat{g}^{(\leq N_0)}(\underline{k})]. \quad (4.3.6)$$

Given this definition for $P_{N_0}[d\Psi]$, it is easy to check that, for a given monomial $\prod_{i=1}^n \hat{\Psi}_{\underline{k}_i,\rho_i}^{\epsilon_i}$ its action yields zero if $|\{i : \epsilon_i = +\}| \neq |\{i : \epsilon_i = -\}|$; otherwise,

$$\int P_{N_0}[d\Psi] \prod_{i=1}^m \hat{\Psi}_{\underline{k}_i,\rho_i}^- \hat{\Psi}_{\underline{p}_i,\rho'_i}^+ = \det[K(\underline{k}_i, \rho_i; \underline{p}_j, \rho'_j)]_{i,j=1}^m \quad (4.3.7)$$

with $K(\underline{k}, \rho; \underline{p}, \rho') = \beta L \delta_{\underline{k},\underline{p}} \hat{g}_{\rho\rho'}^{(\leq N_0)}(\underline{k})$. Eq. (4.3.7) is the Wick's rule, for Grassmann Gaussian fields.

Next, let us introduce the real-space fields as:

$$\Psi_{\underline{x},\rho}^{\pm} = \frac{1}{\beta L} \sum_{\underline{k} \in \mathbb{D}_{\beta,L,N_0}} e^{\mp i \underline{k} \cdot (\underline{x} - \underline{y})} \hat{\Psi}_{\underline{k},\rho}^{\pm} . \quad (4.3.8)$$

One can check that:

$$\int P_{N_0}[d\Psi] \Psi_{\underline{x},\rho}^{-} \Psi_{\underline{y},\rho'}^{+} = g_{\rho\rho'}^{(\leq N_0)}(\underline{x}; \underline{y}), \quad (4.3.9)$$

where the real-space propagator is:

$$g_{\rho\rho'}^{(\leq N_0)}(\underline{x}; \underline{y}) = \sum_{\underline{k} \in \mathbb{D}_{\beta,L,N_0}} e^{i \underline{k} \cdot (\underline{x} - \underline{y})} \hat{g}_{\rho\rho'}^{(\leq N_0)}(\underline{k}). \quad (4.3.10)$$

It is well-known that, as $\underline{x} \neq \underline{y}$, the $N_0 \rightarrow \infty$ limit of the real-space two-point function converges to the Euclidean two-point function of the non-interacting model,

$$\lim_{N_0 \rightarrow \infty} g_{\rho\rho'}^{(\leq N_0)}(\underline{x}; \underline{y}) = \langle \mathbf{T} \gamma_{x_0}(a_{x,\rho}) \gamma_{y_0}(a_{y,\rho'}^*) \rangle_{\beta,L}^0 \equiv g_{\rho\rho'}(\underline{x}; \underline{y}) . \quad (4.3.11)$$

Let us now define the interacting Grassmann field theory. The Grassmann interaction is:

$$V(\Psi) := \lambda \int_{\mathbb{T}_{\beta}^2} dx_0 dy_0 \sum_{\mathbf{x}, \mathbf{y} \in \Lambda_L} n_{\mathbf{x}} \delta(x_0 - y_0) w(x, y) n_{\mathbf{y}} \quad (4.3.12)$$

with $\mathbf{x} = (\underline{x}, \rho)$ and with $n_{\mathbf{x}} = \Psi_{\mathbf{x}}^{+} \Psi_{\mathbf{x}}^{-}$. It should be noted that the factors $-1/2$ are absent in this Grassmann representation; this is related to the fact that:

$$\lim_{N_0 \rightarrow \infty} g_{\rho\rho'}^{(\leq N_0)}((x_0, x); (x_0, y)) = g_{\rho\rho'}((x_0, x); (x_0, y)) - \frac{\delta_{\mathbf{x}, \mathbf{y}}}{2} . \quad (4.3.13)$$

We can rewrite expression (4.3.12) in momentum space as:

$$V(\Psi) := \frac{1}{\beta L} \sum_{\underline{p} \in \mathbb{M}_{\beta,L,N_0}} \lambda \hat{w}(p) \hat{n}_{-\underline{p}} \hat{n}_{\underline{p}} , \quad (4.3.14)$$

where $\mathbb{M}_{\beta,L,N_0} := \{\underline{p} = (p_0, p) \in \frac{2\pi}{\beta} \mathbb{Z} \times B_L \mid \chi_{N_0}(p_0) > 0\}$ and:

$$\hat{n}_{\underline{p}} = \int_0^{\beta} dx_0 \sum_{\mathbf{x} \in \Lambda_L} e^{-i \underline{p} \cdot \underline{x}} n_{\mathbf{x}} = \frac{1}{\beta L} \sum_{\underline{k} \in \mathbb{D}_{\beta,L,N_0}} (\hat{\Psi}_{\underline{k}-\underline{p}}^{+} \hat{J}_{\nu}(k, p) \hat{\Psi}_{\underline{k}}^{-}) . \quad (4.3.15)$$

At last, we introduce the partition function of this Grassmann model as:

$$\mathcal{Z}_{\beta,L,N_0} := \int P_{N_0}[d\Psi] e^{-V(\Psi)} . \quad (4.3.16)$$

We are now ready to define the generating functional of the Grassmann field theory. Let us start by introducing the source terms. Using the short-hand notation:

$$\int_{\beta,L} d\underline{x} := \int_{\mathbb{T}_{\beta}} dx_0 \sum_{x \in \Gamma_L} , \quad (4.3.17)$$

and considering two Grassmann fields $\phi_{\underline{x},\rho}^{\pm}$ and a real valued vector field $A_{\underline{x}}^{\nu}$, we set

$$B(\Psi; \phi) := \int_{\beta,L} d\underline{x} (\phi_{\underline{x}}^{+}, \Psi_{\underline{x}}^{-}) + (\Psi_{\underline{x}}^{+}, \phi_{\underline{x}}^{-}) , \quad C(\Psi; A) := \sum_{\nu=0,1} \int_{\beta,L} d\underline{x} A_{\underline{x}}^{\nu} j_{\nu,\underline{x}} , \quad (4.3.18)$$

with $j_{\nu,\underline{x}}$ the Grassmann counterpart of the lattice 2-current defined in (4.1.33):

$$j_{\nu,\underline{x}} = \frac{1}{\beta L} \sum_{\underline{p} \in \mathbb{M}_{\beta,L,N}} e^{i\underline{p} \cdot \underline{x}} \hat{j}_{\nu,\underline{p}} \quad \hat{j}_{\nu,\underline{p}} = \frac{1}{\beta L} \sum_{\underline{k} \in \mathbb{D}_{\beta,L,N}} (\hat{\Psi}_{\underline{k}-\underline{p}}^+, \hat{j}_{\nu}(k, p) \hat{\Psi}_{\underline{k}}^-). \quad (4.3.19)$$

In order to make sure that the field $\phi^\pm$ also generates a finite dimensional algebra, we will suppose that it is supported for finitely many Fourier modes.

The generating functional of the correlation functions $\mathcal{W}_{\beta,L,N_0}(\phi, A)$ is then given by:

$$\mathcal{W}_{\beta,L,N_0}(\phi, A) = \log \int P_{N_0}[d\Psi] e^{-V(\Psi) + B(\Psi; \phi) + C(\Psi; A)}. \quad (4.3.20)$$

A corollary of the construction that we are going to describe is that, for $\|A\|_\infty, |\lambda|$ small enough, the argument of the logarithm is non-zero, and hence the generating functional is well defined at finite β, L, N_0 .

It is well-known that correlation functions of the Gibbs state of the model in Fock space at non-coinciding space-time points can be represented as functional derivatives of the generating functional (4.3.20), in the limit $N_0 \rightarrow \infty$; see *e.g.* Section 2.4.3 or [42] for a discussion on this point. In particular, we are interested in the following type of momentum-space correlation functions:

$$\langle \mathbf{T} \hat{n}_{\underline{p}_1}; \dots; \hat{n}_{\underline{p}_{m-1}}; \hat{j}_{\nu,\underline{p}_m} \rangle_{\beta,L} = \lim_{N_0 \rightarrow \infty} (\beta L)^m \frac{\partial^m \mathcal{W}_{\beta,L,N_0}(\phi, A)}{\partial \hat{A}_{-\underline{p}_1}^0 \dots \partial \hat{A}_{-\underline{p}_{m-1}}^0 \partial \hat{A}_{-\underline{p}_m}^\nu} \Big|_{A=\phi=0}. \quad (4.3.21)$$

The identity (4.3.21) will allow us to use the properties of Grassmann Gaussian fields in order to investigate the Euclidean correlation functions of our model. Equation (4.3.21) holds in the analyticity domain of the Grassmann theory, which is contained in the analyticity domain of the Fock space model.

4.3.2 Integration of the ultraviolet degrees of freedom

As in all applications of renormalisation group to condensed matter problems, the starting point is the integration of the ultraviolet degrees of freedom. After this step, which will be briefly described in this Section (referring to other works for the technical details), the problem will be reduced to the analysis of a suitable, effective infrared model.

Let $\chi(\cdot)$ be a smooth cutoff function as in (4.3.1). We decompose $\hat{g}^{(\leq N_0)}$ into the sum of an ultraviolet (UV) and infrared (IR) part as:

$$\hat{g}^{(\leq N_0)}(\underline{k}) = \hat{g}^{\text{UV}}(\underline{k}) + \hat{g}^{\text{IR}}(\underline{k}), \quad \hat{g}^{\text{UV}}(\underline{k}) = \frac{\chi_{[0,N_0]}(k_0)}{ik_0 + \hat{H}(\underline{k}) - \mu} \quad (4.3.22)$$

where $\chi_{[0,N_0]}(k_0) = \chi_{N_0}(k_0) - \chi(k_0)$. By the addition principle of Grassmann Gaussian variables, we split the Grassmann field as:

$$\hat{\Psi}_{\underline{k}}^\pm = \hat{\Psi}_{\underline{k}}^{\text{UV},\pm} + \hat{\Psi}_{\underline{k}}^{\text{IR}}, \quad (4.3.23)$$

where $\hat{\Psi}_{\underline{k}}^{\text{UV},\pm}, \hat{\Psi}_{\underline{k}}^{\text{IR}}$ are independent Grassmann fields, with propagators given by, respectively, $\hat{g}^{\text{UV}}(\underline{k})$ and $\hat{g}^{\text{IR}}(\underline{k})$. Thanks to the fact that the cutoff function $\chi_{[0,N_0]}(k_0)$ is vanishing for $|k_0| \leq 1$, the propagator $\hat{g}^{\text{UV}}(\underline{k})$ is non-singular, uniformly in β, L . Let:

$$V(\Psi; A, \phi) = -V(\Psi) + B(\Psi; \phi) + C(\Psi; A), \quad (4.3.24)$$

recall (4.3.20). We then rewrite the generating functional of correlations as:

$$e^{\mathcal{W}_{\beta,L,N_0}(\phi,A)} = \int P_{\text{IR}}[d\Psi^{\text{IR}}] \int P_{\text{UV}}[d\Psi^{\text{UV}}] e^{V(\Psi^{\text{IR}}+\Psi^{\text{UV}};A,\phi)}, \quad (4.3.25)$$

and we proceed with the integration of the ultraviolet field. A standard multiscale analysis for the ultraviolet degrees of freedom yields, see *e.g.* [42]:

$$\int P_{\text{UV}}[d\Psi^{\text{UV}}] e^{V(\Psi^{\text{IR}}+\Psi^{\text{UV}};A,\phi)} = e^{\mathcal{W}_{\beta,L,N_0}^{\text{IR}}(\phi,A)+V^{\text{IR}}(\Psi^{\text{IR}};\phi,A)}, \quad (4.3.26)$$

where the argument of the exponential have the form:

$$V^{\text{IR}}(\Psi^{\text{IR}};\phi,A) = \sum_{\Gamma=(\Gamma_\Psi,\Gamma_\phi,\Gamma_A)} \int_{\beta,L} d\underline{X} d\underline{Y} d\underline{Z} \Psi_{\Gamma_\Psi}^{\text{IR}}(\underline{X}) \phi_{\Gamma_\phi}(\underline{Y}) A_{\Gamma_A}(\underline{Z}) W_{\Gamma}^{\text{IR}}(\underline{X}, \underline{Y}, \underline{Z}), \quad (4.3.27)$$

and:

$$\mathcal{W}_{\beta,L,N_0}^{\text{IR}}(\phi,A) = \sum_{\Gamma=(\Gamma_\phi,\Gamma_A)} \int_{\beta,L} d\underline{Y} d\underline{Z} \phi_{\Gamma_\phi}(\underline{Y}) A_{\Gamma_A}(\underline{Z}) W_{\Gamma}^{\text{IR}}(\underline{Y}, \underline{Z}); \quad (4.3.28)$$

let us explain the meaning of the various objects entering these expressions.

1. The labels $\Gamma_\Psi, \Gamma_\phi, \Gamma_A$ denote the collection of all labels of the fields, integrated and external. That is, if n is the order of the Ψ -monomial, $\Gamma_\Psi = \{(\rho_1, \varepsilon_1), \dots, (\rho_n, \varepsilon_n)\}$, and:

$$\Psi_{\Gamma_\Psi}(\underline{X}) = \prod_{i=1}^n \Psi_{\underline{x}_i, \rho_i}^{\varepsilon_i} \quad (4.3.29)$$

with the understanding that $\underline{X} = (x_1, \dots, x_n)$. If r is the order of the ϕ -monomial, $\Gamma_\phi = \{(\rho_1, \varepsilon_1), \dots, (\rho_r, \varepsilon_r)\}$, and:

$$\Phi_{\Gamma_\phi}(\underline{Y}) = \prod_{i=1}^r \phi_{\underline{y}_i, \rho_i}^{\varepsilon_i}, \quad (4.3.30)$$

with $\underline{Y} = (y_1, \dots, y_r)$. If m is the order of the A -monomial, $\Gamma_A = (\nu_1, \dots, \nu_m)$, and:

$$A_{\Gamma_A}(\underline{Z}) = \prod_{i=1}^m A_{\underline{z}_i}^{\nu_i}, \quad (4.3.31)$$

with $\underline{Z} = (z_1, \dots, z_m)$.

2. The integral runs over all the space-time labels of the monomials determined by Γ .
3. The functions $W_{\Gamma}^{\text{IR}}(\underline{X}, \underline{Y}, \underline{Z})$, with the understanding that $W_{(\emptyset, \Gamma_\phi, \Gamma_A)}^{\text{IR}}(\underline{X}, \underline{Y}, \underline{Z}) \equiv W_{(\Gamma_\phi, \Gamma_A)}^{\text{IR}}(\underline{Y}, \underline{Z})$, also called kernels, are the outcome of the integration of the ultraviolet field. They are analytic in λ for $|\lambda|$ small enough, they are invariant under simultaneous equal translation of all space-time coordinates, and they satisfy the following bound:

$$\int_{\beta,L} D\underline{Q} \prod_{i < j} \| \underline{q}_i - \underline{q}_j \|_{\beta,L}^{n_{ij}} |W_{\Gamma}^{\text{IR}}(\underline{Q})| \leq \beta L C_{\Gamma; \{n_{ij}\}} \quad \forall n_{ij} \in \mathbb{N}, \quad (4.3.32)$$

with $\underline{Q} = (\underline{X}, \underline{Y}, \underline{Z}) = (q_1, \dots, q_{n+m+r})$, where $\|\cdot\|_{\beta,L}$ denotes the periodic space-time norm:

$$\|\underline{q}\|_{\beta,L} = \inf_{(n_0, n_1) \in \mathbb{Z}^2} \|\underline{q} + (n_0\beta, n_1L)\|. \quad (4.3.33)$$

The kernels $W_{\Gamma}^{\text{IR}}(Q)$ play the role of new effective interactions for the infrared field. The analyticity of the effective interactions is a nontrivial result, and it follows from the application of the Brydges-Battle-Federbush (BBF) formula, for fermionic truncated expectations. This formula provides a convergent representation of fermionic perturbation theory; see *e.g.* [36, 45] for a review. Observe that the expressions (4.3.27), (4.3.28) actually involve finitely many monomials Ψ^{IR} and ϕ , due to the fact that these Grassmann fields involve finitely many Fourier modes, and hence generate a finite-dimensional Grassmann algebra. Instead, the sum over the A monomials can be proved to be convergent, for $\|A\|_{\infty}$ small enough. Thus, both (4.3.27), (4.3.28) are well defined.

We are left with the integration of the field $\Psi^{\text{IR}\pm}$. Its propagator is:

$$\hat{g}^{\text{IR}}(\underline{k}) = \frac{\chi(k_0)}{ik_0 + \hat{H}(k) - \mu}, \quad (4.3.34)$$

which is now singular as $\beta, L \rightarrow \infty$, for $k_0 = 0$ and for k equal to one of the Fermi points k_F^{ω} . Let $k_F^{\omega}(\lambda)$ be points in B_L such that $|k_F^{\omega} - k_F^{\omega}(\lambda)| \leq C|\lambda|$, to be fixed later on; they play the role of renormalised Fermi points. It is convenient to rewrite the infrared propagator as:

$$\hat{g}^{\text{IR}}(\underline{k}) = \hat{g}^{\text{IR},1}(\underline{k}) + \hat{g}^{\text{IR},2}(\underline{k}) \quad (4.3.35)$$

where, for $\underline{k}_F^{\omega}(\lambda) = (0, k_F^{\omega}(\lambda))$:

$$\hat{g}^{\text{IR},1}(\underline{k}) = \sum_{\omega=1}^{N_f} \frac{\chi_0(\|\underline{k} - \underline{k}_F^{\omega}(\lambda)\|_{\omega})}{ik_0 + \hat{H}(k) - \mu}, \quad (4.3.36)$$

with: $\chi_0(\cdot) = \chi(\delta(\cdot))$ for $\delta > 0$ small enough; $\|\underline{q}\|_{\omega}^2 = |q_0|^2 + v_{\omega}^2 |q_1|_{\mathbb{T}}^2$; and $\hat{g}^{\text{IR},2}(\underline{k})$ is a smooth, bounded function of $\underline{k} = (k_0, k_1)$. The number δ is chosen so that the cutoffs $\chi_0(\|\underline{k} - \underline{k}_F^{\omega}(\lambda)\|_{\omega})$ have disjoint supports, and so that $\chi_0(\|\underline{k} - \underline{k}_F^{\omega}(\lambda)\|_{\omega}) \neq 0$ implies $\chi(k_0) = 1$. The advantage of the rewriting (4.3.35) is that the cutoffs entering the massless propagator $\hat{g}^{\text{IR},1}(\underline{k})$ are symmetric in k_0 and in $v_{\omega}[k_1 - k_F^{\omega}(\lambda)]$.

We use the addition principle to write:

$$\hat{\Psi}_{\underline{k}}^{\text{IR},\pm} = \hat{\Psi}_{\underline{k}}^{\text{IR},1,\pm} + \hat{\Psi}_{\underline{k}}^{\text{IR},2,\pm}, \quad (4.3.37)$$

and to integrate the field $\hat{\Psi}_{\underline{k}}^{\text{IR},2,\pm}$ via fermionic cluster expansion, thus producing a new effective action for the field $\hat{\Psi}_{\underline{k}}^{\text{IR},1,\pm}$, determined by kernels $W_{\Gamma}^{\text{IR},1}(\cdot)$ satisfying the same bounds as $W_{\Gamma}^{\text{IR}}(\cdot)$, with different numerical constants.

Let us now focus on the massless propagator $\hat{g}^{\text{IR},1}(\underline{k})$. Thanks to Assumption 4.1.1, we know that the spectrum in a window of size Δ around μ is characterised by the local band functions $\{e_{\omega} : I_{\omega} \rightarrow \mathbb{R}\}_{\omega=1,\dots,N_f}$, so that the IR part may be rewritten to be

$$\hat{g}^{\text{IR},1}(\underline{k}) = \sum_{\omega=1}^{N_f} \hat{g}_{\omega}^{\text{IR}}(\underline{k}) P_{\omega}(k), \quad (4.3.38)$$

with $P_{\omega}(k)$ the rank-one projector onto the eigenvector associated with the eigenvalue $e_{\omega}(k)$, namely

$$P_{\omega}(k) := |\xi^{\omega}(k)\rangle \langle \xi^{\omega}(k)| \quad \hat{H}(k) \xi^{\omega}(k) = e_{\omega}(k) \xi^{\omega}(k), \quad (4.3.39)$$

and where the propagators $\hat{g}_\omega^{\text{IR}}(\underline{k})$ are given by:

$$\hat{g}_\omega^{\text{IR}}(\underline{k}) = \frac{\chi_0(\|\underline{k} - \underline{k}_F^\omega(\lambda)\|_\omega)}{ik_0 + e_\omega(k) - \mu}. \quad (4.3.40)$$

By the addition principle, we can represent the field $\hat{\Psi}_{\underline{k}}^{\text{IR1},\pm}$ as:

$$\hat{\Psi}_{\underline{k},\rho}^{\text{IR1},\pm} = \sum_{\omega=1}^{N_f} \hat{\Psi}_{\underline{k},\omega,\rho}^{\text{IR1},\pm} \quad (4.3.41)$$

where the field $\hat{\Psi}_{\underline{k},\omega,\rho}^{\text{IR1},\pm}$ has propagator given by $\hat{g}_\omega^{\text{IR}}(\underline{k})P_{\omega,\rho,\rho'}(k)$. This Gaussian Grassmann field can be conveniently represented as, for all $\underline{k}$ in the support of $\chi_0(\|\underline{k} - \underline{k}_F^\omega\|_\omega)$:

$$\hat{\Psi}_{\underline{k},\omega,\rho}^{\text{IR1},-} = \psi_{\underline{k},\omega}^{\text{IR},-} \xi_\rho^\omega(k), \quad (4.3.42)$$

where the new Grassmann field $\psi_{\underline{k},\omega}^{\text{IR},\pm}$ has propagator given by $\hat{g}_\omega^{\text{IR}}(\underline{k})$ in (4.3.40). The fields $\{\psi_{\underline{k},\omega}^{\text{IR},\pm}\}$ are quasi-particle fields, and they characterise the low-energy excitations of the system.

Let us now represent the effective action in terms of these new fields. Let:

$$\check{\xi}_\rho^\omega(x) = \frac{1}{L} \sum_{k \in B_L} e^{ikx} \chi(|k - k_F^\omega(\lambda)|_{\mathbb{T}}/2) \hat{\xi}_\rho^\omega(k). \quad (4.3.43)$$

Observe that, from (4.3.41), (4.3.42):

$$\Psi_{\underline{x},\rho}^{\text{IR1},-} = \sum_{\omega} (\check{\xi}_\rho^\omega * \psi_\omega^{\text{IR},-})_{\underline{x}}, \quad \Psi_{\underline{x},\rho}^{\text{IR1},+} = \sum_{\omega} (\overline{\check{\xi}_\rho^\omega} * \psi_\omega^{\text{IR},+})_{\underline{x}} \quad (4.3.44)$$

with $*$ denoting spatial convolution:

$$(\check{\xi}_\rho^\omega * \psi_\omega^{\text{IR},-})_{(x_0,x)} = \sum_{z \in \Gamma_L} \check{\xi}_\rho^\omega(x-z) \psi_{(x_0,z),\omega}^{\text{IR},-}. \quad (4.3.45)$$

Since the field $\psi_{\underline{k},\omega}^{\text{IR},\pm}$ has the same support properties of $\hat{g}_\omega^{\text{IR}}(\underline{k})$, in Eq. (4.3.42) we can freely multiply the Bloch function $\xi_\rho^\omega(k)$ by a cutoff function which is equal to 1 in the support of the field; this yields (4.3.44) with $\check{\xi}_\rho^\omega$ as in (4.3.45). A standard integration by parts argument, combined with the smoothness of the cutoff function and of $\hat{\xi}^\omega(k)$ around the ω -Fermi point, shows that:

$$|\check{\xi}_\rho^\omega(x)| \leq \frac{C_n}{1+|x|_L^n} \quad \text{for all } n \in \mathbb{N}. \quad (4.3.46)$$

Thus, we obtained the following rewriting of the generating functional of the correlations:

$$\begin{aligned} e^{\mathcal{W}_{\beta,L,N_0}(\phi,A)} &= e^{\mathcal{W}_{\beta,L,N_0}^{\text{IR1}}(\phi,A)} \int P_{\text{IR1}}[d\Psi^{\text{IR1}}] e^{V^{\text{IR1}}(\Psi^{\text{IR1}};\phi,A)} \\ &= e^{\mathcal{W}_{\beta,L,N_0}^{\text{IR1}}(\phi,A)} \int P_{\text{IR1}}[d\psi^{\text{IR}}] e^{V^{\text{IR1}}(\psi^{\text{IR}}*\xi;\phi,A)}, \end{aligned} \quad (4.3.47)$$

where in the last step we used the representation (4.3.44) of the field Ψ^{IR1} . Let $\{\nu_\omega\}$ be real numbers, such that $|\nu_\omega| \leq C|\lambda|$, to be determined later on. We further add and

subtract a quadratic term in the effective interaction, up to a redefinition of the Gaussian integration:

$$\begin{aligned}
& e^{\mathcal{W}_{\beta,L,N_0}^{\text{IR1}}(\phi,A)} \int P_{\text{IR}}[d\psi^{\text{IR}}] e^{V^{\text{IR1}}(\psi^{\text{IR}}*\xi;\phi,A)} \\
&= e^{\mathcal{W}_{\beta,L,N_0}^{\text{IR1}}(\phi,A)} \int P_{\text{IR}}[d\psi^{\text{IR}}] e^{\sum_{\omega} \nu_{\omega} \int_{\beta,L} d\underline{x} \psi_{\underline{x},\omega}^{\text{IR},+} \psi_{\underline{x},\omega}^{\text{IR},-}} e^{V^{\text{IR1}}(\psi^{\text{IR}}*\xi;\phi,A) - \sum_{\omega} \nu_{\omega} \int_{\beta,L} d\underline{x} \psi_{\underline{x},\omega}^{\text{IR},+} \psi_{\underline{x},\omega}^{\text{IR},-}} \\
&\equiv e^{\mathcal{W}_{\beta,L,N_0}^{(0)}(\phi,A)} \int P_0[d\psi^{(\leq 0)}] e^{V^{(0)}(\psi^{(\leq 0)};\phi,A)},
\end{aligned} \tag{4.3.48}$$

where, setting $\underline{k}' = \underline{k} - \underline{k}_F^{\omega}(\lambda)$, the new field $\psi_{\underline{k}',\omega}^{(\leq 0)\pm}$ has propagator given by:

$$\hat{g}_{\omega}^{(\leq 0)}(\underline{k}') = \frac{\chi_0(\|\underline{k}'\|_{\omega})}{ik_0 + e_{\omega}(k' + k_F^{\omega}(\lambda)) - \mu + \nu_{\omega} \chi_0(\|\underline{k}'\|_{\omega})}. \tag{4.3.49}$$

The variable $\underline{k}'$ expresses the relative shift of the Euclidean quasi-momentum with respect to the Fermi points. Thus, the configuration-space counterpart of the new field is:

$$e^{\mp i \underline{k}_F^{\omega}(\lambda) \cdot \underline{x}} \psi_{\underline{x},\omega}^{(\leq 0)\pm} \quad \text{with} \quad \psi_{\underline{x},\omega}^{(\leq 0)\pm} = \frac{1}{\beta L} \sum_{\underline{k}'} e^{\mp i \underline{k}' \cdot \underline{x}} \psi_{\underline{k}',\omega}^{(\leq 0)\pm}. \tag{4.3.50}$$

From (4.3.49), observe that the new Gaussian integration takes into account the first exponential factor in (4.3.48). The change of normalisation of the Grassmann integration is taken into account by $\mathcal{W}_{\beta,L,N_0}^{(\leq 0)}(\phi,A)$, while the new effective interaction $V^{(0)}(\psi^{(\leq 0)};\phi,A)$ takes into account all the remaining contributions appearing at the exponent. It has the form:

$$V^{(0)}(\psi^{(\leq 0)};\phi,A) = \sum_{\Gamma=(\Gamma_{\psi},\Gamma_{\phi},\Gamma_A)} \int_{\beta,L} d\underline{X} d\underline{Y} d\underline{Z} \psi_{\Gamma_{\psi}}^{(\leq 0)}(\underline{X}) \phi_{\Gamma_{\phi}}(\underline{Y}) A_{\Gamma_A}(\underline{Z}) W_{\Gamma}^{(0)}(\underline{X}, \underline{Y}, \underline{Z}), \tag{4.3.51}$$

where now $\Gamma_{\psi} = \{(\omega_1, \varepsilon_1), \dots, (\omega_n, \varepsilon_n)\}$.

Thanks to the decay properties of $\check{\xi}$, and to the properties of the kernels of the field Ψ^{IR1} , the new kernels are: analytic in λ for $|\lambda|$ small enough; invariant under translation of all space-time arguments; they satisfy the decay estimate:

$$\int_{\beta,L} D\underline{Q} \prod_{i < j} \|\underline{q}_i - \underline{q}_j\|_{\beta,L}^{n_{ij}} |W_{\Gamma}^{(0)}(\underline{Q})| \leq \beta L \tilde{C}_{\Gamma;\{n_{ij}\}} \quad \forall n_{ij} \in \mathbb{N}, \tag{4.3.52}$$

which is the analogue of (4.3.32), with $C_{\{n_{ij}\}} < \tilde{C}_{\{n_{ij}\}} \leq 2C_{\{n_{ij}\}}$ for $|\lambda|$ small enough. Observe that, if $\Gamma_{\psi} \neq \emptyset$, the bound is proportional to $|\lambda|$; actually, it is $|\lambda|^{|\Gamma_{\psi}|/4}$ if $|\Gamma_{\psi}| \geq 4$. All the bounds are uniform in the ultraviolet cutoff N_0 ; from now on, we will not carry on the dependence on N_0 (the limit $N_0 \rightarrow \infty$ will be eventually taken at the end).

Finally, let us comment on the necessity for “adding and subtracting” the quadratic terms proportional to ν_{ω} . These quadratic terms will play the role of counterterms for the renormalisation group iteration, discussed in the next Sections. In particular, they will be chosen to control the flow of the relevant terms, which are the quadratic terms in the fermionic fields. The shifted Fermi points $\underline{k}_F^{\omega}(\lambda)$ will be chosen in order to take into account the shift of the singularity of (4.3.49), due to the presence of ν_{ω} as the denominator.

More precisely, the pair $(k_F^\omega(\lambda), \nu_\omega)$ will satisfy the condition, as $\beta, L \rightarrow \infty$:

$$e_\omega(k_F^\omega(\lambda)) - \mu + \nu_\omega = 0, \quad (4.3.53)$$

and with $|k_F^\omega(\lambda) - k_F^\omega(0)| \leq C|\lambda|$, $|\nu_\omega| \leq C|\lambda|$. For L finite, equation (4.3.53) may not have a solution, due to the condition $k_F^\omega(\lambda) \in B_L$. To accommodate this, we will instead require that the condition (4.3.53) holds with 0 on the right-hand side replaced by $O(1/L)$.

To conclude, we obtained the following rewriting for the generating functional of the correlations:

$$\mathcal{W}_{\beta,L}(\phi, A) = \mathcal{W}_{\beta,L}^{(0)}(\phi, A) + \log \int P_0[d\psi^{(\leq 0)}] e^{V^{(0)}(\psi^{(\leq 0)}; \phi, A)}, \quad (4.3.54)$$

where $\mathcal{W}^{(0)}$, $V^{(0)}$ have good analyticity and locality properties (kernels that decay faster than any power in Euclidean space). Renormalisation group allows to further integrate the massless field $\psi^{(\leq 0)}$, via a renormalised, convergent expansion. This will be the content of the next section.

4.3.3 Renormalisation group analysis

In this section we discuss the multiscale integration of the massless fields $\psi_\omega^{(\leq 0)}$. This is done via a renormalisation group strategy, that has been carried out in a number of earlier references; see *e.g.* [2, 61] for recent, closely related works, and references therein. In this section we will review the construction. This will be needed to set-up the rest of the chapter, since the method will be used to derive sharp estimates on the multi-point current-current correlators, which will play a key role in the asymptotic validity of linear response.

The starting point is the outcome of the integration of the ultraviolet and high-frequency degrees of freedom, Eq. (4.3.54). Due to the singularity of the covariance of the field at the Fermi points, it turns out that the perturbation theory obtained expanding the theory around the Gaussian case is divergent at all orders, in the limit $\beta, L \rightarrow \infty$. In order to obtain a meaningful expansion, we will perform a multiscale analysis, by decomposing the field into a sum of single scale fields. The single scale fields will be integrated in a progressive way, starting from the high energy scales until the lowest energy scales. The covariance and the effective action at a given scale will be defined inductively, via the combination of suitable *localisation* and *renormalisation* operations. This will ultimately allow to exploit *cancellations* in the naive expansion in a systematic way, and to prove analyticity of the correlation functions for $|\lambda| < \lambda_0$ for some λ_0 independent of β, L .

Remark 4.3.1. In the rest of the chapter, we will use the following short-hand notations:

$$\int_{\beta,L} d\underline{k}(\cdots) := \frac{1}{\beta L} \sum_{\underline{k} \in \mathbb{D}_{\beta,L}} (\cdots), \quad \int_{\beta,L} d\underline{p}(\cdots) := \frac{1}{\beta L} \sum_{\underline{p} \in \mathbb{M}_{\beta,L}} (\cdots) \quad (4.3.55)$$

with

$$\mathbb{D}_{\beta,L} = \left\{ \underline{k} \mid k_0 \in \frac{2\pi}{\beta} \left(\mathbb{Z} + \frac{1}{2} \right), \ k \in B_L \right\}, \quad \mathbb{M}_{\beta,L} = \left\{ \underline{p} \mid p_0 \in \frac{2\pi}{\beta} \mathbb{Z}, \ p \in B_L \right\}. \quad (4.3.56)$$

For β, L finite, the summands that we will consider will be supported on a finite number of momenta.

Setting up the multiscale integration

Let $h_\beta := \lfloor \log_2(\pi/\delta\beta) \rfloor$, and let $h \in \mathbb{Z}_-$, $0 \geq h \geq h_\beta$. Suppose that the generating functional can be written as:

$$\mathcal{W}_{\beta,L}(\phi, A) = \mathcal{W}_{\beta,L}^{(h)}(\phi, A) + \log \int P_h[d\psi^{(\leq h)}] e^{V^{(h)}(\sqrt{Z_h}\psi^{(\leq h)}; \phi, A)} \quad (4.3.57)$$

with the following meaning of the various objects entering (4.3.57).

1. The notation $\sqrt{Z_h}\psi^{(\leq h)}$ is a short-hand for $\sqrt{Z_{h,\omega}}\psi_\omega^{(\leq h)}$, meaning that each field $\psi_\omega^{(\leq h)}$ appearing in $V^{(h)}$ is multiplied by a factor $\sqrt{Z_{h,\omega}}$.
2. The Grassmann Gaussian measure $P_h(d\psi^{(\leq h)})$ has covariance given by:

$$\begin{aligned} \int P_h(d\psi^{(\leq h)}) \psi_{\underline{x},\omega}^{(\leq h)-} \psi_{\underline{y},\omega'}^{(\leq h)+} &= \delta_{\omega\omega'} \int_{\beta,L} d\underline{k}' e^{i\underline{k}' \cdot (\underline{x}-\underline{y})} \hat{g}_\omega^{(\leq h)}(\underline{k}') \\ \hat{g}_\omega^{(\leq h)}(\underline{k}') &= \frac{1}{Z_{h,\omega}} \frac{\chi_{h,\omega}(\underline{k}')}{ik_0 + v_{h,\omega}k'} (1 + r_{h,\omega}(\underline{k}')) , \end{aligned} \quad (4.3.58)$$

where

$$\chi_{h,\omega}(\underline{k}') = \chi\left(\sqrt{k_0^2 + v_{h,\omega}^2 |\underline{k}'|_{\mathbb{T}}^2} / (2^h \delta)\right) ; \quad (4.3.59)$$

$\hat{g}_\omega^{(\leq h)}(\underline{k}')$ is the *renormalised propagator* on scale h . The quantities $r_{h,\omega}$, $Z_{h,\omega}$, $v_{h,\omega}$ are analytic functions of λ for $|\lambda| < \lambda_0$ such that, for $a > 0$, and for $\underline{k}'$ in the support of $\chi_{h,\omega}(\underline{k}')$:

$$|d_{k_0}^{n_0} d_{k_1}^{n_1} r_{h,\omega}(\underline{k}')| \leq C 2^{h(a-n_0-n_1)} , \quad |v_{h,\omega} - v_\omega| \leq C|\lambda| , \quad \left| \frac{Z_{h+1,\omega}}{Z_{h,\omega}} \right| \leq e^{c|\lambda|} . \quad (4.3.60)$$

Furthermore, $Z_{h,\omega}$, $v_{h,\omega}$ are real. The constants $Z_{h,\omega}$ are called the wavefunction renormalisations, while the constants $v_{h,\omega}$ are called the effective Fermi velocities, on scale h .

3. The effective interaction on scale h takes the form (suppressing the $(\leq h)$ label on the fermionic fields):

$$V^{(h)}(\psi; \phi, A) = \sum_{\Gamma=(\Gamma_\psi, \Gamma_\phi, \Gamma_A)} \int_{\beta,L} d\underline{X} \psi_{\Gamma_\psi}(\underline{X}_\psi) \phi_{\Gamma_\phi}(\underline{X}_\phi) A_{\Gamma_A}(\underline{X}_A) W_\Gamma^{(h)}(\underline{X}) \quad (4.3.61)$$

with translation-invariant kernels $W_\Gamma^{(h)}$ analytic for $|\lambda| < \bar{\lambda}$; $\mathcal{W}_{\beta,L}^{(h)}$ admits an analogous expansion, with the constraint $\Gamma_\psi = \emptyset$. The kernels satisfy the following estimate, for $P_\sharp = |\Gamma_\sharp|$:

$$\int_{\beta,L} d\underline{X} \prod_{i < j} \|\underline{x}_i - \underline{x}_j\|_{\beta,L}^{n_{ij}} |W_\Gamma^{(h)}(\underline{X})| \leq \beta L C_{\{n_{ij}\}} 2^{h(2-(1/2)P_\psi - P_A - (3/2)P_\phi) - c|\lambda|P_A} . \quad (4.3.62)$$

Our goal will be to determine a recursion relation for the kernels, as functions of the scale parameter h , that allows in particular to check the estimate (4.3.62) on all scales, until h_β . To conclude the Section, we remark that all inductive assumptions are true on scale $h = 0$, as a consequence of the cluster expansion that has been used to integrate the ultraviolet degrees of freedom.

localisation and renormalisation

In this section we define a localisation operator, that allows to extract from the effective interaction on scale h the relevant and marginal contributions; these are terms that a priori might expand under the renormalisation group iteration, and that will be studied separately. Let us define:

$$V_{n,m,l}^{(h)}(\psi; \phi, A) = \sum_{\substack{\Gamma \\ |\Gamma_\psi|=n, |\Gamma_\phi|=m, |\Gamma_A|=l}} \int_{\beta,L} d\underline{X} \psi_{\Gamma_\psi}(\underline{X}_\psi) \phi_{\Gamma_\phi}(\underline{X}_\phi) A_{\Gamma_A}(\underline{X}_A) W_\Gamma^{(h)}(\underline{X}) . \quad (4.3.63)$$

We will be interested in the following contributions to the effective interaction:

$$\begin{aligned} V_{2,0,0}^{(h)}(\psi; \phi, A) &= \sum_{\omega} \int_{\beta,L} d\underline{k}' \hat{\psi}_{\underline{k}',\omega}^+ \hat{\psi}_{\underline{k}',\omega}^- \widehat{W}_{2,0,0;\omega}^{(h)}(\underline{k}') \\ V_{4,0,0}^{(h)}(\psi; \phi, A) &= \sum_{\underline{\omega}} \int_{\beta,L} d\underline{k}'_1 d\underline{k}'_2 d\underline{k}'_3 \hat{\psi}_{\underline{k}'_1,\omega_1}^+ \hat{\psi}_{\underline{k}'_2,\omega_2}^- \hat{\psi}_{\underline{k}'_3,\omega_3}^+ \hat{\psi}_{\underline{k}'_1+\underline{k}'_3-\underline{k}'_2,\omega_4}^- \widehat{W}_{4,0,0;\underline{\omega}}^{(h)}(\underline{k}'_1, \underline{k}'_2, \underline{k}'_3) \\ V_{1,1,0}^{(h)}(\psi; \phi, A) &= \sum_{\omega,\rho} \int_{\beta,L} d\underline{k}' \left(\hat{\phi}_{\underline{k}'+\underline{k}_F,\rho}^+ \hat{\psi}_{\underline{k}',\omega}^- \widehat{W}_{1,1,0;\rho,\omega}^{(h)}(\underline{k}') + \hat{\psi}_{\underline{k}',\omega}^+ \hat{\phi}_{\underline{k}'+\underline{k}_F,\rho}^- \widehat{W}_{1,1,0;\omega,\rho}^{(h)}(\underline{k}') \right) \\ V_{2,0,1}^{(h)}(\psi; \phi, A) &= \sum_{\omega,\nu} \int_{\beta,L} d\underline{k}' d\underline{p} \hat{\psi}_{\underline{k}'+\underline{p},\omega_1}^+ \hat{\psi}_{\underline{k}',\omega_2}^- \hat{A}_{-\underline{p}}^\nu \widehat{W}_{2,0,1;\omega,\nu}^{(h)}(\underline{k}', \underline{p}) , \end{aligned} \quad (4.3.64)$$

for suitable kernels. Let $\underline{0}_\beta^\pm := (\pm\pi/\beta, 0)$. We set:

$$F(\underline{0}_\beta) := \frac{1}{2} \sum_{\sigma=\pm} F(\underline{0}_\beta^\sigma) . \quad (4.3.65)$$

Definition 4.3.2. The localisation operator $\mathfrak{L}$ is a linear operator that acts as follows on $V_{n,m,l}^{(h)}$:

$$\begin{aligned} \mathfrak{L}V_{n,m,l}^{(h)}(\psi, \phi, A) &= 0 \quad \text{if} \quad (n, m, l) \neq (2, 0, 0), (4, 0, 0), (2, 0, 1) \\ \mathfrak{L}V_{n,m,l}^{(h)}(\psi, \phi, A) &= V_{n,m,l}^{(h)} \quad \text{if} \quad (n, m, l) = (1, 1, 0) , \end{aligned} \quad (4.3.66)$$

and for the remaining cases, its action is defined by the following identities for the momentum space kernels:

$$\begin{aligned} \mathfrak{L}\widehat{W}_{2,0,0;\omega}^{(h)}(\underline{k}') &:= \widehat{W}_{2,0,0;\omega}^{(h)}(\underline{0}_\beta) + \underline{k}' \cdot \underline{d}\widehat{W}_{2,0,0;\omega}^{(h)}(\underline{0}_\beta) \\ \mathfrak{L}\widehat{W}_{4,0,0;\underline{\omega}}^{(h)}(\underline{k}'_1, \underline{k}'_2, \underline{k}'_3) &:= \widehat{W}_{4,0,0;\underline{\omega}}^{(h)}(\underline{0}_\beta, \underline{0}_\beta, \underline{0}_\beta) \\ \mathfrak{L}\widehat{W}_{2,0,1;\omega,\nu}^{(h)}(\underline{k}', \underline{p}) &:= \widehat{W}_{2,0,1;\omega,\nu}^{(h)}(\underline{0}_\beta, \underline{k}_F^{\omega_1}(\lambda) - \underline{k}_F^{\omega_2}(\lambda)) , \end{aligned} \quad (4.3.67)$$

with $\underline{d}_0, \underline{d}_1$ the discrete derivatives. Observe that, by momentum conservation and by assumption (4.1.11):

$$\widehat{W}_{4,0,0;\underline{\omega}}^{(h)}(\underline{0}_\beta, \underline{0}_\beta, \underline{0}_\beta) = 0 \quad \text{unless } \omega_1 = \omega_2, \omega_3 = \omega_4 \text{ or } \omega_1 = \omega_4, \omega_2 = \omega_3 . \quad (4.3.68)$$

Furthermore, one may check that $\widehat{W}_{\omega_1\omega_2\omega_3\omega_4}^{(h)}(\underline{k}_1, \underline{k}_2, \underline{k}_3) = -\widehat{W}_{\omega_3\omega_2\omega_1\omega_4}^{(h)}(\underline{k}_3, \underline{k}_2, \underline{k}_1)$, which implies

$$\widehat{W}_{4,0,0;\omega\omega'\omega'\omega}^{(h)}(\underline{0}_\beta, \underline{0}_\beta, \underline{0}_\beta) = -\widehat{W}_{4,0,0;\omega'\omega'\omega\omega}^{(h)}(\underline{0}_\beta, \underline{0}_\beta, \underline{0}_\beta) \quad (4.3.69)$$

so that the case $\omega_1 = \omega_4$, $\omega_2 = \omega_3$ in (4.3.68) can be reduced to the case $\omega_1 = \omega_2$, $\omega_3 = \omega_4$. Next, keeping (4.3.67)-(4.3.69) in mind, let:

$$\begin{aligned} z_{h,0,\omega} &:= -id_0 \widehat{W}_{2,0,0;\omega}^{(h)}(\underline{0}_\beta) & z_{h,1,\omega} &:= d_1 \widehat{W}_{2,0,0;\omega}^{(h)}(\underline{0}_\beta) \\ \tilde{\nu}_{h,\omega} &:= 2^{-h} \widehat{W}_{2,0,0;\omega}^{(h)}(\underline{0}_\beta) & \tilde{\lambda}_{h,\omega\omega'} &:= 2 \widehat{W}_{4,0,0;\omega\omega\omega'\omega'}^{(h)}(\underline{0}_\beta, \underline{0}_\beta, \underline{0}_\beta) , \end{aligned} \quad (4.3.70)$$

so that

$$\begin{aligned} \mathfrak{L}V_{2,0,0}^{(h)}(\psi; \phi, A) &= \sum_{\omega} \int_{\beta, L} d\underline{k}' [2^h \tilde{\nu}_{h,\omega} + iz_{h,0,\omega} k_0 + z_{h,1,\omega} k'_1] \hat{\psi}_{\underline{k}',\omega}^+ \hat{\psi}_{\underline{k}',\omega}^- \\ \mathfrak{L}V_{4,0,0}^{(h)}(\psi; \phi, A) &= \sum_{\omega, \omega'} \int_{\beta, L} d\underline{p} \tilde{\lambda}_{h,\omega\omega'} \hat{n}_{-\underline{p},\omega} \hat{n}_{\underline{p},\omega'} \\ \mathfrak{L}V_{2,0,1}^{(h)}(\psi; \phi, A) &= \sum_{\omega, \omega', \nu} \int_{\beta, L} d\underline{p} d\underline{k}' \widehat{W}_{2,0,1;\omega\omega',\nu}^{(h)}(\underline{0}_\beta, \underline{k}_F^\omega(\lambda) - \underline{k}_F^{\omega'}(\lambda)) \hat{A}_{-\underline{p}}^\nu \hat{\psi}_{\underline{k}'+\underline{p},\omega}^+ \psi_{\underline{k}',\omega'}^- . \end{aligned} \quad (4.3.71)$$

One may check that $\mathfrak{L}$ is idempotent, *i.e.* $\mathfrak{L}^2 = \mathfrak{L}$; defining the *renormalisation* operator $\mathfrak{R} := 1 - \mathfrak{L}$, one immediately has that $\mathfrak{R}^2 = \mathfrak{R}$ as well.

Remark 4.3.3. The constants $\tilde{\nu}_{h,\omega}$, $\tilde{\lambda}_{h,\omega\omega'}$, $z_{h,0,\omega}$, $z_{h,1,\omega}$ are real numbers, since the functional integral defining the partition function $\mathcal{Z}_{\beta,L,N_0}$ in (4.3.16) is invariant under the transformation

$$\Psi^- \rightarrow \Psi^+ \quad \Psi^+ \rightarrow -\Psi^- \quad c \rightarrow \bar{c} \quad k_0 \rightarrow -k_0 , \quad (4.3.72)$$

where c is a generic constant entering in the action. It should be noted that this symmetry can be implemented for any finite N_0 , since the cutoff function χ_{N_0} is even in k_0 .

Next, we absorb the $\underline{k}$ -dependent part of $\mathfrak{L}V_{2,0,0}^{(h)}$ in the covariance of the Gaussian measure P_h , obtaining:

$$P_h[d\psi^{(\leq h)}] e^{V^{(h)}(\sqrt{Z_h} \psi^{(\leq h)}; \phi, A)} = e^{\beta L t_h} \tilde{P}_h[d\psi^{(\leq h)}] e^{\tilde{V}^{(h)}(\sqrt{Z_h} \psi^{(\leq h)}; \phi, A)} ; \quad (4.3.73)$$

here t_h is a normalisation constant, due to the change of covariance of the Grassmann integration. The new covariance is given by

$$\tilde{g}_\omega^{(\leq h)}(\underline{k}') = \frac{1}{\tilde{Z}_{h-1,\omega}(\underline{k}')} \frac{\chi_{h,\omega}(\underline{k}')}{ik_0 + \tilde{v}_{h-1,\omega}(\underline{k}')k'} [1 + \tilde{r}_{h,\omega}(\underline{k}')] , \quad (4.3.74)$$

for a suitable new error term $\tilde{r}_{h,\omega}(\underline{k}')$, and where:

$$\begin{aligned} \tilde{Z}_{h-1,\omega}(\underline{k}') &= Z_{h,\omega} [1 + z_{h,0,\omega} \chi_{h,\omega}(\underline{k}')] \\ \tilde{Z}_{h-1,\omega}(\underline{k}') \tilde{v}_{h-1,\omega}(\underline{k}') &= Z_{h,\omega} [v_{h,\omega} + z_{h,1,\omega} \chi_{h,\omega}(\underline{k}')] . \end{aligned} \quad (4.3.75)$$

The effective potential $\tilde{V}^{(h)}$ has the form

$$\tilde{V}^{(h)}(\psi; \phi, A) = \mathfrak{L}\tilde{V}^{(h)}(\psi) + B^{(h)}(\psi; \phi) + C^{(h)}(\psi; A) + \mathfrak{R}V^{(h)}(\psi; \phi, A) , \quad (4.3.76)$$

with:

$$\begin{aligned}\mathfrak{L}\tilde{V}^{(h)}(\psi) &= \sum_{\omega} 2^h \tilde{\nu}_{h,\omega} \int_{\beta,L} d\underline{k}' \hat{\psi}_{\underline{k}',\omega}^+ \hat{\psi}_{\underline{k}',\omega}^- + \mathfrak{L}V_{4,0,0}^{(h)}(\psi; 0, 0) \\ B^{(h)}(\psi; \phi) &= \sum_{\omega,\rho} \int_{\beta,L} d\underline{k}' \left(\hat{\phi}_{\underline{k}'+\underline{k}_F^\omega(\lambda),\rho}^+ \hat{\psi}_{\underline{k}',\omega}^- \widehat{W}_{1,1,0;\rho,\omega}^{(h)}(\underline{k}') + \hat{\psi}_{\underline{k}',\omega}^+ \hat{\phi}_{\underline{k}'+\underline{k}_F^\omega(\lambda),\rho}^- \widehat{W}_{1,1,0;\omega,\rho}^{(h)}(\underline{k}') \right) \\ C^{(h)}(\psi; A) &= \sum_{\omega,\omega',\nu} \int_{\beta,L} d\underline{k}' d\underline{p} \hat{\psi}_{\underline{k}'+\underline{p},\omega}^+ \hat{\psi}_{\underline{k}',\omega'}^- \hat{A}_{-\underline{p}}^\nu \widehat{W}_{2,0,1;\omega\omega',\nu}^{(h)}(\underline{0}_\beta, \underline{k}_F^\omega(\lambda) - \underline{k}_F^{\omega'}(\lambda)) .\end{aligned}$$

Finally, we set $Z_{h-1,\omega} := \tilde{Z}_{h-1,\omega}(\underline{0}_\beta)$ and we rescale the fields:

$$e^{\beta L t_h} \tilde{P}_h[d\psi^{(\leq h)}] e^{\tilde{V}^{(h)}(\sqrt{Z_h}\psi^{(\leq h)}; \phi, A)} = e^{\beta L t_h} \tilde{P}_h[d\psi^{(\leq h)}] e^{\widehat{V}^{(h)}(\sqrt{Z_{h-1}}\psi^{(\leq h)}; \phi, A)} , \quad (4.3.77)$$

where $\widehat{V}^{(h)}$ can be represented as follows:

$$\widehat{V}^{(h)}(\psi; \phi, A) = \mathfrak{L}\widehat{V}^{(h)}(\psi) + \widehat{B}^{(h)}(\psi; \phi) + \widehat{C}^{(h)}(\psi; A) + \mathfrak{R}\widehat{V}^{(h)}(\psi; \phi, A) , \quad (4.3.78)$$

where

$$\begin{aligned}\mathfrak{L}\widehat{V}^{(h)}(\psi) &= \sum_{\omega} 2^h \nu_{h,\omega} \int_{\beta,L} d\underline{k}' \hat{\psi}_{\underline{k}',\omega}^+ \hat{\psi}_{\underline{k}',\omega}^- + \sum_{\omega,\omega'} \lambda_{h,\omega\omega'} \int_{\beta,L} d\underline{p} \hat{n}_{-\underline{p},\omega} \hat{n}_{\underline{p},\omega'} \\ \widehat{B}^{(h)}(\psi; \phi) &= \sum_{\omega} \int_{\beta,L} d\underline{k}' \left[(\hat{\phi}_{\underline{k}'+\underline{k}_F^\omega(\lambda)}^+, \mathbf{Q}_{h,\omega}^+(\underline{k}')) \hat{\psi}_{\underline{k}',\omega}^- + \hat{\psi}_{\underline{k}',\omega}^+ (\mathbf{Q}_{h,\omega}^-(\underline{k}'), \hat{\phi}_{\underline{k}'+\underline{k}_F^\omega(\lambda)}^-) \right] \\ \widehat{C}^{(h)}(\psi; A) &= \sum_{\omega,\omega',\nu} \int_{\beta,L} d\underline{p} d\underline{k}' \hat{A}_{-\underline{p}}^\nu Z_{h,\omega\omega'}^\nu \hat{\psi}_{\underline{k}'+\underline{p},\omega}^+ \hat{\psi}_{\underline{k}',\omega'}^- ,\end{aligned} \quad (4.3.79)$$

with effective couplings

$$\nu_{h,\omega} := \frac{Z_{h,\omega}}{Z_{h-1,\omega}} \tilde{\nu}_{h,\omega} \quad \lambda_{h,\omega\omega'} := \frac{Z_{h,\omega} Z_{h,\omega'}}{Z_{h-1,\omega} Z_{h-1,\omega'}} \tilde{\lambda}_{h,\omega\omega'} \quad (4.3.80)$$

and

$$\begin{aligned}\mathbf{Q}_{h,\omega,\rho}^+(\underline{k}') &= \sqrt{\frac{Z_{h,\omega}}{Z_{h-1,\omega}}} \widehat{W}_{1,1,0;\rho,\omega}^{(h)}(\underline{k}') , \quad \mathbf{Q}_{h,\omega,\rho}^-(\underline{k}') = \sqrt{\frac{Z_{h,\omega}}{Z_{h-1,\omega}}} \widehat{W}_{1,1,0;\omega,\rho}^{(h)}(\underline{k}') \\ Z_{h,\omega\omega'}^\nu &= \frac{\sqrt{Z_{h,\omega} Z_{h,\omega'}}}{\sqrt{Z_{h-1,\omega} Z_{h-1,\omega'}}} \widehat{W}_{2,0,1;\omega\omega',\nu}^{(h)}(\underline{0}_\beta, \underline{k}_F^\omega(\lambda) - \underline{k}_F^{\omega'}(\lambda)) .\end{aligned} \quad (4.3.81)$$

In the following, we shall use the notation:

$$Z_{h,\omega}^\nu \equiv Z_{h,\omega\omega}^\nu . \quad (4.3.82)$$

Remark 4.3.4. Notice that, since $\mathfrak{L}\widehat{V}_{4,0,0}^{(h)}$ can be written in real space as

$$\sum_{\omega,\omega'} \lambda_{h,\omega\omega'} \int_{\beta,L} d\underline{x} n_{\underline{x},\omega} n_{\underline{x},\omega'} , \quad (4.3.83)$$

we can suppose without loss of generality that:

$$\lambda_{h,\omega\omega} = 0 , \quad \lambda_{h,\omega\omega'} = \lambda_{h,\omega'\omega} . \quad (4.3.84)$$

Single-scale integration

We now perform the single-scale integration that will yield $\mathcal{W}^{(h-1)}, V^{(h-1)}$. In order to do so, we split the propagator as:

$$\begin{aligned}\tilde{g}_\omega^{(\leq h)} &= \hat{g}_\omega^{(\leq h-1)} + g_\omega^{(h)} \\ g_\omega^{(h)}(\underline{k}') &= \frac{1}{\tilde{Z}_{h-1,\omega}(\underline{k}')} \frac{f_{h,\omega}(\underline{k}')}{ik_0 + \tilde{v}_{h-1,\omega}(\underline{k}')k'} [1 + r_{h-1,\omega}(\underline{k}')] \\ \hat{g}_\omega^{(\leq h-1)}(\underline{k}') &= \frac{1}{Z_{h-1,\omega}} \frac{\chi_{h-1,\omega}(\underline{k}')}{ik_0 + v_{h-1,\omega}k'} [1 + r_{h-1,\omega}(\underline{k}')] ,\end{aligned}\tag{4.3.85}$$

with $f_{h,\omega}(\underline{k}') = \chi_{h,\omega}(\underline{k}') - \chi_{h-1,\omega}(\underline{k}')$. In the last identity, we used the fact that, for $\underline{k}'$ in the support of $\chi_{h-1,\omega}(\underline{k}')$:

$$\tilde{Z}_{h-1,\omega}(\underline{k}') = \tilde{Z}_{h-1,\omega}(\underline{0}_\beta) \equiv Z_{h-1,\omega} , \quad \tilde{v}_{h-1,\omega}(\underline{k}') = \tilde{v}_{h-1,\omega}(\underline{0}_\beta) \equiv v_{h-1,\omega} . \tag{4.3.86}$$

recall equation (4.3.75). Denoting with $\tilde{P}_h(d\psi^{(h)})$ the Grassmann Gaussian integration associated with $g^{(h)}$, we use the addition principle to write:

$$\begin{aligned}\int \tilde{P}_h[d\psi^{(\leq h)}] e^{\hat{V}^{(h)}(\sqrt{Z_{h-1}}\psi^{(\leq h)}; \phi, A)} \\ = \int P_{h-1}[d\psi^{(\leq h-1)}] \int \tilde{P}_h[d\psi^{(h)}] e^{\hat{V}^{(h)}(\sqrt{Z_{h-1}}(\psi^{(\leq h-1)} + \psi^{(h)}); \phi, A)} ,\end{aligned}\tag{4.3.87}$$

where $P_{h-1}[d\psi^{(\leq h-1)}]$ is the Grassmann Gaussian integration associated with $\hat{g}_\omega^{(\leq h-1)}(\underline{k}')$. This identity allows to define the effective potential $V^{(h-1)}$ and the effective generating functional $\mathcal{W}^{(h-1)}$ as:

$$\begin{aligned}\mathcal{W}_{\beta,L}^{(h-1)}(\phi, A) + V^{(h-1)}(\sqrt{Z_{h-1}}\psi^{(\leq h-1)}; \phi, A) \\ := \mathcal{W}_{\beta,L}^{(h)}(\phi, A) + \beta L t_h + \log \int \tilde{P}_h[d\psi^{(h)}] e^{\hat{V}^{(h)}(\sqrt{Z_{h-1}}(\psi^{(\leq h-1)} + \psi^{(h)}); \phi, A)} .\end{aligned}\tag{4.3.88}$$

As reviewed in Section 4.3.5, equation (4.3.88) is the starting point for the Gallavotti-Nicolò tree expansion for the effective potential, which will allow to control the flow of the effective potential. A key ingredient for this graphical representation are the bounds for the running coupling constants, whose flow is discussed in the next Section.

4.3.4 Flow of the effective couplings

The above iteration gives rise to a discrete dynamical system for the running coupling parameters, which has the following form:

$$\begin{aligned}\frac{Z_{h-1,\omega}}{Z_{h,\omega}} &= 1 + z_{h,0,\omega} =: 1 + \beta_{h,\omega}^z \\ v_{h-1,\omega} &= \frac{Z_{h,\omega}}{Z_{h-1,\omega}} (v_{h,\omega} + z_{h,1,\omega}) =: v_h + \beta_{h,\omega}^v \\ 2^h \nu_{h,\omega} &= \frac{Z_{h,\omega}}{Z_{h-1,\omega}} \widehat{W}_{2;\omega}^{(h)}(\underline{0}_\beta) =: 2^{h+1} \nu_{h+1,\omega} + 2^{h+1} \beta_{h+1,\omega}^\nu \\ \lambda_{h,\omega\omega'} &= \frac{Z_{h,\omega} Z_{h,\omega'}}{Z_{h-1,\omega} Z_{h-1,\omega'}} \widehat{W}_{4;\omega\omega'\omega'\omega'}^{(h)}(\underline{0}_\beta, \underline{0}_\beta, \underline{0}) =: \lambda_{h+1,\omega\omega'} + \beta_{h+1,\omega\omega'}^\lambda \\ Z_{h,\omega\omega'}^\nu &= Z_{h+1,\omega\omega'}^\nu + \beta_{h+1,\omega\omega'}^{z,\nu} .\end{aligned}\tag{4.3.89}$$

with initial data given by

$$\begin{aligned} Z_{0,\omega} &= 1, & v_{0,\omega} &= v_\omega + O(\lambda), & \nu_{0,\omega} &= \nu_\omega + O(\lambda), \\ \lambda_{0,\omega\omega'} &= -\lambda + O(\lambda^2), & |Z_{0,\omega\omega'}^\nu| &\leq C \end{aligned} \quad (4.3.90)$$

where the higher-order corrections in λ are due to the integration of the ultraviolet degrees of freedom. The vector field:

$$\begin{aligned} \beta_h &= \beta_h(\{Z_k, v_k, \nu_k, \lambda_k, Z_k^\nu\}_{k=h+1}^0) \\ &= (\beta_{h,\omega}^z, \beta_{h,\omega}^v, \beta_{h,\omega}^\nu, \beta_{h,\omega}^\lambda, \beta_{h,\omega}^{z,\nu})_\omega \end{aligned} \quad (4.3.91)$$

defined by (4.3.89) is called the *beta function* of the theory, and it drives the flow of the couplings. Precise estimates on this object are needed in order to safely iterate the equations in (4.3.89) in h . The control of the running coupling constants is the content of the following proposition. The result has been obtained for lattice fermions with two Fermi points in [17, 18], and it has been generalised to multiple Fermi points in [61]. We state the result, without discussing its proof.

Proposition 4.3.5 (Bounds for the running coupling constants.). *There exists $\bar{\lambda} > 0$ such that for $|\lambda| < \bar{\lambda}$ the following is true. There exists a choice of real parameters $\{\nu_\omega\}$, with $\nu_\omega \equiv \nu_\omega(\lambda)$, $|\nu_\omega| \leq C|\lambda|$, such that:*

$$\begin{aligned} \left| \frac{Z_{h-1,\omega}}{Z_{h,\omega}} \right| &\leq e^{c|\lambda|}, & |v_{h,\omega} - v_\omega| &\leq C|\lambda|, & |\lambda_{h,\omega\omega'} - \lambda| &\leq C|\lambda|^2, \\ |Z_{h,\omega}^\nu| &\leq C|Z_{1,\omega}^\nu|, & \left| \frac{Z_{h-1,\omega\omega'}^\nu}{Z_{h,\omega\omega'}^\nu} \right| &\leq e^{c|\lambda|}, \end{aligned} \quad (4.3.92)$$

for some positive constants c, C independent of β, L , for all $\omega, \omega' = 1, \dots, N_f$, $\nu = 0, 1$, and for all $h \geq h_\beta := \lfloor \log(\pi/\beta) \rfloor$.

Remark 4.3.6. It should be noted that the independence of these bounds from β allows to control the dynamical system also in the zero temperature limit $h_\beta \rightarrow -\infty$.

The proof of this result uses the structure of the scaling limit of the system, which describes relativistic, chiral $1 + 1$ dimensional massless fermions. The model describing this system, called the reference model, will be reviewed later on. The boundedness of the flow of the running coupling constants is a deep consequence of the vanishing of the beta function for the reference model, which can be viewed as an instance of the integrability of the scaling limit.

Observe that the wave function renormalisation $Z_{h,\omega}$, has a divergent flow. We have:

$$Z_{h,\omega} \sim 2^{\xi_\omega h}, \quad \text{with } \xi_\omega > 0, \xi_\omega = O(\lambda^2). \quad (4.3.93)$$

As it is evident from the form of the single-scale propagator (4.3.85), the divergence of the wave function renormalisation changes the scaling of the two-point function, and introduces an interaction dependent anomalous exponent. This is the reason for the rescaling of the fields in the multiscale analysis. Moreover, the vertex renormalisations $Z_{h,\omega\omega'}^\nu$ for $\omega \neq \omega'$ might also diverge with an anomalous power law. The fact that $Z_{h,\omega}^\nu$ stays bounded is non-trivial, and follows from the emergent (anomalous) chiral symmetry of the model [16].

Finally, proceeding as in *e.g.* Section 3.4 of [40], it is possible to prove that the external $\phi^\pm$ source term kernels $\mathbf{Q}_{h,\omega}^\pm$ satisfy the recursion relation:

$$\begin{aligned}\mathbf{Q}_{h,\omega}^+(\underline{k}') &= \mathbf{Q}_{h+1,\omega}^+(\underline{k}') - \widehat{W}_{2,0,0;\omega\omega}^{(h)}(\underline{k}') \sum_{h'=h+1}^1 \hat{g}_\omega^{(h')}(\underline{k}') \mathbf{Q}_{h',\omega}^+(\underline{k}') \\ \mathbf{Q}_{h,\omega}^-(\underline{k}') &= \overline{\mathbf{Q}_{h,\omega}^+(\underline{k}')}. \end{aligned} \quad (4.3.94)$$

In particular, for $\underline{k}' \in \text{supp } \hat{g}_\omega^{(h)}$, equation (4.3.94) reduces to

$$\mathbf{Q}_{h,\omega}^+(\underline{k}') = \mathbf{Q}_{1,\omega}^+(\underline{k}') [1 - \widehat{W}_{2,0,0;\omega\omega}^{(h)}(\underline{k}') \hat{g}_\omega^{(h)}(\underline{k}')], \quad (4.3.95)$$

which, thanks to the estimates (4.3.92), and the bounds for $\widehat{W}_{2,0,0;\omega\omega}^{(h)}$ which will be derived in Section 4.3.5, implies that $\mathbf{Q}_{h,\omega}^+ = \mathbf{Q}_{1,\omega}^+(1 + O(\lambda)) = \xi^\omega(1 + O(\lambda))$.

4.3.5 Tree expansion for kernels

In this Section, we recall the *Gallavotti-Nicolò* (GN) *tree* representation of the kernels $W_\Gamma^{(h)}$ contributing to the effective potential $V^{(h)}$ and of the generating functional $\mathcal{W}^{(h)}$. Although this is by now a standard construction (see [36] for a review), we prefer to recall some of its basic elements because they will be useful in deriving the improved estimates in Section 4.3.6. We will discuss how to obtain a tree expansion for the effective potential in terms of trees that extend up to scale 0, which takes as input the effective potential generated after the integration of the ultraviolet degrees of freedom; this is the infrared construction, and it is the technically most demanding one. The effective potential on scale 0 is in turn the outcome of an ultraviolet multiscale analysis, which is much easier to carry out and will be recalled in a second moment.

Employing the fact that the logarithm of a Grassmann Gaussian expectation may be written in terms of cumulants, we may transform equation (4.3.88), upon replacing $h - 1$ with h , by as

$$\begin{aligned}\mathcal{W}_{\beta,L}^{(h)}(\phi, A) + V^{(h)}(\sqrt{Z_h} \psi^{(\leq h)}; \phi, A) \\ = \mathcal{W}_{\beta,L}^{(h+1)}(\phi, A) + \beta L t_h + \sum_{n \geq 0} \frac{(-1)^n}{n!} \mathbb{E}_{h+1}[\underbrace{\widehat{V}^{(h+1)}; \dots; \widehat{V}^{(h+1)}}_{n \text{ times}}] \end{aligned} \quad (4.3.96)$$

where $\mathbb{E}_h[\cdot; \dots; \cdot]$ denotes the *truncated expectation* (or cumulant) with respect to the single-scale Gaussian Grassmann measure $\tilde{P}_h$. Equation (4.3.96) can be iterated over all scales $h + 1, h + 2, \dots$, until $h = 0$ and the result can be graphically expressed as a sum over GN trees.

Let us first introduce some definitions.

1. An *unlabelled* tree is a connected, acyclic¹ finite graph connecting a point r , called the root, with an ordered set of $n \geq 1$ points, the endpoints of the tree, so that r is not a branching point. The number n will be called the *order* of the unlabelled tree; in particular, the root r is not counted as a vertex. We will denote by $V(\tau)$ the set of vertices of the unlabelled tree τ . The unlabelled trees have a natural partial

¹That is, it has no loops.

ordering $\prec$, that is: orienting the tree from the root, we set $v \prec v'$ if there is an oriented path from r to v' that passes through v .

We denote by $V_f(\tau) \subset V(\tau)$ the set of endpoints, also called *leaves* or *final vertices*, of τ . For each vertex v , we will denote by s_v its branching number: in particular, vertices with $s > 1$ will be called the *non-trivial vertices*. Notice that $s_v = 0$ if $v \in V_f(\tau)$.

Two unlabelled trees are identified if they can be superposed by a continuous deformation, which is compatible with the partial ordering of the non-trivial vertices. It is then easy to see that the number of unlabelled trees with n endpoints and only non-trivial vertices is bounded by 4^n .

We shall also consider *labelled trees* (or just trees, in the following), see Fig. 4.2; they are defined by associating suitable labels to the vertices of unlabelled trees, as explained in the following points.

2. Given a labelled tree τ , to each vertex v of the labelled tree τ we associate a scale label h_v in $\{h+1, \dots, 0\}$, as in Fig. 4.2. The scale of the root r is set to be h . Note that if $v_1 \prec v_2$, then $h_{v_1} < h_{v_2}$.
3. There is only one vertex immediately following the root, which will be denoted by v_0 and cannot be an endpoint. Its scale is $h_{v_0} = h+1$.
4. With each endpoint v on scale $h_v = 0$ we associate one of the monomials contributing to $\widehat{V}^{(0)}(\sqrt{Z_{-1}}\psi^{(\leq 0)}; \phi, A)$. Instead, with each leaf v on scale $h_v < 0$ we associate one of the monomials contributing to $\mathfrak{L}\widehat{V}^{(h_v)}(\sqrt{Z_{h_v-1}}\psi^{(\leq h_v)}; \phi, A)$. In particular, a leaf $v \in V_f(\tau)$ will be called:
 - i. of type λ if it is associated with $\mathcal{L}\widehat{V}_{4,0,0}^{(h_v)}$;
 - ii. of type ν if it is associated with a ν_{h_v} coupling constant;
 - iii. of type ϕ if it is associated with a $\widehat{B}^{(h_v)}$ term;
 - iv. of type A if it is associated with a $\widehat{C}^{(h_v)}$ term.
5. We introduce a *field label* f to distinguish the field and external source variables appearing in the monomials associated with the endpoints. The set of field labels associated with the endpoint v will be called $I_v = \{f_1, \dots, f_{|I_v|}\}$ and it will admit a partitioning $I_v = I_v^\psi \cup I_v^\phi \cup I_v^A$.

If f labels a field ψ , we denote by $\underline{x}(f)$ its space-time position, by $\omega(f)$ its quasi-particle label, and by $\epsilon(f)$ its particle-hole label. We have an analogous labelling for the external sources ϕ, A associated to the endpoint v , so that the monomial associated to it is given by

$$\prod_{f \in I_v^\psi} \psi_{\underline{x}(f), \omega(f)}^{(\leq h_v), \epsilon(f)} \prod_{f \in I_v^\phi} \phi_{\underline{x}(f), \rho(f)}^{\epsilon(f)} \prod_{f \in I_v^A} A_{\underline{x}(f)}^{\nu(f)}. \quad (4.3.97)$$

If v is not an endpoint, we shall call I_v the set of field labels associated with the endpoints following the vertex v , namely

$$I_v := \prod_{\substack{v' \in V_f(\tau) \\ v \prec v'}} I_{v'}. \quad (4.3.98)$$

²A tree τ is said to be trivial if it contains only one edge (and hence only one leaf).

We will denote by $Q_{v_i} := \Gamma_v^\psi \cap \Gamma_{v_i}^\psi$ the set of fields external to v subordinated on v_i ; the union of the subsets $\Gamma_{v_i}^\psi \setminus Q_{v_i}$ will constitute the set of fields *internal* to v , and each one of these subsets has to be non-empty if $s_v > 1$.

For external sources, we simply set $\Gamma_v^\phi = I_v^\phi$ and $\Gamma_v^A = I_v^A$.

We shall denote with $\mathfrak{L}_\tau$ the set of all possible choices of external fields subordinated to the tree τ , and we shall indicate its elements, *i.e.* given labellings, with

$$\mathbf{\Gamma} = (\Gamma_v)_v \equiv (\Gamma_v^\psi, \Gamma_v^\phi, \Gamma_v^A)_v .$$

Remark 4.3.7. In order to better explain the outcome of the iteration of (4.3.100) by employment of the external fields labellings $\mathbf{\Gamma}$, let us suppose for a moment that $\mathfrak{R} = 1$: we shall take its action into account in a second moment.

Given an internal vertex $v \notin V_f(\tau)$, let us call τ_v the (unique) sub-tree having v as first vertex, so that the scale of the root of τ_v is $h_v - 1$; we set

$$\begin{aligned} V^{(h_v-1)}[\tau_v] &= \sum_{\mathbf{\Gamma}_v \in \mathcal{L}_{\tau_v}} V_{\mathbf{\Gamma}_v}^{(h_v-1)}[\tau_v] \\ V_{\mathbf{\Gamma}_v}^{(h_v-1)}[\tau_v] &= \int_{\beta, L} d\underline{X}_{\mathbf{\Gamma}_v} \tilde{\psi}_{\mathbf{\Gamma}_v^\psi}^{(\leq h_v-1)} \phi_{\mathbf{\Gamma}_v^\phi} A_{\mathbf{\Gamma}_v^A} W_{\mathbf{\Gamma}_v}^{(h_v-1)}[\underline{X}_{\mathbf{\Gamma}_v}; \tau_v] \end{aligned} \quad (4.3.101)$$

where, as usual, $d\underline{X}_{\mathbf{\Gamma}_v} = \prod_{f \in \mathbf{\Gamma}_v} d\underline{x}(f)$ and

$$\tilde{\psi}_{\mathbf{\Gamma}}^{(\leq h)} := \prod_{f \in \mathbf{\Gamma}} \sqrt{Z_{h, \omega(f)} \psi_{\underline{x}(f), \omega(f)}^{(\leq h), \epsilon(f)}} , \quad \phi_{\mathbf{\Gamma}} := \prod_{f \in \mathbf{\Gamma}} \phi_{\underline{x}(f), \rho(f)}^{\epsilon(f)} , \quad A_{\mathbf{\Gamma}} := \prod_{f \in \mathbf{\Gamma}} A_{\underline{x}(f)}^{\nu(f)} . \quad (4.3.102)$$

Remark 4.3.8. Strictly speaking, it should be noted that the kernels $W_{\mathbf{\Gamma}_v}^{(h_v-1)}[\underline{X}_{\mathbf{\Gamma}_v}; \tau_v]$ that we have defined depend on all the coordinates $\underline{X}_{I_v}$, not just on the coordinates $\underline{X}_{\mathbf{\Gamma}_v}$; with a slight abuse of notation, we explicit only the latter ones because we want to highlight the ones associated to fields external to v .

We shall denote by $\mathbf{\Gamma}_v$ a labelling for the sub-tree τ_v ; hence $\mathbf{\Gamma}_{v_0} \equiv \mathbf{\Gamma}$. The recursion (4.3.100), together with the representation (4.3.101), implies the following recursion for the kernels $W_{\mathbf{\Gamma}}^{(h_v-1)}$:

$$\begin{aligned} W_{\mathbf{\Gamma}_v}^{(h_v-1)}[\underline{X}_{\mathbf{\Gamma}_v}; \tau_v] &= \frac{(-1)^{s_v+1}}{s_v!} \left[\prod_{f \in \mathbf{\Gamma}_v^\psi} \sqrt{\frac{Z_{h_v, \omega(f)}}{Z_{h_v-1, \omega(f)}}} \right] \left[\prod_{i=1}^{s_v} W_{\mathbf{\Gamma}_{v_i}}^{(h_{v_i}-1)}[\underline{X}_{\mathbf{\Gamma}_{v_i}}; \tau_{v_i}] \right] \\ &\quad \cdot \mathbb{E}_{(h_v)} \left[\tilde{\psi}_{\mathbf{\Gamma}_{v_1}^\psi \setminus Q_{v_1}}^{(h_v)} ; \dots ; \tilde{\psi}_{\mathbf{\Gamma}_{v_{s_v}}^\psi \setminus Q_{v_{s_v}}}^{(h_v)} \right] \end{aligned} \quad (4.3.103)$$

where $h_{v_i} = h_v + 1$ by construction, and $\mathbf{\Gamma}_{v_i}$ is the restriction of $\mathbf{\Gamma}_v$ to the sub-tree τ_{v_i} . This iteration stops when v is a leaf, in which case we will have that $W_{\mathbf{\Gamma}_v}^{(h_v-1)} \equiv W_{\mathbf{\Gamma}_v}^{(h_v-1)}$ is equal to one the kernels of the monomials contributing to $\mathfrak{L}\widehat{V}^{(h_v)}(\sqrt{Z_{h_v-1}}\psi^{(\leq h_v)})$; in momentum space, we get

$$\widehat{W}_{\mathbf{\Gamma}_v}^{(h_v-1)}[\underline{K}_{\mathbf{\Gamma}_v}; \tau_v] = \begin{cases} \lambda_{h_v, \omega\omega'} & v \text{ } \lambda\text{-type, } h_v < 0 \\ \nu_{h_v, \omega} & v \text{ } \nu\text{-type, } h_v < 0 \\ Q_{h_v, \omega}^\pm(k') & v \text{ } \phi^\pm\text{-type, } h_v < 0 \\ Z_{h_v, \omega\omega'}^\mu & v \text{ } A^\mu\text{-type, } h_v < 0 \\ \widehat{W}_{\mathbf{\Gamma}_v}^{(0)}[\underline{K}_{\mathbf{\Gamma}_v}] & h_v = 0 . \end{cases} \quad (4.3.104)$$

where as usual $k' = k - k_F^\omega(\lambda)$, and where $W_\Gamma^{(0)}$ indicates a kernel appearing in the monomial appearing in $\hat{V}^{(0)}(\sqrt{Z_{-1}}\psi^{(\leq 0)}; \phi, A)$ parameterised by Γ .

Therefore, noticing that $\tau_{v_0} = \tau$, and writing the overall monomial associated to a given choice of (τ, Γ) , namely

$$V_\Gamma^{(h)}[\tau] = \int_{\beta, L} d\underline{X}_{\Gamma_{v_0}} \tilde{\psi}_{\Gamma_{v_0}^\psi}^{(\leq h)} \phi_{\Gamma_{v_0}^\phi} A_{\Gamma_{v_0}^A} W_{\Gamma_{v_0}}^{(h)}[\underline{X}_{\Gamma_{v_0}}; \tau], \quad (4.3.105)$$

we get

$$W_{\Gamma_v}^{(h)}[\underline{X}_{\Gamma_{v_0}}; \tau] = (-1)^{n+m} L[\tau] \prod_{v \notin V_f(\tau)} \frac{1}{s_v!} \sqrt{\frac{Z_{h_v}(\Gamma_v^\psi)}{Z_{h_v-1}(\Gamma_v^\psi)}} \mathbb{E}_{(h_v)} \left[\tilde{\psi}_{\Gamma_{v_1}^\psi \setminus Q_{v_1}}^{(h_v)}; \dots; \tilde{\psi}_{\Gamma_{v_{s_v}}^\psi \setminus Q_{v_{s_v}}}^{(h_v)} \right] \quad (4.3.106)$$

with the short-hand $Z_h(\Gamma) := \prod_{f \in \Gamma} Z_{h, \omega(f)}$ and with $L[\tau]$ taking into account the contribution of the leaves, namely

$$L[\tau] = \prod_{v \in V_f(\tau)} W_{\Gamma_v}^{(h_v-1)}[\underline{X}_v; \tau_v]. \quad (4.3.107)$$

Comparing (4.3.61) with (4.3.99) and (4.3.99), we have thus obtained the following GN representation for kernels $W_\Gamma^{(h)}$ appearing in $\mathcal{W}_{\beta, L}^{(h)}$ and $V^{(h)}$:

$$W_\Gamma^{(h)}(\underline{X}) = \sum_{n=1}^{\infty} \sum_{\tau \in \mathcal{T}_{h, n}^m} \sum_{\substack{\Gamma \in \mathcal{L}_\tau \\ \Gamma_{v_0} = \Gamma}} \int_{\beta, L} d\underline{X}_{I_{v_0} \setminus \Gamma_{v_0}} W_\Gamma^{(h)}[\underline{X}; \tau]. \quad (4.3.108)$$

This is useful because it allows to prove bounds analogous to (4.3.32) for the kernels $W_\Gamma^{(h)}$, with an explicit h -dependence which is uniform in β, L , as we now illustrate.

Remark 4.3.9. The *Battle-Brydges-Federbush (BBF) formula*, see [24, 56] and references therein, and [36] for a review, allows to express a connected expectation between clusters as a sum over *spanning* trees over these. Given clusters $\{\Gamma_i\}_{i=1}^s$ such that $\sum_i |\Gamma_i| = 2N$, we have that

$$\mathbb{E}[\Psi_{\Gamma_1}; \dots; \Psi_{\Gamma_s}] = \sum_T \sigma_T \left[\prod_{\ell \in T} g_\ell \right] \int d\mu_T(\mathbf{t}) \det[G_T(\mathbf{t})] \quad (4.3.109)$$

where:

- the sum is over so-called anchored (or spanning) trees T , *i.e.* collections of edges in $E(\bigcup_i \Gamma_i)$ that become a tree if each Γ_i is collapsed to a point i ; furthermore, $\sigma_T \in \pm 1$ is a sign associated to T : it will not play any role in the analysis;
- the product is over the oriented edges ℓ composing the anchored tree T , the factor being the propagator g evaluated on it;
- the integration is over parameters $\mathbf{t} = (t_{ij})_{i,j=1}^s$, the probability measure μ_T is supported on a set of $\mathbf{t}$ such that $t_{ij} = \vec{u}_i \cdot \vec{u}_j$, for some family of unit-norm vectors $(\vec{u}_i)_{i=1}^{N-s+1} \subset \mathbb{R}^s$;

- $G_T(\mathbf{t}) \in M_{N-s+1}(\mathbb{C})$ is a matrix whose entries are given by

$$(G_T(\mathbf{t}))_{(i,a)(j,b)} = t_{ij} g(\underline{x}(i,a) - \underline{x}(j,b))$$

where i, j parameterise the clusters and a (resp. b) a half-edge in i (resp. j), such that the oriented edge $((i,a), (j,b))$ does not lie in T .

Employing the BBF formula for the expectations taken at each internal vertex v of the GN tree τ , we may rewrite equation (4.3.106) as

$$\begin{aligned} & |W_{\mathbf{r}}^{(h)}[\underline{X}_{\Gamma_{v_0}}; \tau]| \\ &= \left| L[\tau] \prod_{v \notin V_f(\tau)} \frac{1}{s_v!} \sqrt{\frac{Z_{h_v}(\Gamma_v^\psi)}{Z_{h_v-1}(\Gamma_v^\psi)}} \sum_{T_v} \left[\prod_{\ell \in T_v} Z_{h_v-1, \omega_\ell} g_\ell^{(h_v)} \right] \int d\mu_{T_v}(\mathbf{t}) \det[G_{T_v}^{(h_v)}(\mathbf{t})] \right|. \end{aligned} \quad (4.3.110)$$

Now, recall the bounds on the effective couplings listed in Proposition 4.3.5:

$$\left| \frac{Z_{h,\omega}}{Z_{h-1,\omega}} \right| \leq e^{c|\lambda|}, \quad |v_{h-1,\omega} - v_\omega| \leq C|\lambda|, \quad |\lambda_{h,\omega\omega'}| \leq C|\lambda|, \quad |\nu_{h,\omega}| \leq C|\lambda|.$$

These bounds can be used to prove, following *e.g.* [15], Section 3.14:

$$\begin{aligned} \|\det G_{T_v}^{(h_v)}(\mathbf{t})\|_\infty &\leq C \sum_{i=1}^{s_v} \frac{|\Gamma_{v_i}^\psi|}{2} - \frac{|\Gamma_v^\psi|}{2} - (s_v-1) 2^{h_v} \left[\sum_{i=1}^{s_v} \frac{|\Gamma_{v_i}^\psi|}{2} - \frac{|\Gamma_v^\psi|}{2} - (s_v-1) \right] \\ |Z_{h-1,\omega} \partial_0^{n_0} d_1^{n_1} g_\omega^{(h)}(\underline{x} - \underline{y})| &\leq \frac{C_{n+n_0+n_1} 2^{h(1+n_0+n_1)}}{1 + (2^h \|\underline{x} - \underline{y}\|_{\beta,L})^n}, \quad \forall n \in \mathbb{N}. \end{aligned} \quad (4.3.111)$$

The first bound follows from the Gram-Hadamard inequality for determinants (using that the single-scale propagator admits a Gram representation, Eq. (3.97) of [15]), while the second follows from the smoothness and support properties of the single-scale propagator in momentum space: in particular, it implies $\|g^{(h)}\|_1 \leq c 2^{-h}$. Let $V_f^{od}(\tau) \subseteq V_f(\tau)$ be the set of endpoints of the tree associated with $Z_{h_v, \omega_v \omega'_v}^{\mu_v}$ with $\omega_v \neq \omega'_v$. Using the fact that

$$\sum_{T_v} 1 \leq s_v! C \sum_{i=1}^{s_v} \frac{|\Gamma_{v_i}^\psi|}{2} - \frac{|\Gamma_v^\psi|}{2} - (s_v-1),$$

and defining the intensive L^1 -norm

$$\|W_{\mathbf{r}}^{(h)}[\tau]\|_1 := \frac{1}{\beta L} \int_{\beta, L} d\underline{X}_{I_{v_0}} |W_{\mathbf{r}}^{(h)}[\underline{X}_{\Gamma_{v_0}}; \tau]|, \quad (4.3.112)$$

the bounds (4.3.111) allow to prove (we refer to *e.g.* [36] for the details) that

$$\begin{aligned} \|W_{\mathbf{r}}^{(h)}[\tau]\|_1 &\leq C^{m+m} |\lambda|^n \prod_{v \in V_f^{od}(\tau)} |Z_{h_v, \omega_v \omega'_v}^{\mu_v}| \\ &\quad \cdot 2^{h(2 - \frac{|\Gamma_{v_0}^\psi| + 3|\Gamma_{v_0}^\phi|}{2} - |\Gamma_{v_0}^A|)} \prod_{v \notin V_f(\tau)} 2^{(h_v - h_{v'}) (2 - \frac{|\Gamma_v^\psi| + 3|\Gamma_v^\phi|}{2} - |\Gamma_v^A| + c|\lambda|)}, \end{aligned} \quad (4.3.113)$$

where v' is the vertex immediately preceding v on the tree τ , and hence $h_v - h_{v'} > 0$ (in fact, $h_v - h_{v'} = 1$). This bound however *does not* imply summability in the scale labels of the nontrivial vertices for all Γ , since the factor multiplying $h_v - h_{v'}$ may be positive; this happens when

$$(|\Gamma_v^\psi|, |\Gamma_v^\phi|, |\Gamma_v^A|) = \begin{cases} (2, 0, 0) \\ (4, 0, 0) \\ (2, 0, 1) \end{cases} \quad (4.3.114)$$

If (v, Γ_v) is such that the first case is true, then it is called a *relevant* cluster; for the second or third case, it is called *marginal*; otherwise it is called *irrelevant*.

Remark 4.3.10. Another potentially dangerous case would be $(|\Gamma_v^\psi|, |\Gamma_v^\phi|, |\Gamma_v^A|) = (1, 1, 0)$, but this cannot occur in an internal vertex, since it contributes to the local part, and hence it contributes only to the leaves. Moreover, cases with $|\Gamma_v^\psi| = 0$ also cannot occur in internal vertices other than v_0 .

We take care of this apparent non-summability by reintroducing the action of the operator $\mathfrak{A}$ in each single-scale expectation. Equation (4.3.101) becomes

$$V_{\Gamma_v}^{(h_v-1)}[\tau_v] = \int_{\beta, L} d\underline{X}_{\Gamma_v} \tilde{\psi}_{\Gamma_v^\psi}^{(\leq h_v-1)} \phi_{\Gamma_v^\phi} A_{\Gamma_v^A} \mathfrak{A} W_{\Gamma_v}^{(h_v-1)}[\underline{X}_{\Gamma_v}; \tau_v] \quad (4.3.115)$$

for all internal vertices v . Then, proceeding as in Section 3.14 of [15], one may show that $W_{\Gamma}^{(h)}[\tau]$ admits the improved bound

$$\|W_{\Gamma}^{(h)}[\tau]\|_1 \leq C^{n+m} |\lambda|^n \left[\prod_{v \in V_f^{od}(\tau)} |Z_{h_v, \omega_v \omega_{v'}}^{\mu_v}| \right] 2^{hD_{v_0}} \prod_{v \notin V_f(\tau)} 2^{(h_v - h_{v'})D_v} \quad (4.3.116)$$

with D_v being the (renormalised) *scaling dimension* of cluster (v, Γ_v) , defined as

$$D_v := 2 - \frac{|\Gamma_v^\psi| + 3|\Gamma_v^\phi|}{2} - |\Gamma_v^A| - z_v + c|\lambda| \quad (4.3.117)$$

with the improvement z_v given by

$$z_v = \begin{cases} 2 & \text{if } (|\Gamma_v^\psi|, |\Gamma_v^\phi|, |\Gamma_v^A|) = (2, 0, 0) \\ 1 & \text{if } (|\Gamma_v^\psi|, |\Gamma_v^\phi|, |\Gamma_v^A|) = (4, 0, 0) \\ 1 & \text{if } (|\Gamma_v^\psi|, |\Gamma_v^\phi|, |\Gamma_v^A|) = (2, 0, 1) \\ 0 & \text{otherwise} \end{cases}$$

This ensures that, for $|\lambda|$ small enough, $D_v < 0$ for all internal vertices v , so that summation over (n, τ, Γ) yields (see [36] for details, specifically Lemma A3 and Section A6.1)

$$\|W_{\Gamma}^{(h)}\|_1 \leq C^m 2^{h(2 - \frac{|\Gamma^\psi| + 3|\Gamma^\phi|}{2} - |\Gamma^A| - c|\lambda||\Gamma^A|)} \sum_{n \geq 0} \tilde{C}^n |\lambda|^n \quad (4.3.118)$$

for some constants $C, \tilde{C}$ uniform in β, L , and with $m = |\Gamma^\phi| + |\Gamma^A|$. In a similar way, one can prove that:

$$\begin{aligned} & \frac{1}{\beta L} \int_{\beta, L} d\underline{X} \prod_{i, j \in \Gamma} \|\underline{x}_i - \underline{x}_j\|_{\beta, L}^{n_{ij}} |W_{\Gamma}^{(h)}(\underline{X})| \\ & \leq C_{\{n_{ij}\}}^m 2^{h(2 - \frac{|\Gamma^\psi| + 3|\Gamma^\phi|}{2} - |\Gamma^A| - c|\lambda||\Gamma^A| - \sum_{i, j} n_{ij})} \sum_{n \geq 0} \tilde{C}^n |\lambda|^n. \end{aligned} \quad (4.3.119)$$

Analyticity in λ follows from analyticity of all the effective couplings on scales $\geq h$, and from the convergence of the GN tree expansion. Finally, the bound (4.3.119), combined with equation (4.3.95), also allows to prove that $Q_{h,\omega}^+(\underline{k}') = Q_{1,\omega}^+(\underline{k}')(1 + O(\lambda))$, as anticipated at the end of section 4.3.4. We conclude this section with two important properties of GN trees, both of which will be employed in the next sections.

Remark 4.3.11. (On the external dimension). The term $-c|\lambda||\Gamma^A|$ in the estimate (4.3.119) takes into account the (pessimistic) situation in which all A -endpoints are associated with off-diagonal vertex renormalisations, which diverge with an anomalous exponent; recall (4.3.92). In the application that follows we will be interested in bounding momentum-space correlations where the choice of the external momenta will imply that at most one A -endpoint is off-diagonal.

Remark 4.3.12. (The short memory property). Thanks to the presence of the z_v factors in equation (4.3.116), each branch between two non-trivial vertices $v' \prec v$ comes with a factor $2^{(h_v - h_{v'})D_v}$, where $D_v \leq D = -1 + c|\lambda|$. This implies that *long trees are exponentially suppressed*: if one restricts the sum $\sum_{\tau \in \mathcal{T}_{h,n}}$ to trees having at least one vertex on scale $k > h$, then the final bound is improved by a factor $2^{a(h-k)}$, for $a \in (0, |D|)$. Clearly, the constants appearing will depend on the choice of a . This property is usually referred to as the *short memory property of GN trees*.

Remark 4.3.13. (The continuity property). Consider a tree $\tau \in \mathcal{T}_{h,n}^m$ and denote its value with $\text{Val}(\tau)$. Suppose that $|\text{Val}(\tau)| \leq B_D(\tau)$, where $B_D(\tau)$ is the dimensional bound of the tree, obtained as in (4.3.116), with $D_v \leq D = -1 + c|\lambda|$ for all v . Consider a propagator $\hat{g}_\omega^{(h_v)}$ arising in the truncated expectation associated to a given vertex $v \in \tau$. Suppose that $\hat{g}_\omega^{(h_v)}(\underline{k}')$ is replaced by $\hat{g}_\omega^{(h_v)} + \delta\hat{g}_\omega^{(h_v)}$, with $|\delta\hat{g}_\omega^{(h_v)}(\underline{k}')| \leq C2^{-h_v}2^{ah_v}$ for some $1/2 > a > 0$, and admitting a Gram representation. Let $\text{Val}(\tau) + \delta\text{Val}(\tau)$ be the new value of the tree. The contribution $\delta\text{Val}(\tau)$ can be bounded as follows. Let $\gamma_{r \rightarrow v}$ be the path on τ that connects the root r to v . Then, writing $2^{ah_v} = 2^{ah} \prod_{v \in \gamma_{r \rightarrow v_0}} 2^a$ and using that $D_v \leq D$, we have:

$$|\delta\text{Val}(\tau)| \leq 2^{ah} B_{D+a}(\tau). \quad (4.3.120)$$

in a similar fashion, suppose that $Z_{h_v,\omega}/Z_{h_v-1,\omega}$ is replaced by $Z_{h_v,\omega}/Z_{h_v-1,\omega} + \delta z_{h_v,0,\omega}$, with $|\delta z_{h_v,0,\omega}| \leq C2^{ah_v}$, or suppose that the effective coupling $\lambda_{h_v,\omega\omega'}$ is replaced by $\lambda_{h_v,\omega\omega'} + \delta\lambda_{h_v,\omega\omega'}$, with $|\delta\lambda_{h_v,\omega\omega'}| \leq C2^{ah_v}$. Then, the new value is $\text{Val}(\tau) + \delta\text{Val}(\tau)$, with $\delta\text{Val}(\tau)$ again satisfying the bound (4.3.120). We shall refer to this property as the *continuity property of GN trees*.

Finally, the representation of the effective potential can be made even more explicit by using the tree expansion for the effective potential on scale 0. This tree expansion is similar to the one described above, but much simpler (the ultraviolet regime is superrenormalisable). It is completely standard and largely model independent, see *e.g.* [41, 42]. Since we will use this more explicit representation later on, let us briefly describe it.

In the ultraviolet regime, the single-scale propagator $\hat{g}^{(h)}$ is supported for momenta such that $2^{h-1} \leq |k_0| \leq 2^{h+1}$, $h = 0, 1, \dots, N_0$. In configuration space, it satisfies the estimate:

$$\|g^{(h)}(\underline{x} - \underline{y})\| \leq \frac{C_m}{1 + 2^{hm}|\underline{x}_0 - \underline{y}_0|_\beta^m + |\underline{x} - \underline{y}|_L^m}. \quad (4.3.121)$$

The single-scale propagator is not renormalised; in particular, there is no rescaling of the fields, in contrast to the infrared regime. The trees have endpoints on scale $\leq N_0$. The

endpoints on scale N_0 correspond to the bare quartic interaction or to the bare source terms. The endpoints on scale $h < N_0$ have $|\Gamma_v| = |\Gamma_v^\psi| = 2$; they can actually be “unfolded” in trivial trees with root on scale h and endpoint on scale N_0 . The value of the corresponding kernel is the tadpole graph, evaluated with the propagator $g^{[h, N_0]} := \sum_{h'=h}^{N_0} g^{(h')}$; it is finite uniformly in $h \geq 0$ and N_0 and of order λ . Finally, the scaling dimension associated with a vertex v that is not an endpoint is:

$$D_v = (1 - n_v) \quad (4.3.122)$$

where n_v is the number of endpoints following the vertex v on the tree. This can be written as $n_v = n_v^\psi + n_v^A$, where $n_v^A = |\Gamma_v^A|$ and n_v^ψ takes into account the other two types of endpoints (quadratic or quartic in the fermions, and independent of A). Observe that by construction $D_v \leq -1$: the case $n_v = 1$ and $|\Gamma_v^A| = 0$ cannot take place, since the trivial subtrees are taken into account in the definition of endpoints. With these ingredients, it is clear that the estimate (4.3.116) can be replaced by a similar bound, for trees that are now extended all the way until the scale N_0 , and with scaling dimension equal to (4.3.122) if the vertex is on scale $h_v \geq 0$, or equal to (4.3.117) if the vertex is on scale $h_v < 0$. In the following, we shall denote by $\tilde{\mathcal{T}}_{h,n}^m$ the set of trees that extend in the ultraviolet regimes, with root on scale h , with n endpoints independent of A , and with m external source terms.

4.3.6 The correlation functions

Thanks to the previous Section, we may write the GN expansion for the generating functional $\mathcal{W}_{\beta,L}$. Indeed, we obtain

$$\mathcal{W}_{\beta,L}(\phi, A) = \mathcal{W}_{\beta,L}^{(\text{u.v.})}(\phi, A) + \sum_{h=h_\beta}^0 \sum_{n,m \geq 0} \sum_{\tau \in \mathcal{T}_{h,n}^m} \sum_{\substack{\mathbf{\Gamma} \in \mathcal{L}_\tau \\ \Gamma_{v_0}^\psi = \emptyset}} \mathcal{W}_{\mathbf{\Gamma}}^{(h)}[\phi, A; \tau] \quad (4.3.123)$$

$$\mathcal{W}_{\mathbf{\Gamma}}^{(h)}[\phi, A; \tau] = \int_{\beta,L} d\underline{X}_{I_{v_0}} \phi_{\Gamma_{v_0}^\phi} A_{\Gamma_{v_0}^A} W_{\mathbf{\Gamma}}^{(h)}[\underline{X}_{\Gamma_{v_0}}; \tau],$$

where $\mathcal{W}_{\beta,L}^{(\text{u.v.})}(\phi, A)$ collects the contribution of trees with root on positive scales (purely ultraviolet contributions). In particular, since

$$\langle \mathbf{T} \hat{n}_{\underline{p}_1}; \dots; \hat{n}_{\underline{p}_{m-1}}; \hat{j}_{\mu, \underline{p}_m} \rangle_{\beta,L} = (\beta L)^m \frac{\partial^m \mathcal{W}_{\beta,L}(\phi, A)}{\partial \hat{A}_{-\underline{p}_1}^0 \dots \partial \hat{A}_{-\underline{p}_{m-1}}^0 \partial \hat{A}_{-\underline{p}_m}^\mu} \bigg|_{A=\phi=0}, \quad (4.3.124)$$

we have:

$$\langle \mathbf{T} \hat{n}_{\underline{p}_1}; \dots; \hat{j}_{\mu, \underline{p}_m} \rangle_{\beta,L} = \langle \mathbf{T} \hat{n}_{\underline{p}_1}; \dots; \hat{j}_{\mu, \underline{p}_m} \rangle_{\beta,L}^{(\text{u.v.})} + \langle \mathbf{T} \hat{n}_{\underline{p}_1}; \dots; \hat{j}_{\mu, \underline{p}_m} \rangle_{\beta,L}^{(\text{i.r.})} \quad (4.3.125)$$

where the first term takes into account the purely ultraviolet contributions, and the second term all the other trees:

$$\langle \mathbf{T} \hat{n}_{\underline{p}_1}; \dots; \hat{j}_{\mu, \underline{p}_m} \rangle_{\beta,L}^{(\text{i.r.})} = \sum_{h=h_\beta}^0 \sum_{n=0}^\infty \sum_{\tau \in \mathcal{T}_{h,n}^m} \sum_{\mathbf{\Gamma} \in \mathcal{L}_\tau} \widehat{W}_{\mathbf{\Gamma}}^{(h)}[\tau], \quad (4.3.126)$$

where:

$$\widehat{W}_{\Gamma}^{(h)}[\tau] := \int_{\beta, L} d\underline{X}_{I_{v_0}} W_{\Gamma}^{(h)}[\underline{X}_{\Gamma_{v_0}}; \tau] \prod_{\substack{v \in V_f(\tau) \\ v \text{ A-type}}} e^{-i\underline{p}_v \cdot \underline{x}_v} . \quad (4.3.127)$$

In equation (4.3.126), and until the end of this Section, the set $\mathcal{T}_{h,n}^m$ denotes

$$\mathcal{T}_{h,n}^m = \{ \tau \mid \tau \text{ has: root scale } h; \ n \ \lambda, \nu\text{-leaves, } 0 \ \phi\text{-leaves, } m-1 \ A^0\text{-leaves, } 1 \ A^\mu\text{-leaf} \}$$

and the set $\mathcal{L}_\tau$ parameterises labellings of τ with no fermionic fields ψ at the first vertex $v_0 \in V(\tau)$ and no external fields ϕ , namely

$$\mathcal{L}_\tau := \{ \Gamma \text{ labelling of } \tau \mid \Gamma_{v_0} \equiv \Gamma_{v_0}^A \} .$$

Notice that the labelling Γ is such that an A -type leaf v , having label $-\underline{p}_v$, carries a factor $e^{-i\underline{p}_v \cdot \underline{x}_v}$ ($\underline{x}_v$ being the integration coordinate associated to v), which allows to see (4.3.127) as a Fourier transform.

Remark 4.3.14. Recall that $\underline{p}_i = (\eta_\beta, p_i)$ for $i = 1, \dots, m$, and that $\sum_{i=1}^m \underline{p}_i = 0$, hence $\|\underline{p}_j\| \geq \eta \geq 2^{h_\eta}$ for all $j = 1, \dots, m$. Here $h_\eta := \lfloor \log_2 \eta_\beta \rfloor$, but for β large one may think of it as essentially equal to $\lfloor \log_2 \eta \rfloor$. In what follows, we shall use this identification freely.

Bound for the correlation functions

We wish to find an L^∞ -estimate for $\widehat{W}_{\Gamma}^{(h)}[\tau]$ in terms of η, n, h, τ and Γ .

Proposition 4.3.15 (Bound for density correlators). *Let $m \geq 3$, and let $\underline{p}_i = (\eta_\beta, p_i)$ for all $i = 1, \dots, m-1$, with $|p_i| \leq B|\theta|$. Then, there exists positive constants C, c, ε such that, for β, L large enough, for any $\mu \in \{0, 1\}$:*

$$\frac{1}{\beta L} |\langle \mathbf{T} \hat{n}_{\underline{p}_1}; \dots; \hat{n}_{\underline{p}_{m-1}}; \hat{j}_{\mu, \underline{p}_m} \rangle_{\beta, L}| \leq m! C^m \eta^{2-m} (1 + \mathbf{1}_{m \geq m(\theta)} |\theta|^{\varepsilon/2+c|\lambda|} \eta^{-c|\lambda|}) , \quad (4.3.128)$$

with $m(\theta) = K|\theta|^{-1}$ and $C = \kappa \min_{\omega, \omega': \omega \neq \omega'} |k_F^\omega - k_F^{\omega'}|$ for some $\kappa > 0$.

Remark 4.3.16. For $|\theta| \leq K\eta$, as assumed in Theorem 4.1.5, and choosing η small enough, the last term in (4.3.128) is subleading, and the bound reads:

$$\frac{1}{\beta L} |\langle \mathbf{T} \hat{n}_{\underline{p}_1}; \dots; \hat{n}_{\underline{p}_{m-1}}; \hat{j}_{\mu, \underline{p}_m} \rangle_{\beta, L}| \leq m! C^m \eta^{2-m} . \quad (4.3.129)$$

The proof of the Proposition will use the following lemma.

Lemma 4.3.17 (Existence of a lower bound for the scale of a cluster with few external fermions). *Let τ be a tree with $j > 0$ A -type leaves carrying momenta $\|\underline{p}_i\| \geq \eta_\beta$ and such that $|\sum_{i=0}^j p_{i,0}| \geq \eta_\beta$. Consider a vertex v between the root and the last vertex with $\Gamma_v^A = j$. Then, if $|\Gamma_v^\psi| = 2, 4$, we have that $h_v \geq h_\eta - 4$.*

Proof. Suppose $|\Gamma_v^\psi| = 2$, then by momentum conservation the monomial associated to v has the general form

$$\widehat{W}_{\omega_v, \omega'_v}^{(h_v)}[\underline{k}, \underline{p}_1, \dots, \underline{p}_j; \tau_v] \hat{\psi}_{\underline{k} - \underline{k}_F^{\omega_v} - \underline{p}_{\text{tot}}, \omega_v}^{(\leq h_v), +} \hat{\psi}_{\underline{k} - \underline{k}_F^{\omega'_v}, \omega'_v}^{(\leq h_v), -}$$

with τ_v being the sub-tree with first vertex v , and where $\underline{p}_{\text{tot}} = \sum_{i=1}^j \underline{p}_i$. By construction, we have that

$$\|\underline{k} - \underline{k}_F^{\omega'_v}\|, \|\underline{k} - \underline{k}_F^{\omega_v} - \underline{p}_{\text{tot}}\| \leq 2^{h_v+1},$$

and using $\|\underline{p}_{\text{tot}} - \underline{k}_F^{\omega_v} + \underline{k}_F^{\omega'_v}\| \geq |p_{\text{tot},0}| \geq 2^{h_\eta}$ we obtain

$$2^{h_\eta} - 2^{h_v+1} \leq \|\underline{k}\| \leq 2^{h_v+1} \implies h_v \geq h_\eta - 2.$$

If $|\Gamma_v^\psi| = 4$, we have

$$\widehat{W}_{\omega_v}^{(h_v)}[\underline{k}, \underline{k}', \underline{q}, \underline{p}_1, \dots, \underline{p}_j; \tau_v] \hat{\psi}_{\underline{k} - \underline{k}_F^{\omega_v} - \underline{p}_{\text{tot}}, \omega_v}^{(\leq h_v), +} \hat{\psi}_{\underline{k} - \underline{k}_F^{\omega'_v} + \underline{q}, \omega'_v}^{(\leq h_v), -} \hat{\psi}_{\underline{k}' - \underline{k}_F^{\omega''}, \omega''}^{(\leq h_v), +} \hat{\psi}_{\underline{k}' - \underline{k}_F^{\omega'''}, \omega'''}^{(\leq h_v), -}$$

and reasoning in an analogous way we obtain $\|\underline{q}\| \geq 2^{h_\eta} - 2^{h_v+2}$; the computation above implies then that $h_v \geq h_\eta - 4$. $\square$

Remark 4.3.18. The argument of the above proof has the following corollary. Let $j = 2$ and suppose that $p_{\text{tot},0} = -m\eta_\beta$. Then, the scale h_v of the corresponding quadratic monomial is such that $h_v \geq \lfloor \log_2 m\eta_\beta \rfloor - 2$. This provides a lower bound on the scale of the endpoint corresponding to the external field $\hat{A}_{\underline{p}_m}^\mu$.

Proof. (of Proposition 4.3.15.) Recall the expansion in equation (4.3.126). To begin, it is easy to check that the purely ultraviolet contributions is bounded in η : from the ultraviolet tree expansion, we have:

$$|\langle \mathbf{T} \hat{n}_{\underline{p}_1}; \dots; \hat{j}_{\mu, \underline{p}_m} \rangle_{\beta, L}^{(\text{u.v.})}| \leq m! C^m. \quad (4.3.130)$$

Consider now the second term in (4.3.126). Since by assumption $|p_i| \leq B|\theta|$ for $i = 1, \dots, m-1$, for $|\theta|$ small enough the marginal couplings associated with the fields $\hat{A}_{\underline{p}_i}^{\nu_i}$, $i = 1, \dots, m-1$, correspond to the vertex renormalisations $Z_{h_v, \omega_v \omega_v}^{\mu_v}$, that is with equal fermionic chiralities, by momentum conservation. These running coupling constants have a bounded flow, recall Eq. (4.3.92). Observe that this does not apply to $\hat{A}_{\underline{p}_m}^{\nu_m}$: for large enough m , this external field might multiply fermions with different chiralities. The corresponding vertex renormalisation might have an unbounded flow, (4.3.92); this is ultimately the reason for the last term in the right-hand side of (4.3.128).

Let us write:

$$\langle \mathbf{T} \hat{n}_{\underline{p}_1}; \dots; \hat{j}_{\mu, \underline{p}_m} \rangle_{\beta, L}^{(\text{i.r.})} = \langle \mathbf{T} \hat{n}_{\underline{p}_1}; \dots; \hat{j}_{\mu, \underline{p}_m} \rangle_{\beta, L}^{(\text{i.r.}), (\text{d})} + \langle \mathbf{T} \hat{n}_{\underline{p}_1}; \dots; \hat{j}_{\mu, \underline{p}_m} \rangle_{\beta, L}^{(\text{i.r.}), (\text{od})} \quad (4.3.131)$$

where the first term takes into account GN trees such that all A -endpoints on scale < 0 multiply multiply fermions with the same chirality (“diagonal endpoints”), while the second term collects the GN trees where $\hat{A}_{\underline{p}_m}^{\nu_m}$ multiplies fermions with different chiralities (“off-diagonal endpoints”). Let us start by considering the first term.

As a first simple step, we employ equation (4.3.116) to see that

$$\frac{1}{\beta L} |\widehat{W}_{\mathbf{r}}^{(h)}[\tau]| \leq 2^{h(2-m)} |L[\tau]| \prod_{v \notin V_f(\tau)} 2^{(h_v - h_{v'}) D_v}, \quad (4.3.132)$$

where we used $|\Gamma_{v_0}^A| = m$, and where $L[\tau]$ denotes the contribution of the leaves, namely

$$L[\tau] := \prod_{v \in V_f(\tau)} \text{val}(v) \quad \text{val}(v) = \begin{cases} \lambda_{h_v, \omega \omega'} & v \text{ } \lambda\text{-type, } h_v < 0 \\ \nu_{h_v, \omega} & v \text{ } \nu\text{-type, } h_v < 0 \\ Z_{h_v, \omega}^\mu & v \text{ } A^\mu\text{-type, } h_v < 0 \\ \|W_{\Gamma_v}^{(0)}\|_1 & h_v = 0 . \end{cases} \quad (4.3.133)$$

Therefore, since $D_v < 0$ for all v , we obtain the thesis easily as long as the root scale h is greater or equal to $h_\eta - 4$, in which case (4.3.132), combined with Proposition 4.3.5, immediately yields

$$\frac{1}{\beta L} \sum_{h=h_\eta-4}^0 \sum_{n=0}^\infty \sum_{\tau \in \mathcal{T}_{h,n}^m} \sum_{\Gamma \in \mathcal{L}_\tau} |\widehat{W}_\Gamma^{(h)}[\tau]| \leq m! C^m \eta^{2-m} \sum_{h=h_\eta-4}^0 2^{(2-m)(h-h_\eta+4)} . \quad (4.3.134)$$

Let us now consider the case $h \leq h_\eta - 5$. Let us call v^* the first vertex, encountered starting from the root, such that all external field leaves are in the sub-tree τ_{v^*} whose first vertex is v^* , and let us denote by h^* its scale label; then, for all v in the path $\gamma_{v_0 \rightarrow v^*}$ going from v_0 to v^* , the number of A -fields is constant, so that

$$D_v = 2 - m - \frac{\Gamma_v^\psi}{2} + c|\lambda|$$

which implies

$$\frac{1}{\beta L} |\widehat{W}_\Gamma^{(h)}[\tau]| \leq 2^{h^*(2-m)} |L[\tau]| \prod_{v \in \gamma_{v_0 \rightarrow v^*}} 2^{-(h_v - h_{v'}) (|\Gamma_v^\psi|/2 - c|\lambda|)} \prod_{v \notin \gamma_{v_0 \rightarrow v^*}} 2^{(h_v - h_{v'}) D_v} . \quad (4.3.135)$$

If $h^* \geq h_\eta - 5$, we do not need to obtain anything better than equation (4.3.135), since $|\Gamma_v^\psi| > 0$ for all vertices $v \in \gamma_{v_0 \rightarrow v^*}$ ensures summability over the labellings of τ .

Let us now consider the case $h^* < h_\eta - 5$. In this case, we need to examine carefully the structure of the GN trees contributing to (4.3.126). We notice that each A -type leaf v with label $(\mu_v, \omega_v, -\underline{p}_v)$ attached to the tree on scale < 0 is associated with a monomial (written in momentum space)

$$Z_{h_v, \omega_v}^{\mu_v} \int_{\beta, L} d\underline{k}' \hat{\psi}_{\underline{k}', \omega_v}^{(\leq h_v), +} \hat{\psi}_{\underline{k}' + \underline{p}_v, \omega_v}^{(\leq h_v), -} ,$$

which implies

$$2^{h_v+2} \geq \|\underline{p}_v\| \geq 2^{h_\eta} \implies h_v \geq h_\eta - 2 ; \quad (4.3.136)$$

hence, A -type leaves attach at a scale (roughly) at least h_η : at least one of the ψ fields appearing in the labelling of each A -type leaf needs to be contracted at a scale greater or equal to (roughly) h_η .

All in all, we may summarise these findings as follows: as graphically represented in Figure 4.3, there exists a collection of $1 \leq s \leq m$ disjoint sub-trees with root v^* containing all of the A -type leaves; if we call them $\{\tau_i\}_{i=1}^s$, we have

$$\Gamma^A[\tau_i] \neq \emptyset \quad \forall i , \quad \bigcup_{i=1}^s \Gamma^A[\tau_i] = \Gamma_{v^*}^A = \Gamma_{v_0}^A .$$

Since $h^* < h_\eta - 5$, and since all A -type endpoints attach at scale at least $h_\eta - 2$, each sub-tree τ_i will have $1 \leq \ell_i \leq |\Gamma^A[\tau_i]|$ internal vertices $\{v_{i,j}^{**}\}_{j=1}^{\ell_i} \subseteq V(\tau_i)$, such that:

1. the associated subtrees contain all of the A -type leaves in τ_i , that is

$$\bigcup_{j=1}^{\ell_i} \Gamma_{v_{i,j}^{**}}^A = \Gamma^A[\tau_i] ;$$

2. $h_{i,j}^{**} := h_{v_{i,j}^{**}} = h_\eta - 5$.

Noticing that condition 1 implies

$$\bigcup_{i=1}^s \bigcup_{j=1}^{\ell_i} \Gamma_{v_{i,j}^{**}}^A = \Gamma_{v^*}^A ,$$

we may number these vertices simply as v_j^{**} with $j = 1, \dots, \ell$, and $s \leq \ell \leq m$.

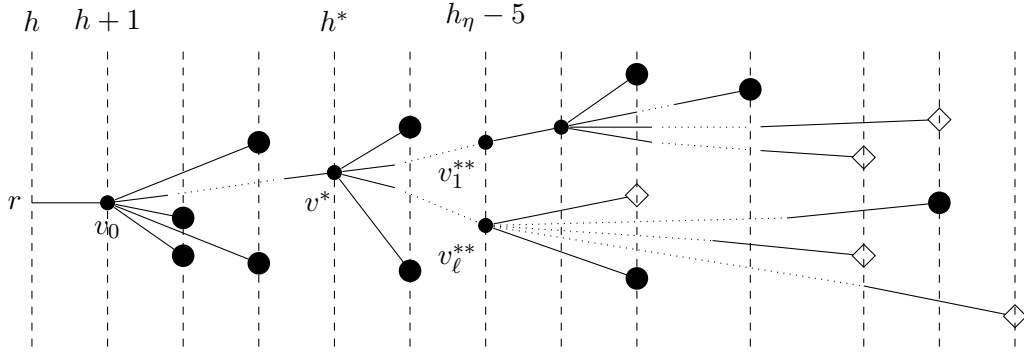


Figure 4.3: A representation of the structure of τ : white diamonds denote A -type leaves, and big circles indicate sub-trees involving only λ or ν type leaves.

Setting $\gamma_{v^* \rightarrow v_j^{**}}$ to be the path from v^* to v_j^{**} , $\gamma_{* \rightarrow **}$ to be the union of all such paths, and by setting $\gamma_{0 \rightarrow **} := \gamma_{v_0 \rightarrow v^*} \cup \gamma_{* \rightarrow **}$, we are led to the following bound for the second product appearing in the left hand side of equation (4.3.135):

$$\begin{aligned} \prod_{v \notin \gamma_{v_0 \rightarrow v^*}} 2^{(h_v - h_{v'}) D_v} &= \prod_{v \in \gamma_{* \rightarrow **}} 2^{(h_v - h_{v'}) D_v} \prod_{v \notin \gamma_{0 \rightarrow **}} 2^{(h_v - h_{v'}) D_v} \\ &= \prod_{j=1}^{\ell} 2^{(h^* - h_j^{**}) |\Gamma_{v_j^{**}}^A|} \prod_{v \in \gamma_{* \rightarrow **}} 2^{(h_v - h_{v'}) (2 - \frac{|\Gamma_v^\psi|}{2} - z_v + c|\lambda|)} \prod_{v \notin \gamma_{0 \rightarrow **}} 2^{(h_v - h_{v'}) D_v} \end{aligned} \quad (4.3.137)$$

where in the second passage we just used telescopic sums to factor out only the external field contribution to D_v . Keeping in mind that $h_j^{**} = h_\eta - 5$ for all $j = 1, \dots, \ell$, and inserting (4.3.137) in (4.3.135), we obtain

$$\begin{aligned} \frac{1}{\beta L} |\widehat{W}_{\mathbf{r}}^{(h)}[\tau]| &\leq |L[\tau]| \prod_{v \in \gamma_{v_0 \rightarrow v^*}} 2^{-(h_v - h_{v'}) (|\Gamma_v^\psi|/2 - c|\lambda|)} \prod_{v \notin \gamma_{0 \rightarrow **}} 2^{(h_v - h_{v'}) D_v} \times \\ &\quad \times 2^{2h^* - m(h_\eta - 5)} \prod_{v \in \gamma_{* \rightarrow **}} 2^{(h_v - h_{v'}) (2 - \frac{|\Gamma_v^\psi|}{2} - z_v + c|\lambda|)}. \end{aligned} \quad (4.3.138)$$

Whereas the first line of (4.3.138) is summable over the scale labels, we see that the product over $v \in \gamma_{* \rightarrow **}$ is possibly not summable in $h_v - h_{v'}$ whenever $|\Gamma_v^\psi| = 2, 4$, since

$$2^{(h_v - h_{v'}) \left(2 - \frac{|\Gamma_v^\psi|}{2} - z_v\right)} = \begin{cases} 2^{h_v - h_{v'}} & \text{if } |\Gamma_v^\psi| = 2, |\Gamma_v^A| > 1 \\ 1 & \text{if } |\Gamma_v^\psi| = 2, |\Gamma_v^A| = 1 \text{ or } |\Gamma_v^\psi| = 4, \end{cases} \geq 1; \quad (4.3.139)$$

this is obviously due to us extracting the contribution relative to the A -type external fields. In order to prove summability in the scale jumps $h_v - h_{v'}$, we would like to show that these instances cannot occur, *i.e.* that

$$\{v \in \gamma_{* \rightarrow **} \mid |\Gamma_v^\psi| \leq 4\} = \emptyset.$$

We proceed by *reductio ad absurdum*: suppose there exists $\bar{v} \in \gamma_{* \rightarrow **}$ such that $|\Gamma_{\bar{v}}^\psi| \leq 4$, then Lemma 4.3.17 would imply $h_{\bar{v}} \geq h_\eta - 4$, but this contradicts $h_v \leq h_\eta - 5$ for all $v \in \gamma_{* \rightarrow **}$. Therefore, we have that

$$\prod_{v \in \gamma_{* \rightarrow **}} 2^{(h_v - h_{v'}) \left(2 - \frac{|\Gamma_v^\psi|}{2} - z_v + c|\lambda|\right)} = \prod_{v \in \gamma_{* \rightarrow **}} 2^{(h_v - h_{v'}) \left(2 - \frac{|\Gamma_v^\psi|}{2} - z_v + c|\lambda|\right)} \mathbf{1}_{|\Gamma_v^\psi| \geq 6}. \quad (4.3.140)$$

Inserting this estimate (4.3.140) into (4.3.138), we are led to

$$\begin{aligned} \frac{1}{\beta L} |\widehat{W}_\Gamma^{(h)}[\tau]| &\leq 2^{5m} |L[\tau]| \prod_{v \in \gamma_{v_0 \rightarrow v^*}} 2^{-(h_v - h_{v'}) |\Gamma_v^\psi|/2} \prod_{v \notin \gamma_{0 \rightarrow **}} 2^{(h_v - h_{v'}) D_v} \times \\ &\quad \times 2^{2h^* - mh_\eta} \prod_{v \in \gamma_{* \rightarrow **}} \left[2^{(h_v - h_{v'}) \left(2 - \frac{|\Gamma_v^\psi|}{2} - z_v + c|\lambda|\right)} \mathbf{1}_{|\Gamma_v^\psi| \geq 6} \right]. \end{aligned} \quad (4.3.141)$$

The first factor in the second line is used to sum over $h^* \leq h_\eta - 5 < h_\eta$, and it allows to obtain the correct dimensional scaling in η . Furthermore, summability over $\Gamma \in \mathcal{L}_\tau$ and $\tau \in \mathcal{T}_{h,n}^m$ is ensured, since all summands involve now factors lesser than 1. In particular, since A -leaves are constrained to attach at scale at least $h_\eta - 2$, choosing any $\varepsilon \in (0, 1 - c|\lambda|)$, we may extract a gain from the minimum depth of τ (which amounts to $h_\eta - 2 - h$) by employing the short-memory property of GN trees; we have

$$\frac{1}{\beta L} \sum'_{\tau \in \mathcal{T}_{h,n}^m} \sum_{\Gamma \in \mathcal{L}_\tau} |\widehat{W}_\Gamma^{(h)}[\tau]| \leq m! C_\varepsilon^{m+n} |\lambda|^n \eta^{2-m} 2^{\varepsilon(h - h_\eta + 2)}, \quad (4.3.142)$$

where the prime denotes the constraint that all A -endpoints on scale < 0 multiply fermions with the same chirality. Summing (4.3.142) over n and combining the resulting bound with (4.3.134), for λ small enough one gets the following estimate for the correlators, for $m \geq 3$:

$$\begin{aligned} &\frac{1}{\beta L} |\langle \mathbf{T} \hat{n}_{p_1}; \dots; \hat{j}_{\mu, p_m} \rangle_{\beta, L}^{(\text{i.r.}), (\text{d})}| \\ &\leq m! \eta^{2-m} \left[C_\varepsilon^m \sum_{h=h_\beta}^{h_\eta-5} 2^{\varepsilon(h - h_\eta + 2)} + \tilde{C}^m \sum_{h=h_\eta-4}^0 2^{(m-2)(h - h_\eta + 4)} \right] \\ &\leq m! C^m \eta^{2-m}, \end{aligned} \quad (4.3.143)$$

for $C = \max\{C_\varepsilon, \tilde{C}\}$.

Consider now the second term in (4.3.131). Recall that this contribution is given by trees with one off-diagonal vertex renormalisation, corresponding to the external field $\hat{A}_{-p_m}^{\nu_m}$. The associated running coupling constant satisfies the bound:

$$|Z_{h_v, \omega_v, \omega'_v}^{\mu_v}| \leq C 2^{-c|\lambda|h_v}. \quad (4.3.144)$$

By Remark 4.3.18, the scale h_v satisfies $h_v \geq \lfloor \log_2 m \eta \rfloor - 2 =: h(m, \eta)$; this scale constraint will ultimately allow to show that the sum of all these trees gives a subleading contribution.

We distinguish three different cases for the scale h of the root: the first is $h \geq h(m, \eta)$; the second is $h(m, \eta) > h \geq h_\eta - 5$; the last is $h \leq h_\eta - 5$. Consider the first case. Denoting by a double prime the condition that there is exactly one off-diagonal endpoint:

$$\frac{1}{\beta L} \sum_{\tau \in \mathcal{T}_{h,n}^m} \sum_{\mathbf{r} \in \mathcal{L}_\tau} |\widehat{W}_{\mathbf{r}}^{(h)}[\tau]| \leq m! C^{m+n} |\lambda|^n 2^{h(2-m-c|\lambda|)} \quad \text{for } h \geq h(m, \eta). \quad (4.3.145)$$

This bound is a direct consequence of (4.3.135), taking into account the bound (4.3.144) for the off-diagonal running coupling constant. Observe that, for $m \geq 3$:

$$\sum_{h \geq h(m, \eta)}^0 2^{h(2-m-c|\lambda|)} \leq C 2^{h(m, \eta)(2-m-c|\lambda|)} \leq K^m m^{2-m-c|\lambda|} \eta^{2-m-c|\lambda|}, \quad (4.3.146)$$

where we used that $h(h, m) \geq \log_2 m + \log_2 \eta - c$. In particular, since $2 - m \leq -1$:

$$\frac{1}{\beta L} \sum_{h \geq h(m, \eta)}^0 \sum_{\tau \in \mathcal{T}_{h,n}^m} \sum_{\mathbf{r} \in \mathcal{L}_\tau} |\widehat{W}_{\mathbf{r}}^{(h)}[\tau]| \leq m! K^{m+n} |\lambda|^n \left(\frac{1}{m}\right)^{1-c|\lambda|} \eta^{2-m-c|\lambda|}. \quad (4.3.147)$$

Next, consider the second case. Here we proceed as in the previous case, with the only addition of using the short-memory property of the GN trees, Remark 4.3.12, to take into account the fact that the off-diagonal endpoint is on scale $\geq h(m, \eta)$, which might be much larger than the root scale h . We have, for $h(m, \eta) > h \geq h_\eta - 5$:

$$\frac{1}{\beta L} \sum_{\tau \in \mathcal{T}_{h,n}^m} \sum_{\mathbf{r} \in \mathcal{L}_\tau} |\widehat{W}_{\mathbf{r}}^{(h)}[\tau]| \leq m! C_\varepsilon^{m+n} |\lambda|^n 2^{h(2-m)} 2^{\varepsilon(h-h(m, \eta))} 2^{-c|\lambda|h(m, \eta)}. \quad (4.3.148)$$

Observe that, for $m \geq 3$:

$$\begin{aligned} \sum_{h=h_\eta}^{h(m, \eta)} 2^{h(2-m)} 2^{\varepsilon(h-h(m, \eta))} &= 2^{h_\eta(2-m)} \sum_{h=h_\eta}^{h(m, \eta)} 2^{(h-h_\eta)(2-m)} 2^{\varepsilon(h-h(m, \eta))} \\ &\leq C^m \eta^{2-m} 2^{(\varepsilon/2)(h_\eta-h(m, \eta))} \\ &\leq K^m \eta^{2-m} \left(\frac{1}{m}\right)^{\varepsilon/2}. \end{aligned} \quad (4.3.149)$$

In particular,

$$\frac{1}{\beta L} \sum_{h=h_\eta-5}^{h(m, \eta)} \sum_{\tau \in \mathcal{T}_{h,n}^m} \sum_{\mathbf{r} \in \mathcal{L}_\tau} |\widehat{W}_{\mathbf{r}}^{(h)}[\tau]| \leq m! K^{m+n} |\lambda|^n \eta^{2-m} \left(\frac{1}{m}\right)^{\varepsilon/2} \left(\frac{1}{\eta m}\right)^{c|\lambda|}. \quad (4.3.150)$$

For the third case, we proceed as in (4.3.142), extracting the short memory factor introduced by the off-diagonal endpoint. We have:

$$\frac{1}{\beta L} \sum_{\tau \in \mathcal{T}_{h,n}^m} \sum_{\mathbf{\Gamma} \in \mathcal{L}_\tau} |\widehat{W}_{\mathbf{\Gamma}}^{(h)}[\tau]| \leq m! C_\varepsilon^{m+n} |\lambda|^n \eta^{2-m} 2^{\varepsilon(h-h(m,\eta))} 2^{-c|\lambda|h(m,\eta)} \quad \text{for } h < h_\eta - 5. \quad (4.3.151)$$

Observe that:

$$\sum_{h \leq h_\eta} 2^{\varepsilon(h-h(m,\eta))} \leq C 2^{\varepsilon(h_\eta-h(m,\eta))} \leq C \left(\frac{1}{m}\right)^\varepsilon. \quad (4.3.152)$$

Therefore, we get:

$$\frac{1}{\beta L} \sum_{h < h_\eta - 5} \sum_{\tau \in \mathcal{T}_{h,n}^m} \sum_{\mathbf{\Gamma} \in \mathcal{L}_\tau} |\widehat{W}_{\mathbf{\Gamma}}^{(h)}[\tau]| \leq m! K^{m+n} |\lambda|^n \eta^{2-m} \left(\frac{1}{m}\right)^\varepsilon \left(\frac{1}{\eta m}\right)^{c|\lambda|}. \quad (4.3.153)$$

The estimate for the second contribution in (4.3.131) is given by the sum of the estimates (4.3.147), (4.3.150), (4.3.153). Next, we observe that, in order for the off-diagonal vertex to be nonzero, we need to have that the spatial component of the momentum $\underline{p}_m$ satisfies $|p_m| \geq c \min_{\omega, \omega': \omega \neq \omega'} |k_F^\omega - k_F^{\omega'}|$, which is only possible for $m \geq K|\theta|^{-1} \equiv m(\theta)$, since $|p_i| \leq B|\theta|$ for all $i = 1, \dots, m-1$. We then have, for $m \geq m(\theta)$:

$$|\langle \mathbf{T} \hat{n}_{\underline{p}_1}; \dots; \hat{j}_{\mu, \underline{p}_m} \rangle_{\beta, L}^{(\text{i.r.}), (\text{od})}| \leq m! C^m \eta^{2-m} |\theta|^{\varepsilon/2 + c|\lambda|} \eta^{-c|\lambda|}. \quad (4.3.154)$$

This bound, combined with (4.3.130) and (4.3.143), concludes the proof of (4.3.128). $\square$

Proposition 4.3.15 provides bounds uniform in β, L large, and with the correct η -dependence, however they are not strong enough to ensure summability of the original (Wick-rotated) Duhamel series, Eq. (4.2.11). Summability could be ensured by assuming that $\|\hat{\mu}\|_1$ is small enough. Remarkably, this is not needed; in the next section, we will improve the m -dependence of these bounds, in a way that will allow to prove summability of the Duhamel series (4.2.11) without any extra assumption on the strength of the external perturbation.

Improved estimates on the Euclidean density-current correlations

Keeping in mind the GN representation (4.3.126), in order to get bounds which are summable in m , we will consider separately the cases with few and many (with respect to m) interactions. Indeed, setting

$$\widehat{W}_{m;n}(\underline{p}_1, \dots, \underline{p}_m) = \sum_{h=h_\beta}^0 \sum_{\tau \in \widetilde{\mathcal{T}}_{h,n}^m} \sum_{\mathbf{\Gamma} \in \mathcal{L}_\tau} \widehat{W}_{\mathbf{\Gamma}}^{(h)}[\tau], \quad (4.3.155)$$

where $\widetilde{\mathcal{T}}_{h,n}^m$ is the set of trees that extend in the ultraviolet regime, recall the end of Section 4.3.5, we rewrite Eq. (4.3.126) as:

$$\begin{aligned} \langle \mathbf{T} \hat{n}_{\underline{p}_1}; \dots; \hat{j}_{\mu, \underline{p}_m} \rangle_{\beta, L} &= \langle \mathbf{T} \hat{n}_{\underline{p}_1}; \dots; \hat{j}_{\mu, \underline{p}_m} \rangle_{\beta, L}^{(\text{u.v.})} + \sum_{n=0}^{n_*} \widehat{W}_{m;n}(\underline{p}_1, \dots, \underline{p}_m) \\ &\quad + \sum_{n=n_*+1}^{\infty} \widehat{W}_{m;n}(\underline{p}_1, \dots, \underline{p}_m), \end{aligned} \quad (4.3.156)$$

with $n_* \equiv n_*(m) := \lfloor \alpha m \rfloor$, $\alpha \in (0, 1)$ constant to be chosen appropriately. Let us start by considering the first sum in (4.3.156). We will show that this term admits an estimate with an improved m -dependence, with respect to the dimensional bounds in Proposition 4.3.15.

Fixing a root scale h , a GN tree $\tau \in \tilde{\mathcal{T}}_{h,n}^m$ and a labelling $\mathbf{\Gamma} \in \mathcal{L}_\tau$, and recalling (4.3.106)-(4.3.115), we can write:

$$\widehat{W}_{\mathbf{\Gamma}}^{(h)}[\tau] = \sum_{\mathbf{T} = \bigcup_{v \notin V_f(\tau)} T_v} \widehat{W}_{\mathbf{\Gamma}}^{(h)}[\tau; \mathbf{T}] \quad (4.3.157)$$

where $\widehat{W}_{\mathbf{\Gamma}}^{(h)}[\tau; \mathbf{T}]$ is constructed via the tree expansion described in Section 4.3.5, keeping fixed the spanning trees T_v 's. Recall that each T_v arises from the use of the BBF formula, Eq. (4.3.109). In the estimate for $\widehat{W}_{\mathbf{\Gamma}}^{(h)}[\tau; \mathbf{T}]$, the propagators associated with the edges of $\mathbf{T}$ are bounded in L^1 norm; in what follows, we shall discuss a modification of this part of the estimate of $\widehat{W}_{\mathbf{\Gamma}}^{(h)}[\tau; \mathbf{T}]$, that will allow us to extract a summable factor in m .

To this end, let us analyse the structure of $\mathbf{T}$ more in detail. Since it is given by the union of the anchored trees $(T_v)_{v \notin V_f(\tau)}$, it follows that $\mathbf{T}$ is itself a tree spanning all endpoints of τ , and $|\mathbf{T}| = m + n - 1$. Since $n \leq n_*(m) < m$ by assumption, we see that any spanning tree $\mathbf{T}$ will have a certain number of *chains* contained in it: by chain we mean the graph that is formed by only connecting A -vertices. If $n = 0$, the only admissible graphs are (closed) chains. In general, if $n \neq 0$, there exists a set of chains $\mathbf{C}_{\mathbf{T}} \subseteq \mathbf{T}$ with N_c connected components such that

$$\mathbf{T} = \mathbf{C}_{\mathbf{T}} \cup (\mathbf{T} \setminus \mathbf{C}_{\mathbf{T}}) \quad \mathbf{C}_{\mathbf{T}} := \prod_{j=1}^{N_c} \mathbf{C}_j, \quad (4.3.158)$$

and where each chain $\mathbf{C}_j$ is obtained by considering the maximum number of consecutive edges between only A -type leaves along $\mathbf{T}$ (see Fig. 4.4 for an example). Thus, the number of propagators in a chain with m_j A -vertices is $|\mathbf{C}_j| = m_j - 1$. It should be noted that the number of chains is bounded as $1 \leq N_c \leq \max\{2n, 1\}$.

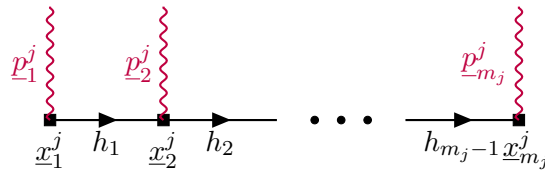


Figure 4.4: An example of a chain $\mathbf{C}_j$ with m_j A -leaves: external momenta injections are denoted by wiggly lines, and the scale of the i -th edge ℓ_i on the chain is indicated with h_i . We have indicated also the position variables associated to each density leaf.

Notice that the number of chains N_c and their lengths $m_j - 1$ depend on the triple $(\tau, \mathbf{\Gamma}, \mathbf{T})$, with the following additional constraint: by construction we have that

$$|\mathbf{C}_{\mathbf{T}}| = \sum_{j=1}^{N_c} |\mathbf{C}_j| \equiv \sum_{j=1}^{N_c} (m_j - 1) = m - N_c,$$

and therefore $|\mathbf{T} \setminus \mathbf{C}_{\mathbf{T}}| = n - 1 + N_c$. Each $\mathbf{C}_j$ may be parameterised (see Fig. 4.4) by the ordered set of leaves connected by it, and these in turn are identified by the ordered external momenta m_j -tuple $(\underline{p}_1^j, \dots, \underline{p}_{m_j}^j)$ and directed edges $(\ell_i)_{i=1}^{m_j-1}$, with $\ell_i \equiv (\underline{x}_i^j, \underline{x}_{i+1}^j)$.

Therefore, the value of a chain may be written (up to the action of $\mathfrak{R}$) as

$$\prod_{\ell \in \mathbf{C}_j} g_\ell^{(h_\ell)} \equiv \prod_{i=1}^{m_j-1} g^{(h_i)}(\underline{x}_i^j - \underline{x}_{i+1}^j) \quad (4.3.159)$$

with $h_i \equiv h_{\ell_i}$. As we will show below, for the special choice of external momenta that we are considering, in which the temporal part of all but one external momenta are the same and equal to η , the presence of “long chains” will introduce a gain in the final estimate which is *factorial* in their length. In turn, this will imply an improvement in the bounds for the kernels $\widehat{W}_{m;n}$ with $n < n_*(m)$, providing overall summability in m .

Lemma 4.3.19 (Bounding chains). *Let $r \in \mathbb{N}$ and let $\mathbf{C}$ be a chain as described above, with r oriented edges labelled as $\ell_1, \dots, \ell_r$, and scale labels $h_1, \dots, h_r$. Let us indicate with $h^{\min} := \min_i h_i$ the lowest scale; then, for any given choice of scales $\{h_i\}_{i=1}^r$ compatible with the values of the external momenta, and for any subset $G \subseteq \mathbf{C}$ containing at least one line with scale $h^{\min}$, we have*

$$\prod_{\ell \in G} 2^{-h_\ell} \leq \left(\frac{C}{\eta}\right)^{|G|-1} \frac{2^{-h^{\min}}}{(|G|-1)!} . \quad (4.3.160)$$

In particular, by picking $G = \mathbf{C}$ we obtain

$$\prod_{i=1}^r 2^{-h_i} \leq \left(\frac{C}{\eta}\right)^{r-1} \frac{2^{-h^{\min}}}{(r-1)!} .$$

Proof. We start by considering the value of the chain in momentum space. We start by assuming that all propagators in the chain are on scale ≤ 0 . We will comment at the end on the case of chains extending in the ultraviolet regime. Recalling (4.3.159), the value of the chain $\mathbf{C}$ (up to the action of $\mathfrak{R}$) is given by

$$\prod_{\ell \in \mathbf{C}} g_\ell^{(h_\ell)} \equiv \prod_{s=1}^r g_{\omega_s}^{(h_s)}(\underline{x}_s - \underline{x}_{s+1}) \quad (4.3.161)$$

with $h_s \equiv h_{\ell_s}$. Observe that the chain is attached to an incoming and an outgoing line; taking into account these lines, the value of the extended chain is:

$$g_{\ell_{in}}^{(h_{in})}(\underline{x}_{in} - \underline{x}_1) \left[\prod_{\ell \in \mathbf{C}} g_\ell^{(h_\ell)} \right] g_{\ell_{out}}^{(h_{out})}(\underline{x}_{r+1} - \underline{x}_{out}) . \quad (4.3.162)$$

Observe that the coordinates $\underline{x}_1, \dots, \underline{x}_{r+1}$, which are the coordinates of the A -leaves attached to the chain, only appear in (4.3.162), and nowhere else in the evaluation of the GN tree. Let us consider (4.3.162), multiplied by the phase factors $e^{-i\underline{p}_i \cdot \underline{x}_i}$ attached to the external fields, $i = 1, \dots, r+1$. Using the Fourier representation for the single-scale propagator

$$g_\omega^{(h)}(\underline{x} - \underline{x}') = \int_{\beta, L} d\underline{k}' e^{i\underline{k}' \cdot (\underline{x} - \underline{x}')} \hat{g}_\omega^{(h)}(\underline{k}') , \quad (4.3.163)$$

and introducing the short-hand

$$\underline{p}_{\leq s} := \sum_{s' \leq s} \underline{p}_{s'} \quad \forall s \in \{1, \dots, r+1\},$$

the Fourier transform of (4.3.162) is:

$$\begin{aligned} & \int_{\beta, L} d\underline{x}_1 \cdots d\underline{x}_{r+1} e^{-i \sum_{s=1}^{r+1} \underline{p}_s \cdot \underline{x}_s} g_{\ell_{in}}^{(h_{in})}(\underline{x}_{in} - \underline{x}_1) \left[\prod_{\ell \in \mathbf{C}} g_{\ell}^{(h_{\ell})} \right] g_{\ell_{out}}^{(h_{out})}(\underline{x}_{r+1} - \underline{x}_{out}) \\ &= e^{-i \underline{p}_{\leq r+1} \cdot \underline{x}_{out}} \int_{\beta, L} d\underline{k}' e^{i \underline{k}' \cdot (\underline{x}_{in} - \underline{x}_{out})} \hat{g}_{\omega_{in}}^{(h_{in})}(\underline{k}') \left[\prod_{s=1}^r \hat{g}_{\omega_s}^{(h_s)}(\underline{k}' + \underline{p}_{\leq s}) \right] \hat{g}_{\omega_{out}}^{(h_{out})}(\underline{k}' + \underline{p}_{\leq r+1}). \end{aligned} \quad (4.3.164)$$

Observe that, by momentum conservation, the momenta of the propagators on the chain are completely determined by the ingoing momentum $\underline{k}'$ and the external momenta of the source terms. By the support properties of the single-scale propagators $\hat{g}^{(h_i)}$ we see that

$$\|\underline{k}' + \underline{p}_{\leq s}\| \geq 2^{h_s-1} \implies 2^{-h_s} \leq \frac{1}{2\|\underline{k}' + \underline{p}_{\leq s}\|}. \quad (4.3.165)$$

In what follows, we shall assume that $p_{i,0} = \eta$ for all $i = 1, \dots, r$. Later we will discuss the adaptation to the case in which one of the momenta among $\underline{p}_1, \dots, \underline{p}_r$ corresponds to $\underline{p}_m$, i.e. it has temporal component equal to $-(m-1)\eta$.

Notice that the first propagator in the chain is supported for the following values of k_0 , $|k_0 + \eta| \leq 2^{h_1+1}$. We will partition this interval as follows. Let us define:

$$\mathcal{A} := \prod_{s=1}^{\bar{n}} \mathcal{A}_s \quad \mathcal{A}_s := \{k_0 \in \mathbb{M}_{\beta} \mid |k_0 + s\eta| < \eta/2\} \quad (4.3.166)$$

with³ $\bar{n} = \min\{r, \lfloor 2^{h_1+1-h_\eta} \rfloor\}$. Then, for all $k_0 \in \mathcal{A}_s$ and for all $l \neq s$, we have:

$$\|\underline{k}' + \underline{p}_{\leq l}\| \geq |k_0 + l\eta| \geq \frac{|l - s|\eta}{2}. \quad (4.3.167)$$

Let us number the lines appearing in G with labels l_α , $\alpha = 1, \dots, |G|$. Consider the estimate for the chain in momentum space, in the support of $\mathcal{A}_s$:

$$1_{\mathcal{A}_s} \prod_{\alpha=1}^{|G|} \frac{1}{\|\underline{k}' + \underline{p}_{\leq l_\alpha}\|}, \quad (4.3.168)$$

for $\mathcal{A}_s$ as in (4.3.166). If $s = l_{\bar{\alpha}}$ for some $\bar{\alpha} = 1, \dots, |G|$, applying (4.3.167) to all the other lines with indices l_α with $\alpha \neq \bar{\alpha}$, we obtain:

$$1_{\mathcal{A}_s} \prod_{\alpha=1}^{|G|} \frac{1}{\|\underline{k}' + \underline{p}_{\leq l_\alpha}\|} \leq \left(\frac{2}{\eta}\right)^{|G|-1} \frac{2^{-h_{l_{\bar{\alpha}}}}}{(\bar{\alpha}-1)! (|G|-\bar{\alpha})!} \leq \left(\frac{4}{\eta}\right)^{|G|-1} \frac{2^{-h^{\min}}}{(|G|-1)!}; \quad (4.3.169)$$

in the first step we used that $|l_\alpha - l_{\bar{\alpha}}| \geq |\alpha - \bar{\alpha}|$, and in the last step we used the binomial theorem combined with the fact that $h_{l_{\bar{\alpha}}} \geq h^{\min}$. Instead, if $s \neq l_\alpha$ for all $\alpha = 1, \dots, |G|$, then $\|\underline{k}' + \underline{p}_{\leq l_\alpha}\| \geq \eta/2$ for k_0 in $\mathcal{A}_s$. We write:

$$1_{\mathcal{A}_s} \prod_{\alpha=1}^{|G|} \frac{1}{\|\underline{k}' + \underline{p}_{\leq l_\alpha}\|} = 1_{\mathcal{A}_s} \left[\prod_{\substack{\alpha=1 \\ l_\alpha \leq s}}^{|G|} \frac{1}{\|\underline{k}' + \underline{p}_{\leq l_\alpha}\|} \right] \left[\prod_{\substack{\alpha=1 \\ l_\alpha > s}}^{|G|} \frac{1}{\|\underline{k}' + \underline{p}_{\leq l_\alpha}\|} \right]. \quad (4.3.170)$$

³Since the width of the intervals is η , the number of disjoint intervals needed to cover the region $|k_0 + \eta| \leq 2^{h_1+1}$ is less than $\lfloor 2^{h_1+1-h_\eta} \rfloor$.

In the first term, we bound the denominator from below as $|k_0 + l_\alpha \eta| \geq \eta(s - l_\alpha - 1/2)$, while in the second term we bound the denominator from below as $|k_0 + l_\alpha \eta| \geq \eta(l_\alpha - s - 1/2)$. Consider the first product, and let us denote by $\bar{\alpha}$ the largest α in the product. We have: $(s - l_\alpha - 1/2) = (s - l_{\bar{\alpha}} - 1/2) + (l_{\bar{\alpha}} - l_\alpha) \geq 1/2 + |\alpha - \bar{\alpha}|$. Concerning the second product, we have $(l_\alpha - s - 1/2) \geq 1/2 + (\alpha - \bar{\alpha})$. Therefore, using these bounds for all $\alpha \neq \bar{\alpha}$, and estimating the term with $\bar{\alpha}$ as

$$\|\underline{k}' + \underline{p}_{\leq l_{\bar{\alpha}}}\| \geq 2^{h_{l_{\bar{\alpha}}}-1} \geq 2^{h^{\min}-1}, \quad (4.3.171)$$

we get:

$$\begin{aligned} 1_{\mathcal{A}_s} \prod_{\alpha=1}^{|G|} \frac{1}{\|\underline{k}' + \underline{p}_{\leq l_\alpha}\|} &\leq \left(\frac{C}{\eta}\right)^{|G|-1} \frac{2^{-h^{\min}}}{(\bar{\alpha}-1)! (|G| - \bar{\alpha})!} \\ &\leq \left(\frac{K}{\eta}\right)^{|G|-1} \frac{2^{-h^{\min}}}{(|G|-1)!}. \end{aligned} \quad (4.3.172)$$

Next, let us consider the case in which $k_0 \in \mathcal{A}^c$. In this case, we have:

$$\|\underline{k}' + \underline{p}_{\leq l}\| \geq |k_0 + l\eta| \geq \begin{cases} (l - 1/2)\eta & \text{if } k_0 \geq -\eta/2 \\ (r + 1/2 - l)\eta & \text{if } k_0 \leq -\eta r - \eta/2, \end{cases} \quad (4.3.173)$$

where the second case holds only if $\bar{n} = r$, *i.e.* if $r < \lfloor 2^{h_1+1-h_\eta} \rfloor$. Let us consider the first case. We apply the bound (4.3.173), together with $l_\alpha - 1/2 \geq \alpha - 1/2 \geq \alpha/2$, to all the lines with $\alpha \neq 1$ and we estimate the line with $\alpha = 1$ by $2^{-h^{\min}}$. We have:

$$1_{\mathcal{A}^c} \prod_{\alpha=1}^{|G|} \frac{1}{\|\underline{k}' + \underline{p}_{\leq l_\alpha}\|} \leq \left(\frac{2}{\eta}\right)^{|G|-1} \frac{2^{-h^{\min}}}{(|G|-1)!}. \quad (4.3.174)$$

Instead, in the second case we bound the propagator with $\alpha = |G|$ by $2^{-h^{\min}}$ and we estimate the product over all the other lines in G using the second of (4.3.173). We obtain:

$$1_{\mathcal{A}^c} \prod_{\alpha=1}^{|G|} \frac{1}{\|\underline{k}' + \underline{p}_{\leq l_\alpha}\|} \leq 2^{-h^{\min}} \prod_{\alpha=1}^{|G|-1} \frac{1}{\eta(r + 1/2 - l_\alpha)} \leq \left(\frac{2}{\eta}\right)^{|G|-1} \frac{2^{-h^{\min}}}{(|G|-1)!}. \quad (4.3.175)$$

Combining the bounds (4.3.169)-(4.3.174) together with equation (4.3.165) only for lines in G , we are then led to

$$\prod_{\alpha=1}^{|G|} 2^{-h_{l_\alpha}} \leq \left(\frac{C}{\eta}\right)^{|G|-1} \frac{2^{-h^{\min}}}{(|G|-1)!} \quad (4.3.176)$$

which is the claim, equation (4.3.160).

Let us now briefly discuss the case in which one of the A -lines carries a momentum $-(m-1)\eta$. Here, we divide the chain G in two parts, called G_1 and G_2 , G_1 being on the left of the A -line carrying momentum $-(m-1)\eta$ and G_2 on the right. We extract the factorial gain separately for each part of the chain. We obtain:

$$\prod_{\alpha=1}^{|G|} 2^{-h_{l_\alpha}} \leq \left(\frac{C}{\eta}\right)^{|G|-1} \frac{2^{-h^{\min}}}{(|G_1|-1)! (|G_2|-1)!} \leq \left(\frac{K}{\eta}\right)^{|G|-1} \frac{2^{-h^{\min}}}{(|G|-1)!}. \quad (4.3.177)$$

To conclude, let us comment on the case of chains extending in the ultraviolet regime. Here, the chirality of the propagators is not defined, so in the above argument the single-scale propagator $g_{\omega_s}^{(h_s)}(\underline{k}')$ is replaced by $g^{(h_s)}(\underline{k}' + \underline{k}_F^\omega)$ with $h_s > 0$. The analysis applies unchanged, using $2^{-h_s} \leq \|k_0 + (\underline{p}_{\leq s})_0\|$. This concludes the proof of Lemma 4.3.19. $\square$

Next, we will use Lemma 4.3.19 to obtain an improved bound for the first term in (4.3.156).

Proposition 4.3.20 (Improved estimate for low perturbative orders). *For any $m \geq 16$,*

$$\frac{1}{\beta L} |\widehat{W}_{m;0}(\underline{p}_1, \dots, \underline{p}_m)| \leq m! C^m \frac{1}{[m/4 - 4]!} \eta^{2-m} (1 + \mathbf{1}_{m \geq m(\theta)} |\theta|^{\varepsilon/2 + c|\lambda|} \eta^{-c|\lambda|}), \quad (4.3.178)$$

and for any $\alpha \in (0, 1/34)$, and any $1 \leq n \leq n_* \equiv \lfloor \alpha m \rfloor$

$$\begin{aligned} & \frac{1}{\beta L} |\widehat{W}_{m;n}(\underline{p}_1, \dots, \underline{p}_m)| \\ & \leq m! C^{m+n} |\lambda|^n \left(\frac{2n}{m - 32n} \right)^{m(1/4 - 8\alpha)} \eta^{2-m} (1 + \mathbf{1}_{m \geq m(\theta)} |\theta|^{\varepsilon/2 + c|\lambda|} \eta^{-c|\lambda|}). \end{aligned} \quad (4.3.179)$$

Proof. The starting point is the identity (4.3.157). Let us focus on the propagators that belong to the chains $\mathbf{C}_j$. One would be tempted to employ Lemma 4.3.19 with $G = \mathbf{C}_j$ to extract a full factorial gain along the chain, namely

$$\prod_{\ell \in \mathbf{C}_j} \|g_\ell^{(h_\ell)}\|_{L^1} \leq C^{m_j} \prod_{i=1}^{m_j-1} 2^{-h_i} \leq \left(\frac{K}{\eta} \right)^{m_j-2} \frac{2^{-h_j^{\min}}}{(m_j - 2)!} \quad (4.3.180)$$

with $h^{\min_j}$ the minimum scale on $\mathbf{C}_j$; however, as it will become clear later in the proof, this will not work, due to the fact that the factor $(1/\eta)^{m_j-2} 2^{-h^{\min}}$ might be much bigger than the factor $\prod_{i=1}^{m_j-1} 2^{-h_i}$. Estimating in this way the propagators that belong to $\mathbf{T}$ would not allow to obtain an estimate for the corresponding GN tree that is summable over the internal scale labels.

In order to deal with this issue, we shall apply the previous reasoning to a proper subset $G^j \subset \mathbf{C}_j$ of *good* lines, defined as follows.

Definition 4.3.21 (Good and bad lines). Let $\widehat{W}_{\mathbf{T}}^{(h)}[\tau; \mathbf{T}]$ as in (4.3.157), and suppose that $\mathbf{C}_j$ is a chain contained in $\mathbf{T}$. We define the set B_v^j of *bad lines* associated to a non-trivial, internal vertex $v \in V(\tau)$ as follows. Let $\Gamma_v^{A,j} \subseteq \Gamma_v^A$ be the set of A -lines corresponding to vertices belonging to the chain $\mathbf{C}_j$ (see Fig. 4.4), contained in the cluster labelled by v . Recall that T_v is the spanning tree of the cluster v .

- a) Suppose that v has no sub-clusters, *i.e.* it is followed only by leaves. If $|T_v \cap \mathbf{C}_j| < 3$, then $B_v^j = T_v \cap \mathbf{C}_j$. Otherwise, $B_v^j = \{\ell, \ell', \ell''\}$ for three arbitrary lines in $T_v \cap \mathbf{C}_j$.
- b) Suppose that v has at least one sub-cluster. The definition is given inductively, as follows:
 - if $|\bigcup_{\bar{v} \succ v} B_{\bar{v}}^j| \geq 3$, then $B_v^j = \emptyset$;
 - if $|\bigcup_{\bar{v} \succ v} B_{\bar{v}}^j| = 3 - k$ and $|T_v \cap \mathbf{C}_j| < k$, then $B_v^j = T_v \cap \mathbf{C}_j$. Otherwise, if $|T_v \cap \mathbf{C}_j| \geq k$ then $B_v^j = \{\ell_1, \dots, \ell_k\}$ for k arbitrary lines in $T_v \cap \mathbf{C}_j$.

We define the set of bad lines on the chain $\mathbf{C}_j$ as:

$$B^j := \bigcup_{v \notin V_f(\tau)} B_v^j, \quad (4.3.181)$$

and the set of *good lines* in the cluster v and on the whole chain as:

$$G_v^j := (T_v \cap \mathbf{C}_j) \setminus B_v^j, \quad G^j := \bigcup_{v \notin V_f(\tau)} G_v^j. \quad (4.3.182)$$

Remark 4.3.22. Observe that if $|G_v^j| \neq 0$ then $|\bigcup_{\bar{v} \succeq v} B_{\bar{v}}^j| \geq 3$.

For the moment, let us ignore the action of the $\mathfrak{R}$ operator. We shall apply an estimate similar to (4.3.180), but only on the good lines. Consider a chain $\mathbf{C}_j$ such that $m_j \geq 5$, and let $s_j \in \{1, \dots, m_j - 1\}$ be a scale minimiser on the chain, *i.e.* $h_{s_j} = h_j^{\min}$. Then:

$$\prod_{\ell \in \mathbf{C}_j} \|g_\ell^{(h_\ell)}\|_{L^1} \leq C^{m_j} \prod_{\ell \in \mathbf{C}_j} 2^{-h_\ell} \leq C^{m_j} \left(\frac{C}{\eta}\right)^{|G^j|-1} \frac{2^{-h_{s_j}}}{(|G^j|-1)!} \prod_{\ell \notin G^j} 2^{-h_\ell}; \quad (4.3.183)$$

we can further rewrite the right-hand side of (4.3.183) as:

$$\prod_{\ell \in \mathbf{C}_j} \|g_\ell^{(h_\ell)}\|_{L^1} \leq C^{m_j} \left[\prod_{\ell \in \mathbf{C}_j} 2^{-h_\ell} \right] \left(\frac{C}{\eta}\right)^{|G^j|-1} \frac{1}{(|G^j|-1)!} \prod_{\substack{\ell \in G^j \\ \ell \neq \ell_{s_j}}} 2^{h_\ell}. \quad (4.3.184)$$

Observe that the line ℓ_{s_j} can be chosen to belong to G^j . This follows from the fact that, since we are assuming that $m_j \geq 5$, the largest cluster (*i.e.* the cluster on the smallest scale) containing the chain $\mathbf{C}_j$ must contain at least one good line. The action of $\mathfrak{R}$ on propagators of a chain is dimensionally equivalent to taking derivatives in real space of the propagators and multiplying them by weights, and it does not hinder the application of Lemma 4.3.19. Its only effect is to change the first factor product in (4.3.184), ensuring that all cluster scaling dimensions are eventually negative.

The rewriting (4.3.184) allows us to recover the standard estimate, up to an extra multiplicative factor. Indeed, we have:

$$\frac{1}{\beta L} |W_{\mathbf{r}}^{(h)}[\tau]| \leq |L[\tau]| C^m \sum_{\mathbf{T}} \left[2^{h(2-m)} \prod_{v \notin V_f(\tau)} \frac{2^{(h_v - h_{v'})D_v}}{s_v!} \right] \prod_{\substack{j=1 \\ m_j \geq 5}}^{N_c} \frac{(C/\eta)^{|G^j|-1}}{(|G^j|-1)!} \prod_{\substack{\ell \in G^j \\ \ell \neq \ell_{s_j}}} 2^{h_\ell}, \quad (4.3.185)$$

where, as usual, D_v indicates the internal dimension of the cluster v , namely

$$\begin{aligned} D_v &= 2 - \frac{|\Gamma_v^\psi|}{2} - |\Gamma_v^A| - z_v + c|\lambda| & \text{if } h_v \leq 0 \\ D_v &= 1 - n_v - |\Gamma_v^A| & \text{if } h_v > 0. \end{aligned} \quad (4.3.186)$$

The last two factors in equation (4.3.185) are the main difference with respect to the previous estimate (4.3.116). Let us focus on the factorial gain. By the multinomial theorem, we have that

$$\prod_{\substack{j=1 \\ m_j \geq 5}}^{N_c} \frac{1}{(|G^j|-1)!} \leq \frac{(N_c - N_c^{\leq 4}) \sum_j^* (|G^j|-1)}{[\sum_j^* (|G^j|-1)]!} \leq \left(\frac{e N_c}{\sum_j^* (|G^j|-1)} \right)^{\sum_j^* (|G^j|-1)} \quad (4.3.187)$$

where $N_c^{\leq 4}$ is the number of chains with $m_j \leq 4$ and the star in the sum enforces the constraint $m_j \geq 5$. We shall derive a lower bound for $|G^j|$ in terms of m_j ; we start by noticing that $|G^j| = m_j - 1 - |B^j|$, so it is enough to find an upper bound for $|B^j|$. The worst possible case is when all m_j leaves are distributed in clusters with $|\Gamma^{A,j}| = 4$, each containing 3 bad lines. Since the number of such clusters is bounded by $\lfloor m_j/4 \rfloor$, we have

$$|B^j| \leq 3\lfloor m_j/4 \rfloor + 3 \quad (4.3.188)$$

where the +3 takes into account the case in which m_j is not divisible by 4. Thus,

$$|G^j| - 1 \geq m_j - 3\lfloor m_j/4 \rfloor - 4.$$

Hence, $\sum_j^* (|G^j| - 1)$ is bounded from below as:

$$\begin{aligned} \sum_{\substack{j=1 \\ m_j \geq 5}}^{N_c} (m_j - 3\lfloor m_j/4 \rfloor - 4) &\geq \sum_{\substack{j=1 \\ m_j \geq 5}}^{N_c} \left(\frac{m_j}{4} - 4 \right) \\ &= \frac{m}{4} - \sum_{\substack{j=1 \\ m_j \leq 4}}^{N_c} \frac{m_j}{4} - 4N_c^{\geq 5} \\ &\geq \frac{m}{4} - N_c^{\leq 4} - 4N_c^{\geq 5} \\ &\geq \frac{m}{4} - 4N_c, \end{aligned} \quad (4.3.189)$$

where $N_c^{\leq b}$ (resp. $N_c^{\geq b}$) indicates the number of chains with length at least (resp. at most) b . The insertion of (4.3.189) into (4.3.187) then yields

$$\prod_{\substack{j=1 \\ m_j \geq 5}}^{N_c} \frac{1}{(|G^j| - 1)!} \leq \left(\frac{4eN_c}{m - 16N_c} \right)^{\sum_j^* (|G^j| - 1)}. \quad (4.3.190)$$

Observe that the number of chains is bounded as $N_c \leq 2n$, and we are assuming that $n \leq \lfloor \alpha m \rfloor$. Thus, for α small enough, the argument of the bracket is less than 1. Hence, we get an upper bound if we replace the exponent by the lower bound in (4.3.189):

$$\prod_{\substack{j=1 \\ m_j \geq 5}}^{N_c} \frac{1}{(|G^j| - 1)!} \leq (4e)^m \left(\frac{N_c}{m - 16N_c} \right)^{\frac{m}{4} - 4N_c}. \quad (4.3.191)$$

We then observe that for $N_c < m/17$ the right-hand side is an increasing function of N_c . Thus, since $N_c \leq 2\lfloor \alpha m \rfloor$, for α small enough we get:

$$\prod_{\substack{j=1 \\ m_j \geq 5}}^{N_c} \frac{1}{(|G^j| - 1)!} \leq f(n; m) := \begin{cases} \frac{1}{\lfloor m/4 - 4 \rfloor!} & n = 0 \\ \left(\frac{2n}{m - 32n} \right)^{\frac{m}{4} - 8\alpha m} & \text{else,} \end{cases} \quad (4.3.192)$$

reducing (4.3.185) to

$$\frac{1}{\beta L} |W_{\mathbf{f}}^{(h)}[\tau]| \leq \tilde{C}^m |L[\tau]| f(n; m) \sum_{\mathbf{T}} \left[2^{h(2-m)} \prod_{v \notin V_f(\tau)} \frac{2^{(h_v - h_{v'})D_v}}{s_v!} \right] \prod_{\substack{j=1 \\ m_j \geq 5}}^{N_c} \prod_{\substack{\ell \in G^j \\ \ell \neq \ell_{s_j}}} 2^{h_\ell - h_\eta}, \quad (4.3.193)$$

with $\tilde{C} = 4eC$. Next, let us consider the last factor in the right-hand side. This factor is a priori dangerous, due to the fact that $h_\ell \geq h_\eta$. We will take care of this bad factor using the negativity of the scaling dimensions. Let us define the set:

$$\mathfrak{G}_v^j := \left\{ \ell \in \bigcup_{v \preceq \bar{v}} T_{\bar{v}} \mid \ell \in \mathbf{C}_j \setminus (B^j \cup \{\ell_{s_j}\}) \right\}, \quad \mathfrak{G}_v = \bigcup_j \mathfrak{G}_v^j. \quad (4.3.194)$$

Observe that $\mathfrak{G}_v^j = \bigcup_{v \preceq \bar{v}} (G_{\bar{v}}^j \setminus \{\ell_{s_j}\})$. Also, observe that, as a consequence of Definition 4.3.21, $|\mathfrak{G}_v^j| = 0$ if $|\Gamma_v^{A,j}| \leq 4$, and in particular $|\mathfrak{G}_v^j| = 0$ if $m_j \leq 4$. In fact, if $|\mathfrak{G}_v^j| \neq 0$ then $|B_v^j| \geq 3$, which means that $|\Gamma_v^{A,j}|$ must be at least 5. Also, observe that $\mathfrak{G}_{v_0} = \bigcup_{j:m_j \geq 5} (G^j \setminus \{\ell_{s_j}\})$.

By a telescopic argument, we rewrite the last factor in (4.3.193) as:

$$\begin{aligned} \prod_{\substack{j=1 \\ m_j \geq 5}}^{N_c} \prod_{\ell \in G^j \setminus \{\ell_{s_j}\}} 2^{h_\ell} &= \prod_{\substack{j=1 \\ m_j \geq 5}}^{N_c} \prod_{v \notin V_f(\tau)} 2^{h_v |G_v^j \setminus \{\ell_{s_j}\}|} \\ &= 2^{h|\mathfrak{G}_{v_0}|} \prod_{v \notin V_f(\tau)} 2^{(h_v - h_{v'})|\mathfrak{G}_v|}, \end{aligned} \quad (4.3.195)$$

which allows to rewrite (4.3.193) as

$$\frac{1}{\beta L} |W_{\mathbf{F}}^{(h)}[\tau]| \leq \tilde{C}^m |L[\tau]| f(n; m) \sum_{\mathbf{T}} \frac{2^{h\tilde{D}_{v_0}}}{\eta^{|\mathfrak{G}_{v_0}|}} \prod_{v \notin V_f(\tau)} \frac{2^{(h_v - h_{v'})\tilde{D}_v}}{s_v!}, \quad (4.3.196)$$

with modified scaling dimension

$$\begin{aligned} \tilde{D}_v &= 2 - \frac{|\Gamma_v^\psi|}{2} - |\Gamma_v^A| + |\mathfrak{G}_v| - z_v + c|\lambda| \quad \text{if } h_v \leq 0 \\ \tilde{D}_v &= 1 - n_v - |\Gamma_v^A| + |\mathfrak{G}_v| \quad \text{if } h_v > 0, \end{aligned} \quad (4.3.197)$$

and modified external dimension

$$\begin{aligned} \tilde{D}_{v_0} &= 2 - m + |\mathfrak{G}_{v_0}| \quad \text{if } h \leq 0 \\ \tilde{D}_{v_0} &= 1 - n - m + |\mathfrak{G}_{v_0}| \quad \text{if } h > 0. \end{aligned} \quad (4.3.198)$$

The key feature that prompted us to extract a factorial gain only from the good lines becomes now clear: since by construction we have that

$$|\mathfrak{G}_v| \leq \Theta(|\Gamma_v^A| - 5) \times (|\Gamma_v^A| - 4), \quad (4.3.199)$$

with Θ the Heaviside function, it follows that $\tilde{D}_v$ is always strictly negative, hence we have preserved summability over the internal scales and labellings of the GN tree. Had we taken as sets of good lines the whole chains, this might not have been the case.

In particular, the external scaling dimension $\tilde{D}_{v_0}$ is also strictly negative: therefore, proceeding as in the proof of Proposition 4.3.15, but with modified scaling dimensions, one is able to show that summation over choices of $(h, \tau, \mathbf{F}, \mathbf{T})$ yields

$$\begin{aligned} \sum_{h=h_\beta}^0 \sum_{\tau \in \tilde{\mathcal{T}}_{h,n}^m} \sum_{\mathbf{F} \in \mathcal{L}_\tau} |L[\tau]| \sum_{\mathbf{T}} \frac{2^{h(2-m+|\mathfrak{G}_{v_0}|)}}{\eta^{|\mathfrak{G}_{v_0}|}} \prod_{v \notin V_f(\tau)} \frac{2^{(h_v - h_{v'})\tilde{D}_v}}{s_v!} \\ \leq C^{m+m} |\lambda|^n \eta^{2-m} (1 + \mathbf{1}_{m \geq m(\theta)} |\theta|^{\varepsilon/2+c|\lambda|} \eta^{-c|\lambda|}). \end{aligned} \quad (4.3.200)$$

The bound (4.3.200) is proved as (4.3.128), replacing $|\Gamma_v^A|$ in the proof of (4.3.128) with $|\Gamma_v^A| - |\mathfrak{G}_v|$. This concludes the proof of Proposition 4.3.20. $\square$

This improved bound is enough to show a summable estimate for the m -point cumulants.

Proposition 4.3.23 (Improved estimate for density correlators). *For $m \geq 16$, there exists $C > 0$, such that for λ sufficiently small (but fixed), and for $\alpha \in (0, 1/34)$, we have that*

$$\begin{aligned} & \frac{1}{\beta L} |\langle \mathbf{T} \hat{n}_{\underline{p}_1}; \dots; \hat{n}_{\underline{p}_{m-1}}; \hat{j}_{\mu, \underline{p}_m} \rangle_{\beta, L}| \\ & \leq m! \eta^{2-m} C^m \left[\frac{1}{[m/4 - 4]!} + |\lambda| \left(\frac{2\alpha}{1 - 32\alpha} \right)^{m(1/4 - 8\alpha)} + \lambda^{\alpha m} \right] R_m(\eta, \theta) \end{aligned} \quad (4.3.201)$$

with

$$R_m(\eta, \theta) = (1 + \mathbf{1}_{m \geq m(\theta)} |\theta|^{\varepsilon/2 + c|\lambda|} \eta^{-c|\lambda|}) . \quad (4.3.202)$$

Proof. The starting point is Eq. (4.3.156). As already commented, the first term in the right-hand side is bounded by $C^m m!$, which is much smaller than the right-hand side of (4.3.201). Then, to estimate the first sum in the right-hand side of (4.3.156) we use Proposition 4.3.20. The contribution with $n = 0$ reproduces the first term in the right-hand side of (4.3.201). The other terms in the sum are bounded as:

$$\frac{1}{\beta L} \left| \sum_{n=1}^{n_*} \widehat{W}_{m;n}(\underline{p}_1, \dots, \underline{p}_m) \right| \leq m! C^m \left(\frac{2\alpha}{1 - 32\alpha} \right)^{m(1/4 - 8\alpha)} \eta^{2-m} R_m(\eta, \theta) \sum_{n=1}^{n_*} C^n |\lambda|^n . \quad (4.3.203)$$

and the thesis is obtained by estimating the sum with $C|\lambda|/(1 - C|\lambda|)$. Finally, consider the second sum on the right-hand side of (4.3.156). Here it is sufficient to proceed as in the proof of Proposition 4.3.15:

$$\begin{aligned} \frac{1}{\beta L} \left| \sum_{n > n_*} \widehat{W}_{m;n}(\underline{p}_1, \dots, \underline{p}_m) \right| & \leq m! C^m \eta^{2-m} R_m(\eta, \theta) \sum_{n > n_*} C^n |\lambda|^n \\ & \leq m! C^{m(1+\alpha)} |\lambda|^{\alpha m} \eta^{2-m} R_m(\eta, \theta) . \end{aligned} \quad (4.3.204)$$

This concludes the proof of the proposition. $\square$

Remark 4.3.24. (i) Recall that the presence of the non-trivial factor $R_m(\eta, \theta)$ is due to the estimate for the GN trees with one off-diagonal A endpoint. If we only restrict to the contribution of these trees, we have:

$$\begin{aligned} & \frac{1}{\beta L} |\langle \mathbf{T} \hat{n}_{\underline{p}_1}; \dots; \hat{n}_{\underline{p}_{m-1}}; \hat{j}_{\mu, \underline{p}_m} \rangle_{\beta, L}^{(\text{i.r.}, (\text{od}))}| \leq C^m m! \eta^{2-m} \\ & \cdot \left[\frac{1}{[m/4 - 4]!} + |\lambda| \left(\frac{2\alpha}{1 - 32\alpha} \right)^{m(1/4 - 8\alpha)} + \lambda^{\alpha m} \right] \mathbf{1}_{m \geq m(\theta)} |\theta|^{\varepsilon/2 + c|\lambda|} \eta^{-c|\lambda|} . \end{aligned} \quad (4.3.205)$$

Plugging this estimate in (4.2.11), we see that the corresponding contribution is summable in m and it is vanishing as $\eta \rightarrow 0^+$, for $|\theta| \leq K\eta$ and for $|\lambda|, \eta$ small enough, as assumed in Theorem 4.1.5.

- (ii) Proposition 4.3.23 allows to prove that the contribution of infrared trees with diagonal A end-points is summable, but so far the bounds are not strong enough to show that the linear response dominates. Actually, without a smallness condition on the perturbation, the sum of the higher-order terms is of the same order of the linear response. In order to prove that the higher-order terms are indeed subleading as $\eta \rightarrow 0$, we will prove a cancellation for the scaling limit of the density-density correlation functions, which is reminiscent of bosonisation.
- (iii) We consider the bound (4.3.201) to be not optimal, but even in this form it will be enough for our purposes. Notice that it reduces to the factorial bounds in Chapter 3 if we set $\lambda = 0$, as expected.

4.4 The reference model

Here we discuss a $1 + 1$ dimensional quantum field theory, called the reference model, that describes the infrared scaling limit of the correlation functions of the lattice model. We will follow the work [61], to which we refer for further details. The key feature of the model is that its correlation functions can be explicitly determined, combining Schwinger-Dyson equations and Ward identities, associated with the (anomalous) chiral symmetry of the QFT. Also, the low-energy behaviour of the lattice correlation functions can be quantitatively captured by the reference model, for a suitable choice of parameters.

Remark 4.4.1. All bare parameters of this reference model QFT should carry a ‘ref’ superscript to distinguish them from the corresponding lattice model ones. In order to avoid cluttering an already heavy notation, we shall omit such superscript for the rest of this section (unless otherwise stated). We will reintroduce it in Section 4.5, when comparing lattice and reference model correlations.

4.4.1 Sets and cutoffs

Let $\beta > 0$, and let $L, \mathfrak{N}_0, \mathfrak{N}_1 \in \mathbb{N}$ such that $\mathfrak{a} := \beta/\mathfrak{N}_0 = L/\mathfrak{N}_1$. We define the space-time lattice of sides β, L and mesh $\mathfrak{a}$ as:

$$\Lambda_{\beta, L, \mathfrak{a}} := \{ \underline{x} = (n_0 \mathfrak{a}, n_1 \mathfrak{a}) \mid 0 \leq n_\mu \leq \mathfrak{N}_\mu - 1 \}. \quad (4.4.1)$$

Also, we define the set of momenta compatible with antiperiodic boundary conditions as:

$$\mathbb{D}_{\beta, L, \mathfrak{a}}^{\mathfrak{a}} := \left\{ \underline{k} = \left(\frac{2\pi}{\beta} \left(m_0 + \frac{1}{2} \right), \frac{2\pi}{L} \left(m_1 + \frac{1}{2} \right) \right) \mid 0 \leq m_\mu \leq \mathfrak{N}_\mu - 1 \right\}. \quad (4.4.2)$$

We will consider Grassmann fields defined on $\Lambda_{\beta, L, \mathfrak{a}}$, satisfying antiperiodic boundary conditions. Let χ be the smooth cutoff function defined in equation (4.3.1). We define its ε -deformation as:

$$\chi^\varepsilon(t) = C_\varepsilon \int_0^\infty ds e^{-|t-s|^2/\varepsilon} \chi(s), \quad (4.4.3)$$

where C_ε is chosen such that $C_\varepsilon \int dt e^{-|t-s|^2/\varepsilon} = 1$. We have that $\lim_{\varepsilon \rightarrow 0} \chi^\varepsilon(t) = \chi(t)$ and $\chi^\varepsilon(t) > 0$ for all t . Furthermore, given $v_\omega \in \mathbb{R} \setminus \{0\}$ for $\omega = 1, \dots, N_f$, we define the norm:

$$|\underline{k}|_\omega := \sqrt{k_0^2 + v_\omega^2 k_1^2} \quad (4.4.4)$$

where the collection $\{v_\omega\}_\omega$ will play the role of bare velocities for the reference model. The set $\mathbb{D}_{\beta,L,\mathbf{a}}^{\mathbf{a}}$ is naturally equipped with periodic boundary conditions. Thus, it is convenient to adapt this norm to $\mathbb{D}_{\beta,L,\mathbf{a}}^{\mathbf{a}}$ by introducing:

$$\|\underline{k}\|_\omega := \inf_{a_1, a_2 \in \mathbb{Z}} |\underline{k} - a_1 \underline{G}_1 - a_2 \underline{G}_2|_\omega, \quad (4.4.5)$$

with $\underline{G}_1 = (2\pi/\mathbf{a}, 0)$, $\underline{G}_2 = (0, 2\pi/\mathbf{a})$. Finally, we shall introduce a function encoding an ultraviolet cutoff, as follows:

$$\chi_N^{\omega,\varepsilon}(\underline{k}) := \chi^\varepsilon(2^{-N}\|\underline{k}\|_\omega). \quad (4.4.6)$$

Observe that the function $\chi_N^\omega(\underline{k}) = \lim_{\varepsilon \rightarrow 0} \chi_N^{\omega,\varepsilon}(\underline{k})$ is supported for momenta $\underline{k}$ such that $\|\underline{k}\|_\omega \leq \frac{3}{2}2^N$. The limit $N \rightarrow \infty$ will be referred to as the ultraviolet limit.

4.4.2 Fields and propagators

To each $\underline{k} \in \mathbb{D}_{\beta,L,\mathbf{a}}^{\mathbf{a}}$, and for each $\omega = 1, \dots, N_f$, we associate Grassmann variables $\hat{\psi}_{\underline{k},\omega}^\pm$. They are extended periodically to the whole $\frac{2\pi}{\beta}(\mathbb{Z} + 1/2) \times \frac{2\pi}{L}(\mathbb{Z} + 1/2)$ as:

$$\hat{\psi}_{\underline{k}+a_1\underline{G}_1+a_2\underline{G}_2,\omega}^\pm := \hat{\psi}_{\underline{k},\omega}^\pm \quad \forall (a_1, a_2) \in \mathbb{Z}^2. \quad (4.4.7)$$

The configuration-space Grassmann field is then:

$$\psi_{\underline{x},\omega}^\pm := \frac{1}{\beta L} \sum_{\underline{k} \in \mathbb{D}_{\beta,L,\mathbf{a}}^{\mathbf{a}}} e^{\mp i \underline{k} \cdot \underline{x}} \hat{\psi}_{\underline{k},\omega}^\pm, \quad (4.4.8)$$

and we notice that it satisfies (β, L) -antiperiodicity in space-time. The relation (4.4.8) can be inverted as:

$$\hat{\psi}_{\underline{k},\omega}^\pm = \mathbf{a}^2 \sum_{\underline{x} \in \Lambda_{\beta,L,\mathbf{a}}} e^{\pm i \underline{k} \cdot \underline{x}} \psi_{\underline{x},\omega}^\pm. \quad (4.4.9)$$

The momentum-space propagator is set to be, for $\underline{k} \in \mathbb{D}_{\beta,L,\mathbf{a}}^{\mathbf{a}}$,

$$\hat{g}_{\omega,\varepsilon,\mathbf{a}}^{(\leq N)}(\underline{k}) := \frac{1}{Z_\omega} \frac{\chi_N^{\omega,\varepsilon}(\underline{k})}{\mathcal{D}_{\omega,\mathbf{a}}(\underline{k})}, \quad \mathcal{D}_{\omega,\mathbf{a}}(\underline{k}) := \frac{i}{\mathbf{a}} \sin(\mathbf{a}k_0) + \frac{v_\omega}{\mathbf{a}} \sin(\mathbf{a}k_1), \quad (4.4.10)$$

where Z_ω, v_ω are free parameters, to be suitably chosen later on. Observe that the denominator in (4.4.10) vanishes for $\underline{k} = (0, 0), (\pi/\mathbf{a}, 0), (0, \pi/\mathbf{a}), (\pi/\mathbf{a}, \pi/\mathbf{a})$ (and their translations by $\underline{G}_i$). However, for $\mathbf{a} > 0$ these points do not belong $\mathbb{D}_{\beta,L,\mathbf{a}}^{\mathbf{a}}$, which means that the propagator is well-defined. Let us define:

$$\hat{g}_\omega^{(\leq N)}(\underline{k}) := \lim_{\varepsilon \rightarrow 0} \lim_{\mathbf{a} \rightarrow 0} \hat{g}_{\omega,\varepsilon,\mathbf{a}}^{(\leq N)}(\underline{k}) = \frac{1}{Z_\omega} \frac{\chi_N^\omega(\underline{k})}{ik_0 + v_\omega k_1}. \quad (4.4.11)$$

This limiting propagator is only singular at $(0, 0)$ (and its translations). For non-zero $\varepsilon, \mathbf{a}$, the propagator is arbitrarily small at $(\pi/\mathbf{a}, 0), (0, \pi/\mathbf{a}), (\pi/\mathbf{a}, \pi/\mathbf{a})$, provided $\mathbf{a}$ is small enough. Thus, in the regime $\mathbf{a} \ll \varepsilon$, the main singular contribution is due to the point $(0, 0)$. Finally, the real-space propagator is:

$$g_{\omega,\varepsilon,\mathbf{a}}^{(\leq N)}(\underline{x}) := \frac{1}{\beta L} \sum_{\underline{k} \in \mathbb{D}_{\beta,L,\mathbf{a}}^{\mathbf{a}}} e^{i \underline{k} \cdot \underline{x}} \hat{g}_{\omega,\varepsilon,\mathbf{a}}^{(\leq N)}(\underline{k}), \quad \forall \underline{x} \in \mathbf{a}\mathbb{Z}^2. \quad (4.4.12)$$

4.4.3 Generating functional of correlations

Let $P_N[d\psi]$ be the Grassmann Gaussian integration with the propagators $\hat{g}_{\omega,\varepsilon,\mathfrak{a}}^{(\leq N)}(\underline{k})$:

$$\int P_N[d\psi] \hat{\psi}_{\underline{k},\omega}^- \hat{\psi}_{\underline{q},\omega'}^+ = \beta L \delta_{\underline{k},\underline{q}} \delta_{\omega,\omega'} \hat{g}_{\omega,\varepsilon,\mathfrak{a}}^{(\leq N)}(\underline{k}) ; \quad (4.4.13)$$

equivalently,

$$\int P_N[d\psi] \psi_{\underline{x},\omega}^- \psi_{\underline{y},\omega'}^+ = \delta_{\omega,\omega'} g_{\omega,\varepsilon,\mathfrak{a}}^{(\leq N)}(\underline{x} - \underline{y}) . \quad (4.4.14)$$

The interaction of the Grassmann QFT is defined as:

$$V(\psi) := \frac{\mathfrak{a}^4}{2} \sum_{\underline{x}, \underline{y} \in \Lambda_{\beta,L,\mathfrak{a}}} \sum_{\omega, \omega'} \lambda_{\omega\omega'} Z_{\omega} Z_{\omega'} v(\underline{x} - \underline{y}) n_{\underline{x},\omega} n_{\underline{y},\omega'} \quad (4.4.15)$$

with $\lambda_{\omega\omega'} \in \mathbb{R}$, $\lambda_{\omega\omega'} = \lambda_{\omega'\omega}$, $\lambda_{\omega\omega} = 0$. The function v satisfies the periodicity condition:

$$v(\underline{x} + a_0 \beta \underline{e}_0 + a_1 L \underline{e}_1) = v(\underline{x}) \quad \forall (a_0, a_1) \in \mathbb{Z}^2, \quad (4.4.16)$$

and also $\hat{v}(0) = 1$. In equation (4.4.15), $n_{\underline{x},\omega} = \psi_{\underline{x},\omega}^+ \psi_{\underline{x},\omega}^-$ is the Grassmann counterpart of the density operator, as usual. Let $\mathbb{D}_{\beta,L}^{\mathfrak{p}}$ be the set of momenta compatible with periodic boundary conditions:

$$\mathbb{D}_{\beta,L}^{\mathfrak{p}} := \left\{ \underline{p} = \left(\frac{2\pi}{\beta} n_0, \frac{2\pi}{L} n_1 \right) \mid (n_0, n_1) \in \mathbb{Z}^2 \right\}. \quad (4.4.17)$$

Then, we can write:

$$v(\underline{x}) = \frac{1}{\beta L} \sum_{\underline{p} \in \mathbb{D}_{\beta,L}^{\mathfrak{p}}} e^{i\underline{p} \cdot \underline{x}} \hat{v}(\underline{p}), \quad \hat{v}(\underline{p}) = \mathfrak{a}^2 \sum_{\underline{x} \in \Lambda_{\beta,L,\mathfrak{a}}} e^{-i\underline{p} \cdot \underline{x}} v(\underline{x}). \quad (4.4.18)$$

We will suppose the function $\hat{v}$ to be the restriction to $\mathbb{D}_{\beta,L}^{\mathfrak{p}}$ of an even, smooth and compactly supported function on $\mathbb{R}^2$, so that

$$\|\underline{x}\|^n |v(\underline{x})| \leq C_n \quad \forall n \in \mathbb{N}, \quad (4.4.19)$$

where $\|\underline{x}\| := \inf_{a_1, a_2 \in \mathbb{Z}} |\underline{x} - a_1 L \underline{e}_1 - a_2 L \underline{e}_2|$. In Fourier space, equation (4.4.15) reads:

$$V(\psi) = \frac{1}{2\beta L} \sum_{\underline{p} \in \mathbb{D}_{\beta,L}^{\mathfrak{p}}} \sum_{\omega, \omega'} \lambda_{\omega\omega'} Z_{\omega} Z_{\omega'} \hat{v}(\underline{p}) \hat{n}_{-\underline{p},\omega} \hat{n}_{\underline{p},\omega'} \quad (4.4.20)$$

where

$$\hat{n}_{\underline{p},\omega} = \mathfrak{a}^2 \sum_{\underline{x} \in \Lambda_{\beta,L,\mathfrak{a}}} e^{-i\underline{p} \cdot \underline{x}} n_{\underline{x},\omega} = \frac{1}{\beta L} \sum_{\underline{k} \in \mathbb{D}_{\beta,L,\mathfrak{a}}^{\mathfrak{a}}} \hat{\psi}_{\underline{k}-\underline{p},\omega}^+ \hat{\psi}_{\underline{k},\omega}^-. \quad (4.4.21)$$

Let us define the source terms as:

$$\begin{aligned} B(\psi; \phi) &:= \sum_{\omega=1}^{N_f} \mathfrak{a}^2 \sum_{\underline{x} \in \Lambda_{\beta,L,\mathfrak{a}}} \left[\psi_{\underline{x},\omega}^+ \mathbf{Q}_{\omega} \phi_{\underline{x},\omega}^- + \phi_{\underline{x},\omega}^+ \bar{\mathbf{Q}}_{\omega} \psi_{\underline{x},\omega}^- \right] \\ C(\psi; A) &:= \sum_{\omega=1}^{N_f} \sum_{\mu=0}^1 \mathfrak{a}^2 \sum_{\underline{x} \in \Lambda_{\beta,L,\mathfrak{a}}} Z_{\omega}^{\mu} A_{\underline{x},\omega}^{\mu} n_{\underline{x},\omega} = \sum_{\omega=1}^{N_f} \sum_{\mu=0}^1 \frac{1}{\beta L} \sum_{\underline{p} \in \mathbb{D}_{\beta,L}^{\mathfrak{p}}} Z_{\omega}^{\mu} \hat{A}_{-\underline{p},\omega}^{\mu} \hat{n}_{\underline{p},\omega} \end{aligned} \quad (4.4.22)$$

where the external fields $A_{\underline{x},\omega}, \phi_{\underline{x},\omega}^{\pm}$ take the form

$$A_{\underline{x},\omega}^{\mu} = \frac{1}{\beta L} \sum_{\underline{p} \in \mathbb{D}_{\beta,L}^p} e^{i\underline{p} \cdot \underline{x}} \hat{A}_{\underline{p},\omega}^{\mu}, \quad \phi_{\underline{x},\omega}^{\pm} := \frac{1}{\beta L} \sum_{\underline{k} \in \mathbb{D}_{\beta,L,a}^a} e^{\mp i\underline{k} \cdot \underline{x}} \hat{\phi}_{\underline{k},\omega}^{\pm}. \quad (4.4.23)$$

Remark 4.4.2. As far as correlations of the reference model are concerned, the constants $Q_{\omega}^{\pm}$ and Z_{ω}^{μ} can be set to 1 without loss of generality. They will become relevant, and will be chosen specifically, when we will perform the comparison of reference model correlations with the corresponding ones of the lattice model, see Propositions 4.5.2-4.5.3 below.

Finally, we define the partition function with source terms as

$$\mathcal{Z}_{\beta,L,a,\varepsilon,N}(\phi, A) := \int P_N[d\psi] e^{-V(\psi) + B(\psi; \phi) + C(\psi; A)}, \quad (4.4.24)$$

and also the generating functional of correlations as

$$\mathcal{W}_{\beta,L,a,\varepsilon,N}(\phi, A) := \log \frac{\mathcal{Z}_{\beta,L,a,\varepsilon,N}(\phi, A)}{\mathcal{Z}_{\beta,L,a,\varepsilon,N}(0, 0)}. \quad (4.4.25)$$

Then, the m -point density-current correlation functions are set to be

$$\prod_{i=1}^m [Z_{\omega_i}^{\mu_i}] \langle \hat{n}_{\underline{p}_1, \omega_1}; \dots; \hat{n}_{\underline{p}_m, \omega_m} \rangle_{\beta,L,a,\varepsilon,N} := (\beta L)^m \frac{\partial^m \mathcal{W}_{\beta,L,a,\varepsilon,N}(\phi, A)}{\partial \hat{A}_{-\underline{p}_1, \omega_1}^{\mu_1} \dots \partial \hat{A}_{-\underline{p}_m, \omega_m}^{\mu_m}} \Big|_{A=\phi=0}, \quad (4.4.26)$$

with $\underline{p}_m := -\sum_{i=1}^{m-1} \underline{p}_i$. At fixed cutoffs, all expressions are well-defined, for $|\lambda_{\omega\omega'}|$ small enough, thanks to the convergence of fermionic cluster expansion. Renormalisation group methods are then used to prove a result that is uniform in all cutoffs.

4.4.4 Sketch of RG analysis of the reference model

The generating functional admits a multiscale expansion similar to the one presented for the lattice model in Section 4.3, but now the scales will run from h_{β} to N ; we shall distinguish two different regimes, according to whether the scale is positive or negative.

Ultraviolet regime. We rewrite the total propagator as its infre-red part plus the sum of single (positive) scale propagators $\hat{g}_{\omega,\varepsilon,a}^{(h)}(\underline{k})$, that is

$$\hat{g}_{\omega,\varepsilon,a}^{(\leq N)}(\underline{k}) = \hat{g}_{\omega,\varepsilon,a}^{[h_{\beta},0]}(\underline{k}) + \sum_{h=1}^N \hat{g}_{\omega,\varepsilon,a}^{(h)}(\underline{k}), \quad \hat{g}_{\omega,\varepsilon,a}^{(h)}(\underline{k}) = \frac{1}{Z_{\omega}} \frac{\chi_h^{\omega,\varepsilon}(\underline{k}) - \chi_{h-1}^{\omega,\varepsilon}(\underline{k})}{\mathcal{D}_{\omega,a}(\underline{k})}, \quad (4.4.27)$$

and we iteratively integrate them out, obtaining

$$\mathcal{W}_{\beta,L,a,\varepsilon,N}(\phi, A) = \mathcal{W}_{\beta,L,a,\varepsilon,N}^{(h)}(\phi, A) + \log \int P_h[d\psi^{(\leq h)}] e^{V^{(h)}(\sqrt{Z}\psi^{(\leq h)}; \phi, A)} \quad (4.4.28)$$

with P_h being the Grassmann Gaussian measure associated to $\{\hat{g}_{\omega,\varepsilon,a}^{(\leq h)}\}_{\omega}$ and $\sqrt{Z}\psi^{(\leq h)}$ a short-hand for $\sqrt{Z_{\omega}}\psi_{\omega}^{(\leq h)}$. The effective interaction $V^{(h)}$ takes the form

$$V^{(h)}(\psi; \phi, A) = \sum_{\Gamma=(\Gamma_{\psi}, \Gamma_{\phi}, \Gamma_A)} \int_{\beta,L} d\underline{X} \psi_{\Gamma_{\psi}}(\underline{X}_{\psi}) \phi_{\Gamma_{\phi}}(\underline{X}_{\phi}) A_{\Gamma_A}(\underline{X}_A) W_{\Gamma}^{(h)}(\underline{X}) \quad (4.4.29)$$

and $\mathcal{W}_{\beta,L,\mathbf{a},\varepsilon,h,N}^{(h)}$ admits an analogous expansion. For $k \in \mathbb{N}$, if we define the intensive weighted k -norms

$$\|W_{\Gamma}^{(h)}\|_{1,k} := \sum_{\substack{\{m_{ij}\}_{i,j=1}^{|\Gamma|} \\ \sum_{ij} m_{i,j}=k}} \frac{\mathbf{a}^{2|\Gamma|}}{|\Lambda_{\beta,L,\mathbf{a}}|} \sum_{\underline{X}} |W_{\Gamma}^{(h)}(\underline{X})| \prod_{i < j} \|\underline{x}_i - \underline{x}_j\|_{\beta,L}^{m_{ij}}, \quad (4.4.30)$$

then, for the case of relevant and marginal kernels, these norms satisfy bounds which are better than the standard ones given in (4.3.62).

Proposition 4.4.3 (Improved estimates for UV kernels, Proposition 8.1 of [61]). *Let us call $\lambda_{\max} := \max_{\omega,\omega'} |\lambda_{\omega\omega'}|$. Then, for $\lambda_{\max}$ small enough uniformly in all cutoff parameters, for all $h \in [0, N]$, there exists $C > 0$ independent of $\beta, L, \mathbf{a}, \varepsilon, N$, such that:*

$$\|W_{\Gamma}^{(h)} - W_{\Gamma}^{(N)}\|_{1,k} \leq C \lambda_{\max} 2^{-kh} 2^{h(D_{\Gamma} - \vartheta_{\Gamma})} \left[\prod_{f \in \Gamma_A} |Z_{\omega(f)}^{\mu(f)}| \right] \left[\prod_{f \in \Gamma_{\phi}} |Q_{\omega(f)}^{\varepsilon(f)}| \right], \quad (4.4.31)$$

where D_{Γ} is the scaling dimension of $W_{\Gamma}^{(j)}$, namely

$$D_{\Gamma} := 2 - \frac{|\Gamma_{\psi}|}{2} - \frac{3|\Gamma_{\phi}|}{2} - |\Gamma_A|$$

and ϑ_{Γ} is given by

$$\vartheta_{\Gamma} = \begin{cases} 2 & \text{if } (|\Gamma_{\psi}|, |\Gamma_{\phi}|, |\Gamma_A|) = (2, 0, 0) \\ 1 & \text{if } (|\Gamma_{\psi}|, |\Gamma_{\phi}|, |\Gamma_A|) = (4, 0, 0), (1, 0, 1), (2, 0, 1) \\ 0 & \text{otherwise.} \end{cases} \quad (4.4.32)$$

These improved estimates suggest, following [10], that it is convenient to localise the relevant and marginal kernels wholly: indeed, for any $h \geq 0$, we set

$$\mathfrak{L}V_{n,m,l}^{(h)}(\psi, \phi, A) := \begin{cases} V_{n,m,l}^{(h)} & \text{if } (n, m, l) = (2, 0, 0), (4, 0, 0), (1, 0, 1), (2, 0, 1) \\ 0 & \text{else.} \end{cases} \quad (4.4.33)$$

Reintroducing temporarily the ‘ref’ superscript, we see that (4.4.27) implies that the bare couplings $(Z_{\omega}^{\text{ref}}, v_{\omega}^{\text{ref}})$ do not need to be updated for positive scales. The relevant and marginal kernels are controlled by (4.4.31), and all scaling dimensions in the GN tree expansion are negative.

Infrared regime. For negative scales $h \in [h_{\beta}, -1]$, one proceeds in the same exact fashion as with the infrared leg of the lattice model described in Section 4.3; one localises relevant and marginal kernels according to Definition 4.3.2, thus obtaining effective couplings $(Z_{h,\omega}^{\text{ref}}, v_{h,\omega}^{\text{ref}}, \lambda_{h,\omega\omega'}^{\text{ref}}, Z_{h,\omega}^{\mu,\text{ref}})$, but no coupling $\nu_{h,\omega}^{\text{ref}}$ is created due to the reference model propagator being odd. The flow is then controlled thanks to the following Proposition, which is a consequence of Theorem 8.5 of [61], in turn based on the analysis for the case of two chiralities [17, 18].

Proposition 4.4.4 (Bounds for the reference model running coupling constants.). *There exists $\bar{\lambda} > 0$ such that for all $\lambda_{\max} := \max_{\omega, \omega'} |\lambda_{\omega\omega'}^{\text{ref}}| < \bar{\lambda}$*

$$\begin{aligned} \left| \frac{Z_{h-1, \omega}^{\text{ref}}}{Z_{h, \omega}^{\text{ref}}} \right| &\leq e^{c\lambda_{\max}}, & |v_{h, \omega}^{\text{ref}} - v_{\omega}^{\text{ref}}| &\leq C\lambda_{\max}, \\ |Z_{h, \omega}^{\mu, \text{ref}}| &\leq C|Z_{\omega}^{\mu, \text{ref}}|, & |\lambda_{h, \omega\omega'}^{\text{ref}} - \lambda_{\omega\omega'}^{\text{ref}}| &\leq C\lambda_{\max}^2, \end{aligned} \quad (4.4.34)$$

for some $c, C > 0$ independent of $\beta, L, \mathfrak{a}, \varepsilon, h, N$, for all $\omega, \omega' = 1, \dots, N_f$, $\mu = 0, 1$, and for all $h_{\beta} \leq h < 0$.

Remark 4.4.5. Notice that the independence of these bounds from β allows to control the dynamical system also in the zero temperature limit $h_{\beta} \rightarrow -\infty$.

All in all, we may summarise these findings for the ultraviolet and infrared regimes in a GN tree expansion for the generating functional of the reference model: we have,

$$\begin{aligned} \mathcal{W}_{\beta, L, \mathfrak{a}, \varepsilon, N}(\phi, A) &= \sum_{h=h_{\beta}}^N \sum_{n, m \geq 0} \sum_{\tau \in \mathcal{T}_{h, n}^{m, \text{ref}}} \sum_{\substack{\Gamma \in \mathcal{L}_{\tau} \\ \Gamma_{v_0}^{\psi} = \emptyset}} \mathcal{W}_{\Gamma}^{(h)}[\phi, A; \tau] \\ \mathcal{W}_{\Gamma}^{(h)}[\phi, A; \tau] &= \int_{\beta, L, \mathfrak{a}} d\underline{X}_{I_{v_0}} \phi_{\Gamma_{v_0}^{\phi}} A_{\Gamma_{v_0}^A} W_{\Gamma}^{(h)}[\underline{X}_{\Gamma_{v_0}}; \tau], \end{aligned} \quad (4.4.35)$$

with

$$\int_{\beta, L, \mathfrak{a}} d\underline{X}_{\Gamma} := \prod_{f \in \Gamma} \mathfrak{a}^2 \sum_{\underline{x}(f) \in \Lambda_{\beta, L, \mathfrak{a}}},$$

and where $\mathcal{T}_{h, m}^{m, \text{ref}}$ denotes the set of GN trees with root scale h , m external field leaves and n leaves of type λ that can attach on any scale between h and N . The set of labellings $\mathcal{L}_{\tau}$ respects the two different localisation protocols for positive and negative scales. The tree-dependent kernels $W_{\Gamma}^{(h)}$ satisfy the GN-type estimate

$$\|W_{\Gamma}^{(h)}[\tau]\|_{1,0} \leq C^{n+m} \lambda_{\max}^n 2^{h(2 - \frac{3}{2}|\Gamma_{v_0}^{\phi}| - |\Gamma_{v_0}^A|)} \prod_{v \notin V_f(\tau)} 2^{(h_v - h_{v'})D_v}, \quad (4.4.36)$$

with internal dimension

$$D_v = \begin{cases} 2 - \frac{|\Gamma_v^{\psi}|}{2} - \frac{3|\Gamma_v^{\phi}|}{2} - |\Gamma_v^A| & \text{if } h_v \geq 0 \\ 2 - \frac{|\Gamma_v^{\psi}|}{2} - \frac{3|\Gamma_v^{\phi}|}{2} - |\Gamma_v^A| - z_v + c\lambda_{\max} & \text{if } h_v \leq -1. \end{cases}$$

We stress that D_v is always strictly negative for all vertices $v \notin V_f(\tau)$: while this is manifest for vertices at negative scale, it is true also for the case $h_v \geq 0$, since thanks to (4.4.33) we have localised all the relevant and marginal kernels.

4.4.5 Ward identities for the m -point correlation functions

In this section we will prove identities for the m -point correlation functions of the reference model, that allow to prove the vanishing of the density-density correlations with $m \geq 3$ after removing all cutoffs. This will follow from the anomalous chiral symmetry

of the QFT, to be discussed here. In what follows, we shall use the following short-hand notation:

$$\begin{aligned}\langle \cdot \rangle_N &\equiv \langle \cdot \rangle_{\beta, L, \mathbf{a}, \varepsilon, N} \\ \mathcal{Z}_N(\phi, A) &\equiv \mathcal{Z}_{\beta, L, \mathbf{a}, \varepsilon, N}(\phi, A) \\ \mathcal{W}_N(\phi, A) &\equiv \mathcal{W}_{\beta, L, \mathbf{a}, \varepsilon, N}(\phi, A).\end{aligned}\tag{4.4.37}$$

We notice that, thanks to the presence of the antiperiodic boundary conditions, the propagator is not singular for $\mathbf{a} > 0$, and the Grassmann algebra is finite dimensional.

Let us consider the chiral local gauge transformation:

$$\psi_{\underline{x}, \omega}^{\pm} \mapsto e^{\pm i \alpha_{\omega}(\underline{x})} \psi_{\underline{x}, \omega}^{\pm} \tag{4.4.38}$$

with $\{\alpha_{\omega}(\underline{x})\}$ a collection of functions on $(\mathbf{a}\mathbb{Z})^2$ such that $\alpha_{\omega}(\underline{x} + n_1 L \underline{e}_1 + n_2 L \underline{e}_2) = \alpha_{\omega}(\underline{x})$ for all chiralities ω . Then, it is possible to check that, given any monomial $Q(\psi^+, \psi^-)$, denoting with $Q_{\alpha}(\psi^+, \psi^-)$ the same monomial under the transformation (4.4.38):

$$\int d\psi Q(\psi^+, \psi^-) = \int d\psi Q_{\alpha}(\psi^+, \psi^-). \tag{4.4.39}$$

Thus, by linearity and by finiteness of the Grassmann algebra, for any function $f(\psi)$ we have:

$$\int d\psi f(\psi) = \int d\psi f_{\alpha}(\psi). \tag{4.4.40}$$

Let:

$$\begin{aligned}\Delta_{\underline{p}, \omega}(\psi) &:= \frac{1}{\beta L} \sum_{\underline{k} \in \mathbb{D}_{\beta, L, \mathbf{a}}^{\mathbf{a}}} \hat{\psi}_{\underline{k} - \underline{p}, \omega}^+ \Delta_{\omega}(\underline{k}, \underline{p}) \hat{\psi}_{\underline{k}, \omega}^- \\ \Delta_{\omega}(\underline{k}, \underline{p}) &:= -Z_{\omega} \frac{\mathcal{D}_{\omega, \mathbf{a}}(\underline{k} - \underline{p})}{\chi_N^{\omega, \varepsilon}(\underline{k} - \underline{p})} + Z_{\omega} \frac{\mathcal{D}_{\omega, \mathbf{a}}(\underline{k})}{\chi_N^{\omega, \varepsilon}(\underline{k})} + Z_{\omega} \mathcal{D}_{\omega, \mathbf{a}}(\underline{p}),\end{aligned}\tag{4.4.41}$$

with $\mathcal{D}_{\omega, \mathbf{a}}(\underline{k}) = i\mathbf{a}^{-1} \sin(\mathbf{a}k_0) + v_{\omega} \mathbf{a}^{-1} \sin(\mathbf{a}k_1)$. Applying the identity (4.4.40) to the partition function, and differentiating with respect to $\hat{\alpha}_{\omega}(\underline{p})$, we get:

$$0 = \int P_N[d\psi] e^{-V(\psi) + C(\psi; A)} \left[Z_{\omega} \mathcal{D}_{\omega, \mathbf{a}}(\underline{p}) \hat{n}_{\underline{p}, \omega} - \Delta_{\underline{p}, \omega}(\psi) \right], \tag{4.4.42}$$

where P_N is the Grassmann integration given by:

$$\begin{aligned}P_N[d\psi] &\propto d\psi e^{-K_N(\psi)} \\ K_N(\psi) &= \frac{1}{\beta L} \sum_{\omega} \sum_{\underline{k} \in \mathbb{D}_{\beta, L, \mathbf{a}}^{\mathbf{a}}} \hat{\psi}_{\underline{k}, \omega}^+ \left[\hat{g}_{\omega, \varepsilon, \mathbf{a}}^{(\leq N)}(\underline{k}) \right]^{-1} \hat{\psi}_{\underline{k}, \omega}^-, \end{aligned}\tag{4.4.43}$$

with

$$g_{\omega, \varepsilon, \mathbf{a}}^{(\leq N)}(\underline{k}) = \frac{1}{Z_{\omega}} \frac{\chi_N^{\omega, \varepsilon}(\underline{k})}{\mathcal{D}_{\omega, \mathbf{a}}(\underline{k})}, \quad \chi_N^{\omega, \varepsilon}(\underline{k}) = \chi^{\varepsilon}(2^{-N} \|\underline{k}\|_{\omega}). \tag{4.4.44}$$

From now on, we shall use the short-hand notation $\hat{g}_{\omega}^{(\leq N)}(\underline{k}) \equiv \hat{g}_{\omega, \varepsilon, \mathbf{a}}^{(\leq N)}(\underline{k})$.

Proposition 4.4.6 (Ward identity for the density-density correlators). *The following identity holds true:*

$$\begin{aligned} & Z_{\omega_1} \mathcal{D}_{\omega_1, \mathbf{a}}(\underline{p}_1) \langle \hat{n}_{\underline{p}_1, \omega_1}; \dots; \hat{n}_{\underline{p}_m, \omega_m} \rangle_N \\ &= -\mathcal{D}_{\omega_1, \mathbf{a}}(\underline{p}_1) \mathfrak{B}_{\omega_1}^N(\underline{p}_1) \sum_{\bar{\omega}} \lambda_{\omega_1 \bar{\omega}} Z_{\bar{\omega}} \hat{v}(\underline{p}_1) \langle \hat{n}_{\underline{p}_1, \bar{\omega}}; \dots; \hat{n}_{\underline{p}_m, \omega_m} \rangle_N \\ &+ \frac{1}{\beta L} \sum_{\underline{k} \in \mathbb{D}_{\beta, L, \mathbf{a}}^{\mathbf{a}}} \Delta_{\omega_1}(\underline{k}, \underline{p}_1) \hat{g}_{\omega_1}^{(\leq N)}(\underline{k} - \underline{p}_1) \hat{g}_{\omega_1}^{(\leq N)}(\underline{k}) \mathcal{R}_{\omega_1}^N(\underline{k}; \underline{\mathbf{p}}_m), \end{aligned} \quad (4.4.45)$$

where $\mathfrak{B}_{\omega}^N(\underline{p})$ is the anomalous bubble diagram,

$$\mathfrak{B}_{\omega}^N(\underline{p}) := -\frac{1}{\beta L} \frac{Z_{\omega}}{\mathcal{D}_{\omega, \mathbf{a}}(\underline{p})} \sum_{\underline{k} \in \mathbb{D}_{\beta, L, \mathbf{a}}^{\mathbf{a}}} \Delta_{\omega}(\underline{k}, \underline{p}) \hat{g}_{\omega}^{(\leq N)}(\underline{k} - \underline{p}) \hat{g}_{\omega}^{(\leq N)}(\underline{k}), \quad (4.4.46)$$

$\underline{\mathbf{p}}_m = (\underline{p}_1, \dots, \underline{p}_m)$, and the error term $\mathcal{R}_{\omega}^N$ satisfies the estimate, for $m \geq 3$:

$$\frac{1}{\beta L} \sup_{\underline{k}: c_1 2^N \leq \|\underline{k}\|_{\omega_1} \leq c_2 2^N} |\mathcal{R}_{\omega}^N(\underline{k}; \underline{\mathbf{p}}_m)| \leq C_{\beta, m} 2^{-\gamma N}, \quad (4.4.47)$$

where $\gamma > 0$ and the constant $C_{\beta, m}$ is independent of $\mathbf{a}, \varepsilon, N$, provided $\mathbf{a}, \varepsilon$ and $\mathbf{a}/\varepsilon$ are small enough.

Corollary 4.4.7 (Vanishing of the density-density correlators). *Let $m \geq 3$. Then, there exists $\gamma > 0$ such that density-density correlators satisfy the following estimate:*

$$\lim_{\varepsilon \rightarrow 0} \lim_{\mathbf{a} \rightarrow 0} \frac{1}{\beta L} |\langle \hat{n}_{\underline{p}_1, \omega_1}; \dots; \hat{n}_{\underline{p}_m, \omega_m} \rangle_N| \leq C_{\beta, m} N 2^{-\gamma N} \quad (4.4.48)$$

with a positive constant $C_{\beta, m}$ independent of N .

Remark 4.4.8. Notice that the bounds (4.4.47), (4.4.48) are not optimal in their β -dependence. With quite some extra effort, one could establish an estimate uniform in β , and not uniform in the external momenta. This improvement will however not be needed in our application, where the external momenta are bounded, and where the $N \rightarrow \infty$ limit is performed at β fixed.

Proof of Corollary 4.4.7. Let us start by estimating the last term in the right-hand side of (4.4.45). Since $\mathcal{R}_{\omega}^N$ is bounded uniformly in $\mathbf{a}, \varepsilon$ small (see Remark B.3), we may take the limit $\mathbf{a} \rightarrow 0$ and then $\varepsilon \rightarrow 0$ as well. In this limit, the function $\Delta_{\omega_1} \hat{g}_{\omega_1}^{(\leq N)} \hat{g}_{\omega_1}^{(\leq N)}$ becomes, by (4.4.41) and (4.4.44):

$$\frac{\chi_N^{\omega_1}(\underline{k} + \underline{p}_1) - \chi_N^{\omega_1}(\underline{k} - \underline{p}_1)}{Z_{\omega_1}^2 D_{\omega_1}(\underline{k})} \chi_N^{\omega_1}(\underline{k}) \quad (4.4.49)$$

with $D_{\omega_1}(\underline{k}) := ik_0 + v_{\omega_1} k_1$ for $\underline{k} \in \mathbb{D}_{\beta, L}^{\mathbf{a}} = \frac{2\pi}{\beta}(\mathbb{Z} + \frac{1}{2}) \times \frac{2\pi}{L}(\mathbb{Z} + \frac{1}{2})$, so that

$$\begin{aligned} & \frac{1}{\beta L} \sum_{\underline{k} \in \mathbb{D}_{\beta, L, \mathbf{a}}^{\mathbf{a}}} \Delta_{\omega_1}(\underline{k}, \underline{p}_1) \hat{g}_{\omega_1}^{(\leq N)}(\underline{k} - \underline{p}_1) \hat{g}_{\omega_1}^{(\leq N)}(\underline{k}) \mathcal{R}_{\omega_1}^N(\underline{k}; \underline{\mathbf{p}}_m) \\ & \xrightarrow{\mathbf{a}, \varepsilon \rightarrow 0} \frac{1}{\beta L} \sum_{\underline{k} \in \mathbb{D}_{\beta, L}^{\mathbf{a}}} \frac{\chi_N^{\omega_1}(\underline{k} + \underline{p}_1) - \chi_N^{\omega_1}(\underline{k} - \underline{p}_1)}{Z_{\omega_1}^2 D_{\omega_1}(\underline{k})} \chi_N^{\omega_1}(\underline{k}) \mathcal{R}_{\omega_1}^N(\underline{k}; \underline{\mathbf{p}}_m), \end{aligned} \quad (4.4.50)$$

where with a slight abuse of notation we renamed $\mathcal{R}_\omega^N$ its $\mathfrak{a}, \varepsilon \rightarrow 0$ limit.

Thanks to the properties of the cutoff function $\chi_N^{\omega_1}$, we notice that, for fixed $\underline{p}_1 \neq \underline{0}$, the function (4.4.49) is supported on very high momenta, *i.e.* for $c_1 2^N \leq \|\underline{k}\|_{\omega_1} \leq c_2 2^N$. Hence, we have:

$$\begin{aligned} & \left| \frac{1}{\beta L} \sum_{\underline{k} \in \mathbb{D}_{\beta, L}^{\mathfrak{a}}} \frac{\chi_N^{\omega_1}(\underline{k} + \underline{p}_1) - \chi_N^{\omega_1}(\underline{k} - \underline{p}_1)}{Z_{\omega_1}^2 D_{\omega_1}(\underline{k})} \chi_N^{\omega_1}(\underline{k}) \mathcal{R}_\omega^N(\underline{k}; \underline{\mathbf{p}}_m) \right| \\ & \leq C_{\beta, m} 2^{-\gamma N} \frac{1}{\beta L} \sum_{\underline{k} \in \mathbb{D}_{\beta, L}^{\mathfrak{a}}} \left| \frac{\chi_N^{\omega_1}(\underline{k} + \underline{p}_1) - \chi_N^{\omega_1}(\underline{k} - \underline{p}_1)}{Z_{\omega_1}^2 D_{\omega_1}(\underline{k})} \chi_N^{\omega_1}(\underline{k}) \right| \\ & \leq \tilde{C}_{\beta, m} 2^{-\gamma N} N, \end{aligned} \quad (4.4.51)$$

where in the first step we employed (4.4.47); the last factor N comes from the logarithmic divergence of the sum. Therefore, dividing left-hand side and right-hand side of (4.4.45) by $Z_{\omega_1} D_{\omega_1}(\underline{p}_1)$, we have:

$$\lim_{\varepsilon \rightarrow 0} \lim_{\mathfrak{a} \rightarrow 0} \left| \sum_{\tilde{\omega}} (T_N^{-1}(\underline{p}_1))_{\omega_1 \tilde{\omega}} \langle \hat{n}_{\underline{p}_1, \tilde{\omega}}; \dots; \hat{n}_{\underline{p}_m, \omega_m} \rangle_N \right| \leq \frac{\tilde{C}_{\beta, m}}{\|\underline{p}_1\|_{\omega_1}} 2^{-\gamma N} N, \quad (4.4.52)$$

where $T_N^{-1}(\underline{p})$ is the matrix:

$$(T_N^{-1}(\underline{p}))_{\omega \omega'} = \delta_{\omega, \omega'} + \mathfrak{B}_\omega^N(\underline{p}) Z_\omega^{-1} \lambda_{\omega \omega'} Z_{\omega'} \hat{v}(\underline{p}). \quad (4.4.53)$$

The final bound (4.4.48) follows from the invertibility of the matrix $T_N^{-1}(\underline{p})$ defined in (4.4.53), for $\lambda_{\max} := \max_{\omega, \omega'} |\lambda_{\omega \omega'}|$ small enough. $\square$

We refer the reader to Appendix B for the proof of Proposition 4.4.6. The identity (4.4.48) will play a key role in the proof of exactness of linear response for the lattice model. In the bosonisation approach to the chiral Luttinger liquid, the identity (4.4.48) as $N \rightarrow \infty$ is a (formal) consequence of the fact that the density operator is described by (the derivative of) a bosonic Gaussian free field. In our setting, this identity is proved rigorously starting from a regularised QFT. The advantage of this setting is that it is robust enough to be used to derive consequences about the original lattice model, at low energies. This will be discussed in the next section.

To conclude this section, we report briefly the structure of the density-density and vertex correlation functions, when all cutoffs are removed and when β, L are sent to infinity.

Proposition 4.4.9 (Density-density correlation and vertex function; Proposition 9.11 of [61]). *Let us set*

$$\langle \cdot \rangle = \lim_{\beta, L \rightarrow \infty} \lim_{N \rightarrow \infty} \lim_{\varepsilon \rightarrow 0} \lim_{\mathfrak{a} \rightarrow 0} \langle \cdot \rangle_N;$$

then, for any $\underline{p} \neq \underline{0}$, the density two-point function takes the form

$$\langle \hat{n}_{\underline{p}, \omega}; \hat{n}_{-\underline{p}, \omega'} \rangle = T_{\omega \omega'}(\underline{p}) \frac{\mathfrak{B}_{\omega'}^N(\underline{p})}{Z_{\omega'}^2}, \quad (4.4.54)$$

and the vertex function is given by

$$\langle \hat{n}_{\underline{p}, \omega}; \hat{\psi}_{\underline{k}, \omega'}^-; \hat{\psi}_{\underline{k} + \underline{p}, \omega'}^+ \rangle = \frac{T_{\omega \omega'}(\underline{p})}{Z_\omega D_\omega(\underline{p})} [\langle \hat{\psi}_{\underline{k}, \omega'}^- \hat{\psi}_{\underline{k}, \omega'}^+ \rangle - \langle \hat{\psi}_{\underline{k} + \underline{p}, \omega'}^- \hat{\psi}_{\underline{k} + \underline{p}, \omega'}^+ \rangle]. \quad (4.4.55)$$

Here

$$\mathfrak{B}_\omega(\underline{p}) = \lim_{\beta, L \rightarrow \infty} \lim_{N \rightarrow \infty} \lim_{\varepsilon \rightarrow 0} \lim_{\mathfrak{a} \rightarrow 0} \mathfrak{B}_\omega^N(\underline{p}) = \frac{1}{4\pi|v_\omega|} \frac{-ip_0 + v_\omega p_1}{ip_0 + v_\omega p_1}, \quad (4.4.56)$$

and

$$T(\underline{p}) := \lim_{\beta, L \rightarrow \infty} \lim_{N \rightarrow \infty} \lim_{\varepsilon \rightarrow 0} \lim_{\mathfrak{a} \rightarrow 0} T_N(\underline{p}) = \frac{1}{1 + \mathfrak{B}(\underline{p})\Lambda_Z \hat{v}(\underline{p})}, \quad (4.4.57)$$

where

$$\Lambda_Z := Z^{-1}\Lambda Z,$$

with $Z = \text{diag } Z_\omega$, and $\Lambda_{\omega\omega'} = \lambda_{\omega\omega'}$.

Proof. We refer to Section 9 of [61] for the proof. It should be noted that the expression (4.4.56) of the relativistic bubble $\mathfrak{B}$ takes a (slightly) different form with respect to the one in [61], due to different Fourier conventions adopted in [70], on which this chapter is based. $\square$

4.5 Proof of Main Theorem

4.5.1 Matching the lattice and reference models

In this section we shall prove that lattice correlators are well-approximated by the corresponding ones in the reference model, with summable bounds on the deviations.

Remark 4.5.1. In this section, we shall denote the reference model expectation, having already performed the $\mathfrak{a} \rightarrow 0$ and $\varepsilon \rightarrow 0$ limits, with $\langle \cdot \rangle_{\beta, L, N}^{\text{ref}}$.

Proposition 4.5.2 (Comparison of correlations). *Let β, L, N be large enough. Calling λ the bare coupling of the lattice model, there exists $\bar{\lambda} > 0$ independent of β, L such that, for all $|\lambda| < \bar{\lambda}$, one may find a choice of the bare couplings of the reference model, satisfying*

$$\begin{aligned} |v_\omega^{\text{ref}} - v_\omega| &= C|\lambda|, & |\lambda_{\omega\omega'}^{\text{ref}} - \lambda| &= C|\lambda|^2, \\ |Z_\omega^{\text{ref}} - 1| &= C|\lambda|, & |Z_\omega^{\mu, \text{ref}} - Z_\omega^\mu| &= C|\lambda|, \end{aligned} \quad (4.5.1)$$

for all ω, ω' , such that the following points hold.

Let $m \geq 3$, $\underline{p}_i = (\eta_\beta, p_i)$ for $i = 1, \dots, m-1$, and set $\underline{p}_m = -\sum_{i=1}^{m-1} \underline{p}_i$. Assume that $|p_i| \leq B|\theta|$, and that $|\theta| \leq K\eta$, as in Theorem 4.1.5. Then, for η small enough, the lattice density-current m -point correlations can be written as

$$\begin{aligned} \langle \mathbf{T} \hat{n}_{\underline{p}_1}; \dots; \hat{n}_{\underline{p}_{m-1}}; \hat{j}_{\mu, \underline{p}_m} \rangle_{\beta, L} &= \sum_{\underline{\omega}} \left[\prod_{i=1}^{m-1} Z_{\omega_i}^{0, \text{ref}} \right] Z_{\omega_m}^{\mu, \text{ref}} \langle \hat{n}_{\underline{p}_1, \omega_1}; \dots; \hat{n}_{\underline{p}_m, \omega_m} \rangle_{\beta, L, N}^{\text{ref}} \\ &\quad + R_{m; \mu}^{\beta, L, N}(\underline{p}_1, \dots, \underline{p}_m); \end{aligned} \quad (4.5.2)$$

the error term admits the following estimate:

$$\frac{1}{\beta L} |R_{m; \mu}^{\beta, L, N}(\underline{p}_1, \dots, \underline{p}_m)| \leq m! C^m \eta^{2-m+\gamma} F(m; \lambda, \alpha), \quad (4.5.3)$$

where

$$F(m; \lambda, \alpha) := \begin{cases} 1 & \text{if } m < 16 \\ \frac{1}{[m/4 - 4]!} + |\lambda| \left(\frac{2\alpha}{1 - 32\alpha} \right)^{(1/4 - 8\alpha)m} + |\lambda|^{\alpha m} & \text{if } m \geq 16 \end{cases} \quad (4.5.4)$$

with $\gamma \in (0, 1)$ and any $\alpha \in (0, 1/34)$, and C uniform in β, L, N .

Proof. As in the proof of Proposition 4.3.15, we start by writing:

$$\begin{aligned} \langle \mathbf{T}\hat{n}_{\underline{p}_1}; \dots; \hat{j}_{\mu, \underline{p}_m} \rangle_{\beta, L} &= \langle \mathbf{T}\hat{n}_{\underline{p}_1}; \dots; \hat{j}_{\mu, \underline{p}_m} \rangle_{\beta, L}^{(\text{u.v.})} \\ &\quad + \langle \mathbf{T}\hat{n}_{\underline{p}_1}; \dots; \hat{j}_{\mu, \underline{p}_m} \rangle_{\beta, L}^{(\text{i.r.}), (\text{d})} + \langle \mathbf{T}\hat{n}_{\underline{p}_1}; \dots; \hat{j}_{\mu, \underline{p}_m} \rangle_{\beta, L}^{(\text{i.r.}), (\text{od})}; \end{aligned} \quad (4.5.5)$$

as noticed already in (4.3.130), (4.3.205), the first and the last term already satisfy the bound (4.5.3), for η small enough. Let us focus on the second term. We begin by recalling the GN expansion of the density correlations, Eq. (4.3.156):

$$\langle \mathbf{T}\hat{n}_{\underline{p}_1}; \dots; \hat{j}_{\mu, \underline{p}_m} \rangle_{\beta, L}^{(\text{i.r.}), (\text{d})} = \sum_{h=h_\beta}^0 \sum_{n=0}^{\infty} \sum'_{\tau \in \tilde{\mathcal{T}}_{h,n}^m} \sum_{\Gamma \in \mathcal{L}_\tau} \widehat{W}_\Gamma^{(h)}[\tau] \quad (4.5.6)$$

with $\tilde{\mathcal{T}}_{h,n}^m$ the set of scale-labelled GN trees extending up to the scale of the UV cutoff, with root-scale h ; $m-1$ leaves of A^0 -type, and one A^μ -leaf; $\mathcal{L}_\tau$ the set of labellings of such trees selecting momenta $\underline{p}_i$ on such leaves, as described in equations (4.3.126)-(4.3.127). The prime in the tree sum enforces the constraint that all A -leaves on scale < 0 are diagonal, that is they involve the same fermionic chiralities.

Splitting $\tilde{\mathcal{T}}_{h,n}^m$ into $\mathcal{T}_{h,n}^{m, (\leq -1)}$, the set of labelled trees τ with all leaves at scales ≤ -1 , and $\tilde{\mathcal{T}}_{h,n}^{m, [0, N_0]}$, accounting for trees having at least one endpoint on a scale between 0 and N_0 , we may rewrite (4.5.6) as

$$\begin{aligned} \langle \mathbf{T}\hat{n}_{\underline{p}_1}; \dots; \hat{j}_{\mu, \underline{p}_m} \rangle_{\beta, L}^{(\text{i.r.}), (\text{d})} &= \sum_{h=h_\beta}^{-1} \sum_{n=0}^{\infty} \sum'_{\tau \in \mathcal{T}_{h,n}^{m, (\leq -1)}} \sum_{\Gamma \in \mathcal{L}_\tau} \widehat{W}_\Gamma^{(h)}[\tau] + E_{m; \mu, 1}^{\beta, L}(\underline{p}_1, \dots, \underline{p}_m) \\ E_{m; \mu, 1}^{\beta, L}(\underline{p}_1, \dots, \underline{p}_m) &= \sum_{h=h_\beta}^0 \sum_{n=0}^{\infty} \sum'_{\tau \in \tilde{\mathcal{T}}_{h,n}^{m, [0, N_0]}} \sum_{\Gamma \in \mathcal{L}_\tau} \widehat{W}_\Gamma^{(h)}[\tau]. \end{aligned} \quad (4.5.7)$$

Now, we substitute the first term on right-hand side with the reference model correlation, obtaining

$$\begin{aligned} \langle \mathbf{T}\hat{n}_{\underline{p}_1}; \dots; \hat{j}_{\mu, \underline{p}_m} \rangle_{\beta, L}^{(\text{i.r.})} &= \sum_{\underline{\omega}} \left[\prod_{i=1}^{m-1} Z_{\omega_i}^{0, \text{ref}} \right] Z_{\omega_m}^{\mu, \text{ref}} \langle \hat{n}_{\underline{p}_1, \omega_1}; \dots; \hat{n}_{\underline{p}_m, \omega_m} \rangle_{\beta, L, N}^{\text{ref}} \\ &\quad + \sum_{i=1}^3 E_{m; \mu, i}^{\beta, L, N}(\underline{p}_1, \dots, \underline{p}_m) \end{aligned} \quad (4.5.8)$$

where (observe that the sum over the GN trees of the reference model only involves diagonal A -leaves):

$$\begin{aligned} E_{m; \mu, 2}^{\beta, L, N}(\underline{p}_1, \dots, \underline{p}_m) &:= \sum_{h=h_\beta}^{-1} \sum_{n=0}^{\infty} \sum'_{\tau \in \mathcal{T}_{h,n}^{m, (\leq -1)}} \sum_{\Gamma \in \mathcal{L}_\tau} \left(\widehat{W}_\Gamma^{(h)}[\tau] - \widehat{W}_\Gamma^{(h), \text{ref}}[\tau] \right) \\ E_{m; \mu, 3}^{\beta, L, N}(\underline{p}_1, \dots, \underline{p}_m) &:= - \sum_{h=0}^N \sum_{n=0}^{\infty} \sum'_{\tau \in \tilde{\mathcal{T}}_{h,n}^{m, [0, N]}} \sum_{\Gamma \in \mathcal{L}_\tau} \widehat{W}_\Gamma^{(h), \text{ref}}[\tau]. \end{aligned} \quad (4.5.9)$$

Notice that $E_{m; \mu, 1}^{\beta, L, N} \equiv E_{m; \mu, 1}^{\beta, L}$ does not depend on the reference model, hence it is independent of the reference model cutoff N ; also it will satisfy estimates which are uniform

in the lattice model ultraviolet cutoff N_0 . Furthermore, here $\mathcal{T}_{h,n}^{m,[0,N]}$ indicates the set of reference model GN trees which have at least one leaf attaching on a scale between 0 and N . Finally, since no counterterm is needed for the reference model, *i.e.* $\nu_{h,\omega}^{\text{ref}} = 0$, we use the convention $\widehat{W}_{\Gamma}^{(h),\text{ref}}[\tau] = 0$ whenever there is at least one leaf of type ν .

Thus, our goal is to estimate:

$$R_{m;\mu}^{\beta,L,N} := \sum_{i=1}^3 E_{m;\mu,i}^{\beta,L,N} . \quad (4.5.10)$$

We follow the reasoning in (4.3.156) by splitting this remainder into a term with few leaves of λ or ν type, and one with many:

$$\begin{aligned} R_{m;\mu}^{\beta,L,N} &= \sum_{n=0}^{n_*} R_{m,n;\mu}^{\beta,L,N} + \sum_{n=n_*+1}^{\infty} R_{m,n;\mu}^{\beta,L,N} \equiv R_{m,\leq n_*;\mu}^{\beta,L,N} + R_{m,>n_*;\mu}^{\beta,L,N} \\ R_{m,n;\mu}^{\beta,L,N} &= \sum_{i=1}^3 E_{m,n;\mu,i}^{\beta,L,N} , \end{aligned} \quad (4.5.11)$$

with $E_{m,n;\mu,i}^{\beta,L,N}$ collecting the contribution to $E_{m,n;\mu}^{\beta,L,N}$ coming from trees with n leaves of type λ, ν , and with $n_* = \lfloor \alpha m \rfloor$, for $\alpha \in (0, 1/34)$ to be chosen in the proof of the main Theorem.

Consider the $i = 1$ error term. Since trees τ contributing to $E_{m,n;\mu,1}^{\beta,L,N}$ have at least one vertex on scale 0, by the short-memory property of GN trees the associated kernels will have an extra factor $2^{\gamma(h-0)}$ at the root, for $\gamma \in (0, 1)$; combining this scaling gain with Proposition 4.3.20, we are led to

$$\frac{1}{\beta L} |E_{m,n;\mu,1}^{\beta,L,N}(\underline{p}_1, \dots, \underline{p}_m)| \leq m! \eta^{2-m+\gamma} C_{\gamma}^m \begin{cases} \frac{1}{\lfloor m/4 - 4 \rfloor!} & n = 0 \\ C_{\gamma}^m |\lambda|^n \left(\frac{2n}{m - 32n} \right)^{m/4-8n} & 0 < n \leq n_* , \end{cases} \quad (4.5.12)$$

which implies (for $|\lambda|$ small enough)

$$\frac{1}{\beta L} |E_{m,\leq n_*;\mu,1}^{\beta,L,N}(\underline{p}_1, \dots, \underline{p}_m)| \leq m! \eta^{2-m+\gamma} C_{\gamma}^m \left[\frac{1}{\lfloor m/4 - 4 \rfloor!} + |\lambda| \left(\frac{2\alpha}{1 - 32\alpha} \right)^{(1/4-8\alpha)m} \right] . \quad (4.5.13)$$

Analogously, one obtains

$$\frac{1}{\beta L} |E_{m,>n_*;\mu,1}^{\beta,L,N}(\underline{p}_1, \dots, \underline{p}_m)| \leq m! \eta^{2-m+\gamma} C_{\gamma}^m |\lambda|^{\alpha m} , \quad (4.5.14)$$

which proves the claim for $E_{m;\mu,1}^{\beta,L,N}$.

One may reason analogously for the case $i = 3$: indeed, we know that trees contributing to $E_{m,n;\mu,3}^{\beta,L,N}$ have at least one leaf at a non-negative scale, so let us call such scale $h_* \geq 0$; by the short-memory property of GN trees we have a gain $2^{\gamma(h-h_*)}$ at the root. Repeating the procedure⁴ outlined in (4.5.12) we may obtain an estimate analogous to (4.5.13) (respectively, (4.5.14)) for $E_{m,\leq n_*;\mu,3}^{\beta,L,N}$ (respectively, $E_{m,>n_*;\mu,3}^{\beta,L,N}$).

⁴To be very precise, in the reference model, for A -leaves attaching at positive scales, the corresponding propagator $g_{\text{ref}}^{(h_i)}$ appearing on a chain, see (4.3.164), is also multiplied by a kernel $W_{2,0,1}^{(h_i)}$; this is due to the localisation operator $\mathfrak{L}$ acting as the identity on marginal kernels at positive scales. Nevertheless, this does not play any role in the proof of the bounds we need.

Finally, we move on to the $i = 2$ error term. We start by noticing that the splitting (4.5.8) obtained above does not require a specific choice of the bare parameters of the reference model, but in order to interpret $E_{m;\mu,2}^{\beta,L,N}$ as an error term we need to tune such bare parameters in the infrared region: indeed, we choose $(Z_\omega^{\text{ref}}, v_\omega^{\text{ref}}, \lambda_{\omega\omega'}^{\text{ref}}, Q_\omega^{\text{ref}}, Z_\omega^{\mu,\text{ref}})$ so that

$$Z_{h\beta,\omega}^{\text{ref}} = Z_{h\beta,\omega} \quad v_{h\beta,\omega}^{\text{ref}} = v_{h\beta,\omega} \quad \lambda_{h\beta,\omega\omega'}^{\text{ref}} = \lambda_{h\beta,\omega\omega'} , \quad (4.5.15)$$

and

$$Q_{h\beta,\omega}^{\pm,\text{ref}} = Q_{h\beta,\omega}^\pm \quad Z_{h\beta,\omega}^{\mu,\text{ref}} = Z_{h\beta,\omega}^\mu . \quad (4.5.16)$$

To do this, one can use the implicit function theorem, since all effective couplings are non-constant, differentiable functions of their bare counterparts. Therefore, thanks to Propositions 4.3.5 and 4.4.4, we obtain that

$$\begin{aligned} \left| \frac{Z_{h,\omega}^{\text{ref}}}{Z_{h-1,\omega}^{\text{ref}}} - \frac{Z_{h,\omega}}{Z_{h-1,\omega}} \right| &\leq C|\lambda|^{2\gamma h} & |v_{h,\omega}^{\text{ref}} - v_{h,\omega}| &\leq C|\lambda|^{2\gamma h} \\ |\lambda_{h,\omega\omega'}^{\text{ref}} - \lambda_{h,\omega\omega'}| &\leq C|\lambda|^{2\gamma h} & |Z_{h,\omega}^{\mu,\text{ref}} - Z_{h,\omega}^\mu| &\leq C|\lambda|^{2\gamma h} \end{aligned} \quad (4.5.17)$$

for all $h = h_\beta, \dots, -1$, with $\gamma \in (0, 1)$, and C uniform in β, L, N . Recall also that

$$|\nu_{h,\omega}| \leq C2^{\gamma h} ; \quad (4.5.18)$$

furthermore, comparing the lattice and reference model single-scale propagators, we can write $\hat{g}_\omega^{(h)} = \hat{g}_\omega^{(h),\text{ref}} + \delta\hat{g}_\omega^{(h)}$, and one can check that the deviation satisfies the bound

$$|\delta\hat{g}_\omega^{(h)}(\underline{k}')| \leq C2^{\gamma h}2^{-h} . \quad (4.5.19)$$

Since kernels contributing to $E_{m,n;\mu,2}^{\beta,L,N}$ are given by differences of the lattice and reference model ones, an estimate for them boils down to the control of bounds of the kind (4.5.17)-(4.5.19). Hence, by the continuity property of GN trees, estimates for the contributions to $E_{m,n;\mu,2}^{\beta,L,N}$ carry an extra $2^{\gamma h}$ at the root, so that, proceeding as for the case of $E_{m;\mu,1}^{\beta,L,N}$, one obtains

$$\begin{aligned} \frac{1}{\beta L} |E_{m,\leq n_*;\mu,2}^{\beta,L,N}(\underline{p}_1, \dots, \underline{p}_m)| &\leq m! \eta^{2-m+\gamma} C_\gamma^m \left[\frac{1}{[m/4-4]!} + |\lambda| \left(\frac{2\alpha}{1-32\alpha} \right)^{(1/4-8\alpha)m} \right] \\ \frac{1}{\beta L} |E_{m,>n_*;\mu,2}^{\beta,L,N}(\underline{p}_1, \dots, \underline{p}_m)| &\leq m! \eta^{2-m+\gamma} C_\gamma^m |\lambda|^{\alpha m} . \end{aligned} \quad (4.5.20)$$

Adding together all the bounds above produces the claimed estimate for $|R_{m;\mu}^{\beta,L,N}|$. $\square$

Next, we report for later use the comparisons of the 2-point and vertex functions.

Proposition 4.5.3 (Comparison of 2-point and vertex functions). *Under the same hypotheses of Proposition 4.5.2, we may compare the 2-point and vertex functions of the lattice model to the corresponding reference model quantities.*

Lattice 2-point function. *There exist $Q^{\pm,\text{ref}}$ satisfying*

$$Q_\omega^{-,\text{ref}} = \overline{Q_\omega^{+,\text{ref}}} , \quad \|Q_\omega^{+,\text{ref}}\|^2 := \sum_{\rho \in S_M} |Q_{\omega,\rho}^{+,\text{ref}}|^2 = 1 + O(\lambda) ,$$

such that, for $\underline{k}' := \underline{k} - \underline{k}_{F,L}^{\omega,\lambda}$ small enough we have:

$$\langle \mathbf{T} \hat{a}_{\underline{k},\rho} \hat{a}_{\underline{k},\rho'}^* \rangle_{\beta,L} = Q_{\omega,\rho}^{+,\text{ref}} Q_{\omega,\rho'}^{-,\text{ref}} \langle \hat{\psi}_{\underline{k}',\omega}^- \hat{\psi}_{\underline{k}',\omega}^+ \rangle_{\beta,L,N}^{\text{ref}} + R_{2;\omega,\rho\rho'}^{\beta,L,N}(\underline{k}') . \quad (4.5.21)$$

with the remainder term being more regular for small $\underline{k}'$, namely

$$\frac{1}{\beta L} |R_{2;\omega,\rho\rho'}^{\beta,L,N}(\underline{k}')| \leq C_\gamma \|\underline{k}'\|^{\gamma-1} \quad (4.5.22)$$

with $\gamma \in (0, 1)$, and C_γ independent of β, L, N .

Lattice vertex function. Furthermore, for $\|\underline{k}'\|$ small enough, and $\|\underline{k}' + \underline{p}\| = \|\underline{k}'\|$, we have that:

$$\begin{aligned} \langle \mathbf{T} \hat{j}_{\underline{\mu},\underline{p}}; \hat{a}_{\underline{k},\rho}; \hat{a}_{\underline{k}+\underline{p},\rho'}^* \rangle_{\beta,L} &= \sum_{\omega'} Z_{\omega'}^{\mu,\text{ref}} Q_{\omega,\rho}^{+,\text{ref}} Q_{\omega,\rho'}^{-,\text{ref}} \langle \hat{n}_{\underline{p},\omega'}; \hat{\psi}_{\underline{k}',\omega}^-; \hat{\psi}_{\underline{k}'+\underline{p},\omega}^+ \rangle_{\beta,L,N}^{\text{ref}} \\ &\quad + R_{\mu;\omega,\rho\rho'}^{\beta,L,N}(\underline{p}, \underline{k}') , \end{aligned} \quad (4.5.23)$$

with the error term again being more regular, that is

$$\frac{1}{\beta L} |R_{\mu;\omega,\rho\rho'}^{\beta,L,N}(\underline{p}, \underline{k}')| \leq C_\gamma \|\underline{k}'\|^{\gamma-2} , \quad (4.5.24)$$

with $\gamma \in (0, 1)$ and C_γ independent of β, L, N .

Density-current correlation function. Finally, we have that

$$\langle \mathbf{T} \hat{n}_{\underline{p}}; \hat{j}_{\underline{\mu},-\underline{p}} \rangle_{\beta,L} = \sum_{\omega,\omega'} Z_{\omega}^{0,\text{ref}} Z_{\omega'}^{\mu,\text{ref}} \langle \hat{n}_{\underline{p},\omega}; \hat{n}_{-\underline{p},\omega'} \rangle_{\beta,L,N}^{\text{ref}} + R_{2;\mu}^{\beta,L,N}(\underline{p}) \quad (4.5.25)$$

where $R_{2;\mu}^{\beta,L,N}$ is Hölder-continuous at $\underline{0}$, for an exponent $\gamma > 0$; that is,

$$\frac{1}{\beta L} |R_{2;\mu}^{\beta,L,N}(\underline{p}) - R_{2;\mu}^{\beta,L,N}(\underline{0})| \leq C \|\underline{p}\|^\gamma , \quad (4.5.26)$$

with C independent of β, L, N .

Proof. We refer to Proposition 6.1 of [2], where a similar statement was proven for the case of a single, spin-degenerate edge mode on a single boundary of a $2d$ topological insulator, and Proposition 10.2 of [61], which generalised the former to the case of arbitrarily many edge modes. In any case, the proof strategy is similar to Proposition 4.5.2, yet simpler: since the number of points is at most three, one does not have to distinguish between the $n \leq n_*$ and $n > n_*$ cases, and dimensional bounds analogous to Proposition 4.3.15 (combined with the short-memory and continuity properties of GN trees) are enough. $\square$

4.5.2 Evaluation of the linear response

In this section, we shall evaluate explicitly the linear term ($m = 2$) in the expansion for $\chi_\nu^{\beta,L}$, for β, L large. A similar computation has been performed in [61], for the edge conductance of $2d$ topological insulators. We have

$$\chi_\nu^{\beta,L,\text{lin}}(x; \eta, \theta) = -\frac{1}{L} \sum_{p \in B_L} \hat{\mu}_\theta(-p) e^{-ipx} \frac{1}{\beta L} \langle \mathbf{T} \hat{n}_{\underline{p}}; \hat{j}_{\nu,-\underline{p}} \rangle_{\beta,L} ; \quad (4.5.27)$$

noticing that equations (4.5.25)-(4.5.26) prove boundedness of the density-current 2-point correlation uniformly in β, L , and N , and recalling that the perturbation μ is given by the L -periodisation of a compactly supported smooth function μ_∞ , we may rewrite (4.5.27) as

$$\chi_\nu^{\beta,L,\text{lin}}(x; \eta, \theta) = \chi_\nu^{\text{lin}}(x; \theta, \eta) + O(\beta^{-1}) + O(L^{-1}), \quad (4.5.28)$$

with χ_ν^{lin} encoding the zero-temperature, infinite-volume response:

$$\chi_\nu^{\text{lin}}(x; \eta, \theta) = - \int_{\mathbb{R}} \hat{\mu}_\infty(-p/\theta) e^{-ipx} \langle \mathbf{T} \hat{n}_{\underline{p}}; \hat{j}_{\nu, -\underline{p}} \rangle \frac{dp}{2\pi\theta} \quad (4.5.29)$$

with $\underline{p} = (\eta, p)$, and with $\langle \mathbf{T} \hat{n}_{\underline{p}}; \hat{j}_{\nu, -\underline{p}} \rangle = \lim_{\beta \rightarrow \infty} \lim_{L \rightarrow \infty} \langle \mathbf{T} \hat{n}_{\underline{p}}; \hat{j}_{\nu, -\underline{p}} \rangle_{\beta, L} / (\beta L)$ being the zero-temperature, infinite volume correlation function.

Proposition 4.5.4 (Final form of linear response). *Let $\theta = a\eta$. For $\nu = 0, 1$, the linear response χ_ν^{lin} of the system, i.e. the $m = 2$ contribution in the Duhamel expansion for $\chi_\nu^{\beta,L}$, is given by*

$$\chi_\nu^{\text{lin}}(x; \eta, \theta) = - \int_{\mathbb{R}} \hat{\mu}_\infty(q) e^{i\theta qx} (\vec{1}, \mathcal{K}^\nu(q) \vec{1}) \frac{dq}{2\pi} + O(\eta^\gamma) \quad (4.5.30)$$

for some $\gamma \in (0, 1)$, $(\cdot, \cdot)$ being the scalar product on $\mathbb{R}^{N_f}$ and with $\vec{1}$ being the vector with all components equal to 1. The matrix $\mathcal{K}^\nu$ given by

$$\mathcal{K}^\nu(q) = \tilde{\mathcal{K}}(q) \frac{v_\nu^{\text{ref}}}{2\pi|v^{\text{ref}}|} \frac{v^{\text{ref}} q}{-i/a + v^{\text{ref}} q} \mathfrak{K}^\nu, \quad (4.5.31)$$

with $v_\nu^{\text{ref}} = \delta_{\nu,0} \mathbb{1} + \delta_{\nu,1} v^{\text{ref}}$, and $\tilde{\mathcal{K}}, \mathfrak{K}^\nu$ defined to be

$$\begin{aligned} \tilde{\mathcal{K}}(q) &= \left[1 - \frac{1}{4\pi|v^{\text{ref}}|} \Lambda \right] \frac{1}{1 + \mathfrak{B}(-1/a, q) \Lambda} \\ \mathfrak{K}^0 &\equiv \mathbb{1} \\ \mathfrak{K}^1 &\equiv \text{sgn } v^{\text{ref}} \frac{1 + (4\pi|v^{\text{ref}}|)^{-1} \Lambda}{1 - (4\pi|v^{\text{ref}}|)^{-1} \Lambda} \text{sgn } v^{\text{ref}}. \end{aligned} \quad (4.5.32)$$

Proof. Since equations (4.5.25)-(4.5.26) hold uniformly in β, L, N , we may write

$$\begin{aligned} \langle \mathbf{T} \hat{n}_{\underline{p}}; \hat{j}_{\mu, -\underline{p}} \rangle &= \sum_{\omega, \omega'} Z_\omega^{0,\text{ref}} Z_{\omega'}^{\nu,\text{ref}} \langle \hat{n}_{\underline{p}, \omega}; \hat{n}_{-\underline{p}, \omega'} \rangle^{\text{ref}} + R_{2;\nu}(\underline{0}) + \tilde{R}_{2;\nu}(\underline{p}) \\ \tilde{R}_{2;\nu}(\underline{p}) &= R_{2;\nu}(\underline{p}) - R_{2;\nu}(\underline{0}), \quad |\tilde{R}_{2;\nu}(\underline{p})| \leq C \|\underline{p}\|^\gamma. \end{aligned} \quad (4.5.33)$$

Next, we determine $R_{2;\nu}(\underline{0})$ in terms of the reference model correlations. The lattice continuity equation (3.1.30) implies the following lattice Ward identity,

$$i\partial_{x_0} \langle \mathbf{T} n_{\underline{x}}; j_{\nu, \underline{y}} \rangle + \text{d}_x \langle \mathbf{T} j_{\underline{x}}; j_{\nu, \underline{y}} \rangle = i\delta(x_0 - y_0) \langle [n_x, j_{\nu, y}] \rangle, \quad (4.5.34)$$

where as usual $\mathcal{O}_{\underline{x}} := \gamma_{x_0}(\mathcal{O}_x)$, and where the contact term on the right-hand side arises due to the definition of time-ordering. Taking the Fourier transform on both sides, we obtain:

$$p_0 \langle \mathbf{T} \hat{n}_{\underline{p}}; \hat{j}_{\nu, -\underline{p}} \rangle + (1 - e^{-ip}) \langle \mathbf{T} \hat{j}_{\underline{p}}; \hat{j}_{\nu, -\underline{p}} \rangle = i \sum_{x \in \mathbb{Z}} e^{-ipx} \langle [n_x, j_{\nu, 0}] \rangle. \quad (4.5.35)$$

Evaluating (4.5.35) at $\underline{p} = (p_0, 0)$, and using that $[\sum_{x \in \mathbb{Z}} n_x, j_{\nu, 0}] = [\mathcal{N}, j_{\nu, 0}] = 0$, we are then led to

$$\langle \mathbf{T} \hat{n}_{(p_0, 0)}; \hat{j}_{\nu, (-p_0, 0)} \rangle = 0 .$$

Hence, by (4.5.33), we obtain

$$R_{2; \nu}(\underline{0}) = - \sum_{\omega, \omega'} Z_{\omega}^{0, \text{ref}} Z_{\omega'}^{\nu, \text{ref}} \lim_{p_0 \rightarrow 0} \langle \hat{n}_{(p_0, 0), \omega}; \hat{n}_{(-p_0, 0), \omega'} \rangle^{\text{ref}} . \quad (4.5.36)$$

We set $S_2^{\text{ref}}(p_0, p) \in \mathcal{M}_{N_f}(\mathbb{C})$ to be the matrix of the reference model density correlations, *i. e.*

$$S_{2; \omega \omega'}^{\text{ref}}(p_0, p) := \langle \hat{n}_{\underline{p}, \omega}; \hat{n}_{-\underline{p}, \omega'} \rangle^{\text{ref}} . \quad (4.5.37)$$

By Proposition 4.4.9, it takes the form

$$S_2^{\text{ref}}(p_0, p) = \frac{1}{1 + \hat{v}(\underline{p}) \mathfrak{B}(\underline{p}) \Lambda_Z} \frac{\mathfrak{B}(\underline{p})}{Z^2} = \frac{1}{1 + \mathfrak{B}(\underline{p}) \Lambda_Z} \frac{\mathfrak{B}(\underline{p})}{Z^2} + O(\|\underline{p}\|^\gamma) \quad (4.5.38)$$

with

$$\mathfrak{B}(\underline{p}) = \frac{1}{4\pi|v^{\text{ref}}|} \frac{-ip_0 + v^{\text{ref}} p}{ip_0 + v^{\text{ref}} p} , \quad \Lambda_Z = Z^{-1} \Lambda Z \quad (4.5.39)$$

and $v^{\text{ref}} = \text{diag } v_{\omega}^{\text{ref}}$, $\Lambda = (\lambda_{\omega \omega'}^{\text{ref}})_{\omega \omega'}$, and $Z = \text{diag } Z_{\omega}^{\text{ref}}$. In the second step we approximated the reference model interaction $\hat{v}(\underline{p})$ with $\hat{v}(\underline{0}) \equiv 1$; also, we used the convenient short-hand of writing $AB^{-1} \equiv \frac{A}{B}$ whenever A, B commute.

Next, realising that $\mathfrak{B}(\varepsilon \underline{p}) = \mathfrak{B}(\underline{p})$ for all $\varepsilon \in \mathbb{R}^\times$, recalling that $\theta = a\eta$, and reparameterising $\underline{p} = \theta \underline{q}$, with $\underline{q} = (a^{-1}, q)$, we get

$$\langle \mathbf{T} \hat{n}_{\theta \underline{q}}; \hat{j}_{\nu, -\theta \underline{q}} \rangle = (\vec{Z}^{0, \text{ref}}, \mathcal{Q}(q; a) \vec{Z}^{\nu, \text{ref}}) + O(\eta^\gamma (1 + a^2 q^2)^{\gamma/2}) \quad (4.5.40)$$

with $\vec{Z}^{\nu, \text{ref}}$ the vector in $\mathbb{R}^{N_f}$ with components $Z_{\omega}^{\nu, \text{ref}}$, and $(\cdot, \cdot)$ the standard scalar product on $\mathbb{R}^{N_f}$, and with the matrix-valued function $\mathcal{Q}$ being

$$\begin{aligned} \mathcal{Q}(q; a) &= \frac{1}{1 + \mathfrak{B}(1, aq) \Lambda_Z} \frac{\mathfrak{B}(1, aq)}{Z^2} - \frac{1}{1 + \mathfrak{B}(1, 0) \Lambda_Z} \frac{\mathfrak{B}(1, 0)}{Z^2} \\ &= \frac{1}{1 + \mathfrak{B}(1, aq) \Lambda_Z} \frac{v^{\text{ref}} aq}{i + v^{\text{ref}} aq} \frac{1}{1 - (4\pi|v^{\text{ref}}|)^{-1} \Lambda_Z} \frac{1}{2\pi|v^{\text{ref}}| Z^2} . \end{aligned} \quad (4.5.41)$$

Then, inserting of (4.5.40) into (4.5.29), and using the fact that $\hat{\mu}_\infty$ is of Schwartz class, we are led to

$$\chi_\nu^{\text{lin}}(x; \eta, \theta) = - \int_{\mathbb{R}} \hat{\mu}_\infty(q) e^{i\theta q x} (\vec{Z}^{0, \text{ref}}, \mathcal{Q}(-q; a) \vec{Z}^{\nu, \text{ref}}) \frac{dq}{2\pi} + O(\eta^\gamma) . \quad (4.5.42)$$

In the case in which $a(\eta) := \theta/\eta = o(1)$, we see that $\|\mathcal{Q}(q; a)\| = O(a)$, hence the response is trivial, namely

$$\chi_\nu^{\text{lin}}(x; \eta, \theta) = O(a) + O(\eta^\gamma) ;$$

therefore we may focus on the Euler scaling case, *i. e.* we shall assume a to be constant.

We shall now evaluate $\vec{Z}^{\nu, \text{ref}}$ in terms of the other bare parameters of the reference model, namely Z , v^{ref} , and Λ .

Computing the vertex renormalisations. To avoid cluttering of the notation, we set, for $\mu = 0, 1$:

$$\begin{aligned}\Sigma_{\rho\rho'}(\underline{k}) &:= \langle \mathbf{T} \hat{a}_{\underline{k},\rho}; \hat{a}_{\underline{k},\rho'}^* \rangle & S_{1,2;\mu,\rho\rho'}(\underline{p}, \underline{k}) &:= \langle \mathbf{T} \hat{J}_{\mu,\underline{p}}; a_{\underline{k},\rho}; a_{\underline{k}+\underline{p},\rho'}^* \rangle \\ \Sigma_{\omega}^{\text{ref}}(\underline{k}') &:= \langle \hat{\psi}_{\underline{k}',\omega}^-; \hat{\psi}_{\underline{k}',\omega}^+ \rangle^{\text{ref}} & S_{1,2;\omega\omega'}^{\text{ref}}(\underline{p}, \underline{k}') &:= \langle \hat{n}_{\underline{p}}; \hat{\psi}_{\underline{k}',\omega}^-; \hat{\psi}_{\underline{k}'+\underline{p},\omega}^+ \rangle^{\text{ref}}\end{aligned}\quad (4.5.43)$$

The lattice vertex function satisfies the following lattice vertex Ward identity:

$$\sum_{\mu=0}^1 p_{\mu} Y_{\mu}(p) S_{1,2;\mu,\rho\rho'}(\underline{p}, \underline{k}) = \Sigma_{\rho\rho'}(\underline{k}) - \Sigma_{\rho\rho'}(\underline{k} + \underline{p}) . \quad (4.5.44)$$

with $Y_0(p) \equiv i$, $p_1 = p$, and $Y_1(p) = (1 - e^{-ip})/ip = 1 + O(p)$. By (4.5.23), for $\underline{k} = \underline{k}_F^{\omega}(\lambda) + \underline{k}'$:

$$S_{1,2;\mu,\rho\rho'}(\underline{p}, \underline{k}) = \sum_{\omega'} Z_{\omega'}^{\mu,\text{ref}} Q_{\omega,\rho}^{+,\text{ref}} Q_{\omega,\rho'}^{-,\text{ref}} S_{1,2;\omega'\omega}^{\text{ref}}(\underline{p}, \underline{k}') + R_{\mu,\omega,\rho\rho'}(\underline{p}, \underline{k}) , \quad (4.5.45)$$

and, recalling (4.5.21):

$$\Sigma_{\rho\rho'}(\underline{k}) = Q_{\omega,\rho}^{+,\text{ref}} Q_{\omega,\rho'}^{-,\text{ref}} \Sigma_{\omega}^{\text{ref}}(\underline{k}') + R_{2;\omega,\rho\rho'}(\underline{k}) . \quad (4.5.46)$$

Since $\|Q_{\omega}^{+,\text{ref}}\| = 1 + O(\lambda)$, we may multiply both sides of (4.5.44) by $Q_{\omega,\rho}^{-,\text{ref}} Q_{\omega,\rho'}^{+,\text{ref}} / \|Q_{\omega}^{+,\text{ref}}\|^4$, and we may sum over all ρ, ρ' , obtaining:

$$\begin{aligned}\sum_{\mu=0}^1 p_{\mu} Y_{\mu}(p) \left[\sum_{\omega'} Z_{\omega'}^{\mu,\text{ref}} S_{1,2;\omega'\omega}^{\text{ref}}(\underline{p}, \underline{k}') + R_{\mu,\omega}(\underline{p}, \underline{k}') \right] = \\ \Sigma_{\omega}^{\text{ref}}(\underline{k}') - \Sigma_{\omega}^{\text{ref}}(\underline{k}' + \underline{p}) + R_{2;\omega}(\underline{k}') - R_{2;\omega}(\underline{k}' + \underline{p}) ,\end{aligned}\quad (4.5.47)$$

where we introduced the notations:

$$\begin{aligned}R_{\mu,\omega}(\underline{p}, \underline{k}') &:= \frac{1}{\|Q_{\omega}^{+,\text{ref}}\|^4} \sum_{\rho,\rho'} R_{\mu,\omega,\rho\rho'}(\underline{p}, \underline{k}') Q_{\omega,\rho}^{-,\text{ref}} Q_{\omega,\rho'}^{+,\text{ref}} \\ R_{2;\omega}(\underline{k}') &:= \frac{1}{\|Q_{\omega}^{+,\text{ref}}\|^4} \sum_{\rho,\rho'} R_{2;\omega,\rho\rho'}(\underline{k}') Q_{\omega,\rho}^{-,\text{ref}} Q_{\omega,\rho'}^{+,\text{ref}} .\end{aligned}\quad (4.5.48)$$

Next, recall the explicit expression for the vertex function of the reference model, equation (4.4.55):

$$S_{1,2;\omega'\omega}^{\text{ref}}(\underline{p}, \underline{k}') = \frac{T_{\omega'\omega}(\underline{p})}{Z_{\omega'}^{\text{ref}} D_{\omega}(\underline{p})} [\Sigma_{\omega'}^{\text{ref}}(\underline{k}') - \Sigma_{\omega'}^{\text{ref}}(\underline{k}' + \underline{p})] . \quad (4.5.49)$$

Recalling the estimates (4.5.22), (4.5.24) for the error terms, and choosing $\|\underline{k}'\| = \kappa$, $\|\underline{k}' + \underline{p}\| = \kappa$, $\|\underline{p}\| = O(\kappa)$, we see that in (4.5.47) the R -terms give contributions that are suppressed by a factor κ^{γ} with respect to the other quantities. Hence, we get:

$$\sum_{\omega'} D_{\omega'}^Z(\underline{p}) \frac{T_{\omega'\omega}(\underline{p})}{Z_{\omega'}^{\text{ref}} D_{\omega}(\underline{p})} [\Sigma_{\omega'}^{\text{ref}}(\underline{k}') - \Sigma_{\omega'}^{\text{ref}}(\underline{k}' + \underline{p})] = \Sigma_{\omega}^{\text{ref}}(\underline{k}') - \Sigma_{\omega}^{\text{ref}}(\underline{k}' + \underline{p}) + O(\kappa^{\gamma-1}) \quad (4.5.50)$$

with

$$D_{\omega'}^Z(\underline{p}) := ip_0 Z_{\omega'}^{0,\text{ref}} + p Z_{\omega'}^{1,\text{ref}} . \quad (4.5.51)$$

Dividing both sides of (4.5.50) by $[\Sigma_\omega^{\text{ref}}(\underline{k}') - \Sigma_\omega^{\text{ref}}(\underline{k}' + \underline{p})]/[Z_\omega^{\text{ref}} D_\omega(\underline{p})] = O(\kappa^{-2})$, we find:

$$\sum_{\omega'} D_{\omega'}^Z(\underline{p}) T_{\omega'\omega}(\underline{p}) = Z_\omega^{\text{ref}} D_\omega(\underline{p}) + O(\kappa^{1+\gamma}). \quad (4.5.52)$$

Recalling the expression (4.4.53) for $T(\underline{p})$, we have that

$$\lim_{p_0 \rightarrow 0} \lim_{p \rightarrow 0} T(\underline{p}) = \frac{1}{1 - (4\pi|v^{\text{ref}}|)^{-1} \Lambda_Z}, \quad \lim_{p \rightarrow 0} \lim_{p_0 \rightarrow 0} T(\underline{p}) = \frac{1}{1 + (4\pi|v^{\text{ref}}|)^{-1} \Lambda_Z}. \quad (4.5.53)$$

Therefore, evaluation of equation (4.5.52) by taking $\underline{p} \rightarrow 0$ horizontally and vertically implies the following two equalities:

$$\left(\frac{1}{1 - (4\pi|v^{\text{ref}}|)^{-1} \Lambda_Z} \right)^T \vec{Z}^{0,\text{ref}} = \vec{Z}^{\text{ref}}, \quad \left(\frac{1}{1 + (4\pi|v^{\text{ref}}|)^{-1} \Lambda_Z} \right)^T \vec{Z}^{1,\text{ref}} = v^{\text{ref}} \vec{Z}^{\text{ref}}, \quad (4.5.54)$$

with $\vec{Z}^{\text{ref}}$ the vector of components Z_ω^{ref} . Inverting, we obtain, for $\mu = 0, 1$,

$$\vec{Z}^{\mu,\text{ref}} = [1 - (-1)^\mu \Lambda_Z^T (4\pi|v^{\text{ref}}|)^{-1}] v_\mu^{\text{ref}} \vec{Z}^{\text{ref}}, \quad (4.5.55)$$

where $v_0^{\text{ref}} := \mathbb{1}_{N_f}$ and $v_1^{\text{ref}} := v^{\text{ref}}$.

Final form of the response. Substituting equation (4.5.55) into (4.5.42), we obtain, for $\nu = 0, 1$,

$$\chi_\nu^{\text{lin}}(x; \eta, \theta) = - \int_{\mathbb{R}} \hat{\mu}_\infty(q) e^{i\theta qx} (\vec{Z}^{\text{ref}}, \mathcal{K}_Z^\nu(q) \vec{Z}^{\text{ref}}) \frac{dq}{2\pi} + O(\eta^\gamma), \quad (4.5.56)$$

with $\mathcal{K}_Z^\nu$ being

$$\begin{aligned} \mathcal{K}_Z^\nu(q) &= \left[1 - \frac{1}{4\pi|v^{\text{ref}}|} \Lambda_Z \right] \mathcal{Q}(-q; a) \left[1 - (-1)^\nu \Lambda_Z^T \frac{1}{4\pi|v^{\text{ref}}|} \right] v_\nu^{\text{ref}} \\ &= \frac{1}{Z} \left[1 - \frac{1}{4\pi|v^{\text{ref}}|} \Lambda \right] Z \mathcal{Q}(-q; a) Z \left[1 - (-1)^\nu \Lambda \frac{1}{4\pi|v^{\text{ref}}|} \right] \frac{v_\nu^{\text{ref}}}{Z}. \end{aligned} \quad (4.5.57)$$

Here, we employed the symmetry of Λ and the fact that Z commutes with v^{ref} (since both are diagonal) to bring the conjugation by Z from just Λ to the whole expression. Now we will show that $Z \mathcal{Q} Z$ is independent of Z : indeed, since $\mathfrak{B}$ is diagonal, we have

$$\begin{aligned} Z \mathcal{Q}(-q) Z &= Z \frac{1}{1 + \mathfrak{B}(-1, aq) \Lambda_Z} \frac{v^{\text{ref}} q}{-i/a + v^{\text{ref}} q} \frac{1}{1 - (4\pi|v^{\text{ref}}|)^{-1} \Lambda_Z} \frac{1}{2\pi|v^{\text{ref}}| Z^2} Z \\ &= \frac{1}{1 + \mathfrak{B}(-1, aq) \Lambda} \frac{v^{\text{ref}} q}{-i/a + v^{\text{ref}} q} \frac{1}{1 - (4\pi|v^{\text{ref}}|)^{-1} \Lambda} \frac{1}{2\pi|v^{\text{ref}}|}. \end{aligned} \quad (4.5.58)$$

Inserting (4.5.57) and (4.5.58) into (4.5.56), we obtain at last

$$\chi_\nu^{\text{lin}}(x) = - \int_{\mathbb{R}} \hat{\mu}_\infty(q) e^{i\theta qx} (\vec{1}, \mathcal{K}^\nu(q) \vec{1}) \frac{dq}{2\pi} + O(\eta^\gamma), \quad (4.5.59)$$

with $\vec{1}$ being the vector with all components equal to 1, and $\mathcal{K}^\nu$ a matrix-valued function depending only on v^{ref} and Λ , given as

$$\begin{aligned}\mathcal{K}^0(q) &= \left[1 - \frac{1}{4\pi|v^{\text{ref}}|}\Lambda\right] \frac{1}{1 + \mathfrak{B}(-1/a, q)\Lambda} \frac{1}{2\pi|v^{\text{ref}}|} \frac{v^{\text{ref}}q}{-i/a + v^{\text{ref}}q} \\ \mathcal{K}^1(q) &= \left[1 - \frac{1}{4\pi|v^{\text{ref}}|}\Lambda\right] \frac{1}{1 + \mathfrak{B}(-1/a, q)\Lambda} \frac{\text{sgn } v^{\text{ref}}}{2\pi} \frac{v^{\text{ref}}q}{-i/a + v^{\text{ref}}q} \mathfrak{K} \\ \mathfrak{K} &\equiv \text{sgn } v^{\text{ref}} \frac{1 + (4\pi|v^{\text{ref}}|)^{-1}\Lambda}{1 - (4\pi|v^{\text{ref}}|)^{-1}\Lambda} \text{sgn } v^{\text{ref}}.\end{aligned}\quad (4.5.60)$$

The proof is completed by recognising $\tilde{\mathcal{K}}$, $\mathfrak{K}^\nu$ and v_ν^{ref} in the expression above. $\square$

4.5.3 Proof of Theorem 4.1.5

Proof of Theorem 4.1.5. We begin by recalling the expansion for $\chi_\nu^{\beta,L}$, namely

$$\begin{aligned}\chi_\nu^{\beta,L}(x; \eta, \theta) &= - \sum_{m \geq 2} \frac{(-\theta)^{m-2}}{(m-1)!} \frac{1}{L^{m-1}} \sum_{p \in B_L^{m-1}} \prod_{i=1}^{m-1} [\hat{\mu}_\theta(-p_i)] e^{ip_m x} \\ &\quad \cdot \frac{\langle \mathbf{T} \hat{n}_{\underline{p}_1}; \dots; \hat{n}_{\underline{p}_{m-1}}; \hat{j}_{\nu, \underline{p}_m} \rangle_{\beta, L}}{\beta L} + E_\nu^{\beta, L}(x; \eta, \theta)\end{aligned}\quad (4.5.61)$$

with $\underline{p}_i = (\eta_\beta, p_i)$, $\underline{p}_m = -\sum_{i=1}^{m-1} \underline{p}_i$, and with $\hat{\mu}_\theta(\cdot) := \hat{\mu}(\cdot/\theta)/\theta$. Thanks to (4.2.12), we know that $E_\nu^{\beta, L}$ is bounded as

$$|E_\nu^{\beta, L}(x; \eta, \theta)| \leq \frac{C}{\eta^3 \beta}.$$

Let us focus on the corrections to the linear order $m = 2$. We estimate them as

$$\sum_{m \geq 3} \frac{\theta^{m-2}}{(m-1)!} \frac{1}{L^{m-1}} \sum_{p \in B_L^{m-1}} \prod_{i=1}^{m-1} [\hat{\mu}_\theta(-p_i)] \frac{|\langle \mathbf{T} \hat{n}_{\underline{p}_1}; \dots; \hat{j}_{\nu, \underline{p}_m} \rangle_{\beta, L}|}{\beta L}. \quad (4.5.62)$$

Thanks to Proposition 4.5.2, we rewrite the m -point correlation functions in terms of the corresponding ones for the reference model:

$$\begin{aligned}\langle \hat{n}_{\underline{p}_1}; \dots; \hat{j}_{\nu, \underline{p}_m} \rangle_{\beta, L} &= \sum_{\underline{\omega}} \left[\prod_{i=1}^{m-1} Z_{\omega_i}^{0, \text{ref}} \right] Z_{\omega_m}^{\nu, \text{ref}} \langle \hat{n}_{\underline{p}_1, \omega_1}; \dots; \hat{n}_{\underline{p}_m, \omega_m} \rangle_{\beta, L, N}^{\text{ref}} \\ &\quad + R_{m; \nu}^{\beta, L, N}(\underline{p}_1, \dots, \underline{p}_m).\end{aligned}\quad (4.5.63)$$

By Corollary 4.4.7, the reference model density correlations are vanishing when the UV cutoff is removed, that is, uniformly in N large,

$$\frac{1}{\beta L} \left| \left[\prod_{i=1}^{m-1} Z_{\omega_i}^{0, \text{ref}} \right] Z_{\omega_m}^{\nu, \text{ref}} \langle \hat{n}_{\underline{p}_1, \omega_1}; \dots; \hat{n}_{\underline{p}_m, \omega_m} \rangle_{\beta, L, N}^{\text{ref}} \right| \leq C_{\beta, m} N 2^{-\gamma N}, \quad (4.5.64)$$

with $\gamma > 0$ and $C_{\beta, m}$ depending on β, η , but independent of $\underline{\omega}, L, N$; also, since the bounds in Proposition 4.5.2 for $R_{m; \nu}^{\beta, L, N}$ are uniform in N large, by combining these latter with (4.5.64), in the $N \rightarrow \infty$ limit equation (4.5.63) yields

$$\frac{1}{\beta L} |\langle \mathbf{T} \hat{n}_{\underline{p}_1}; \dots; \hat{j}_{\nu, \underline{p}_m} \rangle_{\beta, L}| \leq m! C^m \eta^{2-m+\gamma} F(m; \lambda, \alpha) \quad (4.5.65)$$

for some $\gamma \in (0, 1 - c|\lambda|)$, and for any $\alpha \in (0, 1/34)$, with C uniform in β, L large. Here, F is given by (4.5.4), namely

$$F(m; \lambda, \alpha) := \begin{cases} 1 & \text{if } m < 16 \\ \frac{1}{[m/4 - 4]!} + |\lambda| \left(\frac{2\alpha}{1 - 32\alpha} \right)^{(1/4 - 8\alpha)m} + |\lambda|^{\alpha m} & \text{if } m \geq 16. \end{cases} \quad (4.5.66)$$

By employing (4.5.65), we therefore get the following estimate for (4.5.62):

$$\begin{aligned} \sum_{m \geq 3} \frac{\theta^{m-2}}{(m-1)!} \frac{1}{L^{m-1}} \sum_{\{p_i\} \in B_L^{m-1}} \prod_{i=1}^{m-1} [|\hat{\mu}_\theta(-p_i)|] \frac{|\langle \mathbf{T} \hat{n}_{\underline{p}_1}; \dots; \hat{j}_{\nu, \underline{p}_m} \rangle_{\beta, L}|}{\beta L} \\ \leq \sum_{m \geq 3} m C^m \eta^\gamma F(m; \lambda, \alpha) a^{m-2} \frac{1}{L^{m-1}} \sum_{\{p_i\} \in B_L^{m-1}} \prod_{i=1}^{m-1} \frac{|\hat{\mu}(-p_i/\theta)|}{\theta} \\ \leq \eta^\gamma \sum_{m \geq 3} m C^m F(m; \lambda, \alpha) a^{m-2} \|\hat{\mu}_\infty\|_{L^1}^{m-1} \end{aligned} \quad (4.5.67)$$

where we recall that $a := \theta/\eta$, and where in the first step we employed (4.5.65).

Now, if a is small enough, we do not need the improvement guaranteed by $F(\cdot; \lambda, \alpha)$ for large inputs, since the sum over m is convergent. The smallness of the sum is guaranteed by the η^γ factor. Instead, for more general values of a , the aforementioned improvement is crucial to perform the sum. For given a , we choose $\alpha \in (0, 1/34)$ small enough and $|\lambda|$ small enough so that:

$$2C \|\hat{\mu}_\infty\|_{L^1} a \left(\frac{2\alpha}{1 - 32\alpha} \right)^{1/4 - 8\alpha} < 1, \quad 2C \|\hat{\mu}_\infty\|_{L^1} a |\lambda|^\alpha < 1. \quad (4.5.68)$$

With this choice, we obtain:

$$\sum_{m \geq 3} \frac{\theta^{m-2}}{(m-1)!} \frac{1}{L^{m-1}} \sum_{\{p_i\} \in B_L^{m-1}} \prod_{i=1}^{m-1} [|\hat{\mu}_\theta(-p_i)|] \frac{|\langle \mathbf{T} \hat{n}_{\underline{p}_1}; \dots; \hat{j}_{\nu, \underline{p}_m} \rangle_{\beta, L}|}{\beta L} \leq \tilde{C}_{\lambda, \alpha} \eta^\gamma \quad (4.5.69)$$

for some $\tilde{C}_{\lambda, \alpha}$ independent of β, L . This estimate for the higher-order terms in the Duhamel expansion, together with Proposition 4.5.4, concludes the proof of Theorem 4.1.5. $\square$

Appendix A

Grassmann Algebras

In this Appendix we recall some basic concepts of Grassmann calculus. For a reference, see the book [74] or the review [36].

Definition A.1 (Grassmann algebra). Let A be a finite set of indices. We denote by $\text{Gr}(A, \Psi)$ the Grassmann $\mathbb{C}$ -algebra generated by A , defined by taking the free $\mathbb{C}$ -algebra of generators $\{\Psi_\alpha^-, \Psi_\alpha^+\}_{\alpha \in A}$, to which we refer as Grassmann variables, and imposing the following relations:

$$\{\Psi_\alpha^\varepsilon, \Psi_{\alpha'}^{\varepsilon'}\} := \Psi_\alpha^\varepsilon \Psi_{\alpha'}^{\varepsilon'} + \Psi_{\alpha'}^{\varepsilon'} \Psi_\alpha^\varepsilon = 0, \quad \forall (\alpha, \varepsilon), (\alpha', \varepsilon') \in A \times \{\pm\}; \quad (\text{A.1})$$

in particular we have that $\Psi_\alpha^\varepsilon \Psi_\alpha^\varepsilon = 0$ for all $\alpha \in A$, $\varepsilon = \pm$.

It is easy to check that $\text{Gr}(A, \Psi)$ is isomorphic, as algebra, to the exterior algebra over $\mathbb{C}^{2|A|}$, and therefore $\dim_{\mathbb{C}} \text{Gr}(A, \Psi) = 4^{|A|}$. Also, there is an evident notion of degree for monomials in $\text{Gr}(A, \Psi)$, going from a minimum of 0 (for constants) to a maximum of $2|A|$.

Next, given $(\alpha, \varepsilon) \in A \times \{\pm\}$, we indicate by

$$\int d\Psi_\alpha^\varepsilon : \text{Gr}(A, \Psi) \rightarrow \text{Gr}(A, \Psi)$$

the unique map on our Grassmann algebra determined by the following conditions:

1. it is a linear map, hence it is enough to give its action on monomials;
2. for any $(\alpha', \varepsilon') \in A \times \{\pm\}$ we have

$$\int d\Psi_\alpha^\varepsilon 1 = 0, \quad \int d\Psi_\alpha^\varepsilon \Psi_{\alpha'}^{\varepsilon'} = \delta_{\alpha, \alpha'} \delta_{\varepsilon, \varepsilon'};$$

3. it satisfies the graded Leibniz condition: for any monomials $F(\Psi)$, $G(\Psi)$ in $\text{Gr}(A, \Psi)$

$$\int d\Psi_\alpha^\varepsilon F(\Psi)G(\Psi) = \left[\int d\Psi_\alpha^\varepsilon F(\Psi) \right] G(\Psi) + (-1)^{\deg F} F(\Psi) \left[\int d\Psi_\alpha^\varepsilon G(\Psi) \right],$$

with $\deg F$ denoting the degree of F .

Notice that these are maps of degree -1 , *i.e.* they send degree k monomials to monomials of degree at most $k - 1$, for all $k = 1, \dots, 2|A|$. Also, one may check that any two such maps anti-commute, that is

$$\int d\Psi_\alpha^\varepsilon \circ \int d\Psi_{\alpha'}^{\varepsilon'} = - \int d\Psi_{\alpha'}^{\varepsilon'} \circ \int d\Psi_\alpha^\varepsilon, \quad (\text{A.2})$$

and we shall write their composition using the short-hand

$$\int d\Psi_\alpha^\varepsilon d\Psi_{\alpha'}^{\varepsilon'} := \int d\Psi_\alpha^\varepsilon \circ \int d\Psi_{\alpha'}^{\varepsilon'} .$$

Definition A.2 (Berezin integration). We will call *Berezin integral* the linear functional

$$\int d\Psi : \text{Gr}(A, \Psi) \rightarrow \mathbb{C}$$

that to each polynomial $F \in \text{Gr}(A, \Psi)$ assigns

$$\int d\Psi F(\Psi) := \int \prod_{\alpha \in A} [d\Psi_\alpha^+ d\Psi_\alpha^-] F(\Psi) .$$

Notice that this is indeed a complex number, as the degree of Berezin integration is $-2|A|$. In fact, the Berezin integral is non-zero only on monomials of maximum degree, namely $2|A|$, for whom we have

$$\int d\Psi \prod_{\alpha \in A} \Psi_\alpha^- \Psi_\alpha^+ = 1 , \quad (\text{A.3})$$

and by (A.1)-(A.2) we have that any other monomial obtained by an even¹ (resp. odd) permutation of the one in (A.3) will be mapped to 1 (resp. -1).

The reason for calling it integral is that it behaves like a Grassmann analogue of the Lebesgue integral. Taking inspiration from complex gaussian probability measures, which are defined by multiplying the Lebesgue measure with a gaussian weight parameterised by the covariance, and setting², for any $F \in \text{Gr}(A, \Psi)$,

$$e^{F(\Psi)} := \sum_{n \geq 0} \frac{(F(\Psi))^n}{n!} ,$$

we give the following Definition.

Definition A.3 (Grassmann Gaussian integration). Given $K \in \text{GL}(|A|, \mathbb{C})$, we associate to it the Grassmann Gaussian measure $P_{K^{-1}}$ of covariance K^{-1} , namely the functional over $\text{Gr}(A, \Psi)$ given by

$$\mathbb{E}_K[F(\Psi)] \equiv \int P_{K^{-1}}[d\Psi] F(\Psi) := \frac{\int d\Psi e^{-\sum_{\alpha, \beta \in A} \Psi_\alpha^+ K_{\alpha\beta} \Psi_\beta^-} F(\Psi)}{\int d\Psi e^{-\sum_{\alpha, \beta \in A} \Psi_\alpha^+ K_{\alpha\beta} \Psi_\beta^-}} ,$$

for any $F \in \text{Gr}(A, \Psi)$.

The Definition is well-posed, thanks to the following lemma, which proves that the denominator is non-vanishing.

¹A permutation is called even (odd) if it can be obtained by an even (odd) number of elementary exchanges.

²Notice that the summand becomes definitively zero due to (A.1), therefore the sum is over finitely many n .

Lemma A.4. *The normalisation factor, also called (free) partition function, is*

$$\int d\Psi e^{-\sum_{\alpha,\beta \in A} \Psi_{\alpha}^{+} K_{\alpha\beta} \Psi_{\beta}^{-}} = \det K .$$

Proof. Consider the first point of the statement. Developing the exponential, we see that the Berezin integral selects only degree $2|A|$ monomials

$$\int d\Psi e^{-\sum_{\alpha,\beta \in A} \Psi_{\alpha}^{+} K_{\alpha\beta} \Psi_{\beta}^{-}} = \frac{1}{|A|!} \sum_{\substack{\alpha_1, \dots, \alpha_{|A|} \\ \beta_1, \dots, \beta_{|A|}}} \int d\Psi \prod_{i=1}^{|A|} [\Psi_{\beta_i}^{-} K_{\alpha_i \beta_i} \Psi_{\alpha_i}^{+}] . \quad (\text{A.4})$$

More specifically the only non-zero contributions come from choices of $\alpha_1, \dots, \alpha_{|A|}$ all different from each other, and the same can be said for $\beta_1, \dots, \beta_{|A|}$; therefore, for a given monomial, there is a bijection between the two index $|A|$ -tuples. In particular, at fixed indices, let us call $\sigma \in S_{|A|}$ the permutation such that $\alpha_i = \sigma(\beta_i)$ for each $i = 1, \dots, |A|$, then we have that

$$\prod_{i=1}^{|A|} \Psi_{\beta_i}^{-} \Psi_{\alpha_i}^{+} \equiv \prod_{i=1}^{|A|} \Psi_{\beta_i}^{-} \Psi_{\sigma(\beta_i)}^{+} = \text{sgn}(\sigma) \prod_{i=1}^{|A|} \Psi_{\beta_i}^{-} \Psi_{\beta_i}^{+} = \text{sgn}(\sigma) \prod_{\beta \in A} \Psi_{\beta}^{-} \Psi_{\beta}^{+}$$

where in the second passage we employed anticommutativity and in the last one we permuted the indices $\{\beta_i\}$ to order them according to A , and we used the fact that the objects being permuted are of degree two, hence they commute, so that no extra sign is acquired. Plugging this outcome in Equation (A.4), we obtain

$$\begin{aligned} \int d\Psi e^{-\sum_{\alpha,\beta \in A} \Psi_{\alpha}^{+} K_{\alpha\beta} \Psi_{\beta}^{-}} &= \frac{1}{|A|!} \sum_{\substack{\alpha_1, \dots, \alpha_{|A|} \\ \beta_1, \dots, \beta_{|A|}}} \int d\Psi \prod_{i=1}^{|A|} [\Psi_{\beta_i}^{-} K_{\alpha_i \beta_i} \Psi_{\alpha_i}^{+}] \\ &= \frac{1}{|A|!} \sum_{\beta_1, \dots, \beta_{|A|}} \sum_{\sigma \in S_{|A|}} \text{sgn}(\sigma) \int d\Psi \prod_{\beta \in A} \Psi_{\beta}^{-} K_{\sigma(\beta)\beta} \Psi_{\beta}^{+} \\ &= \frac{|A|!}{|A|!} \sum_{\sigma \in S_{|A|}} \text{sgn}(\sigma) \prod_{\beta \in A} K_{\sigma(\beta)\beta} = \det K \end{aligned}$$

where in the second passage we employed the fact that the number of possible choices of $\beta_1, \dots, \beta_{|A|}$ is $|A|!$, and in the last passage we recognised the usual formula for the determinant. $\square$

Proposition A.5. *The following points hold true.*

1. *The measure $P_{K^{-1}}$ has vanishing moments of odd order, that is, for all $n \in \mathbb{N}$,*

$$\mathbb{E}_K \left[\prod_{i=1}^{2n+1} \Psi_{\gamma_i}^{\varepsilon_i} \right] = 0 ,$$

for all choices of $(\gamma_i, \varepsilon_i) \in A \times \{\pm\}$. In particular, the measure is centered: taking $n = 0$,

$$\mathbb{E}_K[\Psi_{\gamma}^{\varepsilon}] = 0 .$$

2. The non-zero second moments are given by

$$\mathbb{E}_K[\Psi_\alpha^- \Psi_\beta^+] = (K^{-1})_{\alpha\beta}$$

for a given pair $\alpha, \beta \in A$;

3. All cumulants of order greater than 2 are zero; this is equivalent to Wick's rule for the moments of the measure $P_{K^{-1}}$: given $n \in \mathbb{N}$,

$$\mathbb{E}_K \left[\prod_{i=1}^n \Psi_{\alpha_i}^- \Psi_{\beta_i}^+ \right] = \sum_{\pi \in S_n} \text{sgn}(\pi) \prod_{i=1}^n (K^{-1})_{\alpha_i \beta_{\pi(i)}} ,$$

for any choice of indices $\{\alpha_i\}_{i=1}^n, \{\beta_i\}_{i=1}^n$.

Proof. The first point is straightforward to show: since all of the monomials contributing to

$$e^{-\sum_{\alpha, \beta \in A} \Psi_\alpha^+ K_{\alpha\beta} \Psi_\beta^-} \prod_{i=1}^{2n+1} \Psi_{\gamma_i}^{\varepsilon_i}$$

have odd degree, its Berezin integral is zero.

We turn to the remaining claims. Using the short-hand $(F, G) := \sum_\alpha F_\alpha G_\alpha$, we introduce the cumulant generating function in $\text{Gr}(A, \Xi)$

$$\mathcal{W}[\Xi] := \log \int P_{K^{-1}}[d\Psi] e^{(\Psi^+, \Xi^-) + (\Xi^+, \Psi^-)}$$

and, using Definition A.3 and Lemma A.4, by completing the square we have

$$\mathcal{W}[\Xi] = \log \left[\frac{e^{(\Xi^+, K^{-1}\Xi^-)}}{\det K} \int d\Psi e^{-(\Psi^+ - \Xi^+(K^{-1})^*, K(\Psi^- - K^{-1}\Xi^-))} \right] = (\Xi^+, K^{-1}\Xi^-) ,$$

where in the last passage we used (Grassmann) translation invariance of the Berezin integral.

Cumulants, also known as connected expectations, are given by derivatives of the generating functional $\mathcal{W}$, namely

$$\mathbb{E}_K[\Psi_{\alpha_1}^-; \dots; \Psi_{\alpha_n}^-; \Psi_{\beta_1}^+; \dots; \Psi_{\beta_n}^+] := \frac{(-1)^n \partial^n \mathcal{W}[\Xi]}{\partial \Xi_{\alpha_1}^+ \dots \partial \Xi_{\alpha_n}^+ \partial \Xi_{\beta_1}^- \dots \partial \Xi_{\beta_n}^-} \Big|_{\Xi=0} . \quad (\text{A.5})$$

Since $\mathcal{W}$ is quadratic in Ξ , it follows immediately that only cumulants of order 2 are non-zero, and point two follows from $P_{K^{-1}}$ being centered:

$$\mathbb{E}_K[\Psi_\alpha^- \Psi_\beta^+] = \mathbb{E}_K[\Psi_\alpha^-; \Psi_\beta^+] = \frac{\partial^2 \mathcal{W}[\Xi]}{\partial \Xi_\beta^- \partial \Xi_\alpha^+} \Big|_{\Xi=0} = (K^{-1})_{\alpha\beta} . \quad (\text{A.6})$$

Finally, the second part of the last point, namely Wick's rule for Grassmann variables, follows from writing a given moment in terms of cumulants of order smaller or equal than its degree: as one may check in various ways, *e.g.* by induction on n , we have

$$\mathbb{E}_K[\Psi_{\alpha_1}^- \Psi_{\beta_1}^+ \dots \Psi_{\alpha_n}^- \Psi_{\beta_n}^+] = \sum_P \text{sgn}(P) \prod_{J \in P} \mathbb{E}_K[\Psi_{j_1}^{\varepsilon_{j_1}}; \dots; \Psi_{j_{|J|}}^{\varepsilon_{j_{|J|}}}] ,$$

where P is a partition of the set $\{\alpha_1, \beta_1, \dots, \alpha_n, \beta_n\}$ into ordered subsets, J is an element of the partition P , $J = \{j_1, \dots, j_{|J|}\}$, and $\text{sgn}(P)$ is the sign of the permutation $\sigma \in S_{2n}$ needed to bring the list $(\alpha_1, \beta_1, \dots, \alpha_n, \beta_n)$ to P .

In particular, since only cumulants of order 2 are non-zero, this rewriting reduces to a sum over all pairings of the indices $\{\alpha_i\}_i$ with the indices $\{\beta_i\}_i$, with the $\text{sgn}(P)$ reducing to the sign of the permutation $\pi \in S_n$ given by the pairings:

$$\mathbb{E}_K[\Psi_{\alpha_1}^- \Psi_{\beta_1}^+ \dots \Psi_{\alpha_n}^- \Psi_{\beta_n}^+] = \sum_{\pi \in S_n} \text{sgn}(\pi) \prod_{i=1}^n \langle \Psi_{\alpha_i}^- \Psi_{\beta_{\pi(i)}}^+ \rangle .$$

It is then enough to evaluate the expression above using (A.6). This concludes the proof of the proposition. $\square$

Appendix B

Proof of Proposition 4.4.6

Recall Eq. (4.4.42),

$$0 = \frac{1}{Z_N(A)} \int P_N[d\psi] e^{-V(\psi) + \tilde{C}(\psi; A)} \left[Z_\omega \mathcal{D}_{\omega, \mathbf{a}}(\underline{p}) \hat{n}_{\underline{p}, \omega} - \Delta_{\underline{p}, \omega}(\psi) \right], \quad (\text{B.1})$$

where the source term is:

$$\tilde{C}(\psi; A) := \frac{1}{\beta L} \sum_{\omega=1}^{N_f} \sum_{\underline{p} \in \mathbb{D}_{\beta, L}^{\mathbf{p}}} \hat{A}_{-\underline{p}, \omega} \hat{n}_{\underline{p}, \omega}.$$

Differentiating with respect to $\hat{A}_{-\underline{p}_i, \omega_i}$, $i = 2, \dots, m$ and setting $A = 0$ we get:

$$Z_{\omega_1} \mathcal{D}_{\omega, \mathbf{a}}(\underline{p}_1) \langle \hat{n}_{\underline{p}_1, \omega_1}; \dots; \hat{n}_{\underline{p}_m, \omega_m} \rangle_N = \langle \Delta_{\underline{p}_1, \omega_1}(\psi); \hat{n}_{\underline{p}_2, \omega_2}; \dots; \hat{n}_{\underline{p}_m, \omega_m} \rangle_N. \quad (\text{B.2})$$

The next lemma provides a useful rewriting of the right-hand side of (B.2).

Lemma B.1 (Schwinger-Dyson equation for density correlation functions). *Let $m \geq 3$. For any collection of momenta $\{\underline{p}_i \neq \underline{0}\}_{i=1}^m$ such that $p_{i,0} = p_{1,0} \neq 0$ for $i = 1, \dots, m-1$, $\underline{p}_m$ fixed by momentum conservation, and for any choice of chiralities $\{\omega_i\}_{i=1}^m$, we have:*

$$\begin{aligned} & \langle \Delta_{\underline{p}_1, \omega_1}(\psi); \hat{n}_{\underline{p}_2, \omega_2}; \dots; \hat{n}_{\underline{p}_m, \omega_m} \rangle_N \\ &= - \mathcal{D}_{\omega_1, \mathbf{a}}(\underline{p}_1) \mathfrak{B}_{\omega_1}^N(\underline{p}_1) \hat{v}(\underline{p}_1) \sum_{\bar{\omega}} \lambda_{\omega_1 \bar{\omega}} Z_{\bar{\omega}} \langle \hat{n}_{\underline{p}_1, \bar{\omega}}; \hat{n}_{\underline{p}_2, \omega_2}; \dots; \hat{n}_{\underline{p}_m, \omega_m} \rangle_N \\ &+ \frac{1}{\beta L} \sum_{\underline{k} \in \mathbb{D}_{\beta, L, \mathbf{a}}^{\mathbf{a}}} \Delta_{\omega_1}(\underline{k}, \underline{p}_1) \hat{g}_{\omega_1}^{(\leq N)}(\underline{k} - \underline{p}_1) \hat{g}_{\omega_1}^{(\leq N)}(\underline{k}) \mathcal{R}_{\omega}^N(\underline{k}; \mathbf{p}_m), \end{aligned} \quad (\text{B.3})$$

where:

$$\begin{aligned} \mathcal{R}_{\omega}^N(\underline{k}; \mathbf{p}_m) &:= \sum_{\bar{\omega}, \bar{\omega}'} \lambda_{\omega_1 \bar{\omega}} \lambda_{\omega_1 \bar{\omega}'} Z_{\bar{\omega}} Z_{\bar{\omega}'} \frac{Z_{\omega_1}^2}{(\beta L)^2} \sum_{\underline{q}, \underline{q}'} \hat{v}_s(\underline{q}) \hat{v}_s(\underline{q}') \mathcal{F}_{\omega}^{2, m+1}(\underline{k}, \underline{q}, \underline{q}'; \mathbf{p}_m) \\ &- \sum_{i=2}^m \delta_{\omega_1, \omega_i} \sum_{\bar{\omega}} \lambda_{\omega_1 \bar{\omega}} Z_{\bar{\omega}} \frac{Z_{\omega_1}^2}{\beta L} \sum_{\underline{q}} \hat{v}_s(\underline{q}) \mathcal{F}_{\omega; i}^{2, m-1}(\underline{k}, \underline{q}; \mathbf{p}_m) \\ &+ \sum_{\substack{i, j=2 \\ i \neq j}}^m \delta_{\omega_1, \omega_i} \delta_{\omega_1, \omega_j} Z_{\omega_1}^2 \mathcal{F}_{\omega; i, j}^{2, m-3}(\underline{k}; \mathbf{p}_m), \end{aligned} \quad (\text{B.4})$$

with $\mathbf{p}_m = (p_1, \dots, p_m)$ and $\hat{v}_s(q) := [\hat{v}(q) + \hat{v}(-q)]/2$ the even part of the interaction potential. The functions $\mathcal{F}$ are given by:

$$\begin{aligned} \mathcal{F}_{\omega}^{2,m+1}(\underline{k}, \underline{q}, \underline{q}'; \mathbf{p}_m) &:= \langle \hat{\psi}_{\underline{k}-\underline{p}_1+\underline{q}, \omega_1}^+ \hat{\psi}_{\underline{k}-\underline{q}', \omega_1}^- \hat{n}_{\underline{q}, \bar{\omega}} \hat{n}_{\underline{q}', \bar{\omega}'}; \hat{n}_{\underline{p}_2, \omega_2}; \dots; \hat{n}_{\underline{p}_m, \omega_m} \rangle_N, \\ \mathcal{F}_{\omega; i}^{2,m-1}(\underline{k}, \underline{q}; \mathbf{p}_m) &:= \langle \hat{\psi}_{\underline{k}-\underline{p}_1+\underline{q}, \omega_1}^+ \hat{\psi}_{\underline{k}+\underline{p}_i, \omega_1}^- \hat{n}_{\underline{q}, \bar{\omega}}; \hat{n}_{\underline{p}_2, \omega_2}; \dots; \widehat{\hat{n}_{\underline{p}_i, \omega_i}}; \dots; \hat{n}_{\underline{p}_m, \omega_m} \rangle_N \\ &\quad + \langle \hat{\psi}_{\underline{k}+\underline{q}, \omega_1}^- \hat{\psi}_{\underline{k}-\underline{p}_1+\underline{p}_i, \omega_1}^+ \hat{n}_{\underline{q}, \bar{\omega}}; \hat{n}_{\underline{p}_2, \omega_2}; \dots; \widehat{\hat{n}_{\underline{p}_i, \omega_i}}; \dots; \hat{n}_{\underline{p}_m, \omega_m} \rangle_N, \\ \mathcal{F}_{\omega; i, j}^{2,m-3}(\underline{k}; \mathbf{p}_m) &:= \langle \hat{\psi}_{\underline{k}-\underline{p}_1-\underline{p}_i, \omega_1}^+ \hat{\psi}_{\underline{k}+\underline{p}_j, \omega_1}^-; \hat{n}_{\underline{p}_2, \omega_2}; \dots; \widehat{\hat{n}_{\underline{p}_i, \omega_i}}; \dots; \widehat{\hat{n}_{\underline{p}_j, \omega_j}}; \dots; \hat{n}_{\underline{p}_m, \omega_m} \rangle_N, \end{aligned} \quad (\text{B.5})$$

with the wide hat indicating omission of a density.

Proof. We can write $\langle \Delta_{\underline{p}_1, \omega_1}(\psi); \hat{n}_{\underline{p}_2, \omega_2}; \dots; \hat{n}_{\underline{p}_m, \omega_m} \rangle_N$ as follows:

$$\begin{aligned} &\langle \Delta_{\underline{p}_1, \omega_1}(\psi); \hat{n}_{\underline{p}_2, \omega_2}; \dots; \hat{n}_{\underline{p}_m, \omega_m} \rangle_N \\ &= \frac{(\beta L)^m \partial^m}{\partial \hat{B}_{-\underline{p}_1, \omega_1} \partial \hat{A}_{-\underline{p}_2, \omega_2} \dots \partial \hat{A}_{-\underline{p}_m, \omega_m}} \frac{1}{\mathcal{Z}_N(A)} \int P_N[d\psi] e^{-V(\psi) + \tilde{C}(\psi; A) + \Delta(\psi; B)} \Big|_{A, B=0} \\ &= \frac{1}{\beta L} \sum_{\underline{k} \in \mathbb{D}_{\beta, L, \alpha}^a} \Delta_{\omega_1}(\underline{k}, \underline{p}_1) \frac{(\beta L)^{m-1} \partial^{m-1}}{\prod_{i=2}^m \partial \hat{A}_{-\underline{p}_i, \omega_i}} \frac{1}{\mathcal{Z}_N(A)} \int P_N[d\psi] e^{-V(\psi) + \tilde{C}(\psi; A)} \hat{\psi}_{\underline{k}-\underline{p}_1, \omega_1}^+ \hat{\psi}_{\underline{k}, \omega_1}^- \Big|_{A=0} \end{aligned} \quad (\text{B.6})$$

where we used (4.4.41) and where

$$\Delta(\psi; B) := \frac{1}{\beta L} \sum_{\omega=1}^N \sum_{\underline{p} \in \mathbb{D}_{\beta, L}^p} \hat{B}_{-\underline{p}, \omega} \Delta_{\underline{p}, \omega}(\psi), \quad (\text{B.7})$$

with $\hat{B}_{\underline{p}, \omega}$ external fields defined as $\hat{A}_{\underline{p}, \omega}$ (see (4.4.23)).

From the definition of the Grassmann measure given by (4.4.43), we have:

$$\frac{1}{(\beta L)^2} e^{-K_N(\psi)} \hat{\psi}_{\underline{k}-\underline{p}_1, \omega_1}^+ \hat{\psi}_{\underline{k}, \omega_1}^- = \hat{g}_{\omega_1}^{(\leq N)}(\underline{k} - \underline{p}_1) \hat{g}_{\omega_1}^{(\leq N)}(\underline{k}) \frac{\partial^2}{\partial \hat{\psi}_{\underline{k}, \omega_1}^+ \partial \hat{\psi}_{\underline{k}-\underline{p}_1, \omega_1}^-} e^{-K_N(\psi)} \quad (\text{B.8})$$

so that, integrating by parts the argument of the derivatives in (B.6), we obtain:

$$\begin{aligned} &\frac{1}{\mathcal{Z}_N(A)} \int P_N[d\psi] e^{-V(\psi) + \tilde{C}(\psi; A)} \hat{\psi}_{\underline{k}-\underline{p}_1, \omega_1}^+ \hat{\psi}_{\underline{k}, \omega_1}^- \\ &= \hat{g}_{\omega_1}^{(\leq N)}(\underline{k} - \underline{p}_1) \hat{g}_{\omega_1}^{(\leq N)}(\underline{k}) \frac{(\beta L)^2}{\mathcal{Z}_N(A)} \int P_N[d\psi] \frac{\partial^2}{\partial \hat{\psi}_{\underline{k}-\underline{p}_1, \omega_1}^- \partial \hat{\psi}_{\underline{k}, \omega_1}^+} e^{-V(\psi) + \tilde{C}(\psi; A)} \\ &= \hat{g}_{\omega_1}^{(\leq N)}(\underline{k} - \underline{p}_1) \hat{g}_{\omega_1}^{(\leq N)}(\underline{k}) \frac{(\beta L)^2}{\mathcal{Z}_N(A)} \int P_N[d\psi] e^{-V(\psi) + \tilde{C}(\psi; A)} \times \\ &\quad \times \left[\frac{\partial^2 [V(\psi) - \tilde{C}(\psi; A)]}{\partial \hat{\psi}_{\underline{k}-\underline{p}_1, \omega_1}^- \partial \hat{\psi}_{\underline{k}, \omega_1}^+} + \frac{\partial [V(\psi) - \tilde{C}(\psi; A)]}{\partial \hat{\psi}_{\underline{k}-\underline{p}_1, \omega_1}^-} \frac{\partial [V(\psi) - \tilde{C}(\psi; A)]}{\partial \hat{\psi}_{\underline{k}, \omega_1}^+} \right]. \end{aligned} \quad (\text{B.9})$$

Let us consider the term in (B.9) with the second derivative of $V(\psi) - \tilde{C}(\psi; A)$. From (4.4.20), (4.4.21), (4.4.22), and employing the fact that $\lambda_{\omega\omega} = 0$, we find:

$$\frac{\partial^2 [V(\psi) - \tilde{C}(\psi; A)]}{\partial \hat{\psi}_{\underline{k}-\underline{p}_1, \omega_1}^- \partial \hat{\psi}_{\underline{k}, \omega_1}^+} = \frac{Z_{\omega_1}}{(\beta L)^2} \left[\sum_{\bar{\omega}} \lambda_{\omega_1 \bar{\omega}} Z_{\bar{\omega}} \hat{v}(\underline{p}_1) \hat{n}_{\underline{p}_1, \bar{\omega}} - \hat{A}_{\underline{p}_1, \omega_1} \right]. \quad (\text{B.10})$$

Using this identity in (B.9) and differentiating with respect to $\{\hat{A}_{-p_i, \omega_i}\}_{i=2}^m$, the contribution of the first term in the right-hand side of (B.9) to (B.6) is:

$$Z_{\omega_1} \hat{g}_{\omega_1}^{(\leq N)}(\underline{k} - \underline{p}_1) \hat{g}_{\omega_1}^{(\leq N)}(\underline{k}) \hat{v}(\underline{p}_1) \sum_{\bar{\omega}} \lambda_{\omega_1 \bar{\omega}} Z_{\bar{\omega}} \langle \hat{n}_{\underline{p}_1, \bar{\omega}}; \hat{n}_{\underline{p}_2, \omega_2}; \dots; \hat{n}_{\underline{p}_m, \omega_m} \rangle_N. \quad (\text{B.11})$$

Observe that, since the momenta $\underline{p}_i$ are all nonzero, by momentum conservation we see that the term corresponding to $\hat{A}_{\underline{p}_1, \omega_1}$ in (B.10) does not contribute to the result. The identity (B.3) follows after inserting (B.11) in (B.6), recalling the definition (4.4.46) for the anomalous bubble diagram $\mathcal{B}_{\omega_1}^N$, and setting:

$$\begin{aligned} \mathcal{R}_{\omega}^N(\underline{k}; \underline{\mathbf{p}}_m) &:= \frac{(\beta L)^{m-1} \partial^{m-1}}{\prod_{i=2}^m \partial \hat{A}_{-p_i, \omega_i}} \frac{(\beta L)^2}{Z_N(A)} \int P_N[d\psi] e^{-V(\psi) + \tilde{C}(\psi; A)} \times \\ &\quad \times \frac{\partial[V(\psi) - \tilde{C}(\psi; A)]}{\partial \hat{\psi}_{\underline{k} - \underline{p}_1, \omega_1}^-} \frac{\partial[V(\psi) - \tilde{C}(\psi; A)]}{\partial \hat{\psi}_{\underline{k}, \omega_1}^+} \Big|_{A=0}. \end{aligned} \quad (\text{B.12})$$

The remaining part of the proof is the explicit evaluation of (B.12). We proceed as follows: first, an explicit computation yields

$$\begin{aligned} &\frac{\partial[V(\psi) - \tilde{C}(\psi; A)]}{\partial \hat{\psi}_{\underline{k} - \underline{p}_1, \omega_1}^-} \\ &= \frac{Z_{\omega_1}}{(\beta L)^2} \sum_{\underline{q} \in \mathbb{D}_{\beta, L}^{\mathbf{p}}} \left[\sum_{\bar{\omega}} \lambda_{\omega_1 \bar{\omega}} Z_{\bar{\omega}} \frac{\hat{v}(\underline{q})}{2} \left(\hat{\psi}_{\underline{k} - \underline{p}_1 - \underline{q}, \omega_1}^+ \hat{n}_{-\underline{q}, \bar{\omega}} + \hat{n}_{\underline{q}, \bar{\omega}} \hat{\psi}_{\underline{k} - \underline{p}_1 + \underline{q}, \omega_1}^+ \right) - \hat{A}_{-\underline{q}, \omega_1} \hat{\psi}_{\underline{k} - \underline{p}_1 - \underline{q}, \omega_1}^+ \right] \\ &= \frac{Z_{\omega_1}}{(\beta L)^2} \sum_{\underline{q} \in \mathbb{D}_{\beta, L}^{\mathbf{p}}} \left[\sum_{\bar{\omega}} \lambda_{\omega_1 \bar{\omega}} Z_{\bar{\omega}} \hat{v}_s(\underline{q}) \hat{\psi}_{\underline{k} - \underline{p}_1 + \underline{q}, \omega_1}^+ \hat{n}_{\underline{q}, \bar{\omega}} - \hat{A}_{-\underline{q}, \omega_1} \hat{\psi}_{\underline{k} - \underline{p}_1 - \underline{q}, \omega_1}^+ \right], \end{aligned} \quad (\text{B.13})$$

where $\hat{v}_s(\underline{q}) := [\hat{v}(\underline{q}) + \hat{v}(-\underline{q})]/2$, and

$$\begin{aligned} &\frac{\partial[V(\psi) - \tilde{C}(\psi; A)]}{\partial \hat{\psi}_{\underline{k}, \omega_1}^+} \\ &= \frac{Z_{\omega_1}}{(\beta L)^2} \sum_{\underline{q} \in \mathbb{D}_{\beta, L}^{\mathbf{p}}} \left[\sum_{\bar{\omega}} \lambda_{\omega_1 \bar{\omega}} Z_{\bar{\omega}} \frac{\hat{v}(\underline{q})}{2} \left(\hat{\psi}_{\underline{k} + \underline{q}, \omega_1}^- \hat{n}_{-\underline{q}, \bar{\omega}} + \hat{n}_{\underline{q}, \bar{\omega}} \hat{\psi}_{\underline{k} - \underline{q}, \omega_1}^- \right) - \hat{A}_{-\underline{q}, \omega_1} \hat{\psi}_{\underline{k} + \underline{q}, \omega_1}^- \right] \\ &= \frac{Z_{\omega_1}}{(\beta L)^2} \sum_{\underline{q} \in \mathbb{D}_{\beta, L}^{\mathbf{p}}} \left[\sum_{\bar{\omega}} \lambda_{\omega_1 \bar{\omega}} Z_{\bar{\omega}} \hat{v}_s(\underline{q}) \hat{\psi}_{\underline{k} - \underline{q}, \omega_1}^- \hat{n}_{\underline{q}, \bar{\omega}} - \hat{A}_{-\underline{q}, \omega_1} \hat{\psi}_{\underline{k} + \underline{q}, \omega_1}^- \right]. \end{aligned} \quad (\text{B.14})$$

Second, multiplying (B.13) with (B.14), inserting the result in (B.12), deriving with respect to the external fields $\{\hat{A}_{-p_i, \omega_i}\}_{i=2}^m$ and setting $A = 0$, we are led to a host of different terms contributing to $\mathcal{R}_{\omega}^N$:

1. the first term of (B.13) with the first one of (B.14) gives

$$\sum_{\bar{\omega}, \bar{\omega}'} \lambda_{\omega_1 \bar{\omega}} \lambda_{\omega_1 \bar{\omega}'} Z_{\bar{\omega}} Z_{\bar{\omega}'} \frac{Z_{\omega_1}^2}{(\beta L)^2} \sum_{\underline{q}, \underline{q}'} \hat{v}_s(\underline{q}) \hat{v}_s(\underline{q}') \mathcal{F}_{\omega}^{2, m+1}(\underline{k}, \underline{q}, \underline{q}'; \underline{\mathbf{p}}_m) \quad (\text{B.15})$$

with

$$\mathcal{F}_{\underline{\omega}}^{2,m+1}(\underline{k}, \underline{q}, \underline{q}'; \underline{\mathbf{p}}_m) := \langle \hat{\psi}_{\underline{k}-\underline{p}_1+\underline{q}, \omega_1}^+ \hat{\psi}_{\underline{k}-\underline{q}', \omega_1}^- \hat{n}_{\underline{q}, \bar{\omega}} \hat{n}_{\underline{q}', \bar{\omega}'}; \hat{n}_{\underline{p}_2, \omega_2}; \dots; \hat{n}_{\underline{p}_m, \omega_m} \rangle_N;$$

2. the first (resp. second) term of (B.13) with the second (resp. first) one of (B.14) produces

$$-\sum_{i=2}^m \delta_{\omega_1, \omega_i} \sum_{\bar{\omega}} \lambda_{\omega_1 \bar{\omega}} Z_{\bar{\omega}} \frac{Z_{\omega_1}^2}{\beta L} \sum_{\underline{q}} \hat{v}_s(\underline{q}) \mathcal{F}_{\underline{\omega}; i}^{2,m-1}(\underline{k}, \underline{q}; \underline{\mathbf{p}}_m) \quad (\text{B.16})$$

with

$$\begin{aligned} \mathcal{F}_{\underline{\omega}; i}^{2,m-1}(\underline{k}, \underline{q}; \underline{\mathbf{p}}_m) &:= \langle \hat{\psi}_{\underline{k}-\underline{p}_1+\underline{q}, \omega_1}^+ \hat{\psi}_{\underline{k}+\underline{p}_i, \omega_1}^- \hat{n}_{\underline{q}, \bar{\omega}}; \hat{n}_{\underline{p}_2, \omega_2}; \dots; \widehat{\hat{n}_{\underline{p}_i, \omega_i}}; \dots; \hat{n}_{\underline{p}_m, \omega_m} \rangle_N \\ &\quad + \langle \hat{\psi}_{\underline{k}+\underline{q}, \omega_1}^- \hat{\psi}_{\underline{k}-\underline{p}_1+\underline{p}_i, \omega_1}^+ \hat{n}_{\underline{q}, \bar{\omega}}; \hat{n}_{\underline{p}_2, \omega_2}; \dots; \widehat{\hat{n}_{\underline{p}_i, \omega_i}}; \dots; \hat{n}_{\underline{p}_m, \omega_m} \rangle_N \end{aligned}$$

where the wide hat indicates omission of the corresponding density;

3. finally, the second term of (B.13) with the second one of (B.14) yields

$$Z_{\omega_1}^2 \sum_{\substack{i,j=2 \\ i \neq j}}^m \delta_{\omega_1, \omega_i} \delta_{\omega_1, \omega_j} \mathcal{F}_{\underline{\omega}; i, j}^{2,m-3}(\underline{k}; \underline{\mathbf{p}}_m) \quad (\text{B.17})$$

with

$$\mathcal{F}_{\underline{\omega}; i, j}^{2,m-3}(\underline{k}; \underline{\mathbf{p}}_m) := \langle \hat{\psi}_{\underline{k}-\underline{p}_1-\underline{p}_i, \omega_1}^+ \hat{\psi}_{\underline{k}+\underline{p}_j, \omega_1}^-; \hat{n}_{\underline{p}_2, \omega_2}; \dots; \widehat{\hat{n}_{\underline{p}_i, \omega_i}}; \dots; \widehat{\hat{n}_{\underline{p}_j, \omega_j}}; \dots; \hat{n}_{\underline{p}_m, \omega_m} \rangle_N.$$

This concludes the proof of Lemma B.1. $\square$

The next lemma establishes the bound for the remainder $\mathcal{R}_{\underline{\omega}}^N$.

Lemma B.2. *There exist $\gamma > 0$, $C > 0$ such that for all $\underline{\omega}$, and for $\underline{p}_i$ as in Lemma B.1, with the supplementary requirement that $|p_i| \leq C|\theta|$ and $p_{i,0} = \eta$ for all $i \leq m-1$, we have:*

$$\frac{1}{\beta L} \sup_{c_1 2^N \leq \|\underline{k}\|_{\omega_1} \leq c_2 2^N} |\mathcal{R}_{\underline{\omega}}^N(\underline{k}; \underline{\mathbf{p}}_m)| \leq C^m m! \beta^{m+2} 2^{-\gamma N}. \quad (\text{B.18})$$

Note that, from Lemma B.2 and from definition of $\mathfrak{R}_{\underline{\omega}}^N$ (see (B.4)), the bound (4.4.47) immediately follows.

Proof. In order to use the GN tree estimates introduced earlier, we rewrite all of the $\mathcal{F}$ in terms of *connected* correlation functions, keeping in mind the following three rules:

1. momentum conservation must hold in each connected correlation function separately;
2. $\langle \hat{n}_{\underline{q}} \rangle_N = 0$ for all $\underline{q} \in \mathbb{D}_{\beta, L}^p$;
3. $\langle \hat{\psi}^\pm; \hat{n}; \dots; \hat{n} \rangle_N = 0$ by the fact that expectations of odd fermionic monomials vanish.

Thus, we obtain

$$\begin{aligned}
\mathcal{F}_{\omega}^{2,m+1}(\underline{k}, \underline{q}, \underline{q}'; \underline{\mathbf{p}}_m) = & \langle \hat{\psi}_{\underline{k}-\underline{p}_1+\underline{q}, \omega_1}^+; \hat{\psi}_{\underline{k}-\underline{q}', \omega_1}^-; \hat{n}_{\underline{q}, \bar{\omega}}; \hat{n}_{\underline{q}', \bar{\omega}'}; \hat{n}_{\underline{p}_2, \omega_2}; \cdots; \hat{n}_{\underline{p}_m, \omega_m} \rangle_N \\
& + \delta_{\underline{q}, \underline{p}_1-\underline{q}'} \langle \hat{\psi}_{\underline{k}-\underline{q}', \omega_1}^+; \hat{\psi}_{\underline{k}-\underline{q}', \omega_1}^- \rangle_N \langle \hat{n}_{\underline{p}_1-\underline{q}', \bar{\omega}}; \hat{n}_{\underline{q}', \bar{\omega}'}; \hat{n}_{\underline{p}_2, \omega_2}; \cdots; \hat{n}_{\underline{p}_m, \omega_m} \rangle_N \\
& + \delta_{\underline{q}', \underline{p}_1} \langle \hat{\psi}_{\underline{k}-\underline{p}_1+\underline{q}, \omega_1}^+; \hat{\psi}_{\underline{k}-\underline{p}_1, \omega_1}^-; \hat{n}_{\underline{q}, \bar{\omega}} \rangle_N \langle \hat{n}_{\underline{p}_1, \bar{\omega}'}; \hat{n}_{\underline{p}_2, \omega_2}; \cdots; \hat{n}_{\underline{p}_m, \omega_m} \rangle_N \\
& + \delta_{\underline{q}, \underline{p}_1} \langle \hat{\psi}_{\underline{k}, \omega_1}^+; \hat{\psi}_{\underline{k}-\underline{q}', \omega_1}^-; \hat{n}_{\underline{q}', \bar{\omega}'} \rangle_N \langle \hat{n}_{\underline{p}_1, \bar{\omega}}; \hat{n}_{\underline{p}_2, \omega_2}; \cdots; \hat{n}_{\underline{p}_m, \omega_m} \rangle_N \\
& + \delta_{\underline{q}, -\underline{q}'} \langle \hat{n}_{\underline{q}, \bar{\omega}}; \hat{n}_{-\underline{q}, \bar{\omega}'} \rangle_N \langle \hat{\psi}_{\underline{k}-\underline{p}_1+\underline{q}, \omega_1}^+; \hat{\psi}_{\underline{k}+\underline{q}, \omega_1}^-; \hat{n}_{\underline{p}_2, \omega_2}; \cdots; \hat{n}_{\underline{p}_m, \omega_m} \rangle_N
\end{aligned} \tag{B.19}$$

and

$$\begin{aligned}
\mathcal{F}_{\omega; i}^{2,m-1}(\underline{k}, \underline{q}; \underline{\mathbf{p}}_m) = & \langle \hat{\psi}_{\underline{k}-\underline{p}_1+\underline{q}, \omega_1}^+; \hat{\psi}_{\underline{k}+\underline{p}_i, \omega_1}^-; \hat{n}_{\underline{q}, \bar{\omega}}; \hat{n}_{\underline{p}_2, \omega_2}; \cdots; \widehat{\hat{n}_{\underline{p}_i, \omega_i}}; \cdots; \hat{n}_{\underline{p}_m, \omega_m} \rangle_N \\
& + \delta_{\underline{q}, \underline{p}_1+\underline{p}_i} \langle \hat{\psi}_{\underline{k}+\underline{p}_i, \omega_1}^+; \hat{\psi}_{\underline{k}+\underline{p}_i, \omega_1}^- \rangle_N \langle \hat{n}_{\underline{p}_1+\underline{p}_i, \bar{\omega}}; \hat{n}_{\underline{p}_2, \omega_2}; \cdots; \widehat{\hat{n}_{\underline{p}_i, \omega_i}}; \cdots; \hat{n}_{\underline{p}_m, \omega_m} \rangle_N.
\end{aligned} \tag{B.20}$$

Instead, we see that the terms $\mathcal{F}_{\omega; i, j}^{2,m-3}$ are already connected, namely

$$\mathcal{F}_{\omega; i, j}^{2,m-3}(\underline{k}; \underline{\mathbf{p}}_m) = \langle \hat{\psi}_{\underline{k}-\underline{p}_1-\underline{p}_i, \omega_1}^+; \hat{\psi}_{\underline{k}+\underline{p}_j, \omega_1}^-; \hat{n}_{\underline{p}_2, \omega_2}; \cdots; \widehat{\hat{n}_{\underline{p}_i, \omega_i}}; \cdots; \widehat{\hat{n}_{\underline{p}_j, \omega_j}}; \cdots; \hat{n}_{\underline{p}_m, \omega_m} \rangle_N. \tag{B.21}$$

First, consider all of the correlations occurring in (B.19)-(B.20) with no $\hat{\psi}$ field: these involve $m+1$, m , $m-1$ or 2 densities. For the sake of concreteness, let us consider a token example of m -point correlation functions, namely $\langle \hat{n}_{\underline{p}_1-\underline{q}', \bar{\omega}}; \hat{n}_{\underline{q}', \bar{\omega}'}; \hat{n}_{\underline{p}_2, \omega_2}; \cdots; \hat{n}_{\underline{p}_m, \omega_m} \rangle_N$, whose GN-type estimate is

$$\frac{1}{\beta L} |\langle \hat{n}_{\underline{p}_1, \bar{\omega}}; \hat{n}_{\underline{p}_2, \omega_2}; \cdots; \hat{n}_{\underline{p}_m, \omega_m} \rangle_N| \leq \sum_{h=h_\beta}^N \sum_{n=0}^{\infty} \sum_{\tau, \Gamma} \lambda_{\max}^n 2^{h(2-m)} \prod_{v \notin V_f(\tau)} 2^{(h_v - h_{v'}) D_v}; \tag{B.22}$$

summing over $\Gamma \in \mathcal{L}_\tau$, $\tau \in \mathcal{T}_{h, n}^{m, \text{ref}}$, and $n \in \mathbb{N}$, for $\lambda_{\max}$ small enough and for $m \geq 3$, we obtain

$$\frac{1}{\beta L} |\langle \hat{n}_{\underline{p}_1, \bar{\omega}}; \hat{n}_{\underline{p}_2, \omega_2}; \cdots; \hat{n}_{\underline{p}_m, \omega_m} \rangle_N| \leq m! C^m \sum_{h=h_\beta}^N 2^{h(2-m)} \leq C_{\beta, m}, \tag{B.23}$$

with $C_{\beta, m} \propto m! \beta^{m-2}$. One may proceed in the same way for all the other correlators with m or $m+1$ densities. This bound is of course not optimal, since it does not take into account the dependence on the external momenta, but it will be enough for our purposes.

For the density correlator with $m-1$ points appearing in the second row of (B.20), performing the same estimate we obtain

$$\begin{aligned}
& \frac{1}{\beta L} |\langle \hat{n}_{\underline{p}_1+\underline{p}_i, \bar{\omega}}; \hat{n}_{\underline{p}_2, \omega_2}; \cdots; \widehat{\hat{n}_{\underline{p}_i, \omega_i}}; \cdots; \hat{n}_{\underline{p}_m, \omega_m} \rangle_N| \\
& \leq C^m m! \sum_{h=h_\beta}^N 2^{h(3-m)} \leq C^m m! \cdot \begin{cases} \log \beta + N & m = 3 \\ \beta^{m-3} & m > 3 \end{cases}
\end{aligned} \tag{B.24}$$

and analogously for the two-point density correlation function in the last row of (B.20). Therefore, all correlation functions in (B.19)-(B.20) involving only densities may be

bounded by $C^m m! (\beta^{m-1} + N)$. Again, this is of course not optimal, but it will be enough for our purposes.

Consider now the correlation functions appearing in (B.19)-(B.21) and having two $\hat{\psi}$ fields: we notice that, since $c_1 2^N \leq \|k\|_{\omega_1} \leq c_2 2^N$, for N large enough the momenta of these $\hat{\psi}$ fields will be also of order 2^N , since $\|q\|_{\omega_1}, \|q'\|_{\omega_1}, \|p_i\|_{\omega_1}$ are of order 1 in N : recall that the momenta q, q' are in the support of $\hat{v}$, that we are assuming $|p_i| \leq C|\theta|$, $p_{i,0} = \eta$ for all $i = 1, \dots, m-1$, and that, by momentum conservation, $|p_m| \leq C(m-1)(\eta + |\theta|)$. For the sake of concreteness let us consider the first row of (B.19). Its GN expansion is:

$$\langle \hat{\psi}_{\underline{k}-\underline{p}_1+\underline{q}, \omega_1}^+; \hat{\psi}_{\underline{k}-\underline{q}', \omega_1}^-; \hat{n}_{\underline{q}, \bar{\omega}}; \hat{n}_{\underline{q}', \bar{\omega}'}; \hat{n}_{\underline{p}_2, \omega_2}; \dots; \hat{n}_{\underline{p}_m, \omega_m} \rangle_N = \sum_{h=h_\beta}^N \sum_{n=0}^{\infty} \sum_{\tau, \Gamma} \widehat{W}_\Gamma^{(h)}[\tau]. \quad (\text{B.25})$$

Since ϕ -leaves attach only on the (couple of) scales that contain the momentum by which they are labelled, by the discussion above we see that the two ϕ -leaves will necessarily be on scale N (or $N-1$). Therefore, we may extract a short-memory factor at the root, namely

$$\frac{1}{\beta L} \sum_{n \geq 0} \sum_{\tau \in \mathcal{T}_{h,n}^{m+4, \text{ref}}} \sum_{\Gamma \in \mathcal{L}_\tau} |\widehat{W}_\Gamma^{(h)}[\tau]| \leq m! C^m 2^{h(-m-2)} 2^{\tilde{\gamma}(h-N)}; \quad (\text{B.26})$$

summation over the root scale then yields

$$\frac{1}{\beta L} |\langle \hat{\psi}_{\underline{k}-\underline{p}_1+\underline{q}, \omega_1}^+; \hat{\psi}_{\underline{k}-\underline{q}', \omega_1}^-; \hat{n}_{\underline{q}, \bar{\omega}}; \hat{n}_{\underline{q}', \bar{\omega}'}; \hat{n}_{\underline{p}_2, \omega_2}; \dots; \hat{n}_{\underline{p}_m, \omega_m} \rangle_N| \leq C^m m! \beta^{m+2} 2^{-\tilde{\gamma}N}. \quad (\text{B.27})$$

It is enough to proceed in the same way for all other correlators having two $\hat{\psi}$ fields. Combining all of the estimates that we have obtained, we may bound the $\mathcal{F}$ terms as

$$\begin{aligned} |\mathcal{F}_{\underline{\omega}}^{2,m+1}(\underline{k}, \underline{q}, \underline{q}'; \underline{\mathbf{p}}_m)| &\leq (\beta L) C^m m! (\beta^{m+2} + N) 2^{-\tilde{\gamma}N} [1 + \beta L (\delta_{\underline{q}, \underline{p}_1} + \delta_{\underline{q}', \underline{p}_1} + \delta_{\underline{q}, \underline{p}_1} + \delta_{\underline{q}, -\underline{q}'})] \\ |\mathcal{F}_{\underline{\omega}; i}^{2,m-1}(\underline{k}, \underline{q}; \underline{\mathbf{p}}_m)| &\leq (\beta L) C^m m! (\beta^{m+2} + N) 2^{-\tilde{\gamma}N} [1 + \beta L \delta_{\underline{q}, \underline{p}_1 + \underline{p}_i}] \\ |\mathcal{F}_{\underline{\omega}; i, j}^{2,m-3}(\underline{k}, \underline{q}; \underline{\mathbf{p}}_m)| &\leq (\beta L) C^m m! (\beta^{m+2} + N) 2^{-\tilde{\gamma}N} \end{aligned}$$

Insertion of these bounds in equation (B.4) yields the claim (B.18), for some $\gamma < \tilde{\gamma}$. $\square$

Remark B.3. Actually, the bounds for the correlators of the reference model imply that:

$$|\mathcal{R}_{\underline{\omega}}^N(\underline{k}; \underline{\mathbf{p}}_m)| \leq \beta L C^m m! \beta^{m+2} \quad (\text{B.28})$$

for all $\underline{k}$, uniformly in $\mathbf{a}$, ε small (with $\mathbf{a}/\varepsilon$ small enough). This bound is used to take the limits in (4.4.49), (4.4.50).

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